Material Characterization Report

NanoVNA UTN Toolkit

2026-10-23 19:03:44

Measurement Details:

- **Sample:** Muestra3
- **Technique:** Open-ended coaxial probe (liquids)
- **Temperature:** 23.0 °C
- **Reference Liquids:** Distilled Water / Isopropyl Alcohol (IPA)
- **Probe:** 3 mm probe
- **Frequency Range:** 0.005–6.000 GHz, 1200 pts
- **Data sources:**
  - **Open:** imported
  - **Short:** imported (1 spike(s) replaced in the computation: 4885 MHz)
  - **Ref 1:** imported
  - **Ref 2:** imported
  - **DUT:** imported

Contents

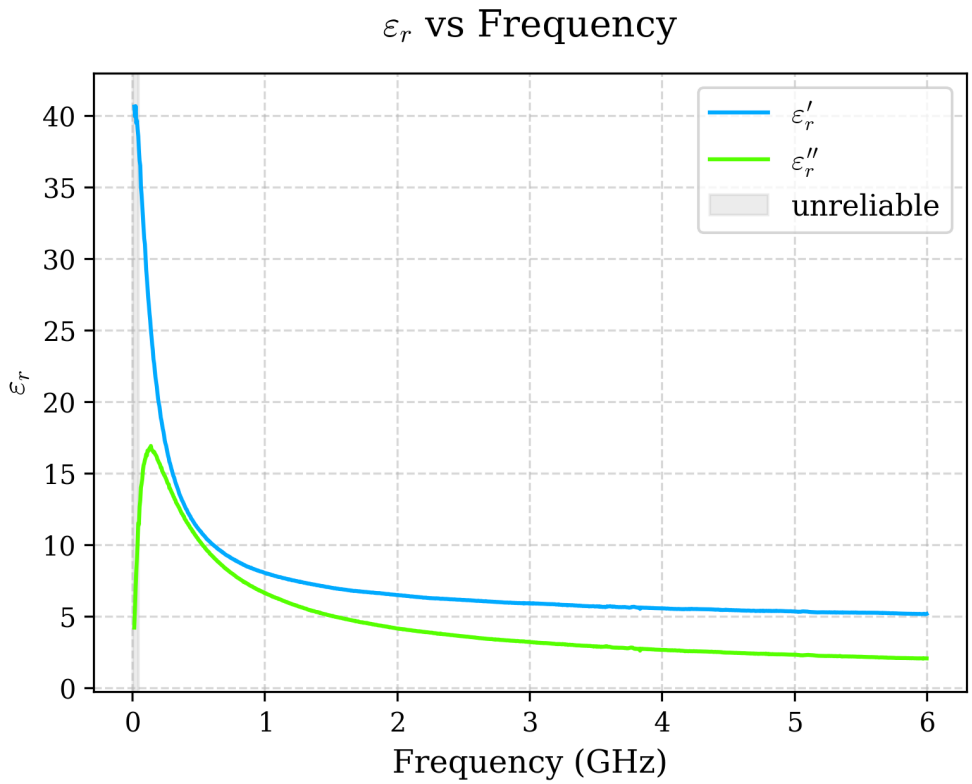
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1 Measurement Results

1.1 Complex Permittivity $\varepsilon_r(f)$



Points without a solution: 2 of 1200 (left blank in the plot and shown as — in the table).

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.005	—	—	—
0.010	—	—	—
0.015	40.6203	4.1962	0.1033
0.020	39.9360	5.5512	0.1390
0.025	40.6488	7.0699	0.1739
0.030	39.4688	8.1552	0.2066
0.035	39.6396	8.9913	0.2268
0.040	39.0312	10.1323	0.2596
0.045	38.5581	11.5009	0.2983
0.050	37.6099	11.4037	0.3032
0.055	36.8302	12.6527	0.3435
0.060	36.4975	13.0986	0.3589
0.065	35.0404	13.9821	0.3990
0.070	34.4782	14.2371	0.4129
0.075	33.6762	14.5762	0.4328
0.080	32.8825	15.1891	0.4619
0.085	32.1256	15.5420	0.4838

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.090	31.3859	15.6868	0.4998
0.095	31.0012	15.9255	0.5137
0.100	30.0684	16.0674	0.5344
0.105	29.2186	16.3167	0.5584
0.110	28.6092	16.2267	0.5672
0.115	27.9285	16.5719	0.5934
0.120	27.2710	16.5316	0.6062
0.125	26.7395	16.6125	0.6213
0.130	26.1334	16.7449	0.6407
0.135	25.6733	16.7244	0.6514
0.140	25.0378	16.8969	0.6749
0.145	24.5386	16.6740	0.6795
0.150	23.9837	16.5969	0.6920
0.155	23.5805	16.5511	0.7019
0.160	22.9420	16.4517	0.7171
0.165	22.6477	16.4818	0.7277
0.170	22.1416	16.3763	0.7396
0.175	21.6926	16.2843	0.7507
0.180	21.3688	16.2225	0.7592
0.185	21.0316	16.1143	0.7662
0.190	20.5723	15.9446	0.7750
0.195	20.1762	15.8644	0.7863
0.200	19.8712	15.7455	0.7924
0.205	19.6411	15.6743	0.7980
0.210	19.2780	15.5709	0.8077
0.215	18.8743	15.3788	0.8148
0.220	18.6702	15.3844	0.8240
0.225	18.3814	15.1668	0.8251
0.230	18.0997	15.0516	0.8316
0.235	17.9004	14.9992	0.8379
0.240	17.6049	14.8634	0.8443
0.245	17.2774	14.7985	0.8565
0.250	17.0744	14.6943	0.8606
0.255	16.9233	14.5947	0.8624
0.260	16.6779	14.4682	0.8675
0.265	16.4424	14.3250	0.8712
0.270	16.2145	14.2473	0.8787
0.275	16.0408	14.2014	0.8853
0.280	15.7892	13.9449	0.8832
0.285	15.6717	13.9201	0.8882
0.290	15.4957	13.8025	0.8907
0.295	15.2925	13.6959	0.8956
0.300	15.1535	13.5951	0.8972
0.305	14.9583	13.5077	0.9030
0.310	14.8035	13.3722	0.9033
0.315	14.7105	13.3013	0.9042

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.320	14.5472	13.1692	0.9053
0.325	14.3639	13.1132	0.9129
0.330	14.2308	12.9510	0.9101
0.335	14.0983	12.8786	0.9135
0.340	13.9784	12.7915	0.9151
0.345	13.8598	12.7264	0.9182
0.350	13.7063	12.6237	0.9210
0.355	13.5433	12.5576	0.9272
0.360	13.4426	12.4344	0.9250
0.365	13.3272	12.3410	0.9260
0.370	13.2273	12.2457	0.9258
0.375	13.0942	12.1529	0.9281
0.380	13.0265	12.1098	0.9296
0.385	12.9208	12.0017	0.9289
0.390	12.8186	11.9594	0.9330
0.395	12.7170	11.8079	0.9285
0.400	12.6111	11.7381	0.9308
0.405	12.5031	11.6784	0.9340
0.410	12.4543	11.5794	0.9298
0.415	12.3514	11.5155	0.9323
0.420	12.2282	11.4623	0.9374
0.425	12.1726	11.3680	0.9339
0.430	12.0782	11.3127	0.9366
0.435	12.0170	11.2177	0.9335
0.440	11.9322	11.1432	0.9339
0.445	11.8048	11.0950	0.9399
0.450	11.7680	11.0226	0.9367
0.455	11.6937	10.9165	0.9335
0.460	11.6264	10.8919	0.9368
0.465	11.5485	10.8013	0.9353
0.470	11.4660	10.7409	0.9368
0.475	11.3915	10.6614	0.9359
0.480	11.3450	10.6085	0.9351
0.485	11.2598	10.5277	0.9350
0.490	11.2126	10.4796	0.9346
0.495	11.1429	10.4149	0.9347
0.500	11.0821	10.3251	0.9317
0.505	11.0305	10.2826	0.9322
0.510	10.9802	10.2269	0.9314
0.515	10.9116	10.1629	0.9314
0.520	10.8842	10.0989	0.9278
0.525	10.8004	10.0415	0.9297
0.530	10.7352	9.9839	0.9300
0.535	10.6995	9.9177	0.9269
0.540	10.6383	9.8709	0.9279
0.545	10.5940	9.8354	0.9284

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.550	10.5419	9.7434	0.9243
0.555	10.4788	9.7045	0.9261
0.560	10.4060	9.6299	0.9254
0.565	10.3814	9.5878	0.9235
0.570	10.3207	9.5358	0.9239
0.575	10.2900	9.4983	0.9231
0.580	10.2426	9.4518	0.9228
0.585	10.2060	9.3663	0.9177
0.590	10.1500	9.3261	0.9188
0.595	10.1030	9.2791	0.9184
0.600	10.0633	9.2444	0.9186
0.605	10.0198	9.1905	0.9172
0.610	9.9782	9.1367	0.9157
0.615	9.9443	9.1094	0.9160
0.620	9.9143	9.0398	0.9118
0.625	9.8630	9.0008	0.9126
0.630	9.8369	8.9529	0.9101
0.635	9.7977	8.9221	0.9106
0.640	9.7383	8.8647	0.9103
0.645	9.7241	8.8073	0.9057
0.650	9.6722	8.7843	0.9082
0.655	9.6395	8.7458	0.9073
0.660	9.6130	8.6933	0.9043
0.665	9.5819	8.6552	0.9033
0.670	9.5383	8.6099	0.9027
0.675	9.5201	8.5664	0.8998
0.680	9.4559	8.5369	0.9028
0.685	9.4474	8.4931	0.8990
0.690	9.4076	8.4437	0.8975
0.695	9.3697	8.3926	0.8957
0.700	9.3294	8.3477	0.8948
0.705	9.2961	8.3337	0.8965
0.710	9.2637	8.2795	0.8938
0.715	9.2395	8.2478	0.8927
0.720	9.2112	8.2189	0.8923
0.725	9.1995	8.1695	0.8880
0.730	9.1663	8.1438	0.8885
0.735	9.1190	8.0987	0.8881
0.740	9.1114	8.0566	0.8842
0.745	9.0686	8.0346	0.8860
0.750	9.0348	7.9996	0.8854
0.755	9.0198	7.9634	0.8829
0.760	8.9884	7.9362	0.8829
0.765	8.9671	7.8958	0.8805
0.770	8.9383	7.8613	0.8795
0.775	8.9286	7.8349	0.8775

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.780	8.8718	7.7872	0.8777
0.785	8.8492	7.7556	0.8764
0.790	8.8511	7.7111	0.8712
0.795	8.8086	7.6865	0.8726
0.800	8.7818	7.6600	0.8723
0.805	8.7608	7.6302	0.8709
0.810	8.7412	7.5966	0.8690
0.815	8.7114	7.5748	0.8695
0.820	8.6748	7.5441	0.8697
0.825	8.6733	7.5103	0.8659
0.830	8.6538	7.4832	0.8647
0.835	8.6094	7.4466	0.8649
0.840	8.5838	7.3864	0.8605
0.845	8.5811	7.3872	0.8609
0.850	8.5510	7.3512	0.8597
0.855	8.5148	7.3490	0.8631
0.860	8.5052	7.2994	0.8582
0.865	8.4689	7.2826	0.8599
0.870	8.4513	7.2590	0.8589
0.875	8.4555	7.2343	0.8556
0.880	8.4325	7.1813	0.8516
0.885	8.3997	7.1650	0.8530
0.890	8.3894	7.1359	0.8506
0.895	8.3589	7.1060	0.8501
0.900	8.3704	7.0851	0.8465
0.905	8.3324	7.0779	0.8494
0.910	8.3172	7.0466	0.8472
0.915	8.2856	7.0030	0.8452
0.920	8.2826	6.9809	0.8428
0.925	8.2626	6.9720	0.8438
0.930	8.2454	6.9372	0.8413
0.935	8.2289	6.8999	0.8385
0.940	8.2037	6.8901	0.8399
0.945	8.1845	6.8609	0.8383
0.950	8.1908	6.8293	0.8338
0.955	8.1594	6.8292	0.8370
0.960	8.1468	6.7820	0.8325
0.965	8.1216	6.7758	0.8343
0.970	8.1186	6.7543	0.8320
0.975	8.1052	6.7226	0.8294
0.980	8.0884	6.7173	0.8305
0.985	8.0829	6.6878	0.8274
0.990	8.0553	6.6544	0.8261
0.995	8.0395	6.6485	0.8270
1.000	8.0280	6.6062	0.8229
1.005	8.0116	6.5913	0.8227

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.010	7.9906	6.5877	0.8244
1.015	7.9594	6.5511	0.8231
1.020	7.9657	6.5344	0.8203
1.025	7.9657	6.5104	0.8173
1.030	7.9433	6.4933	0.8175
1.035	7.9306	6.4717	0.8161
1.040	7.9169	6.4584	0.8158
1.045	7.9053	6.4334	0.8138
1.050	7.8742	6.4116	0.8143
1.055	7.8627	6.3726	0.8105
1.060	7.8664	6.3767	0.8106
1.065	7.8404	6.3426	0.8090
1.070	7.8325	6.3279	0.8079
1.075	7.8103	6.3230	0.8096
1.080	7.8043	6.2922	0.8062
1.085	7.7954	6.2699	0.8043
1.090	7.7818	6.2620	0.8047
1.095	7.7580	6.2444	0.8049
1.100	7.7505	6.2312	0.8040
1.105	7.7335	6.1904	0.8005
1.110	7.7305	6.1899	0.8007
1.115	7.7225	6.1747	0.7996
1.120	7.7188	6.1584	0.7978
1.125	7.7013	6.1319	0.7962
1.130	7.6812	6.1132	0.7959
1.135	7.6577	6.0861	0.7948
1.140	7.6614	6.0700	0.7923
1.145	7.6481	6.0520	0.7913
1.150	7.6443	6.0395	0.7901
1.155	7.6345	6.0136	0.7877
1.160	7.6135	5.9919	0.7870
1.165	7.6036	5.9943	0.7883
1.170	7.5974	5.9694	0.7857
1.175	7.5702	5.9438	0.7852
1.180	7.5579	5.9350	0.7853
1.185	7.5754	5.9085	0.7800
1.190	7.5481	5.8904	0.7804
1.195	7.5416	5.8789	0.7795
1.200	7.5356	5.8641	0.7782
1.205	7.5051	5.8351	0.7775
1.210	7.5082	5.8245	0.7757
1.215	7.4976	5.8059	0.7744
1.220	7.4941	5.7947	0.7732
1.225	7.4880	5.7776	0.7716
1.230	7.4693	5.7696	0.7724
1.235	7.4575	5.7414	0.7699

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.240	7.4501	5.7374	0.7701
1.245	7.4451	5.7216	0.7685
1.250	7.4260	5.6959	0.7670
1.255	7.4144	5.6761	0.7655
1.260	7.4135	5.6617	0.7637
1.265	7.3845	5.6582	0.7662
1.270	7.3986	5.6350	0.7616
1.275	7.3794	5.6176	0.7613
1.280	7.3781	5.6069	0.7599
1.285	7.3656	5.6041	0.7609
1.290	7.3516	5.5770	0.7586
1.295	7.3321	5.5582	0.7581
1.300	7.3285	5.5405	0.7560
1.305	7.3274	5.5310	0.7548
1.310	7.3117	5.5192	0.7548
1.315	7.2998	5.5123	0.7551
1.320	7.3001	5.4807	0.7508
1.325	7.2775	5.4644	0.7509
1.330	7.2741	5.4477	0.7489
1.335	7.2709	5.4287	0.7466
1.340	7.2663	5.4234	0.7464
1.345	7.2534	5.3995	0.7444
1.350	7.2324	5.4013	0.7468
1.355	7.2293	5.3690	0.7427
1.360	7.2288	5.3781	0.7440
1.365	7.2163	5.3572	0.7424
1.370	7.1997	5.3505	0.7431
1.375	7.2035	5.3425	0.7416
1.380	7.1959	5.3201	0.7393
1.385	7.1799	5.3032	0.7386
1.390	7.1693	5.2836	0.7370
1.395	7.1532	5.2881	0.7393
1.400	7.1596	5.2819	0.7377
1.405	7.1485	5.2559	0.7352
1.410	7.1436	5.2568	0.7359
1.415	7.1315	5.2214	0.7322
1.420	7.1099	5.2303	0.7356
1.425	7.1136	5.1987	0.7308
1.430	7.1009	5.1948	0.7316
1.435	7.1051	5.1771	0.7286
1.440	7.0911	5.1633	0.7281
1.445	7.0818	5.1546	0.7279
1.450	7.0745	5.1455	0.7273
1.455	7.0626	5.1413	0.7280
1.460	7.0484	5.1254	0.7272
1.465	7.0399	5.1164	0.7268

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.470	7.0341	5.0926	0.7240
1.475	7.0357	5.0757	0.7214
1.480	7.0277	5.0732	0.7219
1.485	7.0182	5.0582	0.7207
1.490	7.0097	5.0524	0.7208
1.495	7.0031	5.0378	0.7194
1.500	6.9915	5.0250	0.7187
1.505	6.9778	5.0174	0.7191
1.510	6.9718	4.9993	0.7171
1.515	6.9676	5.0081	0.7188
1.520	6.9518	4.9817	0.7166
1.525	6.9435	4.9728	0.7162
1.530	6.9481	4.9565	0.7134
1.535	6.9460	4.9479	0.7123
1.540	6.9245	4.9360	0.7128
1.545	6.9183	4.9285	0.7124
1.550	6.9197	4.9256	0.7118
1.555	6.9120	4.9160	0.7112
1.560	6.8993	4.9000	0.7102
1.565	6.8937	4.8926	0.7097
1.570	6.8882	4.8759	0.7079
1.575	6.8825	4.8819	0.7093
1.580	6.8841	4.8651	0.7067
1.585	6.8738	4.8440	0.7047
1.590	6.8648	4.8361	0.7045
1.595	6.8598	4.8244	0.7033
1.600	6.8519	4.8215	0.7037
1.605	6.8500	4.8086	0.7020
1.610	6.8444	4.7871	0.6994
1.615	6.8374	4.7900	0.7006
1.620	6.8280	4.7828	0.7005
1.625	6.8185	4.7823	0.7014
1.630	6.8135	4.7563	0.6981
1.635	6.8072	4.7533	0.6983
1.640	6.8047	4.7431	0.6970
1.645	6.8005	4.7302	0.6956
1.650	6.7800	4.7183	0.6959
1.655	6.7875	4.7102	0.6940
1.660	6.7759	4.6987	0.6934
1.665	6.7756	4.6975	0.6933
1.670	6.7600	4.6778	0.6920
1.675	6.7551	4.6747	0.6920
1.680	6.7627	4.6782	0.6918
1.685	6.7612	4.6378	0.6859
1.690	6.7504	4.6528	0.6893
1.695	6.7521	4.6328	0.6861

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.700	6.7444	4.6251	0.6858
1.705	6.7313	4.6336	0.6884
1.710	6.7333	4.6006	0.6833
1.715	6.7371	4.5914	0.6815
1.720	6.7153	4.5869	0.6831
1.725	6.7190	4.5700	0.6802
1.730	6.7146	4.5699	0.6806
1.735	6.7124	4.5616	0.6796
1.740	6.7069	4.5659	0.6808
1.745	6.6890	4.5380	0.6784
1.750	6.6976	4.5323	0.6767
1.755	6.6911	4.5344	0.6777
1.760	6.6900	4.5265	0.6766
1.765	6.6907	4.5195	0.6755
1.770	6.6863	4.5041	0.6736
1.775	6.6711	4.5173	0.6771
1.780	6.6726	4.4870	0.6725
1.785	6.6685	4.4918	0.6736
1.790	6.6665	4.4702	0.6706
1.795	6.6549	4.4674	0.6713
1.800	6.6554	4.4606	0.6702
1.805	6.6439	4.4535	0.6703
1.810	6.6479	4.4349	0.6671
1.815	6.6480	4.4191	0.6647
1.820	6.6460	4.4251	0.6658
1.825	6.6358	4.4166	0.6656
1.830	6.6246	4.3970	0.6637
1.835	6.6263	4.3968	0.6635
1.840	6.6152	4.3923	0.6640
1.845	6.6159	4.3852	0.6628
1.850	6.6133	4.3703	0.6608
1.855	6.6074	4.3566	0.6594
1.860	6.6102	4.3520	0.6584
1.865	6.6108	4.3494	0.6579
1.870	6.6023	4.3395	0.6573
1.875	6.5921	4.3395	0.6583
1.880	6.5873	4.3302	0.6574
1.885	6.5806	4.3173	0.6561
1.890	6.5811	4.3050	0.6541
1.895	6.5749	4.2965	0.6535
1.900	6.5647	4.2965	0.6545
1.905	6.5647	4.2894	0.6534
1.910	6.5571	4.2747	0.6519
1.915	6.5445	4.2778	0.6536
1.920	6.5461	4.2629	0.6512
1.925	6.5410	4.2679	0.6525

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.930	6.5439	4.2485	0.6492
1.935	6.5388	4.2414	0.6487
1.940	6.5364	4.2171	0.6452
1.945	6.5187	4.2317	0.6492
1.950	6.5148	4.2134	0.6468
1.955	6.5105	4.2069	0.6462
1.960	6.5148	4.1968	0.6442
1.965	6.5053	4.1905	0.6442
1.970	6.5018	4.1788	0.6427
1.975	6.4997	4.1789	0.6429
1.980	6.4961	4.1657	0.6413
1.985	6.4925	4.1599	0.6407
1.990	6.4802	4.1465	0.6399
1.995	6.4733	4.1430	0.6400
2.000	6.4644	4.1335	0.6394
2.005	6.4802	4.1294	0.6372
2.010	6.4679	4.1320	0.6389
2.015	6.4593	4.1084	0.6360
2.020	6.4555	4.1083	0.6364
2.025	6.4492	4.1066	0.6368
2.030	6.4540	4.0976	0.6349
2.035	6.4465	4.0967	0.6355
2.040	6.4433	4.0834	0.6337
2.045	6.4283	4.0685	0.6329
2.050	6.4322	4.0821	0.6346
2.055	6.4328	4.0762	0.6337
2.060	6.4199	4.0643	0.6331
2.065	6.4241	4.0616	0.6322
2.070	6.4084	4.0453	0.6312
2.075	6.4040	4.0397	0.6308
2.080	6.3987	4.0355	0.6307
2.085	6.4000	4.0256	0.6290
2.090	6.3960	4.0192	0.6284
2.095	6.3846	4.0212	0.6298
2.100	6.3725	4.0189	0.6307
2.105	6.3725	4.0131	0.6298
2.110	6.3785	3.9982	0.6268
2.115	6.3680	3.9989	0.6280
2.120	6.3678	3.9816	0.6253
2.125	6.3595	3.9838	0.6264
2.130	6.3510	3.9733	0.6256
2.135	6.3519	3.9768	0.6261
2.140	6.3440	3.9662	0.6252
2.145	6.3370	3.9567	0.6244
2.150	6.3473	3.9509	0.6225
2.155	6.3330	3.9464	0.6232

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.160	6.3295	3.9371	0.6220
2.165	6.3339	3.9416	0.6223
2.170	6.3197	3.9258	0.6212
2.175	6.3192	3.9329	0.6224
2.180	6.3144	3.9197	0.6208
2.185	6.3125	3.9188	0.6208
2.190	6.3075	3.9091	0.6198
2.195	6.2911	3.8986	0.6197
2.200	6.2929	3.8989	0.6196
2.205	6.2899	3.8882	0.6182
2.210	6.2910	3.8943	0.6190
2.215	6.2866	3.8761	0.6166
2.220	6.2898	3.8761	0.6163
2.225	6.2731	3.8773	0.6181
2.230	6.2686	3.8705	0.6174
2.235	6.2661	3.8645	0.6167
2.240	6.2689	3.8533	0.6147
2.245	6.2548	3.8504	0.6156
2.250	6.2519	3.8448	0.6150
2.255	6.2587	3.8440	0.6142
2.260	6.2498	3.8386	0.6142
2.265	6.2474	3.8412	0.6148
2.270	6.2484	3.8178	0.6110
2.275	6.2387	3.8144	0.6114
2.280	6.2403	3.8032	0.6095
2.285	6.2393	3.8094	0.6105
2.290	6.2299	3.8021	0.6103
2.295	6.2405	3.7991	0.6088
2.300	6.2419	3.7753	0.6048
2.305	6.2338	3.7813	0.6066
2.310	6.2352	3.7632	0.6035
2.315	6.2330	3.7685	0.6046
2.320	6.2208	3.7635	0.6050
2.325	6.2149	3.7493	0.6033
2.330	6.2092	3.7754	0.6080
2.335	6.2107	3.7553	0.6047
2.340	6.2204	3.7451	0.6021
2.345	6.2097	3.7369	0.6018
2.350	6.2103	3.7345	0.6013
2.355	6.2046	3.7284	0.6009
2.360	6.2009	3.7301	0.6015
2.365	6.2011	3.7261	0.6009
2.370	6.2034	3.7154	0.5989
2.375	6.2007	3.7088	0.5981
2.380	6.1956	3.7085	0.5986
2.385	6.1870	3.6988	0.5978

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.390	6.1913	3.7028	0.5981
2.395	6.1806	3.6936	0.5976
2.400	6.1716	3.6877	0.5975
2.405	6.1788	3.6742	0.5947
2.410	6.1746	3.6755	0.5953
2.415	6.1674	3.6739	0.5957
2.420	6.1666	3.6707	0.5953
2.425	6.1670	3.6604	0.5935
2.430	6.1688	3.6522	0.5920
2.435	6.1696	3.6505	0.5917
2.440	6.1613	3.6435	0.5913
2.445	6.1539	3.6410	0.5917
2.450	6.1554	3.6383	0.5911
2.455	6.1488	3.6231	0.5892
2.460	6.1487	3.6250	0.5896
2.465	6.1481	3.6263	0.5898
2.470	6.1447	3.6099	0.5875
2.475	6.1473	3.6064	0.5867
2.480	6.1419	3.6004	0.5862
2.485	6.1364	3.6037	0.5873
2.490	6.1328	3.5940	0.5860
2.495	6.1255	3.5928	0.5865
2.500	6.1342	3.5855	0.5845
2.505	6.1238	3.5769	0.5841
2.510	6.1256	3.5764	0.5838
2.515	6.1208	3.5704	0.5833
2.520	6.1174	3.5671	0.5831
2.525	6.1119	3.5593	0.5824
2.530	6.1091	3.5622	0.5831
2.535	6.1165	3.5553	0.5813
2.540	6.1107	3.5492	0.5808
2.545	6.1062	3.5392	0.5796
2.550	6.1019	3.5382	0.5799
2.555	6.1066	3.5316	0.5783
2.560	6.0984	3.5307	0.5790
2.565	6.0907	3.5253	0.5788
2.570	6.0931	3.5196	0.5776
2.575	6.0898	3.5116	0.5766
2.580	6.0859	3.5087	0.5765
2.585	6.0793	3.5115	0.5776
2.590	6.0795	3.4936	0.5746
2.595	6.0717	3.4974	0.5760
2.600	6.0665	3.4835	0.5742
2.605	6.0621	3.4801	0.5741
2.610	6.0727	3.4807	0.5732
2.615	6.0736	3.4654	0.5706

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.620	6.0599	3.4655	0.5719
2.625	6.0561	3.4553	0.5705
2.630	6.0560	3.4582	0.5710
2.635	6.0616	3.4537	0.5698
2.640	6.0488	3.4439	0.5694
2.645	6.0482	3.4436	0.5694
2.650	6.0361	3.4426	0.5703
2.655	6.0390	3.4293	0.5679
2.660	6.0340	3.4284	0.5682
2.665	6.0382	3.4287	0.5678
2.670	6.0471	3.4156	0.5648
2.675	6.0288	3.4173	0.5668
2.680	6.0281	3.4100	0.5657
2.685	6.0215	3.4078	0.5659
2.690	6.0211	3.4050	0.5655
2.695	6.0195	3.4020	0.5652
2.700	6.0184	3.3948	0.5641
2.705	6.0160	3.3922	0.5639
2.710	6.0077	3.4002	0.5660
2.715	6.0105	3.3884	0.5637
2.720	6.0063	3.3919	0.5647
2.725	6.0007	3.3748	0.5624
2.730	5.9961	3.3679	0.5617
2.735	5.9938	3.3714	0.5625
2.740	5.9995	3.3552	0.5593
2.745	5.9891	3.3706	0.5628
2.750	5.9883	3.3570	0.5606
2.755	5.9804	3.3571	0.5613
2.760	5.9864	3.3597	0.5612
2.765	5.9790	3.3461	0.5596
2.770	5.9797	3.3562	0.5613
2.775	5.9725	3.3350	0.5584
2.780	5.9763	3.3352	0.5581
2.785	5.9658	3.3361	0.5592
2.790	5.9700	3.3281	0.5575
2.795	5.9628	3.3309	0.5586
2.800	5.9581	3.3234	0.5578
2.805	5.9543	3.3272	0.5588
2.810	5.9505	3.3147	0.5570
2.815	5.9463	3.3155	0.5576
2.820	5.9491	3.3129	0.5569
2.825	5.9457	3.3017	0.5553
2.830	5.9493	3.3043	0.5554
2.835	5.9480	3.3056	0.5558
2.840	5.9378	3.2981	0.5554
2.845	5.9327	3.2992	0.5561

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.850	5.9441	3.2861	0.5528
2.855	5.9450	3.2902	0.5534
2.860	5.9394	3.2794	0.5521
2.865	5.9275	3.2781	0.5530
2.870	5.9297	3.2801	0.5532
2.875	5.9259	3.2690	0.5516
2.880	5.9254	3.2749	0.5527
2.885	5.9278	3.2685	0.5514
2.890	5.9153	3.2706	0.5529
2.895	5.9184	3.2564	0.5502
2.900	5.9197	3.2536	0.5496
2.905	5.9149	3.2601	0.5512
2.910	5.9133	3.2598	0.5513
2.915	5.9069	3.2513	0.5504
2.920	5.9163	3.2463	0.5487
2.925	5.9107	3.2413	0.5484
2.930	5.9068	3.2316	0.5471
2.935	5.9064	3.2383	0.5483
2.940	5.9108	3.2363	0.5475
2.945	5.9028	3.2359	0.5482
2.950	5.8956	3.2247	0.5470
2.955	5.9156	3.2333	0.5466
2.960	5.8826	3.2301	0.5491
2.965	5.9061	3.2165	0.5446
2.970	5.8986	3.2096	0.5441
2.975	5.8955	3.2088	0.5443
2.980	5.8990	3.2075	0.5437
2.985	5.9003	3.1982	0.5420
2.990	5.8894	3.2082	0.5447
2.995	5.9022	3.1961	0.5415
3.000	5.8833	3.1915	0.5425
3.005	5.8974	3.1902	0.5410
3.010	5.8761	3.1754	0.5404
3.015	5.8864	3.1751	0.5394
3.020	5.8895	3.1724	0.5386
3.025	5.8686	3.1778	0.5415
3.030	5.8963	3.1560	0.5352
3.035	5.8920	3.1685	0.5378
3.040	5.8795	3.1627	0.5379
3.045	5.8804	3.1510	0.5358
3.050	5.8675	3.1448	0.5360
3.055	5.8796	3.1570	0.5369
3.060	5.8758	3.1399	0.5344
3.065	5.8800	3.1362	0.5334
3.070	5.8650	3.1374	0.5349
3.075	5.8709	3.1332	0.5337

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.080	5.8721	3.1349	0.5339
3.085	5.8575	3.1194	0.5325
3.090	5.8648	3.1249	0.5328
3.095	5.8624	3.1221	0.5326
3.100	5.8584	3.1196	0.5325
3.105	5.8548	3.1091	0.5310
3.110	5.8545	3.1160	0.5322
3.115	5.8547	3.1105	0.5313
3.120	5.8503	3.1152	0.5325
3.125	5.8459	3.1117	0.5323
3.130	5.8456	3.1038	0.5310
3.135	5.8379	3.1012	0.5312
3.140	5.8443	3.0990	0.5303
3.145	5.8383	3.0929	0.5298
3.150	5.8420	3.0907	0.5291
3.155	5.8313	3.0903	0.5300
3.160	5.8387	3.0810	0.5277
3.165	5.8329	3.0863	0.5291
3.170	5.8388	3.0698	0.5258
3.175	5.8347	3.0726	0.5266
3.180	5.8345	3.0638	0.5251
3.185	5.8279	3.0705	0.5269
3.190	5.8324	3.0607	0.5248
3.195	5.8209	3.0548	0.5248
3.200	5.8091	3.0427	0.5238
3.205	5.8117	3.0489	0.5246
3.210	5.8121	3.0501	0.5248
3.215	5.8072	3.0339	0.5224
3.220	5.8202	3.0463	0.5234
3.225	5.7979	3.0504	0.5261
3.230	5.7982	3.0343	0.5233
3.235	5.7926	3.0416	0.5251
3.240	5.7823	3.0368	0.5252
3.245	5.7938	3.0241	0.5220
3.250	5.7703	3.0483	0.5283
3.255	5.7801	3.0316	0.5245
3.260	5.7891	3.0339	0.5241
3.265	5.7737	3.0329	0.5253
3.270	5.7807	3.0057	0.5200
3.275	5.7755	3.0145	0.5220
3.280	5.7807	3.0126	0.5212
3.285	5.7529	3.0126	0.5237
3.290	5.7704	2.9868	0.5176
3.295	5.7602	2.9946	0.5199
3.300	5.7707	2.9958	0.5191
3.305	5.7680	3.0046	0.5209

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.310	5.7592	2.9888	0.5190
3.315	5.7530	2.9856	0.5190
3.320	5.7648	2.9885	0.5184
3.325	5.7623	2.9913	0.5191
3.330	5.7508	2.9786	0.5179
3.335	5.7441	2.9695	0.5170
3.340	5.7518	2.9705	0.5164
3.345	5.7395	2.9668	0.5169
3.350	5.7370	2.9555	0.5152
3.355	5.7440	2.9555	0.5145
3.360	5.7328	2.9504	0.5147
3.365	5.7274	2.9539	0.5157
3.370	5.7208	2.9502	0.5157
3.375	5.7340	2.9480	0.5141
3.380	5.7284	2.9491	0.5148
3.385	5.7143	2.9408	0.5146
3.390	5.7211	2.9350	0.5130
3.395	5.7170	2.9355	0.5135
3.400	5.7052	2.9351	0.5145
3.405	5.7040	2.9278	0.5133
3.410	5.7098	2.9299	0.5131
3.415	5.7023	2.9296	0.5137
3.420	5.7018	2.9234	0.5127
3.425	5.7031	2.9248	0.5128
3.430	5.7022	2.9227	0.5125
3.435	5.7041	2.9249	0.5128
3.440	5.7030	2.9144	0.5110
3.445	5.6944	2.9169	0.5122
3.450	5.6739	2.9138	0.5135
3.455	5.6885	2.9155	0.5125
3.460	5.6862	2.9143	0.5125
3.465	5.6829	2.8996	0.5102
3.470	5.6836	2.9005	0.5103
3.475	5.6912	2.8928	0.5083
3.480	5.6914	2.8848	0.5069
3.485	5.6939	2.8657	0.5033
3.490	5.6764	2.8859	0.5084
3.495	5.6673	2.8904	0.5100
3.500	5.6784	2.8786	0.5069
3.505	5.6773	2.8861	0.5084
3.510	5.6901	2.8547	0.5017
3.515	5.6745	2.8777	0.5071
3.520	5.6702	2.8555	0.5036
3.525	5.6525	2.8705	0.5078
3.530	5.6814	2.8754	0.5061
3.535	5.6633	2.8495	0.5031

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.540	5.6467	2.8446	0.5038
3.545	5.6630	2.8673	0.5063
3.550	5.6390	2.8550	0.5063
3.555	5.6390	2.8541	0.5061
3.560	5.6446	2.8667	0.5079
3.565	5.6548	2.8621	0.5061
3.570	5.6287	2.8643	0.5089
3.575	5.6283	2.8931	0.5140
3.580	5.6558	2.8829	0.5097
3.585	5.6835	2.8950	0.5094
3.590	5.6681	2.8563	0.5039
3.595	5.6706	2.8498	0.5026
3.600	5.6763	2.8610	0.5040
3.605	5.6861	2.8528	0.5017
3.610	5.6921	2.8424	0.4994
3.615	5.6729	2.8327	0.4993
3.620	5.6663	2.8121	0.4963
3.625	5.6732	2.8124	0.4957
3.630	5.6498	2.8016	0.4959
3.635	5.6575	2.8005	0.4950
3.640	5.6698	2.7998	0.4938
3.645	5.6598	2.7926	0.4934
3.650	5.6421	2.7991	0.4961
3.655	5.6360	2.7894	0.4949
3.660	5.6421	2.7976	0.4958
3.665	5.6385	2.7889	0.4946
3.670	5.6318	2.7930	0.4959
3.675	5.6251	2.7816	0.4945
3.680	5.6299	2.7877	0.4952
3.685	5.6334	2.7882	0.4949
3.690	5.6255	2.7868	0.4954
3.695	5.6249	2.7885	0.4957
3.700	5.6322	2.7853	0.4945
3.705	5.6363	2.7837	0.4939
3.710	5.6257	2.7773	0.4937
3.715	5.6316	2.7759	0.4929
3.720	5.6378	2.7623	0.4900
3.725	5.6286	2.7652	0.4913
3.730	5.6204	2.7595	0.4910
3.735	5.6113	2.7476	0.4896
3.740	5.6124	2.7518	0.4903
3.745	5.6055	2.7504	0.4907
3.750	5.5933	2.7612	0.4937
3.755	5.5914	2.7765	0.4966
3.760	5.5948	2.7864	0.4980
3.765	5.6031	2.8000	0.4997

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.770	5.6301	2.8026	0.4978
3.775	5.6477	2.7868	0.4934
3.780	5.6376	2.7920	0.4952
3.785	5.6515	2.7540	0.4873
3.790	5.6539	2.7247	0.4819
3.795	5.6385	2.7115	0.4809
3.800	5.6227	2.7334	0.4861
3.805	5.6162	2.7079	0.4822
3.810	5.6070	2.7111	0.4835
3.815	5.5929	2.6987	0.4825
3.820	5.6061	2.7118	0.4837
3.825	5.5930	2.7055	0.4837
3.830	5.5919	2.7174	0.4859
3.835	5.5047	2.5858	0.4697
3.840	5.5971	2.7180	0.4856
3.845	5.5952	2.7088	0.4841
3.850	5.5930	2.7010	0.4829
3.855	5.5908	2.7154	0.4857
3.860	5.5890	2.7026	0.4835
3.865	5.5840	2.7027	0.4840
3.870	5.5908	2.6989	0.4827
3.875	5.5808	2.6988	0.4836
3.880	5.5873	2.6945	0.4823
3.885	5.5791	2.6914	0.4824
3.890	5.5747	2.6978	0.4839
3.895	5.5800	2.6797	0.4802
3.900	5.5723	2.6879	0.4824
3.905	5.5622	2.6843	0.4826
3.910	5.5753	2.6801	0.4807
3.915	5.5676	2.6803	0.4814
3.920	5.5627	2.6817	0.4821
3.925	5.5665	2.6720	0.4800
3.930	5.5576	2.6716	0.4807
3.935	5.5661	2.6701	0.4797
3.940	5.5556	2.6694	0.4805
3.945	5.5542	2.6586	0.4787
3.950	5.5509	2.6575	0.4787
3.955	5.5536	2.6588	0.4788
3.960	5.5476	2.6523	0.4781
3.965	5.5532	2.6542	0.4780
3.970	5.5437	2.6500	0.4780
3.975	5.5421	2.6499	0.4781
3.980	5.5420	2.6481	0.4778
3.985	5.5405	2.6495	0.4782
3.990	5.5428	2.6437	0.4770
3.995	5.5358	2.6372	0.4764

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.000	5.5349	2.6368	0.4764
4.005	5.5356	2.6368	0.4763
4.010	5.5333	2.6332	0.4759
4.015	5.5324	2.6367	0.4766
4.020	5.5335	2.6295	0.4752
4.025	5.5353	2.6295	0.4751
4.030	5.5250	2.6279	0.4756
4.035	5.5196	2.6282	0.4762
4.040	5.5285	2.6201	0.4739
4.045	5.5302	2.6116	0.4722
4.050	5.5199	2.6139	0.4735
4.055	5.5175	2.6107	0.4732
4.060	5.5154	2.6127	0.4737
4.065	5.5059	2.6022	0.4726
4.070	5.5127	2.6108	0.4736
4.075	5.5102	2.6058	0.4729
4.080	5.5083	2.6153	0.4748
4.085	5.5097	2.6038	0.4726
4.090	5.5049	2.6005	0.4724
4.095	5.4989	2.6083	0.4743
4.100	5.5026	2.6109	0.4745
4.105	5.4983	2.5973	0.4724
4.110	5.4984	2.5967	0.4723
4.115	5.4921	2.5929	0.4721
4.120	5.4997	2.5865	0.4703
4.125	5.4999	2.5938	0.4716
4.130	5.4994	2.5979	0.4724
4.135	5.4903	2.5912	0.4720
4.140	5.4939	2.5724	0.4682
4.145	5.4928	2.5757	0.4689
4.150	5.4845	2.5791	0.4703
4.155	5.4856	2.5755	0.4695
4.160	5.4940	2.5739	0.4685
4.165	5.4803	2.5703	0.4690
4.170	5.4846	2.5606	0.4669
4.175	5.4953	2.5411	0.4624
4.180	5.4944	2.5586	0.4657
4.185	5.4931	2.5505	0.4643
4.190	5.4856	2.5514	0.4651
4.195	5.4890	2.5481	0.4642
4.200	5.5029	2.5394	0.4615
4.205	5.4934	2.5491	0.4640
4.210	5.4828	2.5528	0.4656
4.215	5.5018	2.5443	0.4624
4.220	5.4949	2.5467	0.4635
4.225	5.5102	2.5425	0.4614

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.230	5.4990	2.5428	0.4624
4.235	5.4929	2.5419	0.4628
4.240	5.4944	2.5534	0.4647
4.245	5.4919	2.5478	0.4639
4.250	5.4951	2.5399	0.4622
4.255	5.4954	2.5428	0.4627
4.260	5.4879	2.5422	0.4632
4.265	5.4997	2.5414	0.4621
4.270	5.4906	2.5347	0.4616
4.275	5.4878	2.5360	0.4621
4.280	5.4874	2.5357	0.4621
4.285	5.4858	2.5344	0.4620
4.290	5.4848	2.5326	0.4617
4.295	5.4853	2.5285	0.4610
4.300	5.4839	2.5297	0.4613
4.305	5.4807	2.5293	0.4615
4.310	5.4805	2.5264	0.4610
4.315	5.4794	2.5249	0.4608
4.320	5.4819	2.5241	0.4604
4.325	5.4773	2.5254	0.4611
4.330	5.4809	2.5228	0.4603
4.335	5.4796	2.5201	0.4599
4.340	5.4799	2.5168	0.4593
4.345	5.4726	2.5124	0.4591
4.350	5.4762	2.5152	0.4593
4.355	5.4779	2.5140	0.4589
4.360	5.4705	2.5081	0.4585
4.365	5.4675	2.5063	0.4584
4.370	5.4782	2.5095	0.4581
4.375	5.4675	2.5130	0.4596
4.380	5.4684	2.5072	0.4585
4.385	5.4613	2.5053	0.4587
4.390	5.4658	2.4946	0.4564
4.395	5.4558	2.4947	0.4573
4.400	5.4582	2.4979	0.4576
4.405	5.4456	2.5011	0.4593
4.410	5.4535	2.4920	0.4569
4.415	5.4512	2.4883	0.4565
4.420	5.4518	2.4945	0.4575
4.425	5.4578	2.4899	0.4562
4.430	5.4455	2.4867	0.4566
4.435	5.4370	2.4852	0.4571
4.440	5.4417	2.4830	0.4563
4.445	5.4240	2.4827	0.4577
4.450	5.4347	2.4670	0.4539
4.455	5.4246	2.4739	0.4560

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.460	5.4254	2.4585	0.4531
4.465	5.4290	2.4551	0.4522
4.470	5.4298	2.4605	0.4531
4.475	5.4322	2.4471	0.4505
4.480	5.4436	2.4547	0.4509
4.485	5.4269	2.4467	0.4508
4.490	5.4373	2.4484	0.4503
4.495	5.4327	2.4476	0.4505
4.500	5.4287	2.4460	0.4506
4.505	5.4313	2.4457	0.4503
4.510	5.4145	2.4446	0.4515
4.515	5.4237	2.4487	0.4515
4.520	5.4213	2.4427	0.4506
4.525	5.4248	2.4418	0.4501
4.530	5.4133	2.4361	0.4500
4.535	5.4150	2.4430	0.4512
4.540	5.4163	2.4359	0.4497
4.545	5.4129	2.4316	0.4492
4.550	5.4113	2.4274	0.4486
4.555	5.4050	2.4299	0.4496
4.560	5.4109	2.4285	0.4488
4.565	5.4102	2.4316	0.4495
4.570	5.4033	2.4213	0.4481
4.575	5.4043	2.4235	0.4484
4.580	5.4050	2.4236	0.4484
4.585	5.4013	2.4207	0.4482
4.590	5.4018	2.4179	0.4476
4.595	5.3916	2.4192	0.4487
4.600	5.3941	2.4162	0.4479
4.605	5.3881	2.4117	0.4476
4.610	5.3919	2.4116	0.4473
4.615	5.3847	2.4081	0.4472
4.620	5.3920	2.4068	0.4464
4.625	5.3818	2.4105	0.4479
4.630	5.3866	2.4052	0.4465
4.635	5.3832	2.4008	0.4460
4.640	5.3771	2.3981	0.4460
4.645	5.3859	2.3968	0.4450
4.650	5.3832	2.3943	0.4448
4.655	5.3852	2.3930	0.4444
4.660	5.3854	2.3898	0.4437
4.665	5.3814	2.3857	0.4433
4.670	5.3681	2.3957	0.4463
4.675	5.3693	2.3819	0.4436
4.680	5.3740	2.3844	0.4437
4.685	5.3736	2.3840	0.4436

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.690	5.3807	2.3872	0.4437
4.695	5.3764	2.3697	0.4408
4.700	5.3745	2.3828	0.4434
4.705	5.3589	2.3637	0.4411
4.710	5.3531	2.3767	0.4440
4.715	5.3637	2.3771	0.4432
4.720	5.3662	2.3650	0.4407
4.725	5.3699	2.3716	0.4416
4.730	5.3665	2.3568	0.4392
4.735	5.3642	2.3657	0.4410
4.740	5.3737	2.3650	0.4401
4.745	5.3559	2.3719	0.4429
4.750	5.3690	2.3733	0.4420
4.755	5.3704	2.3688	0.4411
4.760	5.3580	2.3633	0.4411
4.765	5.3817	2.3754	0.4414
4.770	5.3692	2.3674	0.4409
4.775	5.3520	2.3654	0.4420
4.780	5.3638	2.3577	0.4396
4.785	5.3688	2.3815	0.4436
4.790	5.3745	2.3518	0.4376
4.795	5.3609	2.3532	0.4390
4.800	5.3628	2.3568	0.4395
4.805	5.3740	2.3337	0.4343
4.810	5.3594	2.3530	0.4390
4.815	5.3673	2.3359	0.4352
4.820	5.3567	2.3414	0.4371
4.825	5.3681	2.3350	0.4350
4.830	5.3544	2.3362	0.4363
4.835	5.3548	2.3454	0.4380
4.840	5.3599	2.3365	0.4359
4.845	5.3653	2.3338	0.4350
4.850	5.3527	2.3329	0.4358
4.855	5.3615	2.3309	0.4348
4.860	5.3563	2.3310	0.4352
4.865	5.3751	2.3308	0.4336
4.870	5.3631	2.3241	0.4333
4.875	5.3600	2.3290	0.4345
4.880	5.3598	2.3338	0.4354
4.885	5.3541	2.3238	0.4340
4.890	5.3629	2.3285	0.4342
4.895	5.3584	2.3259	0.4341
4.900	5.3585	2.3312	0.4350
4.905	5.3571	2.3281	0.4346
4.910	5.3509	2.3268	0.4348
4.915	5.3517	2.3274	0.4349

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.920	5.3495	2.3249	0.4346
4.925	5.3500	2.3228	0.4342
4.930	5.3412	2.3212	0.4346
4.935	5.3444	2.3201	0.4341
4.940	5.3420	2.3205	0.4344
4.945	5.3372	2.3170	0.4341
4.950	5.3385	2.3138	0.4334
4.955	5.3388	2.3151	0.4336
4.960	5.3368	2.3107	0.4330
4.965	5.3378	2.3065	0.4321
4.970	5.3363	2.3095	0.4328
4.975	5.3427	2.3078	0.4319
4.980	5.3317	2.3059	0.4325
4.985	5.3306	2.2992	0.4313
4.990	5.3332	2.3014	0.4315
4.995	5.3243	2.3086	0.4336
5.000	5.3269	2.3010	0.4320
5.005	5.3262	2.3032	0.4324
5.010	5.3176	2.2979	0.4321
5.015	5.3155	2.3019	0.4331
5.020	5.3145	2.2924	0.4314
5.025	5.3058	2.2856	0.4308
5.030	5.2949	2.2857	0.4317
5.035	5.3091	2.2816	0.4298
5.040	5.3041	2.2742	0.4288
5.045	5.3037	2.2633	0.4267
5.050	5.3080	2.2538	0.4246
5.055	5.3240	2.2623	0.4249
5.060	5.3238	2.2573	0.4240
5.065	5.3326	2.2534	0.4226
5.070	5.3298	2.2630	0.4246
5.075	5.3308	2.2655	0.4250
5.080	5.3236	2.2686	0.4261
5.085	5.3211	2.2679	0.4262
5.090	5.3202	2.2765	0.4279
5.095	5.3196	2.2817	0.4289
5.100	5.3156	2.2815	0.4292
5.105	5.3048	2.2900	0.4317
5.110	5.2968	2.2802	0.4305
5.115	5.2909	2.2768	0.4303
5.120	5.2872	2.2759	0.4305
5.125	5.2831	2.2724	0.4301
5.130	5.2801	2.2680	0.4295
5.135	5.2771	2.2651	0.4292
5.140	5.2733	2.2651	0.4296
5.145	5.2679	2.2553	0.4281

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.150	5.2689	2.2554	0.4281
5.155	5.2661	2.2459	0.4265
5.160	5.2658	2.2413	0.4256
5.165	5.2637	2.2385	0.4253
5.170	5.2625	2.2318	0.4241
5.175	5.2561	2.2311	0.4245
5.180	5.2626	2.2249	0.4228
5.185	5.2602	2.2245	0.4229
5.190	5.2632	2.2241	0.4226
5.195	5.2627	2.2196	0.4218
5.200	5.2653	2.2192	0.4215
5.205	5.2666	2.2173	0.4210
5.210	5.2658	2.2164	0.4209
5.215	5.2660	2.2145	0.4205
5.220	5.2672	2.2153	0.4206
5.225	5.2616	2.2150	0.4210
5.230	5.2666	2.2096	0.4195
5.235	5.2617	2.2146	0.4209
5.240	5.2589	2.2080	0.4199
5.245	5.2655	2.2100	0.4197
5.250	5.2582	2.2146	0.4212
5.255	5.2558	2.2165	0.4217
5.260	5.2644	2.2085	0.4195
5.265	5.2684	2.2177	0.4209
5.270	5.2592	2.2101	0.4202
5.275	5.2529	2.2064	0.4200
5.280	5.2617	2.2123	0.4204
5.285	5.2595	2.2117	0.4205
5.290	5.2530	2.2076	0.4202
5.295	5.2602	2.1995	0.4181
5.300	5.2559	2.2048	0.4195
5.305	5.2684	2.1972	0.4171
5.310	5.2477	2.2050	0.4202
5.315	5.2580	2.2042	0.4192
5.320	5.2449	2.1903	0.4176
5.325	5.2618	2.1990	0.4179
5.330	5.2556	2.1925	0.4172
5.335	5.2556	2.1965	0.4179
5.340	5.2516	2.2024	0.4194
5.345	5.2452	2.2150	0.4223
5.350	5.2610	2.1897	0.4162
5.355	5.2497	2.1920	0.4176
5.360	5.2432	2.1730	0.4144
5.365	5.2648	2.1803	0.4141
5.370	5.2414	2.1812	0.4161
5.375	5.2527	2.1832	0.4156

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.380	5.2526	2.1851	0.4160
5.385	5.2542	2.1868	0.4162
5.390	5.2495	2.1835	0.4159
5.395	5.2501	2.1795	0.4151
5.400	5.2498	2.1834	0.4159
5.405	5.2476	2.1754	0.4146
5.410	5.2485	2.1815	0.4156
5.415	5.2435	2.1696	0.4138
5.420	5.2478	2.1797	0.4154
5.425	5.2459	2.1763	0.4149
5.430	5.2531	2.1746	0.4140
5.435	5.2479	2.1712	0.4137
5.440	5.2470	2.1785	0.4152
5.445	5.2533	2.1726	0.4136
5.450	5.2438	2.1695	0.4137
5.455	5.2478	2.1731	0.4141
5.460	5.2525	2.1654	0.4123
5.465	5.2550	2.1700	0.4129
5.470	5.2384	2.1642	0.4131
5.475	5.2499	2.1571	0.4109
5.480	5.2391	2.1637	0.4130
5.485	5.2408	2.1545	0.4111
5.490	5.2343	2.1643	0.4135
5.495	5.2400	2.1493	0.4102
5.500	5.2383	2.1591	0.4122
5.505	5.2380	2.1599	0.4124
5.510	5.2400	2.1558	0.4114
5.515	5.2432	2.1524	0.4105
5.520	5.2355	2.1507	0.4108
5.525	5.2416	2.1476	0.4097
5.530	5.2381	2.1467	0.4098
5.535	5.2425	2.1417	0.4085
5.540	5.2385	2.1462	0.4097
5.545	5.2329	2.1404	0.4090
5.550	5.2363	2.1398	0.4086
5.555	5.2373	2.1379	0.4082
5.560	5.2355	2.1357	0.4079
5.565	5.2360	2.1357	0.4079
5.570	5.2366	2.1342	0.4075
5.575	5.2369	2.1321	0.4071
5.580	5.2384	2.1361	0.4078
5.585	5.2350	2.1337	0.4076
5.590	5.2366	2.1316	0.4071
5.595	5.2336	2.1298	0.4070
5.600	5.2374	2.1303	0.4068
5.605	5.2369	2.1279	0.4063

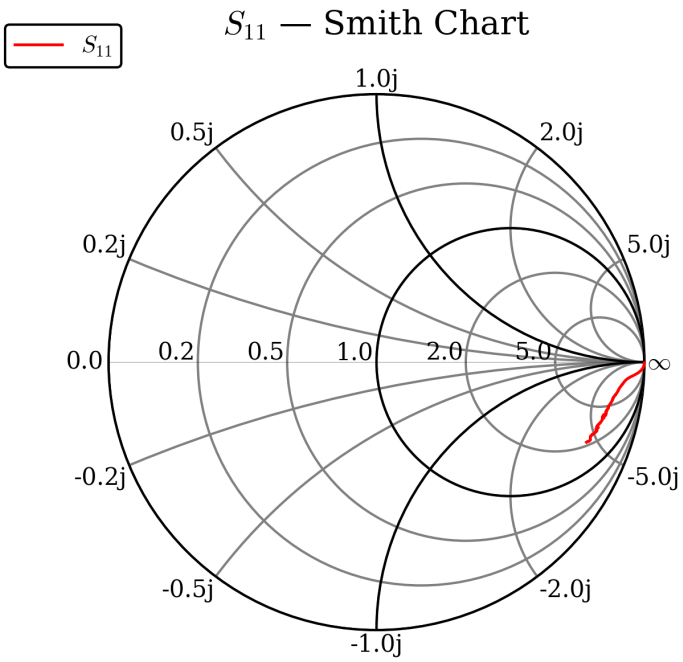
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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.610	5.2344	2.1266	0.4063
5.615	5.2310	2.1134	0.4040
5.620	5.2312	2.1218	0.4056
5.625	5.2337	2.1239	0.4058
5.630	5.2293	2.1185	0.4051
5.635	5.2291	2.1150	0.4045
5.640	5.2280	2.1038	0.4024
5.645	5.2182	2.1099	0.4043
5.650	5.2233	2.1144	0.4048
5.655	5.2188	2.1090	0.4041
5.660	5.2189	2.1133	0.4049
5.665	5.2290	2.1087	0.4033
5.670	5.2109	2.1118	0.4053
5.675	5.2172	2.1022	0.4029
5.680	5.2181	2.1068	0.4037
5.685	5.2205	2.0973	0.4017
5.690	5.2120	2.1023	0.4034
5.695	5.2160	2.1073	0.4040
5.700	5.2102	2.0891	0.4010
5.705	5.2022	2.0912	0.4020
5.710	5.2128	2.0877	0.4005
5.715	5.2069	2.0856	0.4005
5.720	5.2063	2.0911	0.4016
5.725	5.1953	2.0902	0.4023
5.730	5.1967	2.0865	0.4015
5.735	5.1964	2.0834	0.4009
5.740	5.1982	2.0868	0.4015
5.745	5.2033	2.0874	0.4012
5.750	5.1916	2.0837	0.4014
5.755	5.1977	2.0814	0.4004
5.760	5.1896	2.0767	0.4002
5.765	5.1966	2.0738	0.3991
5.770	5.1918	2.0689	0.3985
5.775	5.1928	2.0728	0.3992
5.780	5.1857	2.0726	0.3997
5.785	5.1845	2.0723	0.3997
5.790	5.1802	2.0732	0.4002
5.795	5.1835	2.0728	0.3999
5.800	5.1765	2.0707	0.4000
5.805	5.1777	2.0684	0.3995
5.810	5.1762	2.0687	0.3997
5.815	5.1764	2.0673	0.3994
5.820	5.1741	2.0644	0.3990
5.825	5.1729	2.0664	0.3995
5.830	5.1693	2.0664	0.3998
5.835	5.1711	2.0643	0.3992

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.840	5.1681	2.0642	0.3994
5.845	5.1654	2.0669	0.4001
5.850	5.1680	2.0607	0.3987
5.855	5.1654	2.0598	0.3988
5.860	5.1607	2.0615	0.3995
5.865	5.1676	2.0643	0.3995
5.870	5.1618	2.0548	0.3981
5.875	5.1619	2.0590	0.3989
5.880	5.1601	2.0575	0.3987
5.885	5.1639	2.0511	0.3972
5.890	5.1557	2.0601	0.3996
5.895	5.1540	2.0570	0.3991
5.900	5.1618	2.0505	0.3972
5.905	5.1597	2.0487	0.3971
5.910	5.1574	2.0486	0.3972
5.915	5.1546	2.0513	0.3979
5.920	5.1537	2.0505	0.3979
5.925	5.1571	2.0413	0.3958
5.930	5.1447	2.0483	0.3981
5.935	5.1499	2.0420	0.3965
5.940	5.1452	2.0499	0.3984
5.945	5.1576	2.0384	0.3952
5.950	5.1435	2.0487	0.3983
5.955	5.1491	2.0471	0.3976
5.960	5.1398	2.0415	0.3972
5.965	5.1414	2.0446	0.3977
5.970	5.1503	2.0519	0.3984
5.975	5.1518	2.0268	0.3934
5.980	5.1626	2.0431	0.3958
5.985	5.1368	2.0476	0.3986
5.990	5.1380	2.0452	0.3981
5.995	5.1337	2.0379	0.3970
6.000	5.1545	2.0418	0.3961

1.2 S_{11} — Smith Chart



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.005	0.9999	-0.0028	-0.00
0.010	0.9996	-0.0055	-0.00
0.015	0.9992	-0.0083	-0.01
0.020	0.9985	-0.0109	-0.01
0.025	0.9978	-0.0134	-0.02
0.030	0.9969	-0.0158	-0.03
0.035	0.9958	-0.0182	-0.03
0.040	0.9947	-0.0205	-0.04
0.045	0.9936	-0.0225	-0.05
0.050	0.9924	-0.0245	-0.06
0.055	0.9910	-0.0263	-0.08
0.060	0.9897	-0.0282	-0.09
0.065	0.9883	-0.0296	-0.10
0.070	0.9870	-0.0312	-0.11
0.075	0.9856	-0.0325	-0.12
0.080	0.9842	-0.0339	-0.13
0.085	0.9829	-0.0352	-0.14
0.090	0.9814	-0.0363	-0.16
0.095	0.9802	-0.0374	-0.17
0.100	0.9790	-0.0383	-0.18
0.105	0.9776	-0.0392	-0.19
0.110	0.9764	-0.0401	-0.20
0.115	0.9750	-0.0409	-0.21

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.120	0.9740	-0.0417	-0.22
0.125	0.9729	-0.0424	-0.23
0.130	0.9716	-0.0431	-0.24
0.135	0.9706	-0.0437	-0.25
0.140	0.9694	-0.0443	-0.26
0.145	0.9684	-0.0449	-0.27
0.150	0.9674	-0.0455	-0.28
0.155	0.9665	-0.0460	-0.29
0.160	0.9656	-0.0463	-0.29
0.165	0.9646	-0.0469	-0.30
0.170	0.9637	-0.0473	-0.31
0.175	0.9630	-0.0478	-0.32
0.180	0.9620	-0.0482	-0.33
0.185	0.9612	-0.0486	-0.33
0.190	0.9604	-0.0490	-0.34
0.195	0.9595	-0.0494	-0.35
0.200	0.9588	-0.0498	-0.35
0.205	0.9580	-0.0502	-0.36
0.210	0.9573	-0.0506	-0.37
0.215	0.9565	-0.0509	-0.37
0.220	0.9558	-0.0514	-0.38
0.225	0.9552	-0.0516	-0.39
0.230	0.9545	-0.0518	-0.39
0.235	0.9538	-0.0523	-0.40
0.240	0.9532	-0.0526	-0.40
0.245	0.9526	-0.0530	-0.41
0.250	0.9518	-0.0533	-0.42
0.255	0.9512	-0.0537	-0.42
0.260	0.9506	-0.0539	-0.43
0.265	0.9501	-0.0542	-0.43
0.270	0.9493	-0.0545	-0.44
0.275	0.9489	-0.0547	-0.44
0.280	0.9484	-0.0549	-0.45
0.285	0.9477	-0.0554	-0.45
0.290	0.9472	-0.0557	-0.46
0.295	0.9468	-0.0559	-0.46
0.300	0.9462	-0.0563	-0.46
0.305	0.9457	-0.0565	-0.47
0.310	0.9453	-0.0568	-0.47
0.315	0.9446	-0.0571	-0.48
0.320	0.9442	-0.0573	-0.48
0.325	0.9437	-0.0577	-0.49
0.330	0.9434	-0.0579	-0.49
0.335	0.9428	-0.0582	-0.49
0.340	0.9424	-0.0584	-0.50
0.345	0.9419	-0.0587	-0.50

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.350	0.9415	-0.0589	-0.51
0.355	0.9410	-0.0592	-0.51
0.360	0.9407	-0.0594	-0.51
0.365	0.9403	-0.0597	-0.52
0.370	0.9398	-0.0600	-0.52
0.375	0.9395	-0.0601	-0.52
0.380	0.9389	-0.0606	-0.53
0.385	0.9386	-0.0608	-0.53
0.390	0.9381	-0.0611	-0.54
0.395	0.9379	-0.0614	-0.54
0.400	0.9375	-0.0615	-0.54
0.405	0.9371	-0.0617	-0.55
0.410	0.9368	-0.0620	-0.55
0.415	0.9364	-0.0623	-0.55
0.420	0.9360	-0.0624	-0.56
0.425	0.9357	-0.0628	-0.56
0.430	0.9354	-0.0630	-0.56
0.435	0.9350	-0.0633	-0.56
0.440	0.9348	-0.0635	-0.57
0.445	0.9344	-0.0636	-0.57
0.450	0.9341	-0.0640	-0.57
0.455	0.9338	-0.0642	-0.57
0.460	0.9335	-0.0645	-0.58
0.465	0.9332	-0.0647	-0.58
0.470	0.9328	-0.0648	-0.58
0.475	0.9325	-0.0651	-0.59
0.480	0.9323	-0.0654	-0.59
0.485	0.9320	-0.0656	-0.59
0.490	0.9317	-0.0659	-0.59
0.495	0.9314	-0.0662	-0.60
0.500	0.9312	-0.0663	-0.60
0.505	0.9309	-0.0666	-0.60
0.510	0.9306	-0.0668	-0.60
0.515	0.9304	-0.0671	-0.60
0.520	0.9301	-0.0674	-0.61
0.525	0.9299	-0.0675	-0.61
0.530	0.9297	-0.0677	-0.61
0.535	0.9295	-0.0681	-0.61
0.540	0.9293	-0.0683	-0.61
0.545	0.9290	-0.0685	-0.62
0.550	0.9288	-0.0688	-0.62
0.555	0.9285	-0.0690	-0.62
0.560	0.9284	-0.0691	-0.62
0.565	0.9281	-0.0694	-0.62
0.570	0.9279	-0.0696	-0.63
0.575	0.9277	-0.0699	-0.63

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.580	0.9274	-0.0701	-0.63
0.585	0.9273	-0.0704	-0.63
0.590	0.9271	-0.0705	-0.63
0.595	0.9269	-0.0708	-0.63
0.600	0.9266	-0.0710	-0.64
0.605	0.9264	-0.0713	-0.64
0.610	0.9263	-0.0715	-0.64
0.615	0.9259	-0.0717	-0.64
0.620	0.9259	-0.0720	-0.64
0.625	0.9256	-0.0723	-0.64
0.630	0.9255	-0.0725	-0.65
0.635	0.9252	-0.0728	-0.65
0.640	0.9251	-0.0729	-0.65
0.645	0.9250	-0.0733	-0.65
0.650	0.9247	-0.0734	-0.65
0.655	0.9245	-0.0737	-0.65
0.660	0.9243	-0.0739	-0.66
0.665	0.9242	-0.0741	-0.66
0.670	0.9240	-0.0744	-0.66
0.675	0.9238	-0.0747	-0.66
0.680	0.9236	-0.0748	-0.66
0.685	0.9234	-0.0751	-0.66
0.690	0.9233	-0.0753	-0.66
0.695	0.9232	-0.0755	-0.67
0.700	0.9230	-0.0757	-0.67
0.705	0.9228	-0.0759	-0.67
0.710	0.9227	-0.0762	-0.67
0.715	0.9224	-0.0764	-0.67
0.720	0.9223	-0.0767	-0.67
0.725	0.9220	-0.0770	-0.67
0.730	0.9217	-0.0771	-0.68
0.735	0.9216	-0.0774	-0.68
0.740	0.9216	-0.0777	-0.68
0.745	0.9214	-0.0778	-0.68
0.750	0.9211	-0.0781	-0.68
0.755	0.9210	-0.0784	-0.68
0.760	0.9208	-0.0786	-0.69
0.765	0.9207	-0.0788	-0.69
0.770	0.9205	-0.0791	-0.69
0.775	0.9203	-0.0793	-0.69
0.780	0.9202	-0.0794	-0.69
0.785	0.9199	-0.0797	-0.69
0.790	0.9198	-0.0801	-0.69
0.795	0.9196	-0.0802	-0.70
0.800	0.9194	-0.0805	-0.70
0.805	0.9192	-0.0808	-0.70

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.810	0.9189	-0.0810	-0.70
0.815	0.9188	-0.0812	-0.70
0.820	0.9187	-0.0813	-0.70
0.825	0.9184	-0.0817	-0.71
0.830	0.9183	-0.0819	-0.71
0.835	0.9180	-0.0821	-0.71
0.840	0.9181	-0.0823	-0.71
0.845	0.9177	-0.0826	-0.71
0.850	0.9176	-0.0828	-0.71
0.855	0.9173	-0.0830	-0.71
0.860	0.9171	-0.0833	-0.72
0.865	0.9169	-0.0836	-0.72
0.870	0.9167	-0.0839	-0.72
0.875	0.9164	-0.0841	-0.72
0.880	0.9163	-0.0844	-0.72
0.885	0.9161	-0.0846	-0.72
0.890	0.9161	-0.0849	-0.72
0.895	0.9158	-0.0851	-0.73
0.900	0.9156	-0.0854	-0.73
0.905	0.9154	-0.0856	-0.73
0.910	0.9151	-0.0859	-0.73
0.915	0.9151	-0.0861	-0.73
0.920	0.9147	-0.0864	-0.74
0.925	0.9146	-0.0867	-0.74
0.930	0.9145	-0.0869	-0.74
0.935	0.9143	-0.0873	-0.74
0.940	0.9141	-0.0874	-0.74
0.945	0.9138	-0.0876	-0.74
0.950	0.9137	-0.0880	-0.74
0.955	0.9133	-0.0881	-0.75
0.960	0.9133	-0.0885	-0.75
0.965	0.9131	-0.0887	-0.75
0.970	0.9128	-0.0889	-0.75
0.975	0.9126	-0.0892	-0.75
0.980	0.9124	-0.0894	-0.76
0.985	0.9122	-0.0897	-0.76
0.990	0.9121	-0.0900	-0.76
0.995	0.9117	-0.0902	-0.76
1.000	0.9117	-0.0905	-0.76
1.005	0.9115	-0.0908	-0.76
1.010	0.9112	-0.0910	-0.76
1.015	0.9110	-0.0911	-0.77
1.020	0.9108	-0.0916	-0.77
1.025	0.9107	-0.0918	-0.77
1.030	0.9104	-0.0920	-0.77
1.035	0.9102	-0.0923	-0.77

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.040	0.9100	-0.0925	-0.77
1.045	0.9098	-0.0929	-0.78
1.050	0.9097	-0.0929	-0.78
1.055	0.9096	-0.0932	-0.78
1.060	0.9093	-0.0936	-0.78
1.065	0.9091	-0.0936	-0.78
1.070	0.9090	-0.0940	-0.78
1.075	0.9087	-0.0943	-0.78
1.080	0.9086	-0.0945	-0.79
1.085	0.9084	-0.0947	-0.79
1.090	0.9082	-0.0950	-0.79
1.095	0.9080	-0.0952	-0.79
1.100	0.9078	-0.0954	-0.79
1.105	0.9078	-0.0957	-0.79
1.110	0.9074	-0.0960	-0.80
1.115	0.9073	-0.0963	-0.80
1.120	0.9071	-0.0966	-0.80
1.125	0.9069	-0.0967	-0.80
1.130	0.9068	-0.0969	-0.80
1.135	0.9067	-0.0971	-0.80
1.140	0.9064	-0.0974	-0.80
1.145	0.9065	-0.0976	-0.80
1.150	0.9062	-0.0978	-0.80
1.155	0.9062	-0.0982	-0.80
1.160	0.9060	-0.0983	-0.81
1.165	0.9058	-0.0986	-0.81
1.170	0.9056	-0.0988	-0.81
1.175	0.9055	-0.0990	-0.81
1.180	0.9054	-0.0992	-0.81
1.185	0.9054	-0.0996	-0.81
1.190	0.9052	-0.0997	-0.81
1.195	0.9051	-0.0999	-0.81
1.200	0.9050	-0.1002	-0.81
1.205	0.9048	-0.1002	-0.82
1.210	0.9048	-0.1005	-0.82
1.215	0.9047	-0.1008	-0.82
1.220	0.9045	-0.1009	-0.82
1.225	0.9044	-0.1012	-0.82
1.230	0.9042	-0.1013	-0.82
1.235	0.9041	-0.1016	-0.82
1.240	0.9040	-0.1018	-0.82
1.245	0.9039	-0.1020	-0.82
1.250	0.9038	-0.1022	-0.82
1.255	0.9037	-0.1024	-0.82
1.260	0.9036	-0.1026	-0.82
1.265	0.9035	-0.1026	-0.83

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.270	0.9034	-0.1030	-0.83
1.275	0.9034	-0.1031	-0.83
1.280	0.9032	-0.1034	-0.83
1.285	0.9031	-0.1036	-0.83
1.290	0.9029	-0.1037	-0.83
1.295	0.9030	-0.1038	-0.83
1.300	0.9029	-0.1042	-0.83
1.305	0.9028	-0.1045	-0.83
1.310	0.9026	-0.1045	-0.83
1.315	0.9025	-0.1047	-0.83
1.320	0.9025	-0.1050	-0.83
1.325	0.9026	-0.1051	-0.83
1.330	0.9024	-0.1053	-0.83
1.335	0.9023	-0.1056	-0.83
1.340	0.9022	-0.1057	-0.83
1.345	0.9022	-0.1060	-0.83
1.350	0.9020	-0.1060	-0.84
1.355	0.9020	-0.1062	-0.84
1.360	0.9018	-0.1065	-0.84
1.365	0.9018	-0.1067	-0.84
1.370	0.9017	-0.1066	-0.84
1.375	0.9016	-0.1070	-0.84
1.380	0.9016	-0.1073	-0.84
1.385	0.9015	-0.1074	-0.84
1.390	0.9015	-0.1076	-0.84
1.395	0.9013	-0.1077	-0.84
1.400	0.9011	-0.1080	-0.84
1.405	0.9011	-0.1082	-0.84
1.410	0.9009	-0.1084	-0.84
1.415	0.9009	-0.1086	-0.84
1.420	0.9007	-0.1086	-0.85
1.425	0.9006	-0.1088	-0.85
1.430	0.9005	-0.1092	-0.85
1.435	0.9004	-0.1094	-0.85
1.440	0.9004	-0.1096	-0.85
1.445	0.9002	-0.1099	-0.85
1.450	0.9000	-0.1100	-0.85
1.455	0.8999	-0.1103	-0.85
1.460	0.8999	-0.1105	-0.85
1.465	0.8996	-0.1106	-0.85
1.470	0.8996	-0.1108	-0.85
1.475	0.8994	-0.1112	-0.86
1.480	0.8994	-0.1113	-0.86
1.485	0.8992	-0.1117	-0.86
1.490	0.8990	-0.1118	-0.86
1.495	0.8989	-0.1121	-0.86

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.500	0.8988	-0.1124	-0.86
1.505	0.8986	-0.1125	-0.86
1.510	0.8985	-0.1127	-0.86
1.515	0.8983	-0.1130	-0.86
1.520	0.8982	-0.1133	-0.86
1.525	0.8981	-0.1135	-0.86
1.530	0.8979	-0.1138	-0.87
1.535	0.8977	-0.1141	-0.87
1.540	0.8975	-0.1142	-0.87
1.545	0.8974	-0.1146	-0.87
1.550	0.8971	-0.1148	-0.87
1.555	0.8968	-0.1150	-0.88
1.560	0.8967	-0.1153	-0.88
1.565	0.8965	-0.1157	-0.88
1.570	0.8963	-0.1159	-0.88
1.575	0.8959	-0.1162	-0.88
1.580	0.8957	-0.1164	-0.88
1.585	0.8956	-0.1167	-0.88
1.590	0.8954	-0.1171	-0.89
1.595	0.8952	-0.1173	-0.89
1.600	0.8950	-0.1175	-0.89
1.605	0.8949	-0.1179	-0.89
1.610	0.8947	-0.1181	-0.89
1.615	0.8944	-0.1184	-0.89
1.620	0.8943	-0.1187	-0.89
1.625	0.8940	-0.1189	-0.90
1.630	0.8938	-0.1192	-0.90
1.635	0.8936	-0.1195	-0.90
1.640	0.8934	-0.1197	-0.90
1.645	0.8932	-0.1202	-0.90
1.650	0.8928	-0.1203	-0.91
1.655	0.8927	-0.1207	-0.91
1.660	0.8926	-0.1211	-0.91
1.665	0.8925	-0.1214	-0.91
1.670	0.8922	-0.1215	-0.91
1.675	0.8919	-0.1218	-0.91
1.680	0.8916	-0.1222	-0.92
1.685	0.8915	-0.1225	-0.92
1.690	0.8912	-0.1227	-0.92
1.695	0.8910	-0.1230	-0.92
1.700	0.8908	-0.1233	-0.92
1.705	0.8903	-0.1234	-0.93
1.710	0.8904	-0.1239	-0.93
1.715	0.8900	-0.1241	-0.93
1.720	0.8898	-0.1244	-0.93
1.725	0.8896	-0.1246	-0.93

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.730	0.8894	-0.1248	-0.93
1.735	0.8892	-0.1252	-0.93
1.740	0.8887	-0.1255	-0.94
1.745	0.8888	-0.1258	-0.94
1.750	0.8884	-0.1260	-0.94
1.755	0.8883	-0.1263	-0.94
1.760	0.8879	-0.1266	-0.95
1.765	0.8879	-0.1270	-0.94
1.770	0.8877	-0.1272	-0.95
1.775	0.8874	-0.1273	-0.95
1.780	0.8874	-0.1276	-0.95
1.785	0.8870	-0.1279	-0.95
1.790	0.8870	-0.1281	-0.95
1.795	0.8867	-0.1282	-0.95
1.800	0.8863	-0.1285	-0.96
1.805	0.8862	-0.1287	-0.96
1.810	0.8862	-0.1291	-0.96
1.815	0.8862	-0.1295	-0.96
1.820	0.8857	-0.1296	-0.96
1.825	0.8856	-0.1299	-0.96
1.830	0.8856	-0.1300	-0.96
1.835	0.8854	-0.1304	-0.96
1.840	0.8852	-0.1305	-0.97
1.845	0.8851	-0.1308	-0.97
1.850	0.8850	-0.1309	-0.97
1.855	0.8849	-0.1311	-0.97
1.860	0.8846	-0.1313	-0.97
1.865	0.8844	-0.1315	-0.97
1.870	0.8844	-0.1316	-0.97
1.875	0.8840	-0.1319	-0.98
1.880	0.8839	-0.1321	-0.98
1.885	0.8839	-0.1324	-0.98
1.890	0.8839	-0.1327	-0.98
1.895	0.8838	-0.1328	-0.98
1.900	0.8837	-0.1330	-0.98
1.905	0.8835	-0.1331	-0.98
1.910	0.8834	-0.1332	-0.98
1.915	0.8832	-0.1334	-0.98
1.920	0.8832	-0.1337	-0.98
1.925	0.8830	-0.1339	-0.98
1.930	0.8829	-0.1341	-0.98
1.935	0.8829	-0.1343	-0.98
1.940	0.8829	-0.1346	-0.98
1.945	0.8827	-0.1347	-0.98
1.950	0.8828	-0.1348	-0.98
1.955	0.8828	-0.1350	-0.98

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.960	0.8827	-0.1352	-0.98
1.965	0.8825	-0.1355	-0.98
1.970	0.8825	-0.1356	-0.98
1.975	0.8826	-0.1359	-0.98
1.980	0.8824	-0.1361	-0.99
1.985	0.8823	-0.1360	-0.99
1.990	0.8824	-0.1362	-0.98
1.995	0.8823	-0.1362	-0.99
2.000	0.8824	-0.1365	-0.98
2.005	0.8822	-0.1368	-0.99
2.010	0.8821	-0.1370	-0.99
2.015	0.8820	-0.1371	-0.99
2.020	0.8820	-0.1373	-0.99
2.025	0.8820	-0.1374	-0.99
2.030	0.8819	-0.1375	-0.99
2.035	0.8820	-0.1379	-0.99
2.040	0.8820	-0.1379	-0.99
2.045	0.8818	-0.1381	-0.99
2.050	0.8818	-0.1383	-0.99
2.055	0.8817	-0.1384	-0.99
2.060	0.8817	-0.1387	-0.99
2.065	0.8816	-0.1388	-0.99
2.070	0.8816	-0.1390	-0.99
2.075	0.8814	-0.1391	-0.99
2.080	0.8815	-0.1392	-0.99
2.085	0.8815	-0.1396	-0.99
2.090	0.8815	-0.1397	-0.99
2.095	0.8811	-0.1397	-0.99
2.100	0.8811	-0.1400	-0.99
2.105	0.8811	-0.1401	-0.99
2.110	0.8811	-0.1404	-0.99
2.115	0.8809	-0.1406	-0.99
2.120	0.8809	-0.1410	-0.99
2.125	0.8807	-0.1412	-0.99
2.130	0.8807	-0.1413	-0.99
2.135	0.8805	-0.1415	-1.00
2.140	0.8804	-0.1417	-1.00
2.145	0.8803	-0.1419	-1.00
2.150	0.8803	-0.1424	-1.00
2.155	0.8802	-0.1425	-1.00
2.160	0.8801	-0.1428	-1.00
2.165	0.8797	-0.1431	-1.00
2.170	0.8798	-0.1433	-1.00
2.175	0.8795	-0.1435	-1.00
2.180	0.8795	-0.1438	-1.00
2.185	0.8793	-0.1441	-1.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.190	0.8792	-0.1443	-1.00
2.195	0.8790	-0.1446	-1.00
2.200	0.8787	-0.1450	-1.01
2.205	0.8787	-0.1451	-1.01
2.210	0.8785	-0.1456	-1.01
2.215	0.8784	-0.1458	-1.01
2.220	0.8781	-0.1463	-1.01
2.225	0.8780	-0.1463	-1.01
2.230	0.8778	-0.1466	-1.01
2.235	0.8777	-0.1470	-1.01
2.240	0.8774	-0.1473	-1.02
2.245	0.8773	-0.1475	-1.02
2.250	0.8770	-0.1478	-1.02
2.255	0.8768	-0.1481	-1.02
2.260	0.8767	-0.1485	-1.02
2.265	0.8763	-0.1489	-1.02
2.270	0.8763	-0.1492	-1.02
2.275	0.8761	-0.1495	-1.02
2.280	0.8760	-0.1499	-1.02
2.285	0.8756	-0.1502	-1.03
2.290	0.8754	-0.1504	-1.03
2.295	0.8751	-0.1508	-1.03
2.300	0.8751	-0.1513	-1.03
2.305	0.8749	-0.1518	-1.03
2.310	0.8746	-0.1519	-1.03
2.315	0.8744	-0.1523	-1.04
2.320	0.8742	-0.1525	-1.04
2.325	0.8741	-0.1529	-1.04
2.330	0.8735	-0.1530	-1.04
2.335	0.8735	-0.1535	-1.04
2.340	0.8732	-0.1539	-1.05
2.345	0.8732	-0.1545	-1.04
2.350	0.8730	-0.1548	-1.05
2.355	0.8724	-0.1550	-1.05
2.360	0.8724	-0.1553	-1.05
2.365	0.8722	-0.1556	-1.05
2.370	0.8718	-0.1560	-1.05
2.375	0.8715	-0.1562	-1.06
2.380	0.8715	-0.1566	-1.06
2.385	0.8713	-0.1568	-1.06
2.390	0.8710	-0.1572	-1.06
2.395	0.8709	-0.1576	-1.06
2.400	0.8707	-0.1578	-1.06
2.405	0.8705	-0.1582	-1.06
2.410	0.8702	-0.1585	-1.07
2.415	0.8700	-0.1587	-1.07

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.420	0.8699	-0.1591	-1.07
2.425	0.8696	-0.1594	-1.07
2.430	0.8695	-0.1597	-1.07
2.435	0.8693	-0.1600	-1.07
2.440	0.8690	-0.1602	-1.07
2.445	0.8689	-0.1604	-1.07
2.450	0.8687	-0.1607	-1.08
2.455	0.8687	-0.1609	-1.08
2.460	0.8685	-0.1612	-1.08
2.465	0.8683	-0.1614	-1.08
2.470	0.8681	-0.1617	-1.08
2.475	0.8680	-0.1620	-1.08
2.480	0.8679	-0.1622	-1.08
2.485	0.8676	-0.1623	-1.08
2.490	0.8675	-0.1625	-1.09
2.495	0.8673	-0.1628	-1.09
2.500	0.8671	-0.1630	-1.09
2.505	0.8671	-0.1632	-1.09
2.510	0.8668	-0.1635	-1.09
2.515	0.8667	-0.1637	-1.09
2.520	0.8666	-0.1639	-1.09
2.525	0.8663	-0.1642	-1.09
2.530	0.8661	-0.1643	-1.10
2.535	0.8660	-0.1646	-1.10
2.540	0.8658	-0.1648	-1.10
2.545	0.8658	-0.1650	-1.10
2.550	0.8656	-0.1653	-1.10
2.555	0.8655	-0.1655	-1.10
2.560	0.8653	-0.1656	-1.10
2.565	0.8651	-0.1656	-1.10
2.570	0.8650	-0.1657	-1.10
2.575	0.8652	-0.1658	-1.10
2.580	0.8649	-0.1659	-1.10
2.585	0.8650	-0.1659	-1.10
2.590	0.8652	-0.1660	-1.10
2.595	0.8652	-0.1662	-1.10
2.600	0.8654	-0.1664	-1.10
2.605	0.8654	-0.1666	-1.10
2.610	0.8652	-0.1669	-1.10
2.615	0.8655	-0.1673	-1.10
2.620	0.8655	-0.1673	-1.10
2.625	0.8656	-0.1675	-1.09
2.630	0.8655	-0.1677	-1.09
2.635	0.8654	-0.1678	-1.10
2.640	0.8655	-0.1681	-1.09
2.645	0.8654	-0.1681	-1.10

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.650	0.8654	-0.1682	-1.09
2.655	0.8654	-0.1683	-1.09
2.660	0.8655	-0.1687	-1.09
2.665	0.8654	-0.1687	-1.09
2.670	0.8652	-0.1692	-1.09
2.675	0.8652	-0.1693	-1.09
2.680	0.8653	-0.1694	-1.09
2.685	0.8652	-0.1696	-1.09
2.690	0.8650	-0.1697	-1.10
2.695	0.8650	-0.1700	-1.10
2.700	0.8652	-0.1703	-1.09
2.705	0.8650	-0.1704	-1.09
2.710	0.8650	-0.1705	-1.09
2.715	0.8650	-0.1707	-1.09
2.720	0.8649	-0.1708	-1.09
2.725	0.8651	-0.1710	-1.09
2.730	0.8652	-0.1711	-1.09
2.735	0.8651	-0.1712	-1.09
2.740	0.8651	-0.1713	-1.09
2.745	0.8650	-0.1714	-1.09
2.750	0.8651	-0.1716	-1.09
2.755	0.8649	-0.1716	-1.09
2.760	0.8647	-0.1717	-1.09
2.765	0.8647	-0.1719	-1.09
2.770	0.8646	-0.1720	-1.10
2.775	0.8646	-0.1721	-1.09
2.780	0.8645	-0.1723	-1.10
2.785	0.8643	-0.1723	-1.10
2.790	0.8643	-0.1725	-1.10
2.795	0.8641	-0.1726	-1.10
2.800	0.8641	-0.1727	-1.10
2.805	0.8639	-0.1728	-1.10
2.810	0.8638	-0.1731	-1.10
2.815	0.8637	-0.1732	-1.10
2.820	0.8635	-0.1734	-1.10
2.825	0.8636	-0.1736	-1.10
2.830	0.8634	-0.1739	-1.10
2.835	0.8631	-0.1740	-1.11
2.840	0.8632	-0.1742	-1.10
2.845	0.8630	-0.1744	-1.11
2.850	0.8628	-0.1748	-1.11
2.855	0.8628	-0.1751	-1.11
2.860	0.8628	-0.1755	-1.11
2.865	0.8627	-0.1754	-1.11
2.870	0.8624	-0.1757	-1.11
2.875	0.8625	-0.1757	-1.11

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.880	0.8622	-0.1759	-1.11
2.885	0.8623	-0.1762	-1.11
2.890	0.8623	-0.1762	-1.11
2.895	0.8620	-0.1763	-1.11
2.900	0.8618	-0.1765	-1.11
2.905	0.8616	-0.1766	-1.11
2.910	0.8613	-0.1768	-1.12
2.915	0.8611	-0.1769	-1.12
2.920	0.8609	-0.1774	-1.12
2.925	0.8606	-0.1774	-1.12
2.930	0.8605	-0.1775	-1.12
2.935	0.8601	-0.1780	-1.13
2.940	0.8598	-0.1782	-1.13
2.945	0.8596	-0.1786	-1.13
2.950	0.8597	-0.1788	-1.13
2.955	0.8592	-0.1791	-1.13
2.960	0.8591	-0.1794	-1.13
2.965	0.8590	-0.1797	-1.13
2.970	0.8588	-0.1801	-1.14
2.975	0.8588	-0.1803	-1.13
2.980	0.8587	-0.1806	-1.14
2.985	0.8588	-0.1810	-1.13
2.990	0.8586	-0.1811	-1.14
2.995	0.8585	-0.1816	-1.13
3.000	0.8588	-0.1815	-1.13
3.005	0.8580	-0.1818	-1.14
3.010	0.8582	-0.1818	-1.14
3.015	0.8580	-0.1823	-1.14
3.020	0.8583	-0.1830	-1.13
3.025	0.8581	-0.1825	-1.14
3.030	0.8585	-0.1834	-1.13
3.035	0.8585	-0.1833	-1.13
3.040	0.8585	-0.1832	-1.13
3.045	0.8588	-0.1836	-1.13
3.050	0.8585	-0.1834	-1.13
3.055	0.8588	-0.1836	-1.13
3.060	0.8590	-0.1835	-1.13
3.065	0.8588	-0.1833	-1.13
3.070	0.8588	-0.1831	-1.13
3.075	0.8587	-0.1831	-1.13
3.080	0.8584	-0.1832	-1.13
3.085	0.8583	-0.1829	-1.13
3.090	0.8580	-0.1830	-1.14
3.095	0.8578	-0.1830	-1.14
3.100	0.8575	-0.1828	-1.14
3.105	0.8572	-0.1831	-1.14

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.110	0.8570	-0.1832	-1.15
3.115	0.8567	-0.1834	-1.15
3.120	0.8564	-0.1835	-1.15
3.125	0.8562	-0.1838	-1.15
3.130	0.8561	-0.1841	-1.15
3.135	0.8559	-0.1844	-1.15
3.140	0.8558	-0.1846	-1.16
3.145	0.8557	-0.1851	-1.15
3.150	0.8557	-0.1854	-1.15
3.155	0.8558	-0.1855	-1.15
3.160	0.8559	-0.1859	-1.15
3.165	0.8560	-0.1862	-1.15
3.170	0.8563	-0.1864	-1.15
3.175	0.8560	-0.1868	-1.15
3.180	0.8566	-0.1870	-1.14
3.185	0.8566	-0.1873	-1.14
3.190	0.8568	-0.1871	-1.14
3.195	0.8571	-0.1871	-1.14
3.200	0.8574	-0.1869	-1.13
3.205	0.8575	-0.1869	-1.13
3.210	0.8572	-0.1868	-1.14
3.215	0.8575	-0.1865	-1.13
3.220	0.8571	-0.1865	-1.14
3.225	0.8569	-0.1864	-1.14
3.230	0.8566	-0.1863	-1.14
3.235	0.8565	-0.1865	-1.14
3.240	0.8563	-0.1866	-1.15
3.245	0.8565	-0.1869	-1.14
3.250	0.8561	-0.1869	-1.15
3.255	0.8559	-0.1875	-1.15
3.260	0.8559	-0.1878	-1.15
3.265	0.8563	-0.1876	-1.14
3.270	0.8562	-0.1883	-1.14
3.275	0.8562	-0.1884	-1.14
3.280	0.8563	-0.1887	-1.14
3.285	0.8563	-0.1888	-1.14
3.290	0.8567	-0.1890	-1.14
3.295	0.8565	-0.1893	-1.14
3.300	0.8565	-0.1895	-1.14
3.305	0.8564	-0.1897	-1.14
3.310	0.8568	-0.1898	-1.13
3.315	0.8567	-0.1900	-1.14
3.320	0.8569	-0.1904	-1.13
3.325	0.8568	-0.1905	-1.13
3.330	0.8571	-0.1907	-1.13
3.335	0.8570	-0.1909	-1.13

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.340	0.8572	-0.1912	-1.13
3.345	0.8571	-0.1911	-1.13
3.350	0.8575	-0.1915	-1.12
3.355	0.8574	-0.1916	-1.12
3.360	0.8577	-0.1916	-1.12
3.365	0.8576	-0.1915	-1.12
3.370	0.8577	-0.1918	-1.12
3.375	0.8577	-0.1918	-1.12
3.380	0.8575	-0.1918	-1.12
3.385	0.8578	-0.1917	-1.12
3.390	0.8577	-0.1921	-1.12
3.395	0.8576	-0.1920	-1.12
3.400	0.8576	-0.1922	-1.12
3.405	0.8575	-0.1920	-1.12
3.410	0.8573	-0.1923	-1.12
3.415	0.8570	-0.1924	-1.13
3.420	0.8570	-0.1923	-1.13
3.425	0.8567	-0.1924	-1.13
3.430	0.8565	-0.1925	-1.13
3.435	0.8564	-0.1928	-1.13
3.440	0.8561	-0.1929	-1.13
3.445	0.8558	-0.1931	-1.14
3.450	0.8554	-0.1930	-1.14
3.455	0.8553	-0.1934	-1.14
3.460	0.8550	-0.1938	-1.14
3.465	0.8549	-0.1940	-1.14
3.470	0.8546	-0.1943	-1.15
3.475	0.8543	-0.1948	-1.15
3.480	0.8541	-0.1952	-1.15
3.485	0.8541	-0.1956	-1.15
3.490	0.8542	-0.1961	-1.15
3.495	0.8538	-0.1961	-1.15
3.500	0.8537	-0.1968	-1.15
3.505	0.8538	-0.1966	-1.15
3.510	0.8540	-0.1978	-1.14
3.515	0.8539	-0.1977	-1.15
3.520	0.8542	-0.1979	-1.14
3.525	0.8542	-0.1986	-1.14
3.530	0.8543	-0.1994	-1.14
3.535	0.8550	-0.1989	-1.13
3.540	0.8557	-0.1984	-1.13
3.545	0.8552	-0.1984	-1.13
3.550	0.8554	-0.1974	-1.13
3.555	0.8553	-0.1971	-1.13
3.560	0.8547	-0.1966	-1.14
3.565	0.8542	-0.1965	-1.14

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.570	0.8537	-0.1958	-1.15
3.575	0.8526	-0.1950	-1.16
3.580	0.8518	-0.1952	-1.17
3.585	0.8500	-0.1954	-1.19
3.590	0.8489	-0.1953	-1.20
3.595	0.8480	-0.1961	-1.21
3.600	0.8472	-0.1974	-1.21
3.605	0.8463	-0.1984	-1.22
3.610	0.8464	-0.1998	-1.21
3.615	0.8466	-0.2007	-1.21
3.620	0.8472	-0.2016	-1.20
3.625	0.8470	-0.2018	-1.20
3.630	0.8477	-0.2027	-1.19
3.635	0.8486	-0.2029	-1.18
3.640	0.8490	-0.2031	-1.18
3.645	0.8494	-0.2030	-1.18
3.650	0.8495	-0.2024	-1.18
3.655	0.8495	-0.2024	-1.18
3.660	0.8495	-0.2025	-1.18
3.665	0.8494	-0.2023	-1.18
3.670	0.8492	-0.2021	-1.18
3.675	0.8495	-0.2017	-1.18
3.680	0.8491	-0.2017	-1.18
3.685	0.8490	-0.2017	-1.18
3.690	0.8488	-0.2014	-1.19
3.695	0.8486	-0.2014	-1.19
3.700	0.8483	-0.2013	-1.19
3.705	0.8479	-0.2012	-1.20
3.710	0.8475	-0.2011	-1.20
3.715	0.8470	-0.2009	-1.20
3.720	0.8466	-0.2010	-1.21
3.725	0.8460	-0.2012	-1.21
3.730	0.8451	-0.2012	-1.22
3.735	0.8444	-0.2018	-1.23
3.740	0.8437	-0.2025	-1.23
3.745	0.8433	-0.2035	-1.23
3.750	0.8429	-0.2046	-1.24
3.755	0.8427	-0.2058	-1.23
3.760	0.8433	-0.2072	-1.23
3.765	0.8440	-0.2083	-1.22
3.770	0.8458	-0.2088	-1.20
3.775	0.8471	-0.2087	-1.19
3.780	0.8487	-0.2089	-1.17
3.785	0.8492	-0.2065	-1.17
3.790	0.8486	-0.2054	-1.18
3.795	0.8486	-0.2048	-1.18

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.800	0.8477	-0.2041	-1.19
3.805	0.8472	-0.2037	-1.20
3.810	0.8464	-0.2039	-1.20
3.815	0.8457	-0.2040	-1.21
3.820	0.8451	-0.2044	-1.22
3.825	0.8445	-0.2047	-1.22
3.830	0.8440	-0.2051	-1.22
3.835	0.8440	-0.2071	-1.22
3.840	0.8439	-0.2074	-1.22
3.845	0.8443	-0.2077	-1.21
3.850	0.8443	-0.2079	-1.21
3.855	0.8439	-0.2080	-1.22
3.860	0.8441	-0.2084	-1.22
3.865	0.8439	-0.2088	-1.22
3.870	0.8442	-0.2089	-1.21
3.875	0.8440	-0.2091	-1.21
3.880	0.8440	-0.2093	-1.21
3.885	0.8441	-0.2096	-1.21
3.890	0.8440	-0.2096	-1.21
3.895	0.8441	-0.2101	-1.21
3.900	0.8437	-0.2101	-1.21
3.905	0.8439	-0.2104	-1.21
3.910	0.8437	-0.2107	-1.21
3.915	0.8436	-0.2110	-1.21
3.920	0.8437	-0.2113	-1.21
3.925	0.8436	-0.2119	-1.21
3.930	0.8435	-0.2118	-1.21
3.935	0.8433	-0.2122	-1.21
3.940	0.8432	-0.2125	-1.21
3.945	0.8431	-0.2129	-1.21
3.950	0.8430	-0.2133	-1.21
3.955	0.8429	-0.2138	-1.21
3.960	0.8429	-0.2139	-1.21
3.965	0.8425	-0.2144	-1.22
3.970	0.8427	-0.2145	-1.21
3.975	0.8424	-0.2149	-1.22
3.980	0.8422	-0.2152	-1.22
3.985	0.8420	-0.2155	-1.22
3.990	0.8418	-0.2160	-1.22
3.995	0.8418	-0.2163	-1.22
4.000	0.8415	-0.2168	-1.22
4.005	0.8413	-0.2173	-1.22
4.010	0.8411	-0.2178	-1.22
4.015	0.8408	-0.2180	-1.22
4.020	0.8405	-0.2185	-1.22
4.025	0.8401	-0.2186	-1.23

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.030	0.8401	-0.2191	-1.23
4.035	0.8399	-0.2195	-1.23
4.040	0.8397	-0.2201	-1.23
4.045	0.8394	-0.2205	-1.23
4.050	0.8391	-0.2208	-1.23
4.055	0.8387	-0.2212	-1.24
4.060	0.8384	-0.2217	-1.24
4.065	0.8384	-0.2217	-1.24
4.070	0.8379	-0.2222	-1.24
4.075	0.8376	-0.2228	-1.24
4.080	0.8371	-0.2232	-1.25
4.085	0.8370	-0.2235	-1.25
4.090	0.8366	-0.2238	-1.25
4.095	0.8362	-0.2242	-1.25
4.100	0.8358	-0.2245	-1.26
4.105	0.8351	-0.2250	-1.26
4.110	0.8353	-0.2254	-1.26
4.115	0.8352	-0.2258	-1.26
4.120	0.8348	-0.2268	-1.26
4.125	0.8344	-0.2269	-1.26
4.130	0.8337	-0.2274	-1.27
4.135	0.8332	-0.2279	-1.27
4.140	0.8333	-0.2283	-1.27
4.145	0.8328	-0.2284	-1.27
4.150	0.8323	-0.2294	-1.28
4.155	0.8318	-0.2300	-1.28
4.160	0.8312	-0.2303	-1.28
4.165	0.8306	-0.2305	-1.29
4.170	0.8304	-0.2310	-1.29
4.175	0.8303	-0.2315	-1.29
4.180	0.8296	-0.2315	-1.30
4.185	0.8289	-0.2319	-1.30
4.190	0.8286	-0.2322	-1.30
4.195	0.8284	-0.2326	-1.31
4.200	0.8285	-0.2332	-1.30
4.205	0.8279	-0.2332	-1.31
4.210	0.8275	-0.2334	-1.31
4.215	0.8275	-0.2339	-1.31
4.220	0.8272	-0.2341	-1.31
4.225	0.8269	-0.2344	-1.31
4.230	0.8265	-0.2347	-1.32
4.235	0.8264	-0.2348	-1.32
4.240	0.8262	-0.2351	-1.32
4.245	0.8259	-0.2351	-1.32
4.250	0.8258	-0.2356	-1.32
4.255	0.8257	-0.2357	-1.32

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.260	0.8253	-0.2359	-1.33
4.265	0.8251	-0.2362	-1.33
4.270	0.8251	-0.2364	-1.33
4.275	0.8249	-0.2366	-1.33
4.280	0.8249	-0.2367	-1.33
4.285	0.8248	-0.2369	-1.33
4.290	0.8246	-0.2372	-1.33
4.295	0.8245	-0.2373	-1.33
4.300	0.8244	-0.2374	-1.33
4.305	0.8243	-0.2376	-1.33
4.310	0.8241	-0.2377	-1.33
4.315	0.8240	-0.2378	-1.33
4.320	0.8238	-0.2380	-1.34
4.325	0.8236	-0.2379	-1.34
4.330	0.8234	-0.2383	-1.34
4.335	0.8233	-0.2384	-1.34
4.340	0.8231	-0.2386	-1.34
4.345	0.8232	-0.2387	-1.34
4.350	0.8228	-0.2391	-1.34
4.355	0.8228	-0.2395	-1.34
4.360	0.8229	-0.2395	-1.34
4.365	0.8229	-0.2399	-1.34
4.370	0.8225	-0.2402	-1.34
4.375	0.8227	-0.2404	-1.34
4.380	0.8228	-0.2405	-1.34
4.385	0.8226	-0.2410	-1.34
4.390	0.8228	-0.2411	-1.34
4.395	0.8232	-0.2411	-1.33
4.400	0.8231	-0.2414	-1.33
4.405	0.8232	-0.2414	-1.33
4.410	0.8236	-0.2416	-1.33
4.415	0.8234	-0.2415	-1.33
4.420	0.8237	-0.2417	-1.33
4.425	0.8239	-0.2417	-1.32
4.430	0.8236	-0.2415	-1.33
4.435	0.8234	-0.2416	-1.33
4.440	0.8235	-0.2417	-1.33
4.445	0.8233	-0.2413	-1.33
4.450	0.8233	-0.2417	-1.33
4.455	0.8233	-0.2418	-1.33
4.460	0.8232	-0.2417	-1.33
4.465	0.8230	-0.2416	-1.33
4.470	0.8229	-0.2414	-1.33
4.475	0.8234	-0.2413	-1.33
4.480	0.8233	-0.2415	-1.33
4.485	0.8234	-0.2416	-1.33

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.490	0.8239	-0.2420	-1.32
4.495	0.8241	-0.2420	-1.32
4.500	0.8246	-0.2422	-1.32
4.505	0.8247	-0.2426	-1.31
4.510	0.8253	-0.2426	-1.31
4.515	0.8256	-0.2429	-1.30
4.520	0.8260	-0.2430	-1.30
4.525	0.8262	-0.2429	-1.30
4.530	0.8265	-0.2430	-1.30
4.535	0.8265	-0.2431	-1.29
4.540	0.8265	-0.2429	-1.30
4.545	0.8266	-0.2432	-1.29
4.550	0.8266	-0.2431	-1.29
4.555	0.8268	-0.2431	-1.29
4.560	0.8268	-0.2433	-1.29
4.565	0.8269	-0.2434	-1.29
4.570	0.8272	-0.2434	-1.29
4.575	0.8270	-0.2433	-1.29
4.580	0.8272	-0.2434	-1.29
4.585	0.8272	-0.2435	-1.29
4.590	0.8271	-0.2436	-1.29
4.595	0.8272	-0.2434	-1.29
4.600	0.8271	-0.2436	-1.29
4.605	0.8271	-0.2437	-1.29
4.610	0.8270	-0.2439	-1.29
4.615	0.8269	-0.2439	-1.29
4.620	0.8266	-0.2443	-1.29
4.625	0.8265	-0.2441	-1.29
4.630	0.8265	-0.2445	-1.29
4.635	0.8263	-0.2447	-1.29
4.640	0.8265	-0.2448	-1.29
4.645	0.8261	-0.2451	-1.29
4.650	0.8263	-0.2453	-1.29
4.655	0.8257	-0.2455	-1.30
4.660	0.8261	-0.2456	-1.29
4.665	0.8261	-0.2457	-1.29
4.670	0.8255	-0.2456	-1.30
4.675	0.8256	-0.2459	-1.30
4.680	0.8255	-0.2460	-1.30
4.685	0.8254	-0.2460	-1.30
4.690	0.8255	-0.2465	-1.30
4.695	0.8256	-0.2465	-1.29
4.700	0.8253	-0.2466	-1.30
4.705	0.8250	-0.2467	-1.30
4.710	0.8254	-0.2467	-1.30
4.715	0.8250	-0.2469	-1.30

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.720	0.8249	-0.2473	-1.30
4.725	0.8248	-0.2472	-1.30
4.730	0.8253	-0.2471	-1.29
4.735	0.8251	-0.2467	-1.30
4.740	0.8245	-0.2473	-1.30
4.745	0.8239	-0.2469	-1.31
4.750	0.8233	-0.2472	-1.31
4.755	0.8235	-0.2477	-1.31
4.760	0.8231	-0.2477	-1.31
4.765	0.8224	-0.2483	-1.32
4.770	0.8225	-0.2482	-1.32
4.775	0.8217	-0.2485	-1.33
4.780	0.8216	-0.2491	-1.32
4.785	0.8216	-0.2494	-1.32
4.790	0.8211	-0.2500	-1.33
4.795	0.8211	-0.2499	-1.33
4.800	0.8208	-0.2499	-1.33
4.805	0.8208	-0.2505	-1.33
4.810	0.8209	-0.2499	-1.33
4.815	0.8206	-0.2499	-1.33
4.820	0.8204	-0.2499	-1.33
4.825	0.8207	-0.2500	-1.33
4.830	0.8203	-0.2499	-1.34
4.835	0.8200	-0.2495	-1.34
4.840	0.8202	-0.2496	-1.34
4.845	0.8198	-0.2496	-1.34
4.850	0.8199	-0.2494	-1.34
4.855	0.8201	-0.2490	-1.34
4.860	0.8196	-0.2491	-1.34
4.865	0.8193	-0.2493	-1.35
4.870	0.8195	-0.2494	-1.34
4.875	0.8195	-0.2491	-1.35
4.880	0.8193	-0.2492	-1.35
4.885	0.8193	-0.2490	-1.35
4.890	0.8193	-0.2493	-1.35
4.895	0.8191	-0.2494	-1.35
4.900	0.8191	-0.2495	-1.35
4.905	0.8190	-0.2495	-1.35
4.910	0.8191	-0.2495	-1.35
4.915	0.8191	-0.2497	-1.35
4.920	0.8192	-0.2498	-1.35
4.925	0.8192	-0.2500	-1.35
4.930	0.8193	-0.2500	-1.34
4.935	0.8193	-0.2503	-1.34
4.940	0.8193	-0.2504	-1.34
4.945	0.8194	-0.2505	-1.34

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.950	0.8196	-0.2508	-1.34
4.955	0.8196	-0.2509	-1.34
4.960	0.8196	-0.2512	-1.34
4.965	0.8198	-0.2514	-1.34
4.970	0.8198	-0.2516	-1.34
4.975	0.8197	-0.2519	-1.33
4.980	0.8200	-0.2521	-1.33
4.985	0.8199	-0.2524	-1.33
4.990	0.8201	-0.2528	-1.33
4.995	0.8198	-0.2530	-1.33
5.000	0.8201	-0.2533	-1.33
5.005	0.8199	-0.2538	-1.33
5.010	0.8200	-0.2538	-1.33
5.015	0.8200	-0.2540	-1.33
5.020	0.8201	-0.2543	-1.32
5.025	0.8203	-0.2549	-1.32
5.030	0.8202	-0.2547	-1.32
5.035	0.8203	-0.2552	-1.32
5.040	0.8204	-0.2555	-1.32
5.045	0.8206	-0.2554	-1.32
5.050	0.8204	-0.2559	-1.32
5.055	0.8200	-0.2560	-1.32
5.060	0.8202	-0.2563	-1.32
5.065	0.8201	-0.2563	-1.32
5.070	0.8200	-0.2561	-1.32
5.075	0.8197	-0.2561	-1.32
5.080	0.8195	-0.2558	-1.33
5.085	0.8195	-0.2555	-1.33
5.090	0.8196	-0.2551	-1.33
5.095	0.8195	-0.2549	-1.33
5.100	0.8196	-0.2546	-1.33
5.105	0.8200	-0.2541	-1.33
5.110	0.8199	-0.2540	-1.33
5.115	0.8200	-0.2540	-1.33
5.120	0.8204	-0.2538	-1.32
5.125	0.8206	-0.2538	-1.32
5.130	0.8209	-0.2538	-1.32
5.135	0.8210	-0.2539	-1.32
5.140	0.8214	-0.2538	-1.31
5.145	0.8215	-0.2540	-1.31
5.150	0.8220	-0.2542	-1.31
5.155	0.8222	-0.2544	-1.30
5.160	0.8224	-0.2547	-1.30
5.165	0.8226	-0.2548	-1.30
5.170	0.8227	-0.2550	-1.30
5.175	0.8227	-0.2550	-1.30

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.180	0.8229	-0.2554	-1.29
5.185	0.8228	-0.2556	-1.29
5.190	0.8227	-0.2558	-1.29
5.195	0.8227	-0.2559	-1.29
5.200	0.8226	-0.2562	-1.29
5.205	0.8227	-0.2564	-1.29
5.210	0.8225	-0.2566	-1.29
5.215	0.8226	-0.2569	-1.29
5.220	0.8222	-0.2572	-1.30
5.225	0.8221	-0.2573	-1.30
5.230	0.8221	-0.2576	-1.30
5.235	0.8218	-0.2578	-1.30
5.240	0.8218	-0.2579	-1.30
5.245	0.8215	-0.2582	-1.30
5.250	0.8212	-0.2583	-1.30
5.255	0.8208	-0.2585	-1.30
5.260	0.8206	-0.2587	-1.31
5.265	0.8201	-0.2588	-1.31
5.270	0.8201	-0.2591	-1.31
5.275	0.8197	-0.2591	-1.31
5.280	0.8191	-0.2597	-1.32
5.285	0.8186	-0.2595	-1.32
5.290	0.8185	-0.2599	-1.32
5.295	0.8181	-0.2601	-1.33
5.300	0.8178	-0.2605	-1.33
5.305	0.8177	-0.2613	-1.33
5.310	0.8171	-0.2612	-1.33
5.315	0.8168	-0.2616	-1.33
5.320	0.8169	-0.2615	-1.33
5.325	0.8162	-0.2622	-1.34
5.330	0.8163	-0.2624	-1.34
5.335	0.8158	-0.2632	-1.34
5.340	0.8155	-0.2630	-1.34
5.345	0.8151	-0.2630	-1.35
5.350	0.8152	-0.2636	-1.34
5.355	0.8148	-0.2636	-1.35
5.360	0.8149	-0.2639	-1.34
5.365	0.8143	-0.2643	-1.35
5.370	0.8138	-0.2640	-1.35
5.375	0.8135	-0.2643	-1.36
5.380	0.8131	-0.2644	-1.36
5.385	0.8121	-0.2652	-1.37
5.390	0.8124	-0.2651	-1.37
5.395	0.8119	-0.2651	-1.37
5.400	0.8112	-0.2655	-1.38
5.405	0.8112	-0.2661	-1.37

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.410	0.8105	-0.2662	-1.38
5.415	0.8099	-0.2665	-1.38
5.420	0.8097	-0.2670	-1.39
5.425	0.8094	-0.2675	-1.39
5.430	0.8087	-0.2677	-1.39
5.435	0.8085	-0.2678	-1.39
5.440	0.8081	-0.2684	-1.40
5.445	0.8077	-0.2685	-1.40
5.450	0.8077	-0.2689	-1.40
5.455	0.8074	-0.2690	-1.40
5.460	0.8072	-0.2697	-1.40
5.465	0.8068	-0.2696	-1.40
5.470	0.8069	-0.2698	-1.40
5.475	0.8064	-0.2703	-1.41
5.480	0.8062	-0.2700	-1.41
5.485	0.8060	-0.2703	-1.41
5.490	0.8054	-0.2704	-1.42
5.495	0.8054	-0.2707	-1.41
5.500	0.8047	-0.2707	-1.42
5.505	0.8043	-0.2709	-1.43
5.510	0.8039	-0.2713	-1.43
5.515	0.8035	-0.2717	-1.43
5.520	0.8032	-0.2718	-1.43
5.525	0.8031	-0.2721	-1.43
5.530	0.8030	-0.2724	-1.43
5.535	0.8027	-0.2728	-1.43
5.540	0.8024	-0.2730	-1.44
5.545	0.8023	-0.2732	-1.44
5.550	0.8020	-0.2735	-1.44
5.555	0.8018	-0.2736	-1.44
5.560	0.8017	-0.2739	-1.44
5.565	0.8016	-0.2740	-1.44
5.570	0.8015	-0.2742	-1.44
5.575	0.8013	-0.2744	-1.44
5.580	0.8010	-0.2747	-1.45
5.585	0.8007	-0.2748	-1.45
5.590	0.8003	-0.2749	-1.45
5.595	0.8001	-0.2752	-1.45
5.600	0.7998	-0.2755	-1.45
5.605	0.7995	-0.2757	-1.46
5.610	0.7991	-0.2763	-1.46
5.615	0.7992	-0.2767	-1.45
5.620	0.7986	-0.2771	-1.46
5.625	0.7987	-0.2777	-1.46
5.630	0.7984	-0.2781	-1.46
5.635	0.7984	-0.2785	-1.46

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.640	0.7986	-0.2790	-1.45
5.645	0.7984	-0.2790	-1.46
5.650	0.7984	-0.2796	-1.45
5.655	0.7983	-0.2800	-1.45
5.660	0.7982	-0.2803	-1.45
5.665	0.7981	-0.2806	-1.45
5.670	0.7982	-0.2812	-1.45
5.675	0.7981	-0.2815	-1.45
5.680	0.7979	-0.2823	-1.45
5.685	0.7979	-0.2825	-1.45
5.690	0.7977	-0.2828	-1.45
5.695	0.7979	-0.2833	-1.45
5.700	0.7981	-0.2838	-1.44
5.705	0.7980	-0.2842	-1.44
5.710	0.7980	-0.2845	-1.44
5.715	0.7978	-0.2848	-1.44
5.720	0.7977	-0.2852	-1.44
5.725	0.7979	-0.2854	-1.44
5.730	0.7979	-0.2858	-1.44
5.735	0.7982	-0.2860	-1.43
5.740	0.7981	-0.2867	-1.43
5.745	0.7983	-0.2871	-1.43
5.750	0.7983	-0.2874	-1.43
5.755	0.7982	-0.2876	-1.43
5.760	0.7983	-0.2880	-1.42
5.765	0.7985	-0.2883	-1.42
5.770	0.7983	-0.2884	-1.42
5.775	0.7984	-0.2887	-1.42
5.780	0.7985	-0.2890	-1.42
5.785	0.7984	-0.2894	-1.42
5.790	0.7985	-0.2895	-1.42
5.795	0.7984	-0.2898	-1.42
5.800	0.7984	-0.2899	-1.42
5.805	0.7982	-0.2902	-1.42
5.810	0.7983	-0.2904	-1.42
5.815	0.7983	-0.2906	-1.42
5.820	0.7981	-0.2907	-1.42
5.825	0.7977	-0.2910	-1.42
5.830	0.7976	-0.2911	-1.42
5.835	0.7974	-0.2913	-1.42
5.840	0.7972	-0.2914	-1.42
5.845	0.7970	-0.2916	-1.43
5.850	0.7968	-0.2919	-1.43
5.855	0.7965	-0.2921	-1.43
5.860	0.7962	-0.2921	-1.43
5.865	0.7957	-0.2925	-1.43

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.870	0.7957	-0.2926	-1.43
5.875	0.7951	-0.2929	-1.44
5.880	0.7949	-0.2931	-1.44
5.885	0.7944	-0.2934	-1.44
5.890	0.7942	-0.2938	-1.44
5.895	0.7938	-0.2937	-1.45
5.900	0.7933	-0.2944	-1.45
5.905	0.7935	-0.2946	-1.45
5.910	0.7926	-0.2949	-1.46
5.915	0.7924	-0.2950	-1.46
5.920	0.7920	-0.2953	-1.46
5.925	0.7919	-0.2958	-1.46
5.930	0.7917	-0.2957	-1.46
5.935	0.7912	-0.2961	-1.46
5.940	0.7906	-0.2958	-1.47
5.945	0.7905	-0.2962	-1.47
5.950	0.7897	-0.2961	-1.48
5.955	0.7893	-0.2962	-1.48
5.960	0.7893	-0.2961	-1.48
5.965	0.7886	-0.2963	-1.49
5.970	0.7874	-0.2970	-1.50
5.975	0.7869	-0.2972	-1.50
5.980	0.7862	-0.2971	-1.51
5.985	0.7857	-0.2972	-1.51
5.990	0.7849	-0.2977	-1.52
5.995	0.7847	-0.2978	-1.52
6.000	0.7835	-0.2989	-1.53

2 Measurement Quality

Reliable band: from 45 MHz (S11 error σ = 1.0e-03); the points below it are kept in the plot (shaded, outside the scale) and in the tables, but are not reliable.
Spike removal: on (isolated point, 2nd difference > 6 x local noise and > S11 error budget, not shared with another sweep); replaced 1 point(s): Short at 4885 MHz ($|\Delta S11|$ = 1.4e-03).
Outliers shared by several sweeps, left alone (instrument artefacts cancelled by the calibration): 3835 MHz.

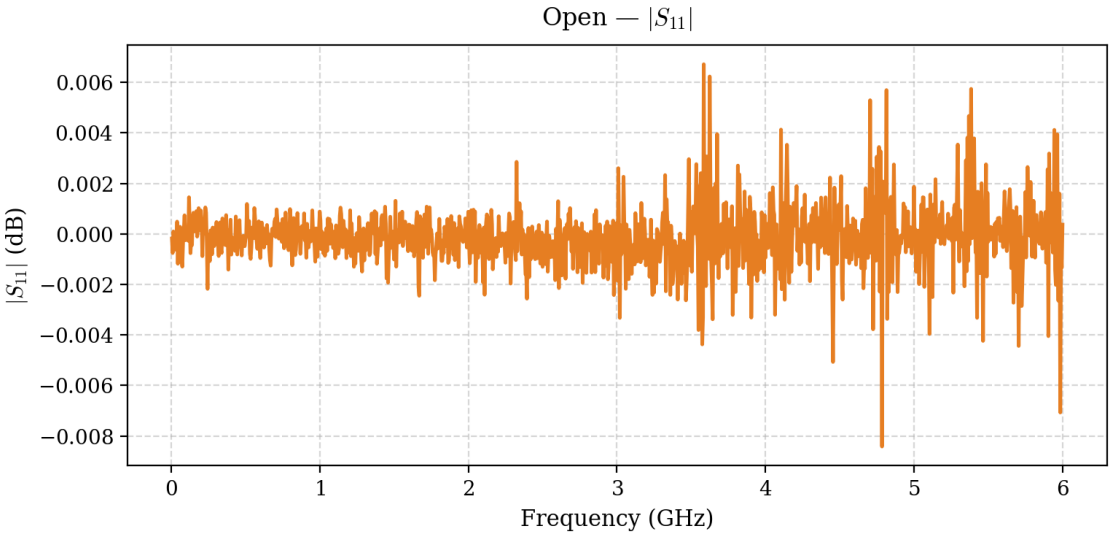
- Unreliable below 45 MHz: 8 point(s) (3 mm probe, S11 error σ = 1.0e-03). There the reference liquids resolve ε_r only to ±17.2 at 5 MHz … ±2.1 at 40 MHz per σ (usable: $\leq$ ±2). Values kept, but left out of the chart scale and of the summary medians; start the sweep at $\geq$ 45 MHz.

- Gn shows no radiation signature even at the top of the sweep (3 GHz – 6 GHz): mean arg(Gn) = -108° (radiation: -90°, median deviation 20°) and $|Gn| \propto f^{0.4}$ (radiation: f^3). Over the whole sweep it holds noise and the inconsistency between the reference liquids instead (its term is ≈7% of the Ref 1 (Distilled Water) admittance), so the full method's radiation correction is empirical here: compare with the simplified cross-check, especially for samples whose ε_r lies outside the range of the reference liquids.

- 2 frequency point(s) excluded: reference liquids nearly indistinguishable ($|S11_ref1 - S11_ref2| < 1e\text{-}02$). Consider starting the sweep above the affected range.

3 Calibration Standard Measurements

3.1 Open



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.005	1.0000	+0.0001	-0.00
0.010	0.9999	-0.0000	-0.00
0.015	1.0000	+0.0001	0.00
0.020	1.0000	+0.0000	-0.00
0.025	0.9999	+0.0000	-0.00
0.030	1.0000	+0.0001	-0.00
0.035	0.9999	+0.0001	-0.00
0.040	1.0001	+0.0000	0.00
0.045	0.9999	+0.0000	-0.00
0.050	1.0000	+0.0000	0.00
0.055	0.9999	+0.0000	-0.00
0.060	1.0000	-0.0001	0.00
0.065	1.0000	+0.0000	0.00
0.070	0.9999	-0.0000	-0.00
0.075	0.9999	-0.0000	-0.00
0.080	1.0000	+0.0001	-0.00
0.085	1.0000	-0.0000	0.00
0.090	1.0001	+0.0001	0.00
0.095	1.0000	+0.0000	-0.00
0.100	1.0000	-0.0000	0.00
0.105	1.0000	+0.0000	0.00
0.110	1.0000	+0.0001	-0.00
0.115	1.0000	+0.0001	-0.00
0.120	1.0002	+0.0001	0.00
0.125	1.0000	-0.0000	0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.130	0.9999	-0.0001	-0.00
0.135	1.0000	-0.0001	-0.00
0.140	0.9999	-0.0000	-0.00
0.145	1.0001	+0.0000	0.00
0.150	1.0001	-0.0000	0.00
0.155	1.0000	+0.0002	0.00
0.160	1.0001	+0.0001	0.00
0.165	1.0000	+0.0000	0.00
0.170	1.0001	+0.0000	0.00
0.175	1.0000	+0.0001	-0.00
0.180	1.0000	-0.0001	0.00
0.185	1.0001	+0.0000	0.00
0.190	1.0000	+0.0001	0.00
0.195	1.0000	+0.0001	0.00
0.200	1.0001	+0.0000	0.00
0.205	1.0001	-0.0001	0.00
0.210	0.9999	+0.0001	-0.00
0.215	1.0000	+0.0000	0.00
0.220	1.0000	-0.0000	0.00
0.225	1.0000	+0.0001	0.00
0.230	1.0001	+0.0001	0.00
0.235	1.0001	-0.0000	0.00
0.240	1.0000	+0.0001	-0.00
0.245	0.9997	+0.0000	-0.00
0.250	1.0000	+0.0002	-0.00
0.255	1.0000	-0.0001	-0.00
0.260	0.9999	+0.0000	-0.00
0.265	1.0000	+0.0000	0.00
0.270	1.0000	+0.0001	0.00
0.275	0.9999	+0.0000	-0.00
0.280	1.0000	+0.0001	0.00
0.285	1.0000	-0.0001	-0.00
0.290	0.9999	+0.0000	-0.00
0.295	1.0000	-0.0000	-0.00
0.300	1.0000	-0.0000	0.00
0.305	1.0001	+0.0001	0.00
0.310	1.0000	+0.0000	-0.00
0.315	1.0000	-0.0000	-0.00
0.320	0.9999	-0.0000	-0.00
0.325	1.0000	+0.0001	0.00
0.330	1.0001	+0.0000	0.00
0.335	0.9999	-0.0001	-0.00
0.340	0.9999	+0.0000	-0.00
0.345	1.0001	-0.0001	0.00
0.350	1.0000	+0.0001	0.00
0.355	1.0000	+0.0001	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.360	1.0000	+0.0000	-0.00
0.365	0.9999	-0.0001	-0.00
0.370	1.0001	-0.0000	0.00
0.375	1.0000	+0.0001	-0.00
0.380	1.0000	+0.0000	0.00
0.385	0.9998	+0.0001	-0.00
0.390	1.0000	-0.0000	-0.00
0.395	1.0000	+0.0000	-0.00
0.400	1.0001	+0.0001	0.00
0.405	0.9999	+0.0001	-0.00
0.410	1.0000	-0.0001	-0.00
0.415	0.9999	+0.0001	-0.00
0.420	0.9999	-0.0000	-0.00
0.425	0.9999	-0.0001	-0.00
0.430	0.9999	-0.0000	-0.00
0.435	1.0001	+0.0000	0.00
0.440	1.0000	+0.0000	-0.00
0.445	0.9999	+0.0000	-0.00
0.450	1.0000	+0.0000	-0.00
0.455	1.0000	+0.0000	0.00
0.460	1.0001	+0.0000	0.00
0.465	0.9999	+0.0001	-0.00
0.470	0.9999	+0.0002	-0.00
0.475	0.9999	+0.0001	-0.00
0.480	1.0000	-0.0000	-0.00
0.485	1.0000	-0.0000	-0.00
0.490	1.0000	+0.0001	-0.00
0.495	0.9999	+0.0001	-0.00
0.500	1.0000	+0.0000	-0.00
0.505	1.0000	-0.0000	-0.00
0.510	1.0001	+0.0001	0.00
0.515	1.0000	+0.0001	0.00
0.520	1.0001	-0.0001	0.00
0.525	1.0001	+0.0000	0.00
0.530	0.9999	+0.0001	-0.00
0.535	0.9999	+0.0001	-0.00
0.540	0.9999	+0.0000	-0.00
0.545	1.0000	+0.0000	0.00
0.550	1.0000	-0.0000	0.00
0.555	1.0000	-0.0001	-0.00
0.560	1.0001	-0.0000	0.00
0.565	1.0001	+0.0000	0.00
0.570	1.0000	+0.0000	-0.00
0.575	1.0000	-0.0001	-0.00
0.580	0.9999	-0.0000	-0.00
0.585	1.0000	+0.0001	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.590	1.0001	-0.0000	0.00
0.595	0.9999	+0.0000	-0.00
0.600	0.9999	-0.0000	-0.00
0.605	1.0000	-0.0000	-0.00
0.610	1.0001	+0.0000	0.00
0.615	0.9999	+0.0001	-0.00
0.620	1.0000	-0.0000	0.00
0.625	0.9999	-0.0000	-0.00
0.630	1.0000	-0.0001	-0.00
0.635	1.0000	+0.0001	-0.00
0.640	1.0000	-0.0000	-0.00
0.645	1.0000	+0.0000	0.00
0.650	1.0000	+0.0000	-0.00
0.655	1.0000	+0.0001	-0.00
0.660	0.9999	-0.0000	-0.00
0.665	0.9999	+0.0000	-0.00
0.670	0.9999	+0.0000	-0.00
0.675	1.0000	-0.0000	-0.00
0.680	1.0000	-0.0000	0.00
0.685	1.0000	+0.0001	0.00
0.690	1.0000	-0.0001	-0.00
0.695	1.0001	+0.0000	0.00
0.700	1.0001	+0.0001	0.00
0.705	1.0000	+0.0000	-0.00
0.710	1.0001	-0.0001	0.00
0.715	1.0001	-0.0000	0.00
0.720	1.0000	+0.0001	0.00
0.725	1.0000	+0.0000	-0.00
0.730	1.0000	+0.0000	0.00
0.735	1.0000	+0.0001	0.00
0.740	1.0000	+0.0000	-0.00
0.745	1.0000	-0.0000	-0.00
0.750	1.0001	+0.0001	0.00
0.755	1.0000	-0.0001	0.00
0.760	0.9999	+0.0000	-0.00
0.765	1.0000	+0.0000	0.00
0.770	1.0001	+0.0000	0.00
0.775	1.0000	-0.0001	0.00
0.780	1.0000	-0.0000	0.00
0.785	0.9999	-0.0000	-0.00
0.790	0.9999	+0.0000	-0.00
0.795	1.0000	-0.0000	-0.00
0.800	1.0001	-0.0001	0.00
0.805	1.0001	+0.0000	0.00
0.810	0.9999	+0.0001	-0.00
0.815	0.9999	+0.0000	-0.00

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
0.820	1.0000	+0.0000	-0.00
0.825	1.0001	+0.0000	0.00
0.830	1.0000	+0.0000	0.00
0.835	0.9999	-0.0000	-0.00
0.840	0.9999	-0.0001	-0.00
0.845	1.0000	-0.0001	0.00
0.850	1.0001	+0.0000	0.00
0.855	1.0000	+0.0000	0.00
0.860	1.0000	+0.0001	0.00
0.865	1.0000	+0.0001	-0.00
0.870	0.9999	+0.0001	-0.00
0.875	1.0000	-0.0000	-0.00
0.880	1.0000	+0.0001	0.00
0.885	1.0000	+0.0001	0.00
0.890	1.0000	+0.0001	-0.00
0.895	1.0001	+0.0000	0.00
0.900	1.0000	-0.0000	-0.00
0.905	1.0000	+0.0000	-0.00
0.910	0.9999	+0.0001	-0.00
0.915	1.0000	+0.0001	0.00
0.920	1.0000	+0.0001	-0.00
0.925	0.9999	+0.0001	-0.00
0.930	1.0000	+0.0001	-0.00
0.935	1.0000	+0.0001	0.00
0.940	1.0000	+0.0001	-0.00
0.945	1.0000	-0.0000	-0.00
0.950	1.0001	+0.0000	0.00
0.955	1.0000	+0.0001	0.00
0.960	1.0000	+0.0000	-0.00
0.965	0.9999	-0.0000	-0.00
0.970	0.9999	+0.0001	-0.00
0.975	1.0000	+0.0001	0.00
0.980	1.0000	+0.0001	-0.00
0.985	0.9999	+0.0000	-0.00
0.990	1.0000	+0.0001	-0.00
0.995	1.0000	+0.0001	0.00
1.000	1.0001	-0.0000	0.00
1.005	1.0001	+0.0001	0.00
1.010	0.9999	-0.0000	-0.00
1.015	1.0000	+0.0000	0.00
1.020	1.0000	+0.0001	-0.00
1.025	0.9999	+0.0001	-0.00
1.030	1.0000	+0.0000	0.00
1.035	0.9999	-0.0001	-0.00
1.040	1.0000	+0.0001	-0.00
1.045	1.0000	+0.0001	0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.050	0.9999	+0.0001	-0.00
1.055	1.0001	+0.0001	0.00
1.060	1.0000	+0.0002	-0.00
1.065	1.0001	+0.0000	0.00
1.070	1.0000	+0.0001	-0.00
1.075	1.0000	+0.0001	0.00
1.080	1.0000	+0.0000	0.00
1.085	1.0000	+0.0000	0.00
1.090	0.9999	+0.0001	-0.00
1.095	0.9999	+0.0001	-0.00
1.100	1.0000	+0.0001	0.00
1.105	0.9999	+0.0000	-0.00
1.110	1.0000	+0.0002	0.00
1.115	1.0000	+0.0002	-0.00
1.120	1.0000	+0.0000	-0.00
1.125	0.9999	-0.0000	-0.00
1.130	0.9999	+0.0002	-0.00
1.135	1.0001	+0.0000	0.00
1.140	1.0001	-0.0000	0.00
1.145	1.0001	+0.0001	0.00
1.150	1.0000	-0.0000	0.00
1.155	0.9999	+0.0001	-0.00
1.160	1.0001	-0.0000	0.00
1.165	1.0000	+0.0000	0.00
1.170	1.0000	+0.0000	0.00
1.175	1.0000	+0.0001	0.00
1.180	0.9999	-0.0000	-0.00
1.185	1.0000	+0.0001	0.00
1.190	1.0000	+0.0000	0.00
1.195	1.0000	+0.0001	-0.00
1.200	0.9999	+0.0001	-0.00
1.205	1.0001	+0.0000	0.00
1.210	0.9999	-0.0000	-0.00
1.215	1.0000	+0.0000	0.00
1.220	0.9999	+0.0000	-0.00
1.225	1.0000	+0.0000	-0.00
1.230	0.9999	+0.0000	-0.00
1.235	1.0000	+0.0001	-0.00
1.240	0.9999	-0.0001	-0.00
1.245	1.0000	+0.0000	-0.00
1.250	1.0000	+0.0001	0.00
1.255	1.0000	+0.0000	-0.00
1.260	0.9999	-0.0001	-0.00
1.265	1.0001	+0.0000	0.00
1.270	1.0000	+0.0001	0.00
1.275	1.0000	-0.0000	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.280	0.9999	+0.0000	-0.00
1.285	0.9998	+0.0000	-0.00
1.290	1.0000	+0.0000	-0.00
1.295	1.0000	+0.0001	0.00
1.300	0.9999	-0.0000	-0.00
1.305	1.0000	-0.0000	0.00
1.310	0.9999	+0.0000	-0.00
1.315	1.0000	+0.0000	-0.00
1.320	1.0000	+0.0001	0.00
1.325	1.0000	-0.0000	0.00
1.330	1.0000	+0.0001	-0.00
1.335	0.9999	+0.0001	-0.00
1.340	0.9999	+0.0001	-0.00
1.345	1.0000	+0.0002	0.00
1.350	1.0001	+0.0000	0.00
1.355	1.0001	+0.0001	0.00
1.360	1.0000	-0.0000	0.00
1.365	1.0000	-0.0000	0.00
1.370	1.0001	+0.0001	0.00
1.375	0.9999	+0.0000	-0.00
1.380	1.0001	+0.0001	0.00
1.385	0.9999	+0.0001	-0.00
1.390	0.9999	-0.0000	-0.00
1.395	1.0000	-0.0000	-0.00
1.400	1.0000	-0.0001	-0.00
1.405	1.0000	+0.0002	0.00
1.410	0.9999	+0.0001	-0.00
1.415	1.0001	+0.0001	0.00
1.420	1.0001	-0.0001	0.00
1.425	1.0000	-0.0000	-0.00
1.430	1.0001	+0.0001	0.00
1.435	1.0000	+0.0002	0.00
1.440	1.0001	+0.0002	0.00
1.445	1.0000	+0.0001	0.00
1.450	0.9998	+0.0000	-0.00
1.455	1.0000	+0.0001	-0.00
1.460	0.9998	+0.0001	-0.00
1.465	0.9999	+0.0001	-0.00
1.470	1.0000	+0.0001	-0.00
1.475	1.0001	-0.0001	0.00
1.480	0.9999	+0.0001	-0.00
1.485	1.0000	-0.0000	-0.00
1.490	1.0000	-0.0000	-0.00
1.495	1.0000	+0.0001	0.00
1.500	0.9999	+0.0001	-0.00
1.505	1.0000	+0.0001	-0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.510	1.0002	+0.0001	0.00
1.515	0.9999	+0.0001	-0.00
1.520	1.0000	+0.0001	-0.00
1.525	1.0000	+0.0001	-0.00
1.530	1.0000	+0.0000	0.00
1.535	1.0000	+0.0001	-0.00
1.540	1.0001	-0.0000	0.00
1.545	1.0000	+0.0002	-0.00
1.550	1.0000	+0.0000	0.00
1.555	1.0001	-0.0000	0.00
1.560	0.9999	+0.0000	-0.00
1.565	1.0000	+0.0001	0.00
1.570	1.0000	+0.0001	0.00
1.575	1.0001	+0.0001	0.00
1.580	1.0000	+0.0001	-0.00
1.585	1.0000	-0.0000	0.00
1.590	1.0000	+0.0000	-0.00
1.595	0.9999	-0.0000	-0.00
1.600	1.0000	+0.0001	0.00
1.605	1.0000	+0.0000	0.00
1.610	1.0000	+0.0001	0.00
1.615	1.0000	+0.0001	0.00
1.620	0.9999	+0.0001	-0.00
1.625	1.0000	+0.0000	0.00
1.630	1.0000	-0.0000	0.00
1.635	0.9999	+0.0001	-0.00
1.640	0.9999	+0.0001	-0.00
1.645	1.0000	+0.0001	-0.00
1.650	1.0001	-0.0000	0.00
1.655	1.0001	+0.0000	0.00
1.660	1.0000	+0.0002	0.00
1.665	0.9998	+0.0001	-0.00
1.670	0.9997	+0.0001	-0.00
1.675	1.0001	+0.0001	0.00
1.680	1.0000	+0.0000	-0.00
1.685	1.0000	-0.0001	0.00
1.690	1.0001	+0.0000	0.00
1.695	0.9999	-0.0000	-0.00
1.700	1.0000	-0.0001	0.00
1.705	1.0001	+0.0001	0.00
1.710	0.9999	-0.0000	-0.00
1.715	1.0000	-0.0001	-0.00
1.720	1.0001	-0.0000	0.00
1.725	1.0001	+0.0001	0.00
1.730	0.9999	-0.0001	-0.00
1.735	1.0001	+0.0001	0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.740	1.0001	+0.0001	0.00
1.745	1.0000	+0.0002	-0.00
1.750	1.0000	-0.0001	0.00
1.755	1.0000	-0.0001	-0.00
1.760	0.9999	-0.0000	-0.00
1.765	0.9999	-0.0000	-0.00
1.770	0.9999	+0.0002	-0.00
1.775	0.9998	+0.0001	-0.00
1.780	0.9999	-0.0000	-0.00
1.785	0.9999	+0.0001	-0.00
1.790	1.0000	+0.0000	0.00
1.795	0.9999	-0.0001	-0.00
1.800	1.0000	-0.0001	-0.00
1.805	1.0001	+0.0001	0.00
1.810	1.0000	-0.0000	-0.00
1.815	1.0000	-0.0000	-0.00
1.820	0.9999	+0.0000	-0.00
1.825	0.9999	-0.0001	-0.00
1.830	0.9999	+0.0000	-0.00
1.835	0.9999	-0.0000	-0.00
1.840	1.0000	+0.0000	0.00
1.845	1.0000	-0.0001	0.00
1.850	1.0000	+0.0001	0.00
1.855	0.9999	-0.0001	-0.00
1.860	1.0001	+0.0001	0.00
1.865	0.9999	+0.0000	-0.00
1.870	0.9999	+0.0001	-0.00
1.875	1.0000	+0.0000	-0.00
1.880	0.9999	-0.0000	-0.00
1.885	1.0001	+0.0000	0.00
1.890	1.0000	-0.0001	0.00
1.895	1.0000	-0.0000	0.00
1.900	0.9999	+0.0001	-0.00
1.905	1.0000	+0.0001	0.00
1.910	0.9999	+0.0000	-0.00
1.915	1.0000	+0.0001	-0.00
1.920	1.0000	+0.0000	-0.00
1.925	1.0000	-0.0000	-0.00
1.930	1.0000	+0.0000	0.00
1.935	0.9998	+0.0000	-0.00
1.940	1.0001	-0.0001	0.00
1.945	1.0000	+0.0001	0.00
1.950	1.0000	+0.0001	0.00
1.955	1.0000	+0.0001	0.00
1.960	1.0000	+0.0000	-0.00
1.965	1.0000	+0.0000	-0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	$\lvert S_{11}\rvert$ **(dB)**
1.970	1.0000	-0.0001	-0.00
1.975	1.0000	+0.0000	-0.00
1.980	1.0001	+0.0000	0.00
1.985	1.0001	-0.0001	0.00
1.990	0.9999	-0.0001	-0.00
1.995	1.0000	+0.0001	-0.00
2.000	1.0001	+0.0001	0.00
2.005	1.0000	+0.0000	-0.00
2.010	0.9999	+0.0002	-0.00
2.015	0.9999	+0.0000	-0.00
2.020	1.0001	+0.0001	0.00
2.025	1.0000	+0.0000	-0.00
2.030	0.9999	-0.0001	-0.00
2.035	1.0001	-0.0001	0.00
2.040	1.0000	-0.0000	0.00
2.045	1.0001	+0.0001	0.00
2.050	0.9998	+0.0001	-0.00
2.055	0.9999	+0.0001	-0.00
2.060	1.0000	+0.0001	0.00
2.065	0.9999	+0.0000	-0.00
2.070	1.0000	-0.0001	0.00
2.075	1.0000	+0.0000	0.00
2.080	1.0000	+0.0000	0.00
2.085	0.9999	-0.0001	-0.00
2.090	1.0000	+0.0000	-0.00
2.095	0.9999	-0.0000	-0.00
2.100	0.9998	+0.0000	-0.00
2.105	1.0000	-0.0000	0.00
2.110	0.9997	+0.0000	-0.00
2.115	1.0000	+0.0001	-0.00
2.120	1.0001	+0.0001	0.00
2.125	1.0001	+0.0001	0.00
2.130	1.0001	+0.0000	0.00
2.135	1.0000	+0.0001	-0.00
2.140	0.9999	-0.0001	-0.00
2.145	1.0000	-0.0000	-0.00
2.150	0.9999	-0.0002	-0.00
2.155	1.0001	-0.0001	0.00
2.160	1.0000	+0.0001	0.00
2.165	0.9998	-0.0000	-0.00
2.170	0.9999	+0.0000	-0.00
2.175	0.9999	+0.0001	-0.00
2.180	0.9999	+0.0000	-0.00
2.185	1.0000	+0.0000	-0.00
2.190	0.9999	+0.0001	-0.00
2.195	1.0000	+0.0001	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.200	0.9999	+0.0001	-0.00
2.205	1.0001	-0.0000	0.00
2.210	1.0000	+0.0001	-0.00
2.215	1.0000	+0.0001	-0.00
2.220	0.9999	+0.0001	-0.00
2.225	1.0000	+0.0002	-0.00
2.230	1.0000	+0.0001	0.00
2.235	1.0000	-0.0001	-0.00
2.240	1.0000	-0.0000	-0.00
2.245	0.9999	+0.0001	-0.00
2.250	0.9999	-0.0000	-0.00
2.255	1.0000	+0.0001	0.00
2.260	0.9999	+0.0002	-0.00
2.265	0.9999	+0.0001	-0.00
2.270	1.0000	-0.0000	0.00
2.275	1.0001	+0.0001	0.00
2.280	1.0001	+0.0000	0.00
2.285	0.9998	+0.0001	-0.00
2.290	0.9999	+0.0000	-0.00
2.295	1.0000	-0.0001	0.00
2.300	1.0000	+0.0000	-0.00
2.305	1.0001	+0.0001	0.00
2.310	1.0000	-0.0001	-0.00
2.315	0.9999	-0.0001	-0.00
2.320	0.9999	+0.0000	-0.00
2.325	1.0003	-0.0001	0.00
2.330	0.9999	+0.0001	-0.00
2.335	1.0000	+0.0001	0.00
2.340	1.0000	-0.0000	-0.00
2.345	1.0001	+0.0002	0.00
2.350	1.0000	+0.0001	0.00
2.355	1.0001	-0.0001	0.00
2.360	0.9999	+0.0000	-0.00
2.365	1.0000	+0.0002	0.00
2.370	0.9999	+0.0001	-0.00
2.375	1.0000	+0.0000	-0.00
2.380	0.9999	+0.0000	-0.00
2.385	0.9998	+0.0002	-0.00
2.390	0.9998	+0.0001	-0.00
2.395	0.9997	+0.0002	-0.00
2.400	1.0000	+0.0001	-0.00
2.405	1.0000	+0.0001	0.00
2.410	0.9998	-0.0001	-0.00
2.415	1.0000	+0.0001	0.00
2.420	0.9999	+0.0001	-0.00
2.425	1.0000	+0.0001	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.430	1.0000	+0.0002	0.00
2.435	1.0000	-0.0000	-0.00
2.440	0.9999	+0.0001	-0.00
2.445	0.9999	-0.0001	-0.00
2.450	0.9999	-0.0001	-0.00
2.455	0.9999	-0.0001	-0.00
2.460	1.0000	-0.0001	0.00
2.465	0.9999	-0.0000	-0.00
2.470	1.0000	+0.0001	-0.00
2.475	1.0000	+0.0001	0.00
2.480	1.0000	+0.0001	-0.00
2.485	0.9999	-0.0000	-0.00
2.490	0.9999	+0.0000	-0.00
2.495	1.0000	+0.0000	0.00
2.500	0.9998	+0.0000	-0.00
2.505	1.0000	-0.0000	-0.00
2.510	0.9999	-0.0000	-0.00
2.515	1.0000	+0.0000	-0.00
2.520	0.9999	+0.0000	-0.00
2.525	1.0000	+0.0002	-0.00
2.530	1.0001	+0.0001	0.00
2.535	0.9998	+0.0000	-0.00
2.540	0.9999	+0.0001	-0.00
2.545	1.0000	-0.0000	-0.00
2.550	1.0000	+0.0001	-0.00
2.555	0.9999	-0.0000	-0.00
2.560	0.9999	+0.0001	-0.00
2.565	0.9999	+0.0000	-0.00
2.570	1.0000	+0.0000	-0.00
2.575	0.9999	-0.0001	-0.00
2.580	0.9999	+0.0000	-0.00
2.585	0.9998	+0.0002	-0.00
2.590	1.0000	+0.0000	0.00
2.595	1.0000	-0.0001	-0.00
2.600	1.0000	-0.0000	-0.00
2.605	1.0001	+0.0002	0.00
2.610	0.9998	-0.0001	-0.00
2.615	1.0001	-0.0002	0.00
2.620	1.0000	-0.0000	0.00
2.625	1.0000	+0.0000	-0.00
2.630	0.9998	-0.0001	-0.00
2.635	1.0000	+0.0001	-0.00
2.640	0.9999	+0.0001	-0.00
2.645	0.9999	-0.0000	-0.00
2.650	1.0000	+0.0002	0.00
2.655	0.9998	+0.0001	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.660	1.0000	+0.0001	0.00
2.665	1.0000	-0.0000	0.00
2.670	1.0000	+0.0002	-0.00
2.675	1.0001	-0.0001	0.00
2.680	1.0000	-0.0000	0.00
2.685	1.0001	+0.0001	0.00
2.690	1.0000	+0.0003	0.00
2.695	0.9999	+0.0001	-0.00
2.700	1.0000	+0.0000	-0.00
2.705	0.9998	+0.0000	-0.00
2.710	1.0000	+0.0001	-0.00
2.715	1.0000	-0.0002	0.00
2.720	0.9999	+0.0000	-0.00
2.725	1.0001	+0.0002	0.00
2.730	1.0000	+0.0003	0.00
2.735	0.9998	+0.0003	-0.00
2.740	1.0000	-0.0000	0.00
2.745	1.0000	+0.0000	-0.00
2.750	0.9998	+0.0002	-0.00
2.755	0.9999	+0.0002	-0.00
2.760	0.9999	+0.0002	-0.00
2.765	1.0000	+0.0001	0.00
2.770	0.9998	+0.0001	-0.00
2.775	1.0000	+0.0001	-0.00
2.780	1.0000	+0.0001	0.00
2.785	0.9999	+0.0001	-0.00
2.790	1.0000	+0.0001	-0.00
2.795	0.9998	+0.0000	-0.00
2.800	1.0000	+0.0001	0.00
2.805	1.0000	-0.0000	-0.00
2.810	1.0000	+0.0002	-0.00
2.815	1.0000	+0.0000	-0.00
2.820	0.9999	+0.0000	-0.00
2.825	1.0000	+0.0002	0.00
2.830	0.9998	+0.0002	-0.00
2.835	0.9999	+0.0001	-0.00
2.840	0.9999	+0.0003	-0.00
2.845	0.9999	+0.0000	-0.00
2.850	1.0001	+0.0001	0.00
2.855	0.9998	+0.0002	-0.00
2.860	1.0001	+0.0000	0.00
2.865	0.9999	+0.0003	-0.00
2.870	1.0000	+0.0001	-0.00
2.875	1.0000	+0.0001	-0.00
2.880	0.9999	+0.0002	-0.00
2.885	1.0000	-0.0001	0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.890	1.0000	+0.0004	-0.00
2.895	0.9999	+0.0002	-0.00
2.900	1.0000	+0.0001	-0.00
2.905	1.0000	+0.0002	-0.00
2.910	0.9998	+0.0001	-0.00
2.915	1.0000	+0.0001	-0.00
2.920	0.9999	+0.0001	-0.00
2.925	1.0000	-0.0001	0.00
2.930	0.9999	+0.0002	-0.00
2.935	1.0001	+0.0001	0.00
2.940	0.9998	+0.0001	-0.00
2.945	0.9998	+0.0000	-0.00
2.950	1.0000	+0.0000	0.00
2.955	1.0000	-0.0001	-0.00
2.960	0.9999	+0.0003	-0.00
2.965	0.9999	+0.0000	-0.00
2.970	0.9999	+0.0001	-0.00
2.975	1.0001	+0.0001	0.00
2.980	0.9997	-0.0002	-0.00
2.985	0.9999	+0.0002	-0.00
2.990	0.9998	+0.0002	-0.00
2.995	1.0000	-0.0000	0.00
3.000	0.9999	+0.0002	-0.00
3.005	0.9998	+0.0003	-0.00
3.010	1.0003	-0.0002	0.00
3.015	1.0000	+0.0002	-0.00
3.020	0.9996	+0.0004	-0.00
3.025	1.0000	+0.0001	0.00
3.030	0.9999	+0.0002	-0.00
3.035	0.9999	-0.0000	-0.00
3.040	0.9999	+0.0002	-0.00
3.045	1.0003	+0.0000	0.00
3.050	1.0001	+0.0001	0.00
3.055	1.0000	+0.0000	0.00
3.060	0.9997	+0.0001	-0.00
3.065	1.0000	-0.0003	0.00
3.070	0.9999	+0.0003	-0.00
3.075	0.9998	+0.0001	-0.00
3.080	0.9998	+0.0001	-0.00
3.085	1.0000	-0.0000	0.00
3.090	1.0000	+0.0001	-0.00
3.095	0.9999	+0.0001	-0.00
3.100	0.9998	-0.0001	-0.00
3.105	0.9999	-0.0001	-0.00
3.110	0.9999	-0.0001	-0.00
3.115	1.0000	-0.0000	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.120	0.9999	+0.0001	-0.00
3.125	1.0001	+0.0001	0.00
3.130	1.0000	+0.0001	0.00
3.135	1.0000	+0.0000	-0.00
3.140	0.9999	+0.0000	-0.00
3.145	1.0000	+0.0001	0.00
3.150	0.9999	+0.0001	-0.00
3.155	0.9999	-0.0000	-0.00
3.160	0.9999	-0.0000	-0.00
3.165	1.0001	+0.0003	0.00
3.170	0.9999	-0.0000	-0.00
3.175	0.9999	+0.0002	-0.00
3.180	1.0000	-0.0001	-0.00
3.185	1.0001	-0.0001	0.00
3.190	1.0000	+0.0002	0.00
3.195	0.9999	+0.0001	-0.00
3.200	0.9999	+0.0002	-0.00
3.205	1.0000	+0.0000	-0.00
3.210	1.0000	+0.0002	-0.00
3.215	1.0000	+0.0000	-0.00
3.220	0.9998	+0.0001	-0.00
3.225	0.9999	+0.0000	-0.00
3.230	1.0001	-0.0002	0.00
3.235	0.9998	-0.0000	-0.00
3.240	0.9997	+0.0001	-0.00
3.245	1.0000	+0.0000	-0.00
3.250	0.9997	+0.0003	-0.00
3.255	0.9999	+0.0002	-0.00
3.260	0.9999	-0.0001	-0.00
3.265	0.9998	-0.0001	-0.00
3.270	1.0000	+0.0000	0.00
3.275	1.0000	+0.0001	0.00
3.280	1.0000	-0.0001	-0.00
3.285	0.9999	+0.0001	-0.00
3.290	0.9999	+0.0001	-0.00
3.295	0.9999	+0.0001	-0.00
3.300	1.0000	-0.0003	-0.00
3.305	1.0001	+0.0001	0.00
3.310	1.0001	+0.0002	0.00
3.315	1.0000	+0.0000	-0.00
3.320	1.0000	+0.0001	-0.00
3.325	1.0003	-0.0000	0.00
3.330	0.9997	-0.0001	-0.00
3.335	1.0002	+0.0001	0.00
3.340	1.0000	-0.0000	0.00
3.345	1.0001	-0.0001	0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.350	0.9999	+0.0002	-0.00
3.355	1.0000	-0.0001	0.00
3.360	0.9997	+0.0003	-0.00
3.365	1.0000	+0.0003	0.00
3.370	1.0000	+0.0001	-0.00
3.375	0.9998	+0.0001	-0.00
3.380	0.9998	+0.0001	-0.00
3.385	0.9999	+0.0001	-0.00
3.390	1.0000	+0.0001	-0.00
3.395	0.9999	-0.0000	-0.00
3.400	0.9999	+0.0003	-0.00
3.405	0.9999	+0.0002	-0.00
3.410	0.9997	+0.0002	-0.00
3.415	0.9999	+0.0004	-0.00
3.420	0.9999	+0.0003	-0.00
3.425	1.0000	+0.0001	0.00
3.430	0.9999	+0.0001	-0.00
3.435	0.9998	+0.0002	-0.00
3.440	1.0000	+0.0002	-0.00
3.445	0.9999	+0.0003	-0.00
3.450	1.0000	+0.0003	-0.00
3.455	0.9998	+0.0003	-0.00
3.460	1.0000	+0.0003	-0.00
3.465	0.9997	+0.0002	-0.00
3.470	0.9998	+0.0003	-0.00
3.475	0.9999	+0.0000	-0.00
3.480	1.0001	+0.0002	0.00
3.485	1.0003	-0.0000	0.00
3.490	1.0000	+0.0002	-0.00
3.495	0.9999	-0.0000	-0.00
3.500	1.0001	+0.0001	0.00
3.505	0.9998	+0.0000	-0.00
3.510	1.0000	+0.0000	0.00
3.515	1.0001	+0.0002	0.00
3.520	1.0000	+0.0003	0.00
3.525	1.0000	+0.0005	0.00
3.530	1.0001	+0.0000	0.00
3.535	1.0003	+0.0001	0.00
3.540	1.0000	+0.0006	0.00
3.545	0.9999	+0.0000	-0.00
3.550	0.9996	+0.0004	-0.00
3.555	1.0002	-0.0002	0.00
3.560	1.0000	+0.0001	-0.00
3.565	1.0003	-0.0004	0.00
3.570	1.0000	+0.0003	-0.00
3.575	0.9995	+0.0005	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.580	0.9996	+0.0006	-0.00
3.585	1.0008	-0.0003	0.01
3.590	1.0000	-0.0000	-0.00
3.595	1.0003	-0.0002	0.00
3.600	0.9998	+0.0003	-0.00
3.605	1.0004	+0.0001	0.00
3.610	1.0003	+0.0003	0.00
3.615	1.0000	+0.0001	0.00
3.620	0.9999	+0.0000	-0.00
3.625	1.0007	-0.0009	0.01
3.630	1.0002	+0.0005	0.00
3.635	0.9999	+0.0001	-0.00
3.640	0.9998	+0.0000	-0.00
3.645	0.9996	+0.0001	-0.00
3.650	1.0001	+0.0005	0.00
3.655	1.0002	+0.0001	0.00
3.660	0.9999	+0.0003	-0.00
3.665	1.0002	+0.0001	0.00
3.670	0.9999	+0.0004	-0.00
3.675	1.0005	+0.0004	0.00
3.680	1.0000	+0.0002	0.00
3.685	0.9998	+0.0003	-0.00
3.690	1.0000	+0.0001	0.00
3.695	0.9999	+0.0000	-0.00
3.700	0.9999	+0.0001	-0.00
3.705	1.0000	-0.0000	0.00
3.710	1.0000	+0.0001	0.00
3.715	0.9999	+0.0000	-0.00
3.720	1.0001	-0.0001	0.00
3.725	0.9999	+0.0001	-0.00
3.730	1.0000	+0.0001	0.00
3.735	1.0001	+0.0002	0.00
3.740	1.0000	+0.0002	-0.00
3.745	0.9998	-0.0000	-0.00
3.750	1.0000	+0.0001	0.00
3.755	1.0001	+0.0000	0.00
3.760	1.0000	-0.0000	0.00
3.765	0.9999	+0.0003	-0.00
3.770	1.0000	+0.0002	0.00
3.775	1.0001	+0.0001	0.00
3.780	0.9996	+0.0011	-0.00
3.785	0.9999	+0.0005	-0.00
3.790	0.9998	+0.0000	-0.00
3.795	1.0000	-0.0001	0.00
3.800	0.9997	-0.0000	-0.00
3.805	0.9999	-0.0003	-0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.810	1.0000	+0.0001	-0.00
3.815	1.0003	-0.0000	0.00
3.820	0.9999	+0.0001	-0.00
3.825	1.0003	+0.0002	0.00
3.830	1.0001	+0.0001	0.00
3.835	1.0000	+0.0001	0.00
3.840	1.0001	-0.0001	0.00
3.845	1.0000	+0.0001	-0.00
3.850	0.9998	-0.0001	-0.00
3.855	0.9999	+0.0003	-0.00
3.860	1.0002	+0.0002	0.00
3.865	0.9999	+0.0005	-0.00
3.870	1.0001	+0.0000	0.00
3.875	0.9999	+0.0001	-0.00
3.880	1.0000	-0.0000	0.00
3.885	0.9999	+0.0001	-0.00
3.890	0.9998	-0.0002	-0.00
3.895	1.0002	-0.0003	0.00
3.900	1.0001	+0.0002	0.00
3.905	0.9996	+0.0002	-0.00
3.910	0.9999	+0.0003	-0.00
3.915	1.0000	+0.0000	-0.00
3.920	0.9998	+0.0001	-0.00
3.925	1.0001	+0.0001	0.00
3.930	1.0001	-0.0000	0.00
3.935	1.0000	+0.0001	-0.00
3.940	1.0000	-0.0000	-0.00
3.945	0.9998	-0.0002	-0.00
3.950	1.0000	+0.0001	-0.00
3.955	0.9998	+0.0001	-0.00
3.960	1.0000	+0.0001	-0.00
3.965	0.9999	-0.0001	-0.00
3.970	0.9998	+0.0002	-0.00
3.975	1.0001	+0.0001	0.00
3.980	1.0000	+0.0001	0.00
3.985	1.0000	+0.0001	0.00
3.990	0.9998	+0.0000	-0.00
3.995	1.0001	+0.0001	0.00
4.000	0.9999	+0.0001	-0.00
4.005	1.0000	+0.0002	0.00
4.010	1.0000	+0.0002	0.00
4.015	0.9999	+0.0001	-0.00
4.020	0.9999	+0.0002	-0.00
4.025	1.0000	-0.0001	-0.00
4.030	1.0002	+0.0003	0.00
4.035	1.0000	+0.0002	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.040	1.0000	+0.0001	0.00
4.045	0.9999	+0.0002	-0.00
4.050	1.0002	-0.0001	0.00
4.055	1.0001	+0.0001	0.00
4.060	1.0001	+0.0002	0.00
4.065	0.9996	+0.0004	-0.00
4.070	0.9999	+0.0003	-0.00
4.075	1.0000	+0.0002	0.00
4.080	0.9999	+0.0001	-0.00
4.085	0.9998	+0.0003	-0.00
4.090	0.9999	+0.0000	-0.00
4.095	1.0000	+0.0003	-0.00
4.100	0.9998	+0.0001	-0.00
4.105	1.0005	+0.0003	0.00
4.110	0.9997	-0.0001	-0.00
4.115	1.0001	+0.0002	0.00
4.120	0.9998	+0.0003	-0.00
4.125	1.0001	-0.0002	0.00
4.130	0.9997	-0.0002	-0.00
4.135	1.0001	+0.0002	0.00
4.140	1.0002	+0.0003	0.00
4.145	1.0004	-0.0006	0.00
4.150	1.0002	+0.0004	0.00
4.155	0.9999	+0.0003	-0.00
4.160	0.9998	-0.0000	-0.00
4.165	1.0001	+0.0001	0.00
4.170	1.0000	+0.0002	-0.00
4.175	0.9998	+0.0002	-0.00
4.180	1.0000	+0.0000	-0.00
4.185	1.0001	+0.0000	0.00
4.190	1.0001	+0.0001	0.00
4.195	1.0001	+0.0001	0.00
4.200	1.0000	+0.0001	0.00
4.205	1.0001	-0.0003	0.00
4.210	1.0002	+0.0002	0.00
4.215	0.9999	+0.0003	-0.00
4.220	1.0000	+0.0001	0.00
4.225	1.0000	+0.0000	-0.00
4.230	0.9999	-0.0000	-0.00
4.235	1.0000	+0.0003	-0.00
4.240	0.9999	+0.0001	-0.00
4.245	1.0001	-0.0003	0.00
4.250	1.0001	+0.0001	0.00
4.255	1.0000	-0.0003	-0.00
4.260	1.0000	+0.0001	-0.00
4.265	1.0002	-0.0001	0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.270	1.0000	+0.0000	-0.00
4.275	1.0000	+0.0001	0.00
4.280	1.0000	+0.0001	-0.00
4.285	0.9999	+0.0001	-0.00
4.290	1.0000	+0.0001	-0.00
4.295	1.0000	+0.0002	0.00
4.300	0.9998	+0.0000	-0.00
4.305	1.0000	+0.0001	0.00
4.310	1.0001	+0.0001	0.00
4.315	1.0000	+0.0001	-0.00
4.320	1.0001	+0.0001	0.00
4.325	1.0000	+0.0002	0.00
4.330	1.0001	+0.0000	0.00
4.335	1.0000	+0.0001	-0.00
4.340	1.0001	+0.0000	0.00
4.345	1.0000	-0.0001	0.00
4.350	1.0000	-0.0001	-0.00
4.355	0.9998	+0.0000	-0.00
4.360	1.0001	+0.0001	0.00
4.365	1.0000	+0.0002	0.00
4.370	0.9999	-0.0000	-0.00
4.375	1.0002	+0.0003	0.00
4.380	1.0001	+0.0000	0.00
4.385	1.0000	+0.0002	0.00
4.390	0.9999	-0.0001	-0.00
4.395	1.0001	+0.0001	0.00
4.400	0.9999	+0.0001	-0.00
4.405	1.0002	-0.0001	0.00
4.410	1.0000	-0.0001	-0.00
4.415	1.0001	-0.0000	0.00
4.420	0.9999	+0.0003	-0.00
4.425	0.9999	+0.0001	-0.00
4.430	1.0000	+0.0001	0.00
4.435	1.0003	+0.0001	0.00
4.440	0.9999	-0.0002	-0.00
4.445	1.0000	+0.0003	0.00
4.450	0.9997	+0.0001	-0.00
4.455	0.9994	+0.0003	-0.01
4.460	0.9998	-0.0000	-0.00
4.465	1.0002	+0.0003	0.00
4.470	1.0000	-0.0002	-0.00
4.475	1.0001	+0.0000	0.00
4.480	0.9999	-0.0000	-0.00
4.485	1.0000	+0.0004	-0.00
4.490	1.0001	+0.0001	0.00
4.495	1.0001	-0.0003	0.00

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.500	1.0000	-0.0001	0.00
4.505	1.0001	+0.0003	0.00
4.510	1.0003	+0.0004	0.00
4.515	0.9998	+0.0002	-0.00
4.520	0.9997	-0.0002	-0.00
4.525	1.0000	-0.0003	0.00
4.530	0.9999	-0.0000	-0.00
4.535	1.0000	-0.0002	0.00
4.540	1.0000	-0.0001	0.00
4.545	0.9999	+0.0000	-0.00
4.550	1.0001	+0.0000	0.00
4.555	1.0000	+0.0001	0.00
4.560	0.9999	+0.0000	-0.00
4.565	0.9999	+0.0001	-0.00
4.570	1.0000	-0.0001	0.00
4.575	1.0000	+0.0001	0.00
4.580	0.9999	+0.0000	-0.00
4.585	0.9999	+0.0001	-0.00
4.590	1.0001	+0.0001	0.00
4.595	1.0000	+0.0000	-0.00
4.600	1.0000	+0.0002	-0.00
4.605	1.0001	+0.0001	0.00
4.610	0.9999	+0.0002	-0.00
4.615	1.0000	+0.0001	-0.00
4.620	1.0000	+0.0001	0.00
4.625	0.9999	-0.0002	-0.00
4.630	1.0000	+0.0001	-0.00
4.635	1.0000	+0.0000	-0.00
4.640	1.0000	+0.0002	-0.00
4.645	0.9998	+0.0003	-0.00
4.650	1.0000	+0.0001	0.00
4.655	1.0001	+0.0003	0.00
4.660	0.9998	-0.0002	-0.00
4.665	0.9999	-0.0002	-0.00
4.670	0.9998	+0.0004	-0.00
4.675	1.0002	+0.0001	0.00
4.680	1.0000	+0.0003	-0.00
4.685	1.0001	+0.0001	0.00
4.690	0.9999	+0.0002	-0.00
4.695	1.0001	+0.0000	0.00
4.700	1.0000	+0.0002	-0.00
4.705	1.0006	-0.0003	0.01
4.710	0.9999	+0.0007	-0.00
4.715	1.0000	+0.0001	0.00
4.720	1.0003	-0.0000	0.00
4.725	0.9996	+0.0003	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.730	1.0000	-0.0000	-0.00
4.735	1.0001	+0.0000	0.00
4.740	1.0002	-0.0001	0.00
4.745	0.9999	+0.0004	-0.00
4.750	1.0004	+0.0001	0.00
4.755	1.0000	-0.0002	-0.00
4.760	1.0003	-0.0001	0.00
4.765	1.0004	-0.0002	0.00
4.770	0.9997	-0.0004	-0.00
4.775	1.0004	+0.0002	0.00
4.780	1.0001	-0.0001	0.00
4.785	0.9990	-0.0002	-0.01
4.790	1.0002	-0.0001	0.00
4.795	1.0001	+0.0001	0.00
4.800	1.0002	-0.0002	0.00
4.805	1.0001	-0.0003	0.00
4.810	1.0001	-0.0002	0.00
4.815	1.0007	-0.0002	0.01
4.820	0.9996	+0.0003	-0.00
4.825	1.0000	+0.0000	-0.00
4.830	0.9997	-0.0000	-0.00
4.835	1.0000	+0.0001	0.00
4.840	1.0001	-0.0001	0.00
4.845	1.0002	-0.0002	0.00
4.850	1.0000	+0.0002	-0.00
4.855	0.9998	+0.0001	-0.00
4.860	1.0000	-0.0001	0.00
4.865	1.0003	+0.0003	0.00
4.870	1.0000	+0.0002	0.00
4.875	1.0000	+0.0001	0.00
4.880	0.9999	-0.0001	-0.00
4.885	1.0000	+0.0001	0.00
4.890	0.9997	-0.0007	-0.00
4.895	0.9999	+0.0003	-0.00
4.900	0.9999	-0.0001	-0.00
4.905	0.9999	-0.0001	-0.00
4.910	1.0000	+0.0002	-0.00
4.915	1.0000	+0.0001	-0.00
4.920	1.0000	+0.0000	-0.00
4.925	1.0000	+0.0000	-0.00
4.930	1.0001	+0.0001	0.00
4.935	1.0000	+0.0001	0.00
4.940	1.0001	-0.0001	0.00
4.945	0.9999	+0.0001	-0.00
4.950	1.0000	-0.0000	0.00
4.955	1.0000	-0.0000	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.960	1.0001	-0.0001	0.00
4.965	0.9999	+0.0000	-0.00
4.970	1.0000	+0.0000	-0.00
4.975	1.0000	-0.0001	0.00
4.980	1.0000	+0.0001	0.00
4.985	0.9999	-0.0001	-0.00
4.990	1.0000	-0.0001	-0.00
4.995	1.0000	+0.0002	-0.00
5.000	1.0002	+0.0002	0.00
5.005	1.0000	+0.0001	-0.00
5.010	1.0000	+0.0001	0.00
5.015	0.9998	-0.0000	-0.00
5.020	1.0000	+0.0001	-0.00
5.025	0.9999	+0.0001	-0.00
5.030	1.0000	+0.0003	0.00
5.035	0.9998	+0.0001	-0.00
5.040	0.9999	+0.0001	-0.00
5.045	1.0000	-0.0001	-0.00
5.050	1.0001	+0.0001	0.00
5.055	0.9999	-0.0001	-0.00
5.060	1.0002	+0.0001	0.00
5.065	1.0002	+0.0003	0.00
5.070	0.9998	+0.0001	-0.00
5.075	0.9999	-0.0001	-0.00
5.080	1.0000	+0.0001	0.00
5.085	1.0000	+0.0003	0.00
5.090	1.0001	+0.0001	0.00
5.095	1.0000	-0.0003	-0.00
5.100	0.9999	-0.0002	-0.00
5.105	0.9995	+0.0000	-0.00
5.110	1.0002	-0.0000	0.00
5.115	0.9999	+0.0001	-0.00
5.120	1.0001	+0.0001	0.00
5.125	0.9997	+0.0000	-0.00
5.130	1.0001	-0.0000	0.00
5.135	1.0000	+0.0004	-0.00
5.140	1.0000	+0.0002	-0.00
5.145	1.0002	+0.0001	0.00
5.150	1.0000	+0.0001	0.00
5.155	1.0001	+0.0000	0.00
5.160	1.0001	-0.0000	0.00
5.165	1.0000	+0.0002	-0.00
5.170	1.0000	+0.0000	0.00
5.175	1.0001	+0.0001	0.00
5.180	1.0000	+0.0001	-0.00
5.185	1.0001	+0.0002	0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.190	1.0001	+0.0001	0.00
5.195	1.0000	+0.0001	-0.00
5.200	1.0000	+0.0001	-0.00
5.205	1.0000	+0.0000	-0.00
5.210	1.0000	+0.0000	0.00
5.215	1.0000	-0.0000	-0.00
5.220	1.0000	+0.0001	0.00
5.225	1.0000	+0.0001	-0.00
5.230	1.0001	+0.0001	0.00
5.235	1.0000	+0.0000	-0.00
5.240	1.0001	+0.0001	0.00
5.245	1.0000	+0.0001	0.00
5.250	1.0000	+0.0000	0.00
5.255	1.0000	+0.0002	-0.00
5.260	1.0001	-0.0002	0.00
5.265	0.9997	-0.0001	-0.00
5.270	1.0001	+0.0002	0.00
5.275	1.0001	+0.0002	0.00
5.280	1.0000	+0.0001	-0.00
5.285	1.0000	+0.0001	-0.00
5.290	1.0002	+0.0000	0.00
5.295	1.0004	+0.0000	0.00
5.300	1.0000	+0.0001	0.00
5.305	0.9999	-0.0003	-0.00
5.310	1.0001	+0.0003	0.00
5.315	1.0000	-0.0001	-0.00
5.320	1.0000	+0.0000	0.00
5.325	1.0000	+0.0001	-0.00
5.330	0.9999	+0.0000	-0.00
5.335	1.0000	+0.0002	-0.00
5.340	0.9998	+0.0002	-0.00
5.345	1.0001	+0.0002	0.00
5.350	1.0003	-0.0002	0.00
5.355	0.9998	+0.0003	-0.00
5.360	1.0004	+0.0001	0.00
5.365	0.9999	-0.0005	-0.00
5.370	1.0005	+0.0002	0.00
5.375	1.0000	+0.0002	-0.00
5.380	1.0002	-0.0002	0.00
5.385	1.0007	+0.0003	0.01
5.390	1.0000	+0.0004	-0.00
5.395	1.0000	-0.0002	0.00
5.400	1.0001	+0.0003	0.00
5.405	1.0004	-0.0001	0.00
5.410	1.0001	+0.0003	0.00
5.415	0.9999	-0.0005	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.420	1.0000	+0.0002	-0.00
5.425	0.9996	-0.0004	-0.00
5.430	1.0002	-0.0001	0.00
5.435	1.0000	-0.0001	0.00
5.440	0.9999	+0.0003	-0.00
5.445	0.9998	-0.0001	-0.00
5.450	1.0001	+0.0003	0.00
5.455	1.0002	-0.0003	0.00
5.460	1.0000	-0.0000	0.00
5.465	0.9995	-0.0002	-0.00
5.470	1.0001	+0.0002	0.00
5.475	1.0000	-0.0000	0.00
5.480	1.0001	+0.0001	0.00
5.485	1.0003	-0.0001	0.00
5.490	0.9998	+0.0003	-0.00
5.495	1.0002	-0.0003	0.00
5.500	1.0000	+0.0001	-0.00
5.505	1.0000	+0.0000	0.00
5.510	1.0000	-0.0001	-0.00
5.515	1.0000	-0.0001	0.00
5.520	1.0000	+0.0001	-0.00
5.525	1.0000	+0.0000	0.00
5.530	1.0000	+0.0000	0.00
5.535	1.0000	-0.0002	0.00
5.540	0.9999	+0.0001	-0.00
5.545	1.0000	+0.0001	0.00
5.550	0.9999	+0.0000	-0.00
5.555	1.0001	+0.0001	0.00
5.560	1.0000	+0.0000	-0.00
5.565	1.0001	-0.0000	0.00
5.570	1.0001	+0.0001	0.00
5.575	1.0001	+0.0000	0.00
5.580	1.0000	-0.0000	-0.00
5.585	1.0000	+0.0000	0.00
5.590	1.0000	+0.0000	0.00
5.595	1.0000	-0.0000	0.00
5.600	0.9999	+0.0000	-0.00
5.605	1.0001	-0.0001	0.00
5.610	1.0002	+0.0001	0.00
5.615	1.0001	+0.0002	0.00
5.620	1.0000	+0.0000	-0.00
5.625	0.9999	+0.0001	-0.00
5.630	1.0000	+0.0001	-0.00
5.635	1.0001	-0.0001	0.00
5.640	1.0001	-0.0000	0.00
5.645	1.0002	+0.0002	0.00

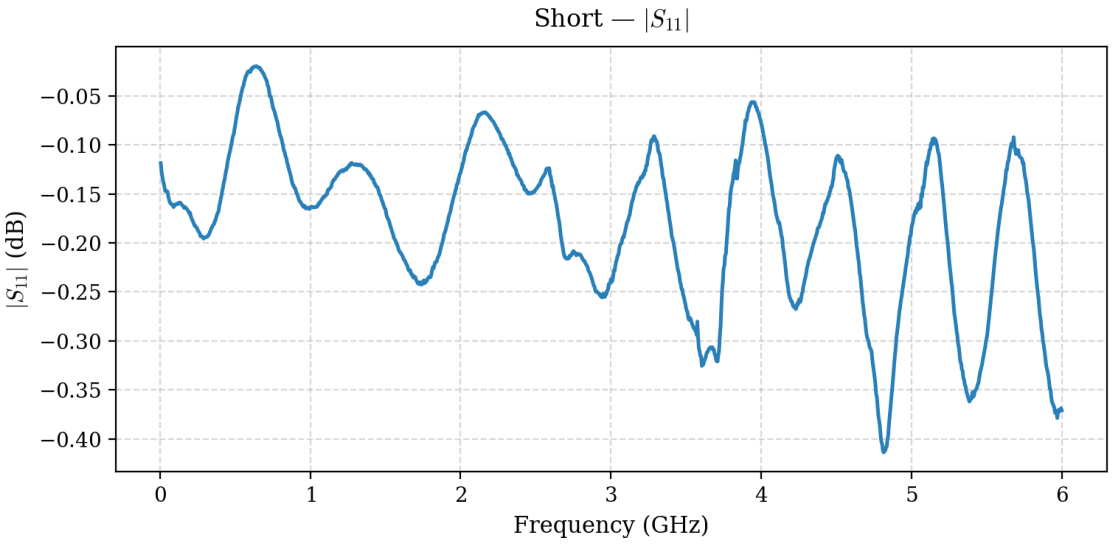
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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.650	1.0001	-0.0001	0.00
5.655	0.9998	-0.0000	-0.00
5.660	0.9999	+0.0002	-0.00
5.665	1.0001	-0.0003	0.00
5.670	0.9997	+0.0004	-0.00
5.675	0.9999	+0.0001	-0.00
5.680	0.9999	+0.0003	-0.00
5.685	1.0000	-0.0003	-0.00
5.690	1.0001	-0.0003	0.00
5.695	0.9997	-0.0002	-0.00
5.700	1.0000	-0.0000	0.00
5.705	0.9995	+0.0003	-0.00
5.710	1.0000	-0.0001	0.00
5.715	1.0001	+0.0004	0.00
5.720	0.9999	-0.0001	-0.00
5.725	0.9997	-0.0001	-0.00
5.730	0.9998	-0.0001	-0.00
5.735	0.9999	+0.0001	-0.00
5.740	1.0000	-0.0000	-0.00
5.745	0.9999	-0.0000	-0.00
5.750	1.0002	-0.0000	0.00
5.755	0.9999	-0.0001	-0.00
5.760	1.0001	+0.0003	0.00
5.765	1.0003	-0.0001	0.00
5.770	1.0002	-0.0000	0.00
5.775	1.0001	-0.0000	0.00
5.780	1.0000	-0.0000	-0.00
5.785	1.0002	+0.0001	0.00
5.790	1.0000	+0.0001	0.00
5.795	0.9999	+0.0000	-0.00
5.800	1.0000	-0.0000	0.00
5.805	1.0002	+0.0002	0.00
5.810	1.0000	-0.0001	-0.00
5.815	0.9999	-0.0001	-0.00
5.820	1.0001	+0.0000	0.00
5.825	1.0001	-0.0001	0.00
5.830	1.0000	+0.0001	0.00
5.835	1.0000	+0.0001	-0.00
5.840	1.0000	+0.0001	-0.00
5.845	1.0000	-0.0001	-0.00
5.850	1.0000	+0.0000	0.00
5.855	1.0001	+0.0001	0.00
5.860	1.0000	+0.0001	-0.00
5.865	0.9999	-0.0000	-0.00
5.870	1.0000	+0.0002	-0.00
5.875	1.0000	+0.0000	-0.00

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Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
5.880	1.0001	+0.0001	0.00
5.885	1.0001	-0.0000	0.00
5.890	0.9999	+0.0002	-0.00
5.895	0.9999	+0.0000	-0.00
5.900	1.0003	+0.0002	0.00
5.905	0.9995	+0.0001	-0.00
5.910	1.0004	+0.0003	0.00
5.915	1.0000	+0.0001	-0.00
5.920	1.0002	-0.0001	0.00
5.925	1.0000	+0.0001	-0.00
5.930	1.0000	+0.0004	0.00
5.935	0.9999	+0.0004	-0.00
5.940	1.0000	+0.0002	-0.00
5.945	1.0005	+0.0001	0.00
5.950	0.9998	-0.0001	-0.00
5.955	0.9998	+0.0003	-0.00
5.960	0.9999	+0.0002	-0.00
5.965	1.0005	+0.0002	0.00
5.970	0.9997	+0.0002	-0.00
5.975	0.9997	-0.0000	-0.00
5.980	1.0002	-0.0005	0.00
5.985	0.9992	+0.0004	-0.01
5.990	0.9999	+0.0003	-0.00
5.995	0.9998	+0.0000	-0.00
6.000	1.0000	-0.0004	0.00

3.2 Short



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.005	-0.9864	+0.0007	-0.12
0.010	-0.9855	+0.0013	-0.13
0.015	-0.9849	+0.0016	-0.13
0.020	-0.9844	+0.0014	-0.14
0.025	-0.9841	+0.0014	-0.14
0.030	-0.9836	+0.0012	-0.14
0.035	-0.9832	+0.0011	-0.15
0.040	-0.9832	+0.0010	-0.15
0.045	-0.9832	+0.0007	-0.15
0.050	-0.9831	+0.0004	-0.15
0.055	-0.9826	+0.0002	-0.15
0.060	-0.9822	+0.0001	-0.16
0.065	-0.9819	-0.0001	-0.16
0.070	-0.9818	-0.0004	-0.16
0.075	-0.9817	-0.0009	-0.16
0.080	-0.9817	-0.0011	-0.16
0.085	-0.9816	-0.0014	-0.16
0.090	-0.9813	-0.0016	-0.16
0.095	-0.9815	-0.0017	-0.16
0.100	-0.9816	-0.0020	-0.16
0.105	-0.9816	-0.0020	-0.16
0.110	-0.9817	-0.0021	-0.16
0.115	-0.9817	-0.0022	-0.16
0.120	-0.9817	-0.0023	-0.16
0.125	-0.9818	-0.0023	-0.16
0.130	-0.9818	-0.0022	-0.16
0.135	-0.9819	-0.0021	-0.16
0.140	-0.9817	-0.0020	-0.16
0.145	-0.9816	-0.0021	-0.16
0.150	-0.9815	-0.0021	-0.16
0.155	-0.9814	-0.0021	-0.16
0.160	-0.9813	-0.0022	-0.16
0.165	-0.9814	-0.0022	-0.16
0.170	-0.9813	-0.0023	-0.16
0.175	-0.9812	-0.0023	-0.17
0.180	-0.9811	-0.0021	-0.17
0.185	-0.9810	-0.0022	-0.17
0.190	-0.9808	-0.0023	-0.17
0.195	-0.9806	-0.0025	-0.17
0.200	-0.9807	-0.0022	-0.17
0.205	-0.9803	-0.0024	-0.17
0.210	-0.9800	-0.0024	-0.18
0.215	-0.9798	-0.0027	-0.18
0.220	-0.9797	-0.0028	-0.18
0.225	-0.9795	-0.0030	-0.18
0.230	-0.9792	-0.0033	-0.18

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.235	-0.9792	-0.0033	-0.18
0.240	-0.9790	-0.0034	-0.18
0.245	-0.9790	-0.0036	-0.18
0.250	-0.9787	-0.0038	-0.19
0.255	-0.9785	-0.0041	-0.19
0.260	-0.9784	-0.0044	-0.19
0.265	-0.9783	-0.0046	-0.19
0.270	-0.9780	-0.0050	-0.19
0.275	-0.9780	-0.0053	-0.19
0.280	-0.9780	-0.0057	-0.19
0.285	-0.9778	-0.0060	-0.19
0.290	-0.9777	-0.0061	-0.20
0.295	-0.9778	-0.0066	-0.19
0.300	-0.9778	-0.0071	-0.19
0.305	-0.9779	-0.0076	-0.19
0.310	-0.9779	-0.0082	-0.19
0.315	-0.9779	-0.0090	-0.19
0.320	-0.9780	-0.0096	-0.19
0.325	-0.9781	-0.0100	-0.19
0.330	-0.9783	-0.0105	-0.19
0.335	-0.9785	-0.0111	-0.19
0.340	-0.9788	-0.0116	-0.19
0.345	-0.9789	-0.0123	-0.18
0.350	-0.9791	-0.0129	-0.18
0.355	-0.9792	-0.0135	-0.18
0.360	-0.9793	-0.0141	-0.18
0.365	-0.9796	-0.0148	-0.18
0.370	-0.9797	-0.0154	-0.18
0.375	-0.9801	-0.0162	-0.17
0.380	-0.9804	-0.0167	-0.17
0.385	-0.9806	-0.0174	-0.17
0.390	-0.9807	-0.0180	-0.17
0.395	-0.9814	-0.0187	-0.16
0.400	-0.9817	-0.0196	-0.16
0.405	-0.9822	-0.0204	-0.15
0.410	-0.9825	-0.0213	-0.15
0.415	-0.9830	-0.0222	-0.15
0.420	-0.9834	-0.0229	-0.14
0.425	-0.9839	-0.0236	-0.14
0.430	-0.9842	-0.0246	-0.14
0.435	-0.9848	-0.0253	-0.13
0.440	-0.9852	-0.0261	-0.13
0.445	-0.9856	-0.0269	-0.12
0.450	-0.9862	-0.0276	-0.12
0.455	-0.9866	-0.0285	-0.11
0.460	-0.9870	-0.0291	-0.11

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.465	-0.9873	-0.0299	-0.11
0.470	-0.9877	-0.0306	-0.10
0.475	-0.9881	-0.0314	-0.10
0.480	-0.9886	-0.0321	-0.10
0.485	-0.9890	-0.0330	-0.09
0.490	-0.9892	-0.0337	-0.09
0.495	-0.9897	-0.0343	-0.09
0.500	-0.9902	-0.0350	-0.08
0.505	-0.9907	-0.0357	-0.08
0.510	-0.9912	-0.0364	-0.07
0.515	-0.9917	-0.0368	-0.07
0.520	-0.9922	-0.0373	-0.06
0.525	-0.9925	-0.0382	-0.06
0.530	-0.9928	-0.0387	-0.06
0.535	-0.9933	-0.0392	-0.05
0.540	-0.9937	-0.0396	-0.05
0.545	-0.9940	-0.0401	-0.05
0.550	-0.9944	-0.0405	-0.04
0.555	-0.9945	-0.0410	-0.04
0.560	-0.9950	-0.0414	-0.04
0.565	-0.9953	-0.0418	-0.03
0.570	-0.9956	-0.0423	-0.03
0.575	-0.9957	-0.0426	-0.03
0.580	-0.9960	-0.0429	-0.03
0.585	-0.9962	-0.0432	-0.03
0.590	-0.9961	-0.0435	-0.03
0.595	-0.9962	-0.0436	-0.02
0.600	-0.9961	-0.0438	-0.03
0.605	-0.9962	-0.0439	-0.02
0.610	-0.9965	-0.0442	-0.02
0.615	-0.9965	-0.0442	-0.02
0.620	-0.9966	-0.0441	-0.02
0.625	-0.9967	-0.0443	-0.02
0.630	-0.9967	-0.0441	-0.02
0.635	-0.9967	-0.0441	-0.02
0.640	-0.9967	-0.0441	-0.02
0.645	-0.9967	-0.0441	-0.02
0.650	-0.9966	-0.0440	-0.02
0.655	-0.9966	-0.0438	-0.02
0.660	-0.9966	-0.0436	-0.02
0.665	-0.9966	-0.0435	-0.02
0.670	-0.9963	-0.0432	-0.02
0.675	-0.9963	-0.0430	-0.02
0.680	-0.9961	-0.0428	-0.03
0.685	-0.9960	-0.0424	-0.03
0.690	-0.9959	-0.0424	-0.03

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.695	-0.9957	-0.0421	-0.03
0.700	-0.9955	-0.0417	-0.03
0.705	-0.9953	-0.0413	-0.03
0.710	-0.9952	-0.0408	-0.03
0.715	-0.9947	-0.0404	-0.04
0.720	-0.9945	-0.0399	-0.04
0.725	-0.9942	-0.0394	-0.04
0.730	-0.9935	-0.0389	-0.05
0.735	-0.9934	-0.0385	-0.05
0.740	-0.9932	-0.0380	-0.05
0.745	-0.9928	-0.0375	-0.06
0.750	-0.9923	-0.0368	-0.06
0.755	-0.9920	-0.0362	-0.06
0.760	-0.9919	-0.0357	-0.07
0.765	-0.9916	-0.0349	-0.07
0.770	-0.9913	-0.0342	-0.07
0.775	-0.9909	-0.0335	-0.07
0.780	-0.9908	-0.0329	-0.08
0.785	-0.9903	-0.0321	-0.08
0.790	-0.9898	-0.0312	-0.08
0.795	-0.9895	-0.0307	-0.09
0.800	-0.9892	-0.0301	-0.09
0.805	-0.9890	-0.0292	-0.09
0.810	-0.9887	-0.0285	-0.10
0.815	-0.9883	-0.0279	-0.10
0.820	-0.9879	-0.0273	-0.10
0.825	-0.9875	-0.0264	-0.11
0.830	-0.9873	-0.0258	-0.11
0.835	-0.9869	-0.0250	-0.11
0.840	-0.9868	-0.0243	-0.11
0.845	-0.9862	-0.0236	-0.12
0.850	-0.9860	-0.0229	-0.12
0.855	-0.9856	-0.0223	-0.12
0.860	-0.9853	-0.0215	-0.13
0.865	-0.9850	-0.0208	-0.13
0.870	-0.9848	-0.0201	-0.13
0.875	-0.9843	-0.0195	-0.14
0.880	-0.9840	-0.0188	-0.14
0.885	-0.9837	-0.0184	-0.14
0.890	-0.9837	-0.0178	-0.14
0.895	-0.9834	-0.0173	-0.14
0.900	-0.9834	-0.0166	-0.14
0.905	-0.9831	-0.0162	-0.15
0.910	-0.9828	-0.0155	-0.15
0.915	-0.9826	-0.0152	-0.15
0.920	-0.9823	-0.0147	-0.15

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
0.925	-0.9821	-0.0141	-0.16
0.930	-0.9819	-0.0137	-0.16
0.935	-0.9819	-0.0133	-0.16
0.940	-0.9816	-0.0129	-0.16
0.945	-0.9816	-0.0125	-0.16
0.950	-0.9814	-0.0121	-0.16
0.955	-0.9813	-0.0120	-0.16
0.960	-0.9814	-0.0119	-0.16
0.965	-0.9813	-0.0117	-0.16
0.970	-0.9811	-0.0115	-0.16
0.975	-0.9812	-0.0112	-0.16
0.980	-0.9812	-0.0111	-0.16
0.985	-0.9811	-0.0108	-0.16
0.990	-0.9811	-0.0110	-0.17
0.995	-0.9811	-0.0109	-0.17
1.000	-0.9812	-0.0110	-0.16
1.005	-0.9813	-0.0108	-0.16
1.010	-0.9813	-0.0108	-0.16
1.015	-0.9812	-0.0111	-0.16
1.020	-0.9813	-0.0114	-0.16
1.025	-0.9813	-0.0114	-0.16
1.030	-0.9812	-0.0116	-0.16
1.035	-0.9813	-0.0118	-0.16
1.040	-0.9813	-0.0121	-0.16
1.045	-0.9813	-0.0124	-0.16
1.050	-0.9815	-0.0131	-0.16
1.055	-0.9817	-0.0134	-0.16
1.060	-0.9816	-0.0138	-0.16
1.065	-0.9818	-0.0143	-0.16
1.070	-0.9819	-0.0147	-0.16
1.075	-0.9821	-0.0150	-0.16
1.080	-0.9822	-0.0156	-0.15
1.085	-0.9822	-0.0163	-0.15
1.090	-0.9825	-0.0169	-0.15
1.095	-0.9824	-0.0173	-0.15
1.100	-0.9825	-0.0177	-0.15
1.105	-0.9826	-0.0185	-0.15
1.110	-0.9826	-0.0192	-0.15
1.115	-0.9830	-0.0197	-0.15
1.120	-0.9832	-0.0203	-0.15
1.125	-0.9831	-0.0210	-0.15
1.130	-0.9833	-0.0218	-0.14
1.135	-0.9833	-0.0226	-0.14
1.140	-0.9835	-0.0233	-0.14
1.145	-0.9838	-0.0241	-0.14
1.150	-0.9840	-0.0247	-0.14

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.155	-0.9842	-0.0254	-0.14
1.160	-0.9841	-0.0262	-0.14
1.165	-0.9843	-0.0267	-0.13
1.170	-0.9844	-0.0275	-0.13
1.175	-0.9845	-0.0282	-0.13
1.180	-0.9845	-0.0290	-0.13
1.185	-0.9848	-0.0297	-0.13
1.190	-0.9848	-0.0305	-0.13
1.195	-0.9849	-0.0310	-0.13
1.200	-0.9849	-0.0315	-0.13
1.205	-0.9850	-0.0324	-0.13
1.210	-0.9850	-0.0330	-0.13
1.215	-0.9849	-0.0337	-0.13
1.220	-0.9849	-0.0342	-0.13
1.225	-0.9850	-0.0348	-0.13
1.230	-0.9850	-0.0354	-0.13
1.235	-0.9851	-0.0358	-0.12
1.240	-0.9850	-0.0362	-0.13
1.245	-0.9852	-0.0366	-0.12
1.250	-0.9852	-0.0371	-0.12
1.255	-0.9853	-0.0376	-0.12
1.260	-0.9855	-0.0379	-0.12
1.265	-0.9854	-0.0382	-0.12
1.270	-0.9857	-0.0386	-0.12
1.275	-0.9857	-0.0386	-0.12
1.280	-0.9856	-0.0388	-0.12
1.285	-0.9855	-0.0389	-0.12
1.290	-0.9855	-0.0390	-0.12
1.295	-0.9855	-0.0388	-0.12
1.300	-0.9855	-0.0386	-0.12
1.305	-0.9856	-0.0385	-0.12
1.310	-0.9855	-0.0387	-0.12
1.315	-0.9855	-0.0385	-0.12
1.320	-0.9854	-0.0382	-0.12
1.325	-0.9855	-0.0383	-0.12
1.330	-0.9855	-0.0380	-0.12
1.335	-0.9855	-0.0376	-0.12
1.340	-0.9854	-0.0372	-0.12
1.345	-0.9854	-0.0368	-0.12
1.350	-0.9853	-0.0364	-0.12
1.355	-0.9852	-0.0358	-0.12
1.360	-0.9851	-0.0351	-0.12
1.365	-0.9851	-0.0344	-0.12
1.370	-0.9851	-0.0338	-0.12
1.375	-0.9850	-0.0331	-0.13
1.380	-0.9849	-0.0324	-0.13

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.385	-0.9847	-0.0316	-0.13
1.390	-0.9849	-0.0307	-0.13
1.395	-0.9846	-0.0297	-0.13
1.400	-0.9846	-0.0286	-0.13
1.405	-0.9844	-0.0277	-0.13
1.410	-0.9843	-0.0267	-0.13
1.415	-0.9843	-0.0258	-0.13
1.420	-0.9842	-0.0246	-0.14
1.425	-0.9838	-0.0235	-0.14
1.430	-0.9838	-0.0221	-0.14
1.435	-0.9836	-0.0209	-0.14
1.440	-0.9835	-0.0198	-0.14
1.445	-0.9835	-0.0185	-0.14
1.450	-0.9833	-0.0173	-0.14
1.455	-0.9831	-0.0160	-0.15
1.460	-0.9827	-0.0146	-0.15
1.465	-0.9826	-0.0135	-0.15
1.470	-0.9824	-0.0121	-0.15
1.475	-0.9821	-0.0106	-0.16
1.480	-0.9820	-0.0092	-0.16
1.485	-0.9817	-0.0078	-0.16
1.490	-0.9814	-0.0068	-0.16
1.495	-0.9812	-0.0051	-0.16
1.500	-0.9811	-0.0037	-0.17
1.505	-0.9806	-0.0023	-0.17
1.510	-0.9803	-0.0010	-0.17
1.515	-0.9800	+0.0003	-0.18
1.520	-0.9799	+0.0015	-0.18
1.525	-0.9797	+0.0030	-0.18
1.530	-0.9793	+0.0043	-0.18
1.535	-0.9790	+0.0058	-0.18
1.540	-0.9787	+0.0071	-0.19
1.545	-0.9785	+0.0083	-0.19
1.550	-0.9782	+0.0095	-0.19
1.555	-0.9778	+0.0107	-0.19
1.560	-0.9776	+0.0118	-0.20
1.565	-0.9775	+0.0129	-0.20
1.570	-0.9773	+0.0139	-0.20
1.575	-0.9770	+0.0149	-0.20
1.580	-0.9768	+0.0157	-0.20
1.585	-0.9765	+0.0167	-0.21
1.590	-0.9762	+0.0175	-0.21
1.595	-0.9761	+0.0185	-0.21
1.600	-0.9759	+0.0191	-0.21
1.605	-0.9758	+0.0199	-0.21
1.610	-0.9756	+0.0207	-0.21

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.615	-0.9755	+0.0215	-0.21
1.620	-0.9752	+0.0223	-0.22
1.625	-0.9750	+0.0230	-0.22
1.630	-0.9749	+0.0237	-0.22
1.635	-0.9747	+0.0242	-0.22
1.640	-0.9745	+0.0247	-0.22
1.645	-0.9743	+0.0252	-0.22
1.650	-0.9740	+0.0254	-0.23
1.655	-0.9737	+0.0257	-0.23
1.660	-0.9737	+0.0261	-0.23
1.665	-0.9737	+0.0263	-0.23
1.670	-0.9733	+0.0263	-0.23
1.675	-0.9732	+0.0263	-0.23
1.680	-0.9732	+0.0265	-0.23
1.685	-0.9729	+0.0265	-0.24
1.690	-0.9730	+0.0262	-0.23
1.695	-0.9730	+0.0259	-0.23
1.700	-0.9729	+0.0257	-0.24
1.705	-0.9726	+0.0254	-0.24
1.710	-0.9725	+0.0252	-0.24
1.715	-0.9723	+0.0249	-0.24
1.720	-0.9724	+0.0243	-0.24
1.725	-0.9722	+0.0237	-0.24
1.730	-0.9724	+0.0231	-0.24
1.735	-0.9725	+0.0226	-0.24
1.740	-0.9722	+0.0218	-0.24
1.745	-0.9725	+0.0209	-0.24
1.750	-0.9724	+0.0198	-0.24
1.755	-0.9728	+0.0190	-0.24
1.760	-0.9725	+0.0180	-0.24
1.765	-0.9725	+0.0169	-0.24
1.770	-0.9728	+0.0162	-0.24
1.775	-0.9727	+0.0151	-0.24
1.780	-0.9730	+0.0139	-0.24
1.785	-0.9731	+0.0131	-0.24
1.790	-0.9733	+0.0121	-0.23
1.795	-0.9733	+0.0108	-0.23
1.800	-0.9733	+0.0094	-0.24
1.805	-0.9736	+0.0082	-0.23
1.810	-0.9740	+0.0072	-0.23
1.815	-0.9744	+0.0061	-0.22
1.820	-0.9743	+0.0048	-0.23
1.825	-0.9746	+0.0038	-0.22
1.830	-0.9750	+0.0023	-0.22
1.835	-0.9750	+0.0010	-0.22
1.840	-0.9751	-0.0004	-0.22

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.845	-0.9754	-0.0015	-0.22
1.850	-0.9756	-0.0029	-0.21
1.855	-0.9758	-0.0041	-0.21
1.860	-0.9760	-0.0054	-0.21
1.865	-0.9763	-0.0066	-0.21
1.870	-0.9767	-0.0079	-0.20
1.875	-0.9770	-0.0091	-0.20
1.880	-0.9773	-0.0104	-0.20
1.885	-0.9777	-0.0115	-0.20
1.890	-0.9781	-0.0129	-0.19
1.895	-0.9783	-0.0140	-0.19
1.900	-0.9785	-0.0151	-0.19
1.905	-0.9789	-0.0162	-0.18
1.910	-0.9792	-0.0172	-0.18
1.915	-0.9795	-0.0182	-0.18
1.920	-0.9800	-0.0193	-0.17
1.925	-0.9803	-0.0203	-0.17
1.930	-0.9805	-0.0213	-0.17
1.935	-0.9808	-0.0221	-0.17
1.940	-0.9813	-0.0227	-0.16
1.945	-0.9816	-0.0238	-0.16
1.950	-0.9819	-0.0244	-0.16
1.955	-0.9823	-0.0251	-0.15
1.960	-0.9825	-0.0257	-0.15
1.965	-0.9829	-0.0263	-0.15
1.970	-0.9831	-0.0270	-0.14
1.975	-0.9835	-0.0273	-0.14
1.980	-0.9837	-0.0278	-0.14
1.985	-0.9841	-0.0283	-0.14
1.990	-0.9844	-0.0286	-0.13
1.995	-0.9848	-0.0289	-0.13
2.000	-0.9849	-0.0289	-0.13
2.005	-0.9853	-0.0289	-0.12
2.010	-0.9856	-0.0289	-0.12
2.015	-0.9860	-0.0290	-0.12
2.020	-0.9862	-0.0288	-0.12
2.025	-0.9868	-0.0285	-0.11
2.030	-0.9871	-0.0282	-0.11
2.035	-0.9875	-0.0281	-0.11
2.040	-0.9875	-0.0277	-0.11
2.045	-0.9880	-0.0273	-0.10
2.050	-0.9882	-0.0267	-0.10
2.055	-0.9884	-0.0263	-0.10
2.060	-0.9888	-0.0258	-0.09
2.065	-0.9891	-0.0250	-0.09
2.070	-0.9893	-0.0244	-0.09

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.075	-0.9897	-0.0235	-0.09
2.080	-0.9901	-0.0228	-0.08
2.085	-0.9903	-0.0220	-0.08
2.090	-0.9905	-0.0213	-0.08
2.095	-0.9906	-0.0204	-0.08
2.100	-0.9909	-0.0195	-0.08
2.105	-0.9912	-0.0184	-0.08
2.110	-0.9914	-0.0174	-0.07
2.115	-0.9916	-0.0163	-0.07
2.120	-0.9919	-0.0151	-0.07
2.125	-0.9919	-0.0139	-0.07
2.130	-0.9920	-0.0130	-0.07
2.135	-0.9920	-0.0116	-0.07
2.140	-0.9922	-0.0104	-0.07
2.145	-0.9922	-0.0092	-0.07
2.150	-0.9922	-0.0079	-0.07
2.155	-0.9923	-0.0068	-0.07
2.160	-0.9923	-0.0056	-0.07
2.165	-0.9923	-0.0043	-0.07
2.170	-0.9923	-0.0031	-0.07
2.175	-0.9921	-0.0020	-0.07
2.180	-0.9921	-0.0006	-0.07
2.185	-0.9920	+0.0006	-0.07
2.190	-0.9918	+0.0018	-0.07
2.195	-0.9918	+0.0030	-0.07
2.200	-0.9916	+0.0043	-0.07
2.205	-0.9915	+0.0055	-0.07
2.210	-0.9913	+0.0066	-0.08
2.215	-0.9913	+0.0077	-0.08
2.220	-0.9912	+0.0088	-0.08
2.225	-0.9909	+0.0095	-0.08
2.230	-0.9908	+0.0106	-0.08
2.235	-0.9906	+0.0115	-0.08
2.240	-0.9904	+0.0124	-0.08
2.245	-0.9903	+0.0132	-0.08
2.250	-0.9900	+0.0139	-0.09
2.255	-0.9899	+0.0147	-0.09
2.260	-0.9898	+0.0155	-0.09
2.265	-0.9895	+0.0162	-0.09
2.270	-0.9892	+0.0167	-0.09
2.275	-0.9890	+0.0173	-0.09
2.280	-0.9890	+0.0177	-0.09
2.285	-0.9888	+0.0183	-0.10
2.290	-0.9886	+0.0185	-0.10
2.295	-0.9884	+0.0190	-0.10
2.300	-0.9883	+0.0194	-0.10

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.305	-0.9881	+0.0197	-0.10
2.310	-0.9878	+0.0198	-0.11
2.315	-0.9877	+0.0200	-0.11
2.320	-0.9873	+0.0199	-0.11
2.325	-0.9873	+0.0200	-0.11
2.330	-0.9869	+0.0200	-0.11
2.335	-0.9868	+0.0199	-0.11
2.340	-0.9865	+0.0200	-0.12
2.345	-0.9863	+0.0198	-0.12
2.350	-0.9860	+0.0195	-0.12
2.355	-0.9859	+0.0189	-0.12
2.360	-0.9857	+0.0183	-0.12
2.365	-0.9856	+0.0176	-0.12
2.370	-0.9854	+0.0170	-0.13
2.375	-0.9849	+0.0161	-0.13
2.380	-0.9849	+0.0155	-0.13
2.385	-0.9848	+0.0145	-0.13
2.390	-0.9846	+0.0137	-0.13
2.395	-0.9845	+0.0126	-0.14
2.400	-0.9844	+0.0118	-0.14
2.405	-0.9842	+0.0105	-0.14
2.410	-0.9839	+0.0094	-0.14
2.415	-0.9838	+0.0081	-0.14
2.420	-0.9838	+0.0069	-0.14
2.425	-0.9833	+0.0056	-0.15
2.430	-0.9834	+0.0043	-0.15
2.435	-0.9833	+0.0028	-0.15
2.440	-0.9831	+0.0012	-0.15
2.445	-0.9830	-0.0003	-0.15
2.450	-0.9829	-0.0018	-0.15
2.455	-0.9830	-0.0035	-0.15
2.460	-0.9829	-0.0052	-0.15
2.465	-0.9829	-0.0068	-0.15
2.470	-0.9829	-0.0084	-0.15
2.475	-0.9830	-0.0100	-0.15
2.480	-0.9830	-0.0115	-0.15
2.485	-0.9830	-0.0133	-0.15
2.490	-0.9830	-0.0151	-0.15
2.495	-0.9831	-0.0167	-0.15
2.500	-0.9832	-0.0183	-0.15
2.505	-0.9833	-0.0201	-0.14
2.510	-0.9834	-0.0218	-0.14
2.515	-0.9834	-0.0234	-0.14
2.520	-0.9835	-0.0251	-0.14
2.525	-0.9835	-0.0266	-0.14
2.530	-0.9838	-0.0282	-0.14

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.535	-0.9840	-0.0297	-0.14
2.540	-0.9841	-0.0311	-0.13
2.545	-0.9842	-0.0327	-0.13
2.550	-0.9843	-0.0341	-0.13
2.555	-0.9844	-0.0355	-0.13
2.560	-0.9845	-0.0369	-0.13
2.565	-0.9846	-0.0383	-0.13
2.570	-0.9849	-0.0394	-0.12
2.575	-0.9850	-0.0403	-0.12
2.580	-0.9849	-0.0411	-0.12
2.585	-0.9850	-0.0417	-0.12
2.590	-0.9849	-0.0423	-0.12
2.595	-0.9841	-0.0433	-0.13
2.600	-0.9839	-0.0439	-0.13
2.605	-0.9833	-0.0451	-0.14
2.610	-0.9826	-0.0460	-0.14
2.615	-0.9827	-0.0470	-0.14
2.620	-0.9819	-0.0482	-0.15
2.625	-0.9815	-0.0495	-0.15
2.630	-0.9809	-0.0506	-0.16
2.635	-0.9805	-0.0518	-0.16
2.640	-0.9799	-0.0528	-0.16
2.645	-0.9793	-0.0539	-0.17
2.650	-0.9786	-0.0547	-0.17
2.655	-0.9782	-0.0556	-0.18
2.660	-0.9775	-0.0567	-0.18
2.665	-0.9766	-0.0576	-0.19
2.670	-0.9759	-0.0586	-0.20
2.675	-0.9753	-0.0599	-0.20
2.680	-0.9747	-0.0610	-0.21
2.685	-0.9741	-0.0622	-0.21
2.690	-0.9737	-0.0634	-0.21
2.695	-0.9734	-0.0647	-0.22
2.700	-0.9734	-0.0656	-0.21
2.705	-0.9731	-0.0666	-0.22
2.710	-0.9731	-0.0676	-0.22
2.715	-0.9731	-0.0682	-0.22
2.720	-0.9730	-0.0690	-0.22
2.725	-0.9733	-0.0690	-0.21
2.730	-0.9733	-0.0693	-0.21
2.735	-0.9733	-0.0696	-0.21
2.740	-0.9734	-0.0697	-0.21
2.745	-0.9736	-0.0697	-0.21
2.750	-0.9738	-0.0698	-0.21
2.755	-0.9737	-0.0699	-0.21
2.760	-0.9735	-0.0698	-0.21

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.765	-0.9735	-0.0696	-0.21
2.770	-0.9733	-0.0695	-0.21
2.775	-0.9735	-0.0694	-0.21
2.780	-0.9735	-0.0692	-0.21
2.785	-0.9734	-0.0694	-0.21
2.790	-0.9734	-0.0692	-0.21
2.795	-0.9735	-0.0690	-0.21
2.800	-0.9734	-0.0689	-0.21
2.805	-0.9732	-0.0689	-0.21
2.810	-0.9730	-0.0684	-0.22
2.815	-0.9729	-0.0685	-0.22
2.820	-0.9728	-0.0683	-0.22
2.825	-0.9728	-0.0682	-0.22
2.830	-0.9725	-0.0678	-0.22
2.835	-0.9724	-0.0675	-0.22
2.840	-0.9723	-0.0672	-0.22
2.845	-0.9722	-0.0670	-0.22
2.850	-0.9718	-0.0667	-0.23
2.855	-0.9717	-0.0664	-0.23
2.860	-0.9715	-0.0663	-0.23
2.865	-0.9713	-0.0661	-0.23
2.870	-0.9711	-0.0658	-0.23
2.875	-0.9708	-0.0658	-0.24
2.880	-0.9705	-0.0657	-0.24
2.885	-0.9702	-0.0654	-0.24
2.890	-0.9701	-0.0653	-0.24
2.895	-0.9697	-0.0651	-0.25
2.900	-0.9695	-0.0650	-0.25
2.905	-0.9695	-0.0649	-0.25
2.910	-0.9694	-0.0649	-0.25
2.915	-0.9693	-0.0648	-0.25
2.920	-0.9692	-0.0648	-0.25
2.925	-0.9692	-0.0650	-0.25
2.930	-0.9690	-0.0653	-0.25
2.935	-0.9689	-0.0654	-0.25
2.940	-0.9688	-0.0657	-0.26
2.945	-0.9688	-0.0658	-0.26
2.950	-0.9692	-0.0661	-0.25
2.955	-0.9689	-0.0665	-0.25
2.960	-0.9687	-0.0671	-0.26
2.965	-0.9689	-0.0675	-0.25
2.970	-0.9690	-0.0678	-0.25
2.975	-0.9694	-0.0683	-0.25
2.980	-0.9694	-0.0688	-0.25
2.985	-0.9700	-0.0692	-0.24
2.990	-0.9700	-0.0699	-0.24

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.995	-0.9701	-0.0702	-0.24
3.000	-0.9705	-0.0711	-0.24
3.005	-0.9701	-0.0713	-0.24
3.010	-0.9702	-0.0721	-0.24
3.015	-0.9706	-0.0728	-0.23
3.020	-0.9713	-0.0733	-0.23
3.025	-0.9712	-0.0742	-0.23
3.030	-0.9718	-0.0752	-0.22
3.035	-0.9721	-0.0761	-0.22
3.040	-0.9723	-0.0774	-0.22
3.045	-0.9726	-0.0783	-0.21
3.050	-0.9726	-0.0794	-0.21
3.055	-0.9732	-0.0802	-0.21
3.060	-0.9735	-0.0817	-0.20
3.065	-0.9736	-0.0828	-0.20
3.070	-0.9736	-0.0837	-0.20
3.075	-0.9739	-0.0850	-0.20
3.080	-0.9741	-0.0861	-0.19
3.085	-0.9740	-0.0872	-0.19
3.090	-0.9743	-0.0879	-0.19
3.095	-0.9745	-0.0889	-0.19
3.100	-0.9748	-0.0900	-0.18
3.105	-0.9752	-0.0908	-0.18
3.110	-0.9754	-0.0917	-0.18
3.115	-0.9754	-0.0926	-0.18
3.120	-0.9758	-0.0934	-0.17
3.125	-0.9759	-0.0942	-0.17
3.130	-0.9764	-0.0950	-0.17
3.135	-0.9768	-0.0959	-0.16
3.140	-0.9771	-0.0970	-0.16
3.145	-0.9773	-0.0976	-0.16
3.150	-0.9776	-0.0983	-0.15
3.155	-0.9779	-0.0990	-0.15
3.160	-0.9780	-0.0996	-0.15
3.165	-0.9783	-0.1001	-0.15
3.170	-0.9786	-0.1007	-0.14
3.175	-0.9786	-0.1010	-0.14
3.180	-0.9789	-0.1017	-0.14
3.185	-0.9788	-0.1019	-0.14
3.190	-0.9790	-0.1024	-0.14
3.195	-0.9791	-0.1029	-0.14
3.200	-0.9793	-0.1036	-0.13
3.205	-0.9793	-0.1037	-0.13
3.210	-0.9792	-0.1040	-0.13
3.215	-0.9796	-0.1048	-0.13
3.220	-0.9795	-0.1050	-0.13

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.225	-0.9799	-0.1053	-0.13
3.230	-0.9798	-0.1063	-0.13
3.235	-0.9804	-0.1061	-0.12
3.240	-0.9805	-0.1063	-0.12
3.245	-0.9812	-0.1063	-0.11
3.250	-0.9816	-0.1067	-0.11
3.255	-0.9818	-0.1066	-0.11
3.260	-0.9823	-0.1064	-0.10
3.265	-0.9830	-0.1067	-0.10
3.270	-0.9834	-0.1062	-0.10
3.275	-0.9832	-0.1061	-0.10
3.280	-0.9836	-0.1056	-0.09
3.285	-0.9839	-0.1058	-0.09
3.290	-0.9839	-0.1052	-0.09
3.295	-0.9835	-0.1048	-0.10
3.300	-0.9836	-0.1044	-0.10
3.305	-0.9834	-0.1041	-0.10
3.310	-0.9831	-0.1035	-0.10
3.315	-0.9824	-0.1031	-0.11
3.320	-0.9824	-0.1018	-0.11
3.325	-0.9822	-0.1016	-0.11
3.330	-0.9817	-0.1005	-0.12
3.335	-0.9811	-0.0998	-0.12
3.340	-0.9807	-0.0985	-0.13
3.345	-0.9799	-0.0977	-0.13
3.350	-0.9799	-0.0964	-0.13
3.355	-0.9793	-0.0955	-0.14
3.360	-0.9789	-0.0943	-0.15
3.365	-0.9785	-0.0928	-0.15
3.370	-0.9779	-0.0915	-0.16
3.375	-0.9776	-0.0903	-0.16
3.380	-0.9770	-0.0893	-0.17
3.385	-0.9766	-0.0882	-0.17
3.390	-0.9760	-0.0866	-0.18
3.395	-0.9756	-0.0855	-0.18
3.400	-0.9752	-0.0841	-0.19
3.405	-0.9745	-0.0829	-0.19
3.410	-0.9743	-0.0817	-0.20
3.415	-0.9741	-0.0803	-0.20
3.420	-0.9737	-0.0793	-0.20
3.425	-0.9732	-0.0780	-0.21
3.430	-0.9727	-0.0768	-0.21
3.435	-0.9723	-0.0755	-0.22
3.440	-0.9718	-0.0744	-0.22
3.445	-0.9714	-0.0733	-0.23
3.450	-0.9712	-0.0722	-0.23

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.455	-0.9709	-0.0708	-0.23
3.460	-0.9703	-0.0697	-0.24
3.465	-0.9701	-0.0690	-0.24
3.470	-0.9696	-0.0675	-0.25
3.475	-0.9696	-0.0666	-0.25
3.480	-0.9691	-0.0657	-0.25
3.485	-0.9689	-0.0647	-0.26
3.490	-0.9685	-0.0638	-0.26
3.495	-0.9683	-0.0631	-0.26
3.500	-0.9678	-0.0619	-0.27
3.505	-0.9675	-0.0613	-0.27
3.510	-0.9674	-0.0603	-0.27
3.515	-0.9668	-0.0595	-0.28
3.520	-0.9667	-0.0587	-0.28
3.525	-0.9670	-0.0572	-0.28
3.530	-0.9670	-0.0555	-0.28
3.535	-0.9667	-0.0556	-0.28
3.540	-0.9664	-0.0554	-0.28
3.545	-0.9662	-0.0550	-0.28
3.550	-0.9660	-0.0546	-0.29
3.555	-0.9659	-0.0544	-0.29
3.560	-0.9657	-0.0541	-0.29
3.565	-0.9659	-0.0539	-0.29
3.570	-0.9653	-0.0535	-0.29
3.575	-0.9667	-0.0544	-0.28
3.580	-0.9650	-0.0537	-0.30
3.585	-0.9637	-0.0538	-0.31
3.590	-0.9634	-0.0542	-0.31
3.595	-0.9629	-0.0539	-0.32
3.600	-0.9628	-0.0536	-0.32
3.605	-0.9618	-0.0529	-0.33
3.610	-0.9619	-0.0523	-0.32
3.615	-0.9620	-0.0523	-0.32
3.620	-0.9623	-0.0524	-0.32
3.625	-0.9625	-0.0527	-0.32
3.630	-0.9628	-0.0526	-0.32
3.635	-0.9631	-0.0527	-0.31
3.640	-0.9634	-0.0526	-0.31
3.645	-0.9633	-0.0532	-0.31
3.650	-0.9636	-0.0533	-0.31
3.655	-0.9637	-0.0535	-0.31
3.660	-0.9637	-0.0535	-0.31
3.665	-0.9638	-0.0536	-0.31
3.670	-0.9637	-0.0540	-0.31
3.675	-0.9638	-0.0543	-0.31
3.680	-0.9635	-0.0546	-0.31

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.685	-0.9636	-0.0542	-0.31
3.690	-0.9631	-0.0539	-0.31
3.695	-0.9629	-0.0532	-0.31
3.700	-0.9625	-0.0526	-0.32
3.705	-0.9624	-0.0512	-0.32
3.710	-0.9624	-0.0501	-0.32
3.715	-0.9633	-0.0484	-0.31
3.720	-0.9639	-0.0465	-0.31
3.725	-0.9657	-0.0450	-0.29
3.730	-0.9668	-0.0434	-0.28
3.735	-0.9690	-0.0432	-0.27
3.740	-0.9709	-0.0423	-0.25
3.745	-0.9719	-0.0431	-0.24
3.750	-0.9734	-0.0430	-0.23
3.755	-0.9736	-0.0432	-0.22
3.760	-0.9748	-0.0433	-0.21
3.765	-0.9751	-0.0428	-0.21
3.770	-0.9763	-0.0423	-0.20
3.775	-0.9775	-0.0422	-0.19
3.780	-0.9788	-0.0424	-0.18
3.785	-0.9802	-0.0424	-0.17
3.790	-0.9806	-0.0422	-0.16
3.795	-0.9819	-0.0420	-0.15
3.800	-0.9826	-0.0420	-0.14
3.805	-0.9830	-0.0421	-0.14
3.810	-0.9834	-0.0415	-0.14
3.815	-0.9841	-0.0413	-0.13
3.820	-0.9846	-0.0407	-0.13
3.825	-0.9850	-0.0407	-0.12
3.830	-0.9859	-0.0400	-0.12
3.835	-0.9839	-0.0388	-0.13
3.840	-0.9839	-0.0374	-0.13
3.845	-0.9851	-0.0363	-0.12
3.850	-0.9858	-0.0356	-0.12
3.855	-0.9859	-0.0343	-0.12
3.860	-0.9865	-0.0334	-0.11
3.865	-0.9869	-0.0319	-0.11
3.870	-0.9876	-0.0307	-0.10
3.875	-0.9881	-0.0293	-0.10
3.880	-0.9888	-0.0279	-0.09
3.885	-0.9896	-0.0266	-0.09
3.890	-0.9901	-0.0252	-0.08
3.895	-0.9911	-0.0240	-0.07
3.900	-0.9910	-0.0224	-0.08
3.905	-0.9914	-0.0206	-0.07
3.910	-0.9921	-0.0191	-0.07

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.915	-0.9923	-0.0175	-0.07
3.920	-0.9928	-0.0158	-0.06
3.925	-0.9929	-0.0140	-0.06
3.930	-0.9933	-0.0131	-0.06
3.935	-0.9934	-0.0115	-0.06
3.940	-0.9934	-0.0099	-0.06
3.945	-0.9935	-0.0083	-0.06
3.950	-0.9934	-0.0067	-0.06
3.955	-0.9935	-0.0050	-0.06
3.960	-0.9933	-0.0035	-0.06
3.965	-0.9929	-0.0020	-0.06
3.970	-0.9929	-0.0007	-0.06
3.975	-0.9925	+0.0011	-0.07
3.980	-0.9923	+0.0027	-0.07
3.985	-0.9921	+0.0042	-0.07
3.990	-0.9917	+0.0055	-0.07
3.995	-0.9914	+0.0071	-0.07
4.000	-0.9910	+0.0087	-0.08
4.005	-0.9906	+0.0099	-0.08
4.010	-0.9902	+0.0113	-0.08
4.015	-0.9896	+0.0125	-0.09
4.020	-0.9892	+0.0137	-0.09
4.025	-0.9886	+0.0146	-0.10
4.030	-0.9883	+0.0159	-0.10
4.035	-0.9878	+0.0169	-0.11
4.040	-0.9872	+0.0181	-0.11
4.045	-0.9867	+0.0191	-0.11
4.050	-0.9860	+0.0198	-0.12
4.055	-0.9853	+0.0206	-0.13
4.060	-0.9846	+0.0213	-0.13
4.065	-0.9841	+0.0216	-0.14
4.070	-0.9836	+0.0221	-0.14
4.075	-0.9829	+0.0227	-0.15
4.080	-0.9823	+0.0229	-0.15
4.085	-0.9819	+0.0230	-0.16
4.090	-0.9814	+0.0231	-0.16
4.095	-0.9807	+0.0232	-0.17
4.100	-0.9802	+0.0233	-0.17
4.105	-0.9799	+0.0234	-0.17
4.110	-0.9795	+0.0232	-0.18
4.115	-0.9792	+0.0232	-0.18
4.120	-0.9793	+0.0232	-0.18
4.125	-0.9789	+0.0232	-0.18
4.130	-0.9783	+0.0229	-0.19
4.135	-0.9778	+0.0232	-0.19
4.140	-0.9775	+0.0232	-0.20

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.145	-0.9764	+0.0230	-0.20
4.150	-0.9759	+0.0226	-0.21
4.155	-0.9752	+0.0223	-0.22
4.160	-0.9745	+0.0217	-0.22
4.165	-0.9737	+0.0203	-0.23
4.170	-0.9728	+0.0193	-0.24
4.175	-0.9724	+0.0176	-0.24
4.180	-0.9718	+0.0165	-0.25
4.185	-0.9712	+0.0151	-0.25
4.190	-0.9709	+0.0134	-0.26
4.195	-0.9707	+0.0117	-0.26
4.200	-0.9705	+0.0100	-0.26
4.205	-0.9705	+0.0082	-0.26
4.210	-0.9701	+0.0062	-0.26
4.215	-0.9699	+0.0041	-0.27
4.220	-0.9699	+0.0021	-0.27
4.225	-0.9700	+0.0000	-0.26
4.230	-0.9697	-0.0021	-0.27
4.235	-0.9699	-0.0045	-0.27
4.240	-0.9703	-0.0068	-0.26
4.245	-0.9701	-0.0093	-0.26
4.250	-0.9706	-0.0112	-0.26
4.255	-0.9707	-0.0137	-0.26
4.260	-0.9704	-0.0158	-0.26
4.265	-0.9706	-0.0181	-0.26
4.270	-0.9712	-0.0203	-0.25
4.275	-0.9713	-0.0228	-0.25
4.280	-0.9719	-0.0252	-0.24
4.285	-0.9722	-0.0276	-0.24
4.290	-0.9726	-0.0300	-0.24
4.295	-0.9728	-0.0322	-0.23
4.300	-0.9732	-0.0346	-0.23
4.305	-0.9737	-0.0370	-0.23
4.310	-0.9739	-0.0394	-0.22
4.315	-0.9742	-0.0416	-0.22
4.320	-0.9744	-0.0439	-0.22
4.325	-0.9745	-0.0463	-0.21
4.330	-0.9748	-0.0487	-0.21
4.335	-0.9750	-0.0509	-0.21
4.340	-0.9752	-0.0530	-0.21
4.345	-0.9754	-0.0552	-0.20
4.350	-0.9756	-0.0575	-0.20
4.355	-0.9757	-0.0594	-0.20
4.360	-0.9758	-0.0619	-0.19
4.365	-0.9760	-0.0640	-0.19
4.370	-0.9761	-0.0660	-0.19

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.375	-0.9765	-0.0681	-0.19
4.380	-0.9765	-0.0700	-0.18
4.385	-0.9764	-0.0720	-0.18
4.390	-0.9765	-0.0740	-0.18
4.395	-0.9767	-0.0757	-0.18
4.400	-0.9768	-0.0772	-0.18
4.405	-0.9769	-0.0789	-0.17
4.410	-0.9773	-0.0810	-0.17
4.415	-0.9771	-0.0826	-0.17
4.420	-0.9775	-0.0841	-0.17
4.425	-0.9779	-0.0854	-0.16
4.430	-0.9778	-0.0874	-0.16
4.435	-0.9780	-0.0887	-0.16
4.440	-0.9780	-0.0899	-0.16
4.445	-0.9782	-0.0918	-0.15
4.450	-0.9782	-0.0927	-0.15
4.455	-0.9785	-0.0945	-0.15
4.460	-0.9785	-0.0960	-0.15
4.465	-0.9788	-0.0978	-0.14
4.470	-0.9786	-0.0994	-0.14
4.475	-0.9790	-0.1015	-0.14
4.480	-0.9792	-0.1033	-0.13
4.485	-0.9800	-0.1049	-0.13
4.490	-0.9804	-0.1059	-0.12
4.495	-0.9807	-0.1074	-0.12
4.500	-0.9811	-0.1082	-0.11
4.505	-0.9811	-0.1090	-0.11
4.510	-0.9812	-0.1093	-0.11
4.515	-0.9809	-0.1096	-0.11
4.520	-0.9810	-0.1101	-0.11
4.525	-0.9804	-0.1102	-0.12
4.530	-0.9803	-0.1100	-0.12
4.535	-0.9807	-0.1099	-0.12
4.540	-0.9802	-0.1096	-0.12
4.545	-0.9798	-0.1089	-0.12
4.550	-0.9794	-0.1092	-0.13
4.555	-0.9794	-0.1089	-0.13
4.560	-0.9793	-0.1085	-0.13
4.565	-0.9789	-0.1081	-0.13
4.570	-0.9787	-0.1076	-0.13
4.575	-0.9785	-0.1072	-0.14
4.580	-0.9777	-0.1066	-0.14
4.585	-0.9777	-0.1060	-0.15
4.590	-0.9772	-0.1056	-0.15
4.595	-0.9767	-0.1051	-0.15
4.600	-0.9762	-0.1045	-0.16

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.605	-0.9756	-0.1038	-0.17
4.610	-0.9751	-0.1035	-0.17
4.615	-0.9744	-0.1029	-0.18
4.620	-0.9738	-0.1024	-0.18
4.625	-0.9734	-0.1019	-0.19
4.630	-0.9725	-0.1013	-0.20
4.635	-0.9717	-0.1009	-0.20
4.640	-0.9711	-0.1005	-0.21
4.645	-0.9704	-0.0999	-0.22
4.650	-0.9699	-0.0993	-0.22
4.655	-0.9688	-0.0989	-0.23
4.660	-0.9683	-0.0987	-0.23
4.665	-0.9674	-0.0981	-0.24
4.670	-0.9666	-0.0982	-0.25
4.675	-0.9655	-0.0981	-0.26
4.680	-0.9649	-0.0979	-0.27
4.685	-0.9640	-0.0981	-0.27
4.690	-0.9636	-0.0982	-0.28
4.695	-0.9626	-0.0986	-0.29
4.700	-0.9619	-0.0986	-0.29
4.705	-0.9614	-0.0983	-0.30
4.710	-0.9612	-0.0988	-0.30
4.715	-0.9608	-0.0990	-0.30
4.720	-0.9605	-0.0984	-0.31
4.725	-0.9601	-0.0984	-0.31
4.730	-0.9601	-0.0978	-0.31
4.735	-0.9598	-0.0975	-0.31
4.740	-0.9590	-0.0968	-0.32
4.745	-0.9584	-0.0960	-0.33
4.750	-0.9578	-0.0954	-0.33
4.755	-0.9572	-0.0945	-0.34
4.760	-0.9565	-0.0943	-0.34
4.765	-0.9554	-0.0936	-0.35
4.770	-0.9551	-0.0935	-0.36
4.775	-0.9540	-0.0926	-0.37
4.780	-0.9530	-0.0933	-0.38
4.785	-0.9524	-0.0937	-0.38
4.790	-0.9518	-0.0935	-0.39
4.795	-0.9506	-0.0942	-0.40
4.800	-0.9499	-0.0951	-0.40
4.805	-0.9495	-0.0962	-0.41
4.810	-0.9487	-0.0977	-0.41
4.815	-0.9483	-0.0990	-0.41
4.820	-0.9483	-0.1004	-0.41
4.825	-0.9483	-0.1011	-0.41
4.830	-0.9484	-0.1033	-0.41

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.835	-0.9484	-0.1046	-0.41
4.840	-0.9490	-0.1057	-0.40
4.845	-0.9494	-0.1069	-0.40
4.850	-0.9505	-0.1081	-0.39
4.855	-0.9512	-0.1098	-0.38
4.860	-0.9520	-0.1108	-0.37
4.865	-0.9530	-0.1116	-0.36
4.870	-0.9537	-0.1125	-0.35
4.875	-0.9546	-0.1138	-0.34
4.880	-0.9555	-0.1146	-0.33
4.885	-0.9564	-0.1168	-0.32
4.890	-0.9574	-0.1162	-0.31
4.895	-0.9581	-0.1169	-0.31
4.900	-0.9593	-0.1178	-0.30
4.905	-0.9599	-0.1183	-0.29
4.910	-0.9605	-0.1188	-0.28
4.915	-0.9612	-0.1192	-0.28
4.920	-0.9620	-0.1197	-0.27
4.925	-0.9625	-0.1205	-0.26
4.930	-0.9631	-0.1210	-0.26
4.935	-0.9635	-0.1215	-0.25
4.940	-0.9641	-0.1220	-0.25
4.945	-0.9646	-0.1226	-0.24
4.950	-0.9652	-0.1233	-0.24
4.955	-0.9656	-0.1238	-0.23
4.960	-0.9663	-0.1240	-0.23
4.965	-0.9667	-0.1248	-0.22
4.970	-0.9673	-0.1253	-0.22
4.975	-0.9677	-0.1256	-0.21
4.980	-0.9686	-0.1261	-0.20
4.985	-0.9689	-0.1268	-0.20
4.990	-0.9698	-0.1266	-0.19
4.995	-0.9698	-0.1271	-0.19
5.000	-0.9707	-0.1270	-0.18
5.005	-0.9711	-0.1270	-0.18
5.010	-0.9714	-0.1270	-0.18
5.015	-0.9719	-0.1274	-0.17
5.020	-0.9723	-0.1267	-0.17
5.025	-0.9726	-0.1274	-0.17
5.030	-0.9725	-0.1266	-0.17
5.035	-0.9730	-0.1270	-0.16
5.040	-0.9730	-0.1257	-0.17
5.045	-0.9735	-0.1268	-0.16
5.050	-0.9730	-0.1260	-0.17
5.055	-0.9737	-0.1267	-0.16
5.060	-0.9732	-0.1262	-0.16

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.065	-0.9747	-0.1258	-0.15
5.070	-0.9744	-0.1265	-0.15
5.075	-0.9756	-0.1257	-0.14
5.080	-0.9754	-0.1261	-0.14
5.085	-0.9762	-0.1252	-0.14
5.090	-0.9770	-0.1255	-0.13
5.095	-0.9780	-0.1245	-0.12
5.100	-0.9787	-0.1246	-0.12
5.105	-0.9785	-0.1234	-0.12
5.110	-0.9794	-0.1224	-0.11
5.115	-0.9799	-0.1210	-0.11
5.120	-0.9806	-0.1200	-0.11
5.125	-0.9810	-0.1182	-0.10
5.130	-0.9818	-0.1168	-0.10
5.135	-0.9819	-0.1149	-0.10
5.140	-0.9828	-0.1131	-0.09
5.145	-0.9824	-0.1106	-0.10
5.150	-0.9833	-0.1089	-0.09
5.155	-0.9833	-0.1067	-0.10
5.160	-0.9837	-0.1042	-0.09
5.165	-0.9835	-0.1020	-0.10
5.170	-0.9835	-0.0994	-0.10
5.175	-0.9831	-0.0970	-0.11
5.180	-0.9831	-0.0943	-0.11
5.185	-0.9828	-0.0918	-0.11
5.190	-0.9822	-0.0891	-0.12
5.195	-0.9815	-0.0867	-0.13
5.200	-0.9808	-0.0840	-0.14
5.205	-0.9804	-0.0815	-0.14
5.210	-0.9797	-0.0790	-0.15
5.215	-0.9790	-0.0765	-0.16
5.220	-0.9781	-0.0740	-0.17
5.225	-0.9773	-0.0716	-0.18
5.230	-0.9765	-0.0692	-0.18
5.235	-0.9758	-0.0670	-0.19
5.240	-0.9753	-0.0647	-0.20
5.245	-0.9745	-0.0625	-0.21
5.250	-0.9735	-0.0605	-0.22
5.255	-0.9727	-0.0583	-0.22
5.260	-0.9721	-0.0563	-0.23
5.265	-0.9713	-0.0542	-0.24
5.270	-0.9708	-0.0526	-0.24
5.275	-0.9699	-0.0508	-0.25
5.280	-0.9690	-0.0489	-0.26
5.285	-0.9680	-0.0473	-0.27
5.290	-0.9675	-0.0456	-0.28

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.295	-0.9669	-0.0443	-0.28
5.300	-0.9662	-0.0425	-0.29
5.305	-0.9658	-0.0410	-0.29
5.310	-0.9649	-0.0392	-0.30
5.315	-0.9646	-0.0379	-0.31
5.320	-0.9638	-0.0367	-0.31
5.325	-0.9631	-0.0350	-0.32
5.330	-0.9629	-0.0342	-0.32
5.335	-0.9623	-0.0326	-0.33
5.340	-0.9621	-0.0316	-0.33
5.345	-0.9614	-0.0308	-0.34
5.350	-0.9612	-0.0289	-0.34
5.355	-0.9606	-0.0282	-0.34
5.360	-0.9605	-0.0270	-0.35
5.365	-0.9600	-0.0262	-0.35
5.370	-0.9597	-0.0251	-0.35
5.375	-0.9594	-0.0242	-0.36
5.380	-0.9592	-0.0229	-0.36
5.385	-0.9589	-0.0221	-0.36
5.390	-0.9591	-0.0211	-0.36
5.395	-0.9592	-0.0206	-0.36
5.400	-0.9594	-0.0197	-0.36
5.405	-0.9601	-0.0189	-0.35
5.410	-0.9596	-0.0181	-0.36
5.415	-0.9596	-0.0178	-0.36
5.420	-0.9601	-0.0172	-0.35
5.425	-0.9603	-0.0164	-0.35
5.430	-0.9605	-0.0159	-0.35
5.435	-0.9606	-0.0157	-0.35
5.440	-0.9610	-0.0154	-0.34
5.445	-0.9611	-0.0151	-0.34
5.450	-0.9617	-0.0152	-0.34
5.455	-0.9622	-0.0153	-0.33
5.460	-0.9626	-0.0149	-0.33
5.465	-0.9629	-0.0152	-0.33
5.470	-0.9633	-0.0156	-0.32
5.475	-0.9639	-0.0155	-0.32
5.480	-0.9643	-0.0161	-0.31
5.485	-0.9648	-0.0161	-0.31
5.490	-0.9652	-0.0165	-0.31
5.495	-0.9658	-0.0167	-0.30
5.500	-0.9662	-0.0170	-0.30
5.505	-0.9667	-0.0174	-0.29
5.510	-0.9676	-0.0177	-0.29
5.515	-0.9683	-0.0180	-0.28
5.520	-0.9692	-0.0186	-0.27

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.525	-0.9699	-0.0194	-0.26
5.530	-0.9708	-0.0195	-0.26
5.535	-0.9717	-0.0202	-0.25
5.540	-0.9726	-0.0209	-0.24
5.545	-0.9735	-0.0217	-0.23
5.550	-0.9743	-0.0223	-0.22
5.555	-0.9752	-0.0231	-0.22
5.560	-0.9760	-0.0237	-0.21
5.565	-0.9767	-0.0244	-0.20
5.570	-0.9775	-0.0251	-0.19
5.575	-0.9783	-0.0259	-0.19
5.580	-0.9789	-0.0264	-0.18
5.585	-0.9796	-0.0270	-0.18
5.590	-0.9803	-0.0277	-0.17
5.595	-0.9808	-0.0282	-0.16
5.600	-0.9812	-0.0284	-0.16
5.605	-0.9818	-0.0289	-0.16
5.610	-0.9824	-0.0294	-0.15
5.615	-0.9831	-0.0297	-0.14
5.620	-0.9835	-0.0298	-0.14
5.625	-0.9842	-0.0301	-0.13
5.630	-0.9852	-0.0302	-0.13
5.635	-0.9852	-0.0303	-0.13
5.640	-0.9861	-0.0305	-0.12
5.645	-0.9862	-0.0309	-0.12
5.650	-0.9867	-0.0315	-0.11
5.655	-0.9873	-0.0316	-0.11
5.660	-0.9878	-0.0317	-0.10
5.665	-0.9882	-0.0323	-0.10
5.670	-0.9882	-0.0318	-0.10
5.675	-0.9882	-0.0321	-0.10
5.680	-0.9889	-0.0320	-0.09
5.685	-0.9878	-0.0321	-0.10
5.690	-0.9875	-0.0324	-0.10
5.695	-0.9870	-0.0322	-0.11
5.700	-0.9874	-0.0323	-0.11
5.705	-0.9872	-0.0322	-0.11
5.710	-0.9874	-0.0323	-0.11
5.715	-0.9868	-0.0324	-0.11
5.720	-0.9866	-0.0325	-0.11
5.725	-0.9866	-0.0326	-0.11
5.730	-0.9865	-0.0320	-0.11
5.735	-0.9864	-0.0327	-0.11
5.740	-0.9859	-0.0319	-0.12
5.745	-0.9859	-0.0321	-0.12
5.750	-0.9850	-0.0314	-0.13

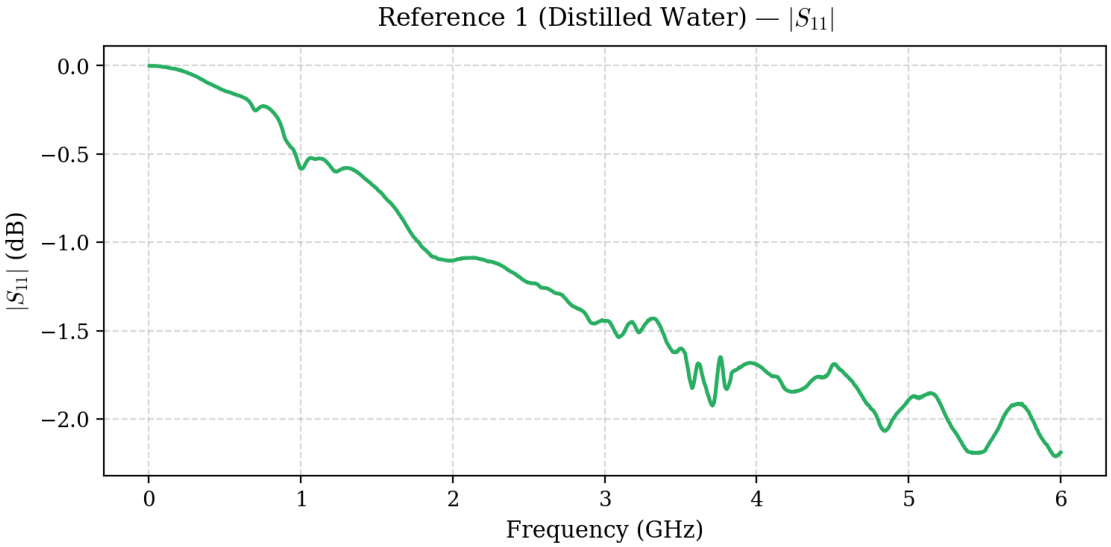
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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.755	-0.9844	-0.0313	-0.13
5.760	-0.9840	-0.0311	-0.14
5.765	-0.9834	-0.0307	-0.14
5.770	-0.9825	-0.0305	-0.15
5.775	-0.9821	-0.0299	-0.15
5.780	-0.9816	-0.0295	-0.16
5.785	-0.9807	-0.0286	-0.17
5.790	-0.9799	-0.0287	-0.17
5.795	-0.9791	-0.0280	-0.18
5.800	-0.9782	-0.0275	-0.19
5.805	-0.9773	-0.0271	-0.20
5.810	-0.9767	-0.0265	-0.20
5.815	-0.9760	-0.0257	-0.21
5.820	-0.9749	-0.0255	-0.22
5.825	-0.9738	-0.0249	-0.23
5.830	-0.9729	-0.0242	-0.24
5.835	-0.9720	-0.0237	-0.24
5.840	-0.9713	-0.0233	-0.25
5.845	-0.9707	-0.0227	-0.26
5.850	-0.9696	-0.0221	-0.27
5.855	-0.9687	-0.0217	-0.27
5.860	-0.9681	-0.0212	-0.28
5.865	-0.9672	-0.0207	-0.29
5.870	-0.9667	-0.0202	-0.29
5.875	-0.9658	-0.0198	-0.30
5.880	-0.9653	-0.0195	-0.30
5.885	-0.9647	-0.0191	-0.31
5.890	-0.9641	-0.0188	-0.32
5.895	-0.9632	-0.0188	-0.32
5.900	-0.9628	-0.0186	-0.33
5.905	-0.9624	-0.0186	-0.33
5.910	-0.9620	-0.0185	-0.34
5.915	-0.9614	-0.0185	-0.34
5.920	-0.9605	-0.0185	-0.35
5.925	-0.9604	-0.0183	-0.35
5.930	-0.9595	-0.0188	-0.36
5.935	-0.9589	-0.0187	-0.36
5.940	-0.9588	-0.0188	-0.36
5.945	-0.9585	-0.0192	-0.37
5.950	-0.9583	-0.0195	-0.37
5.955	-0.9577	-0.0196	-0.37
5.960	-0.9579	-0.0197	-0.37
5.965	-0.9578	-0.0201	-0.37
5.970	-0.9571	-0.0209	-0.38
5.975	-0.9577	-0.0213	-0.37
5.980	-0.9580	-0.0225	-0.37

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Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
5.985	-0.9579	-0.0230	-0.37
5.990	-0.9579	-0.0243	-0.37
5.995	-0.9581	-0.0253	-0.37
6.000	-0.9578	-0.0261	-0.37

3.3 Reference 1 (Distilled Water)



Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
0.005	1.0000	-0.0051	0.00
0.010	0.9998	-0.0101	-0.00
0.015	0.9997	-0.0150	-0.00
0.020	0.9997	-0.0201	-0.00
0.025	0.9995	-0.0252	-0.00
0.030	0.9995	-0.0301	-0.00
0.035	0.9992	-0.0349	-0.00
0.040	0.9991	-0.0401	-0.00
0.045	0.9987	-0.0450	-0.00
0.050	0.9985	-0.0500	-0.00
0.055	0.9981	-0.0550	-0.00
0.060	0.9979	-0.0600	-0.00
0.065	0.9976	-0.0650	-0.00
0.070	0.9972	-0.0699	-0.00
0.075	0.9967	-0.0750	-0.00
0.080	0.9963	-0.0798	-0.00
0.085	0.9958	-0.0848	-0.01
0.090	0.9952	-0.0897	-0.01

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.095	0.9946	-0.0947	-0.01
0.100	0.9942	-0.0996	-0.01
0.105	0.9937	-0.1046	-0.01
0.110	0.9931	-0.1094	-0.01
0.115	0.9925	-0.1144	-0.01
0.120	0.9917	-0.1193	-0.01
0.125	0.9910	-0.1242	-0.01
0.130	0.9902	-0.1292	-0.01
0.135	0.9895	-0.1341	-0.01
0.140	0.9888	-0.1389	-0.01
0.145	0.9879	-0.1440	-0.01
0.150	0.9871	-0.1487	-0.02
0.155	0.9862	-0.1536	-0.02
0.160	0.9855	-0.1583	-0.02
0.165	0.9847	-0.1633	-0.02
0.170	0.9837	-0.1681	-0.02
0.175	0.9829	-0.1730	-0.02
0.180	0.9818	-0.1779	-0.02
0.185	0.9808	-0.1827	-0.02
0.190	0.9797	-0.1876	-0.02
0.195	0.9786	-0.1924	-0.02
0.200	0.9776	-0.1973	-0.02
0.205	0.9763	-0.2020	-0.03
0.210	0.9753	-0.2070	-0.03
0.215	0.9740	-0.2117	-0.03
0.220	0.9730	-0.2166	-0.03
0.225	0.9715	-0.2213	-0.03
0.230	0.9703	-0.2261	-0.03
0.235	0.9689	-0.2308	-0.03
0.240	0.9677	-0.2356	-0.04
0.245	0.9663	-0.2405	-0.04
0.250	0.9650	-0.2453	-0.04
0.255	0.9634	-0.2499	-0.04
0.260	0.9621	-0.2546	-0.04
0.265	0.9606	-0.2594	-0.04
0.270	0.9591	-0.2641	-0.05
0.275	0.9576	-0.2688	-0.05
0.280	0.9561	-0.2735	-0.05
0.285	0.9544	-0.2781	-0.05
0.290	0.9529	-0.2828	-0.05
0.295	0.9513	-0.2874	-0.05
0.300	0.9496	-0.2920	-0.06
0.305	0.9480	-0.2968	-0.06
0.310	0.9463	-0.3013	-0.06
0.315	0.9445	-0.3058	-0.06
0.320	0.9427	-0.3104	-0.07

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.325	0.9410	-0.3149	-0.07
0.330	0.9393	-0.3194	-0.07
0.335	0.9374	-0.3239	-0.07
0.340	0.9355	-0.3286	-0.07
0.345	0.9337	-0.3328	-0.08
0.350	0.9319	-0.3373	-0.08
0.355	0.9299	-0.3418	-0.08
0.360	0.9280	-0.3461	-0.08
0.365	0.9260	-0.3507	-0.09
0.370	0.9240	-0.3551	-0.09
0.375	0.9221	-0.3595	-0.09
0.380	0.9200	-0.3638	-0.09
0.385	0.9179	-0.3681	-0.10
0.390	0.9160	-0.3725	-0.10
0.395	0.9139	-0.3767	-0.10
0.400	0.9121	-0.3810	-0.10
0.405	0.9100	-0.3853	-0.10
0.410	0.9079	-0.3895	-0.11
0.415	0.9058	-0.3936	-0.11
0.420	0.9037	-0.3978	-0.11
0.425	0.9018	-0.4019	-0.11
0.430	0.8996	-0.4060	-0.11
0.435	0.8974	-0.4102	-0.12
0.440	0.8953	-0.4142	-0.12
0.445	0.8932	-0.4183	-0.12
0.450	0.8912	-0.4225	-0.12
0.455	0.8888	-0.4265	-0.12
0.460	0.8866	-0.4304	-0.13
0.465	0.8843	-0.4344	-0.13
0.470	0.8821	-0.4384	-0.13
0.475	0.8798	-0.4422	-0.13
0.480	0.8776	-0.4463	-0.14
0.485	0.8754	-0.4501	-0.14
0.490	0.8732	-0.4541	-0.14
0.495	0.8710	-0.4579	-0.14
0.500	0.8686	-0.4618	-0.14
0.505	0.8664	-0.4656	-0.14
0.510	0.8641	-0.4694	-0.15
0.515	0.8618	-0.4732	-0.15
0.520	0.8596	-0.4770	-0.15
0.525	0.8574	-0.4809	-0.15
0.530	0.8552	-0.4844	-0.15
0.535	0.8529	-0.4881	-0.15
0.540	0.8506	-0.4919	-0.15
0.545	0.8482	-0.4956	-0.15
0.550	0.8457	-0.4993	-0.16

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.555	0.8434	-0.5030	-0.16
0.560	0.8409	-0.5067	-0.16
0.565	0.8386	-0.5103	-0.16
0.570	0.8361	-0.5139	-0.16
0.575	0.8337	-0.5176	-0.16
0.580	0.8313	-0.5212	-0.17
0.585	0.8288	-0.5246	-0.17
0.590	0.8264	-0.5282	-0.17
0.595	0.8241	-0.5317	-0.17
0.600	0.8216	-0.5352	-0.17
0.605	0.8190	-0.5387	-0.17
0.610	0.8164	-0.5422	-0.18
0.615	0.8138	-0.5457	-0.18
0.620	0.8113	-0.5492	-0.18
0.625	0.8086	-0.5526	-0.18
0.630	0.8059	-0.5561	-0.18
0.635	0.8033	-0.5595	-0.18
0.640	0.8006	-0.5628	-0.19
0.645	0.7980	-0.5662	-0.19
0.650	0.7952	-0.5694	-0.19
0.655	0.7922	-0.5726	-0.20
0.660	0.7893	-0.5758	-0.20
0.665	0.7864	-0.5789	-0.21
0.670	0.7832	-0.5819	-0.21
0.675	0.7802	-0.5847	-0.22
0.680	0.7773	-0.5872	-0.23
0.685	0.7742	-0.5895	-0.24
0.690	0.7716	-0.5917	-0.24
0.695	0.7690	-0.5937	-0.25
0.700	0.7669	-0.5961	-0.25
0.705	0.7648	-0.5986	-0.25
0.710	0.7630	-0.6012	-0.25
0.715	0.7612	-0.6043	-0.25
0.720	0.7593	-0.6073	-0.24
0.725	0.7571	-0.6107	-0.24
0.730	0.7549	-0.6142	-0.24
0.735	0.7525	-0.6175	-0.23
0.740	0.7500	-0.6210	-0.23
0.745	0.7474	-0.6244	-0.23
0.750	0.7447	-0.6278	-0.23
0.755	0.7420	-0.6310	-0.23
0.760	0.7390	-0.6341	-0.23
0.765	0.7361	-0.6376	-0.23
0.770	0.7331	-0.6409	-0.23
0.775	0.7298	-0.6440	-0.23
0.780	0.7268	-0.6473	-0.24

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.785	0.7235	-0.6506	-0.24
0.790	0.7203	-0.6536	-0.24
0.795	0.7170	-0.6567	-0.24
0.800	0.7138	-0.6596	-0.25
0.805	0.7104	-0.6626	-0.25
0.810	0.7070	-0.6656	-0.26
0.815	0.7036	-0.6685	-0.26
0.820	0.7000	-0.6715	-0.26
0.825	0.6966	-0.6743	-0.27
0.830	0.6929	-0.6772	-0.27
0.835	0.6894	-0.6800	-0.28
0.840	0.6856	-0.6827	-0.29
0.845	0.6821	-0.6854	-0.29
0.850	0.6782	-0.6881	-0.30
0.855	0.6742	-0.6907	-0.31
0.860	0.6702	-0.6932	-0.32
0.865	0.6662	-0.6956	-0.33
0.870	0.6621	-0.6978	-0.34
0.875	0.6579	-0.6999	-0.35
0.880	0.6537	-0.7017	-0.36
0.885	0.6496	-0.7031	-0.38
0.890	0.6459	-0.7045	-0.39
0.895	0.6423	-0.7056	-0.41
0.900	0.6390	-0.7069	-0.42
0.905	0.6362	-0.7084	-0.43
0.910	0.6332	-0.7101	-0.43
0.915	0.6302	-0.7121	-0.44
0.920	0.6270	-0.7140	-0.44
0.925	0.6238	-0.7157	-0.45
0.930	0.6208	-0.7176	-0.46
0.935	0.6177	-0.7195	-0.46
0.940	0.6146	-0.7217	-0.46
0.945	0.6113	-0.7239	-0.47
0.950	0.6077	-0.7263	-0.47
0.955	0.6039	-0.7283	-0.48
0.960	0.6001	-0.7300	-0.49
0.965	0.5964	-0.7315	-0.50
0.970	0.5926	-0.7330	-0.51
0.975	0.5889	-0.7340	-0.53
0.980	0.5855	-0.7348	-0.54
0.985	0.5823	-0.7353	-0.56
0.990	0.5798	-0.7359	-0.57
0.995	0.5771	-0.7364	-0.58
1.000	0.5753	-0.7372	-0.58
1.005	0.5737	-0.7384	-0.58
1.010	0.5719	-0.7398	-0.58

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.015	0.5702	-0.7416	-0.58
1.020	0.5686	-0.7437	-0.57
1.025	0.5668	-0.7461	-0.57
1.030	0.5648	-0.7488	-0.56
1.035	0.5625	-0.7515	-0.55
1.040	0.5602	-0.7544	-0.54
1.045	0.5575	-0.7572	-0.53
1.050	0.5547	-0.7599	-0.53
1.055	0.5518	-0.7626	-0.53
1.060	0.5488	-0.7651	-0.52
1.065	0.5454	-0.7675	-0.52
1.070	0.5422	-0.7698	-0.52
1.075	0.5390	-0.7720	-0.52
1.080	0.5358	-0.7740	-0.53
1.085	0.5327	-0.7759	-0.53
1.090	0.5297	-0.7777	-0.53
1.095	0.5268	-0.7798	-0.53
1.100	0.5237	-0.7817	-0.53
1.105	0.5209	-0.7838	-0.53
1.110	0.5179	-0.7858	-0.53
1.115	0.5149	-0.7879	-0.53
1.120	0.5119	-0.7901	-0.52
1.125	0.5087	-0.7921	-0.53
1.130	0.5054	-0.7941	-0.53
1.135	0.5023	-0.7958	-0.53
1.140	0.4992	-0.7978	-0.53
1.145	0.4959	-0.7997	-0.53
1.150	0.4925	-0.8015	-0.53
1.155	0.4893	-0.8032	-0.53
1.160	0.4861	-0.8047	-0.54
1.165	0.4828	-0.8062	-0.54
1.170	0.4794	-0.8076	-0.55
1.175	0.4764	-0.8089	-0.55
1.180	0.4732	-0.8102	-0.55
1.185	0.4702	-0.8114	-0.56
1.190	0.4673	-0.8123	-0.56
1.195	0.4645	-0.8132	-0.57
1.200	0.4618	-0.8141	-0.57
1.205	0.4590	-0.8149	-0.58
1.210	0.4563	-0.8157	-0.59
1.215	0.4539	-0.8165	-0.59
1.220	0.4515	-0.8172	-0.60
1.225	0.4492	-0.8183	-0.60
1.230	0.4471	-0.8194	-0.60
1.235	0.4448	-0.8204	-0.60
1.240	0.4428	-0.8217	-0.60

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.245	0.4407	-0.8230	-0.60
1.250	0.4385	-0.8246	-0.59
1.255	0.4362	-0.8261	-0.59
1.260	0.4339	-0.8275	-0.59
1.265	0.4316	-0.8290	-0.59
1.270	0.4292	-0.8306	-0.58
1.275	0.4265	-0.8321	-0.58
1.280	0.4241	-0.8337	-0.58
1.285	0.4212	-0.8350	-0.58
1.290	0.4184	-0.8367	-0.58
1.295	0.4156	-0.8381	-0.58
1.300	0.4129	-0.8394	-0.58
1.305	0.4102	-0.8408	-0.58
1.310	0.4076	-0.8420	-0.58
1.315	0.4048	-0.8433	-0.58
1.320	0.4020	-0.8445	-0.58
1.325	0.3993	-0.8458	-0.58
1.330	0.3966	-0.8468	-0.58
1.335	0.3937	-0.8480	-0.58
1.340	0.3909	-0.8491	-0.59
1.345	0.3880	-0.8500	-0.59
1.350	0.3853	-0.8512	-0.59
1.355	0.3823	-0.8521	-0.59
1.360	0.3794	-0.8531	-0.60
1.365	0.3764	-0.8541	-0.60
1.370	0.3736	-0.8551	-0.60
1.375	0.3707	-0.8559	-0.60
1.380	0.3677	-0.8567	-0.61
1.385	0.3648	-0.8577	-0.61
1.390	0.3618	-0.8586	-0.61
1.395	0.3587	-0.8594	-0.62
1.400	0.3557	-0.8603	-0.62
1.405	0.3525	-0.8611	-0.63
1.410	0.3494	-0.8619	-0.63
1.415	0.3463	-0.8627	-0.63
1.420	0.3431	-0.8635	-0.64
1.425	0.3399	-0.8644	-0.64
1.430	0.3368	-0.8652	-0.64
1.435	0.3335	-0.8661	-0.65
1.440	0.3304	-0.8669	-0.65
1.445	0.3273	-0.8677	-0.65
1.450	0.3240	-0.8684	-0.66
1.455	0.3208	-0.8694	-0.66
1.460	0.3175	-0.8701	-0.67
1.465	0.3143	-0.8707	-0.67
1.470	0.3111	-0.8716	-0.67

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.475	0.3077	-0.8723	-0.68
1.480	0.3044	-0.8731	-0.68
1.485	0.3008	-0.8738	-0.69
1.490	0.2975	-0.8747	-0.69
1.495	0.2940	-0.8754	-0.69
1.500	0.2908	-0.8761	-0.70
1.505	0.2872	-0.8766	-0.70
1.510	0.2836	-0.8773	-0.71
1.515	0.2801	-0.8780	-0.71
1.520	0.2767	-0.8787	-0.71
1.525	0.2731	-0.8795	-0.72
1.530	0.2695	-0.8801	-0.72
1.535	0.2658	-0.8808	-0.72
1.540	0.2621	-0.8813	-0.73
1.545	0.2584	-0.8817	-0.74
1.550	0.2549	-0.8823	-0.74
1.555	0.2509	-0.8827	-0.75
1.560	0.2472	-0.8832	-0.75
1.565	0.2434	-0.8838	-0.76
1.570	0.2396	-0.8844	-0.76
1.575	0.2358	-0.8848	-0.77
1.580	0.2323	-0.8853	-0.77
1.585	0.2284	-0.8861	-0.77
1.590	0.2247	-0.8865	-0.78
1.595	0.2209	-0.8870	-0.78
1.600	0.2173	-0.8874	-0.78
1.605	0.2136	-0.8877	-0.79
1.610	0.2099	-0.8879	-0.80
1.615	0.2061	-0.8883	-0.80
1.620	0.2023	-0.8887	-0.81
1.625	0.1985	-0.8890	-0.81
1.630	0.1947	-0.8892	-0.82
1.635	0.1909	-0.8894	-0.82
1.640	0.1869	-0.8893	-0.83
1.645	0.1832	-0.8893	-0.84
1.650	0.1793	-0.8895	-0.84
1.655	0.1756	-0.8898	-0.85
1.660	0.1720	-0.8896	-0.86
1.665	0.1684	-0.8896	-0.86
1.670	0.1648	-0.8898	-0.87
1.675	0.1612	-0.8896	-0.88
1.680	0.1577	-0.8896	-0.88
1.685	0.1541	-0.8893	-0.89
1.690	0.1507	-0.8891	-0.90
1.695	0.1476	-0.8890	-0.90
1.700	0.1441	-0.8887	-0.91

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.705	0.1408	-0.8886	-0.92
1.710	0.1374	-0.8883	-0.93
1.715	0.1340	-0.8880	-0.93
1.720	0.1310	-0.8878	-0.94
1.725	0.1276	-0.8876	-0.95
1.730	0.1247	-0.8875	-0.95
1.735	0.1215	-0.8870	-0.96
1.740	0.1185	-0.8868	-0.97
1.745	0.1157	-0.8867	-0.97
1.750	0.1128	-0.8866	-0.98
1.755	0.1101	-0.8865	-0.98
1.760	0.1074	-0.8861	-0.99
1.765	0.1048	-0.8860	-0.99
1.770	0.1023	-0.8860	-0.99
1.775	0.0999	-0.8858	-1.00
1.780	0.0976	-0.8855	-1.00
1.785	0.0950	-0.8850	-1.01
1.790	0.0923	-0.8846	-1.02
1.795	0.0899	-0.8843	-1.02
1.800	0.0875	-0.8841	-1.03
1.805	0.0852	-0.8839	-1.03
1.810	0.0829	-0.8838	-1.03
1.815	0.0808	-0.8836	-1.04
1.820	0.0785	-0.8834	-1.04
1.825	0.0766	-0.8831	-1.05
1.830	0.0747	-0.8830	-1.05
1.835	0.0725	-0.8826	-1.06
1.840	0.0706	-0.8824	-1.06
1.845	0.0688	-0.8822	-1.06
1.850	0.0667	-0.8815	-1.07
1.855	0.0648	-0.8815	-1.07
1.860	0.0628	-0.8809	-1.08
1.865	0.0608	-0.8811	-1.08
1.870	0.0590	-0.8809	-1.08
1.875	0.0569	-0.8809	-1.08
1.880	0.0551	-0.8809	-1.08
1.885	0.0535	-0.8812	-1.08
1.890	0.0522	-0.8810	-1.09
1.895	0.0504	-0.8807	-1.09
1.900	0.0486	-0.8804	-1.09
1.905	0.0470	-0.8803	-1.09
1.910	0.0452	-0.8804	-1.09
1.915	0.0434	-0.8804	-1.10
1.920	0.0419	-0.8805	-1.10
1.925	0.0403	-0.8804	-1.10
1.930	0.0389	-0.8805	-1.10

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.935	0.0374	-0.8804	-1.10
1.940	0.0359	-0.8804	-1.10
1.945	0.0345	-0.8804	-1.10
1.950	0.0330	-0.8803	-1.10
1.955	0.0316	-0.8803	-1.10
1.960	0.0302	-0.8802	-1.10
1.965	0.0288	-0.8803	-1.10
1.970	0.0274	-0.8801	-1.10
1.975	0.0262	-0.8803	-1.10
1.980	0.0248	-0.8804	-1.10
1.985	0.0235	-0.8802	-1.10
1.990	0.0220	-0.8804	-1.10
1.995	0.0206	-0.8805	-1.10
2.000	0.0190	-0.8805	-1.10
2.005	0.0175	-0.8807	-1.10
2.010	0.0159	-0.8808	-1.10
2.015	0.0145	-0.8809	-1.10
2.020	0.0130	-0.8810	-1.10
2.025	0.0112	-0.8813	-1.10
2.030	0.0095	-0.8815	-1.09
2.035	0.0079	-0.8815	-1.10
2.040	0.0061	-0.8816	-1.09
2.045	0.0044	-0.8817	-1.09
2.050	0.0026	-0.8817	-1.09
2.055	0.0007	-0.8820	-1.09
2.060	-0.0012	-0.8819	-1.09
2.065	-0.0031	-0.8821	-1.09
2.070	-0.0052	-0.8821	-1.09
2.075	-0.0073	-0.8821	-1.09
2.080	-0.0095	-0.8822	-1.09
2.085	-0.0117	-0.8821	-1.09
2.090	-0.0140	-0.8821	-1.09
2.095	-0.0163	-0.8821	-1.09
2.100	-0.0188	-0.8820	-1.09
2.105	-0.0212	-0.8820	-1.09
2.110	-0.0237	-0.8820	-1.09
2.115	-0.0261	-0.8819	-1.09
2.120	-0.0286	-0.8818	-1.09
2.125	-0.0312	-0.8818	-1.09
2.130	-0.0341	-0.8816	-1.09
2.135	-0.0366	-0.8814	-1.09
2.140	-0.0393	-0.8815	-1.09
2.145	-0.0419	-0.8814	-1.09
2.150	-0.0445	-0.8812	-1.09
2.155	-0.0473	-0.8809	-1.09
2.160	-0.0501	-0.8806	-1.09

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.165	-0.0529	-0.8804	-1.09
2.170	-0.0558	-0.8800	-1.09
2.175	-0.0585	-0.8798	-1.09
2.180	-0.0613	-0.8795	-1.09
2.185	-0.0641	-0.8791	-1.10
2.190	-0.0671	-0.8790	-1.10
2.195	-0.0700	-0.8785	-1.10
2.200	-0.0729	-0.8783	-1.10
2.205	-0.0759	-0.8778	-1.10
2.210	-0.0788	-0.8775	-1.10
2.215	-0.0817	-0.8770	-1.10
2.220	-0.0846	-0.8766	-1.10
2.225	-0.0877	-0.8761	-1.11
2.230	-0.0906	-0.8756	-1.11
2.235	-0.0936	-0.8752	-1.11
2.240	-0.0967	-0.8750	-1.11
2.245	-0.0995	-0.8745	-1.11
2.250	-0.1025	-0.8742	-1.11
2.255	-0.1056	-0.8737	-1.11
2.260	-0.1083	-0.8733	-1.11
2.265	-0.1114	-0.8728	-1.11
2.270	-0.1141	-0.8723	-1.11
2.275	-0.1170	-0.8716	-1.12
2.280	-0.1199	-0.8711	-1.12
2.285	-0.1228	-0.8707	-1.12
2.290	-0.1255	-0.8700	-1.12
2.295	-0.1283	-0.8695	-1.12
2.300	-0.1312	-0.8689	-1.12
2.305	-0.1339	-0.8684	-1.12
2.310	-0.1369	-0.8677	-1.13
2.315	-0.1394	-0.8669	-1.13
2.320	-0.1423	-0.8662	-1.13
2.325	-0.1449	-0.8656	-1.13
2.330	-0.1474	-0.8649	-1.14
2.335	-0.1503	-0.8643	-1.14
2.340	-0.1526	-0.8635	-1.14
2.345	-0.1553	-0.8627	-1.14
2.350	-0.1577	-0.8620	-1.15
2.355	-0.1601	-0.8612	-1.15
2.360	-0.1625	-0.8603	-1.15
2.365	-0.1646	-0.8598	-1.16
2.370	-0.1667	-0.8589	-1.16
2.375	-0.1690	-0.8584	-1.16
2.380	-0.1711	-0.8576	-1.16
2.385	-0.1732	-0.8570	-1.17
2.390	-0.1754	-0.8564	-1.17

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.395	-0.1773	-0.8557	-1.17
2.400	-0.1792	-0.8551	-1.17
2.405	-0.1813	-0.8544	-1.18
2.410	-0.1831	-0.8536	-1.18
2.415	-0.1849	-0.8531	-1.18
2.420	-0.1866	-0.8523	-1.18
2.425	-0.1882	-0.8517	-1.19
2.430	-0.1899	-0.8508	-1.19
2.435	-0.1914	-0.8503	-1.19
2.440	-0.1929	-0.8498	-1.20
2.445	-0.1945	-0.8490	-1.20
2.450	-0.1958	-0.8483	-1.20
2.455	-0.1969	-0.8476	-1.21
2.460	-0.1982	-0.8471	-1.21
2.465	-0.1995	-0.8464	-1.21
2.470	-0.2007	-0.8459	-1.22
2.475	-0.2018	-0.8453	-1.22
2.480	-0.2033	-0.8448	-1.22
2.485	-0.2044	-0.8443	-1.22
2.490	-0.2055	-0.8436	-1.23
2.495	-0.2066	-0.8434	-1.23
2.500	-0.2080	-0.8429	-1.23
2.505	-0.2090	-0.8426	-1.23
2.510	-0.2101	-0.8422	-1.23
2.515	-0.2111	-0.8417	-1.23
2.520	-0.2123	-0.8415	-1.23
2.525	-0.2133	-0.8411	-1.23
2.530	-0.2144	-0.8408	-1.23
2.535	-0.2155	-0.8405	-1.23
2.540	-0.2166	-0.8404	-1.23
2.545	-0.2174	-0.8400	-1.23
2.550	-0.2183	-0.8396	-1.23
2.555	-0.2192	-0.8392	-1.24
2.560	-0.2200	-0.8386	-1.24
2.565	-0.2204	-0.8382	-1.24
2.570	-0.2209	-0.8375	-1.25
2.575	-0.2213	-0.8371	-1.25
2.580	-0.2216	-0.8365	-1.26
2.585	-0.2221	-0.8364	-1.26
2.590	-0.2225	-0.8361	-1.26
2.595	-0.2229	-0.8361	-1.26
2.600	-0.2234	-0.8358	-1.26
2.605	-0.2241	-0.8356	-1.26
2.610	-0.2246	-0.8354	-1.26
2.615	-0.2253	-0.8353	-1.26
2.620	-0.2258	-0.8350	-1.26

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.625	-0.2263	-0.8346	-1.26
2.630	-0.2268	-0.8342	-1.26
2.635	-0.2273	-0.8339	-1.27
2.640	-0.2280	-0.8335	-1.27
2.645	-0.2285	-0.8331	-1.27
2.650	-0.2291	-0.8328	-1.27
2.655	-0.2297	-0.8323	-1.28
2.660	-0.2303	-0.8317	-1.28
2.665	-0.2309	-0.8314	-1.28
2.670	-0.2317	-0.8307	-1.29
2.675	-0.2321	-0.8305	-1.29
2.680	-0.2327	-0.8302	-1.29
2.685	-0.2330	-0.8300	-1.29
2.690	-0.2336	-0.8298	-1.29
2.695	-0.2341	-0.8296	-1.29
2.700	-0.2344	-0.8294	-1.29
2.705	-0.2349	-0.8292	-1.29
2.710	-0.2350	-0.8290	-1.29
2.715	-0.2357	-0.8284	-1.30
2.720	-0.2362	-0.8282	-1.30
2.725	-0.2369	-0.8276	-1.30
2.730	-0.2374	-0.8268	-1.31
2.735	-0.2383	-0.8261	-1.31
2.740	-0.2391	-0.8253	-1.32
2.745	-0.2402	-0.8247	-1.32
2.750	-0.2409	-0.8239	-1.33
2.755	-0.2419	-0.8230	-1.33
2.760	-0.2432	-0.8221	-1.34
2.765	-0.2444	-0.8213	-1.34
2.770	-0.2456	-0.8205	-1.35
2.775	-0.2467	-0.8196	-1.35
2.780	-0.2481	-0.8189	-1.35
2.785	-0.2493	-0.8181	-1.36
2.790	-0.2508	-0.8173	-1.36
2.795	-0.2522	-0.8168	-1.36
2.800	-0.2533	-0.8161	-1.37
2.805	-0.2550	-0.8155	-1.37
2.810	-0.2563	-0.8148	-1.37
2.815	-0.2576	-0.8143	-1.37
2.820	-0.2590	-0.8136	-1.37
2.825	-0.2603	-0.8127	-1.38
2.830	-0.2617	-0.8123	-1.38
2.835	-0.2630	-0.8116	-1.38
2.840	-0.2644	-0.8112	-1.38
2.845	-0.2657	-0.8103	-1.38
2.850	-0.2670	-0.8096	-1.39

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.855	-0.2682	-0.8089	-1.39
2.860	-0.2694	-0.8082	-1.39
2.865	-0.2705	-0.8074	-1.40
2.870	-0.2715	-0.8066	-1.40
2.875	-0.2724	-0.8057	-1.41
2.880	-0.2733	-0.8046	-1.41
2.885	-0.2745	-0.8035	-1.42
2.890	-0.2758	-0.8022	-1.43
2.895	-0.2773	-0.8009	-1.44
2.900	-0.2786	-0.7996	-1.44
2.905	-0.2803	-0.7985	-1.45
2.910	-0.2817	-0.7975	-1.46
2.915	-0.2836	-0.7966	-1.46
2.920	-0.2853	-0.7958	-1.46
2.925	-0.2870	-0.7952	-1.46
2.930	-0.2887	-0.7944	-1.46
2.935	-0.2900	-0.7940	-1.46
2.940	-0.2914	-0.7937	-1.46
2.945	-0.2929	-0.7934	-1.46
2.950	-0.2940	-0.7931	-1.45
2.955	-0.2953	-0.7929	-1.45
2.960	-0.2960	-0.7928	-1.45
2.965	-0.2971	-0.7926	-1.45
2.970	-0.2978	-0.7925	-1.45
2.975	-0.2985	-0.7925	-1.44
2.980	-0.2992	-0.7925	-1.44
2.985	-0.2998	-0.7925	-1.44
2.990	-0.3000	-0.7921	-1.44
2.995	-0.3003	-0.7916	-1.45
3.000	-0.3008	-0.7915	-1.45
3.005	-0.3044	-0.7903	-1.44
3.010	-0.3045	-0.7902	-1.44
3.015	-0.3047	-0.7900	-1.44
3.020	-0.3049	-0.7899	-1.45
3.025	-0.3046	-0.7900	-1.45
3.030	-0.3048	-0.7895	-1.45
3.035	-0.3048	-0.7890	-1.45
3.040	-0.3047	-0.7885	-1.46
3.045	-0.3047	-0.7874	-1.47
3.050	-0.3048	-0.7868	-1.48
3.055	-0.3049	-0.7859	-1.48
3.060	-0.3048	-0.7849	-1.49
3.065	-0.3054	-0.7837	-1.50
3.070	-0.3057	-0.7827	-1.51
3.075	-0.3065	-0.7819	-1.52
3.080	-0.3070	-0.7808	-1.53

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.085	-0.3079	-0.7797	-1.53
3.090	-0.3092	-0.7787	-1.54
3.095	-0.3105	-0.7783	-1.54
3.100	-0.3117	-0.7779	-1.53
3.105	-0.3131	-0.7779	-1.53
3.110	-0.3143	-0.7777	-1.53
3.115	-0.3156	-0.7774	-1.52
3.120	-0.3168	-0.7774	-1.52
3.125	-0.3180	-0.7778	-1.51
3.130	-0.3188	-0.7780	-1.51
3.135	-0.3196	-0.7788	-1.50
3.140	-0.3202	-0.7797	-1.48
3.145	-0.3210	-0.7800	-1.48
3.150	-0.3213	-0.7808	-1.47
3.155	-0.3217	-0.7814	-1.46
3.160	-0.3218	-0.7817	-1.46
3.165	-0.3219	-0.7820	-1.46
3.170	-0.3220	-0.7822	-1.45
3.175	-0.3220	-0.7825	-1.45
3.180	-0.3214	-0.7829	-1.45
3.185	-0.3216	-0.7824	-1.45
3.190	-0.3213	-0.7819	-1.46
3.195	-0.3210	-0.7813	-1.47
3.200	-0.3215	-0.7803	-1.47
3.205	-0.3214	-0.7792	-1.48
3.210	-0.3218	-0.7784	-1.49
3.215	-0.3221	-0.7772	-1.50
3.220	-0.3232	-0.7760	-1.51
3.225	-0.3243	-0.7753	-1.51
3.230	-0.3258	-0.7749	-1.51
3.235	-0.3269	-0.7750	-1.50
3.240	-0.3281	-0.7750	-1.50
3.245	-0.3291	-0.7754	-1.49
3.250	-0.3305	-0.7752	-1.49
3.255	-0.3315	-0.7758	-1.48
3.260	-0.3325	-0.7760	-1.47
3.265	-0.3338	-0.7761	-1.46
3.270	-0.3347	-0.7762	-1.46
3.275	-0.3355	-0.7765	-1.45
3.280	-0.3364	-0.7764	-1.45
3.285	-0.3374	-0.7770	-1.44
3.290	-0.3383	-0.7764	-1.44
3.295	-0.3394	-0.7765	-1.44
3.300	-0.3401	-0.7767	-1.43
3.305	-0.3404	-0.7766	-1.43
3.310	-0.3414	-0.7763	-1.43

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.315	-0.3421	-0.7760	-1.43
3.320	-0.3425	-0.7759	-1.43
3.325	-0.3434	-0.7754	-1.43
3.330	-0.3437	-0.7752	-1.43
3.335	-0.3442	-0.7747	-1.43
3.340	-0.3448	-0.7740	-1.44
3.345	-0.3457	-0.7729	-1.45
3.350	-0.3462	-0.7719	-1.45
3.355	-0.3467	-0.7710	-1.46
3.360	-0.3474	-0.7698	-1.47
3.365	-0.3478	-0.7688	-1.48
3.370	-0.3490	-0.7674	-1.48
3.375	-0.3494	-0.7660	-1.49
3.380	-0.3506	-0.7644	-1.50
3.385	-0.3515	-0.7628	-1.52
3.390	-0.3526	-0.7614	-1.52
3.395	-0.3537	-0.7595	-1.54
3.400	-0.3549	-0.7577	-1.55
3.405	-0.3560	-0.7560	-1.56
3.410	-0.3576	-0.7542	-1.57
3.415	-0.3591	-0.7527	-1.58
3.420	-0.3607	-0.7515	-1.58
3.425	-0.3621	-0.7497	-1.59
3.430	-0.3637	-0.7481	-1.60
3.435	-0.3654	-0.7468	-1.60
3.440	-0.3671	-0.7452	-1.61
3.445	-0.3693	-0.7434	-1.62
3.450	-0.3713	-0.7422	-1.62
3.455	-0.3729	-0.7410	-1.62
3.460	-0.3748	-0.7402	-1.62
3.465	-0.3768	-0.7396	-1.62
3.470	-0.3784	-0.7383	-1.62
3.475	-0.3801	-0.7377	-1.62
3.480	-0.3815	-0.7372	-1.62
3.485	-0.3836	-0.7370	-1.61
3.490	-0.3849	-0.7370	-1.60
3.495	-0.3859	-0.7365	-1.60
3.500	-0.3866	-0.7365	-1.60
3.505	-0.3870	-0.7360	-1.60
3.510	-0.3877	-0.7355	-1.60
3.515	-0.3879	-0.7348	-1.61
3.520	-0.3877	-0.7343	-1.61
3.525	-0.3879	-0.7332	-1.62
3.530	-0.3875	-0.7328	-1.63
3.535	-0.3867	-0.7303	-1.66
3.540	-0.3866	-0.7279	-1.68

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.545	-0.3866	-0.7253	-1.70
3.550	-0.3873	-0.7224	-1.73
3.555	-0.3877	-0.7190	-1.76
3.560	-0.3894	-0.7159	-1.78
3.565	-0.3909	-0.7134	-1.79
3.570	-0.3936	-0.7102	-1.81
3.575	-0.3960	-0.7071	-1.83
3.580	-0.4000	-0.7054	-1.82
3.585	-0.4046	-0.7048	-1.80
3.590	-0.4077	-0.7051	-1.78
3.595	-0.4106	-0.7063	-1.76
3.600	-0.4126	-0.7081	-1.73
3.605	-0.4139	-0.7100	-1.70
3.610	-0.4137	-0.7118	-1.69
3.615	-0.4130	-0.7127	-1.68
3.620	-0.4117	-0.7129	-1.69
3.625	-0.4102	-0.7136	-1.69
3.630	-0.4091	-0.7128	-1.70
3.635	-0.4074	-0.7115	-1.73
3.640	-0.4067	-0.7098	-1.74
3.645	-0.4064	-0.7082	-1.76
3.650	-0.4063	-0.7063	-1.78
3.655	-0.4066	-0.7043	-1.80
3.660	-0.4070	-0.7029	-1.81
3.665	-0.4075	-0.7013	-1.82
3.670	-0.4080	-0.6994	-1.83
3.675	-0.4083	-0.6971	-1.85
3.680	-0.4090	-0.6956	-1.86
3.685	-0.4095	-0.6939	-1.88
3.690	-0.4106	-0.6917	-1.89
3.695	-0.4123	-0.6893	-1.90
3.700	-0.4140	-0.6870	-1.92
3.705	-0.4166	-0.6848	-1.92
3.710	-0.4195	-0.6827	-1.92
3.715	-0.4231	-0.6814	-1.92
3.720	-0.4272	-0.6800	-1.91
3.725	-0.4314	-0.6805	-1.88
3.730	-0.4369	-0.6803	-1.85
3.735	-0.4404	-0.6828	-1.80
3.740	-0.4438	-0.6855	-1.76
3.745	-0.4451	-0.6885	-1.73
3.750	-0.4457	-0.6924	-1.69
3.755	-0.4450	-0.6956	-1.66
3.760	-0.4432	-0.6982	-1.65
3.765	-0.4403	-0.6999	-1.65
3.770	-0.4369	-0.6998	-1.67

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.775	-0.4339	-0.6980	-1.70
3.780	-0.4311	-0.6954	-1.74
3.785	-0.4307	-0.6915	-1.78
3.790	-0.4317	-0.6874	-1.81
3.795	-0.4343	-0.6844	-1.82
3.800	-0.4370	-0.6820	-1.83
3.805	-0.4394	-0.6804	-1.83
3.810	-0.4425	-0.6794	-1.82
3.815	-0.4452	-0.6787	-1.81
3.820	-0.4479	-0.6784	-1.80
3.825	-0.4503	-0.6782	-1.79
3.830	-0.4526	-0.6788	-1.77
3.835	-0.4534	-0.6816	-1.74
3.840	-0.4549	-0.6811	-1.73
3.845	-0.4559	-0.6808	-1.73
3.850	-0.4572	-0.6803	-1.73
3.855	-0.4583	-0.6795	-1.73
3.860	-0.4599	-0.6791	-1.72
3.865	-0.4613	-0.6782	-1.72
3.870	-0.4627	-0.6775	-1.72
3.875	-0.4642	-0.6768	-1.72
3.880	-0.4657	-0.6761	-1.71
3.885	-0.4677	-0.6755	-1.71
3.890	-0.4691	-0.6742	-1.71
3.895	-0.4708	-0.6734	-1.71
3.900	-0.4726	-0.6725	-1.70
3.905	-0.4743	-0.6716	-1.70
3.910	-0.4761	-0.6708	-1.70
3.915	-0.4782	-0.6698	-1.69
3.920	-0.4798	-0.6687	-1.69
3.925	-0.4818	-0.6675	-1.69
3.930	-0.4834	-0.6667	-1.69
3.935	-0.4855	-0.6652	-1.69
3.940	-0.4869	-0.6644	-1.68
3.945	-0.4889	-0.6630	-1.68
3.950	-0.4906	-0.6620	-1.68
3.955	-0.4925	-0.6606	-1.68
3.960	-0.4942	-0.6593	-1.68
3.965	-0.4959	-0.6579	-1.68
3.970	-0.4975	-0.6566	-1.68
3.975	-0.4991	-0.6552	-1.69
3.980	-0.5012	-0.6536	-1.68
3.985	-0.5028	-0.6523	-1.69
3.990	-0.5046	-0.6508	-1.69
3.995	-0.5061	-0.6491	-1.69
4.000	-0.5079	-0.6477	-1.69

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.005	-0.5095	-0.6459	-1.70
4.010	-0.5114	-0.6442	-1.70
4.015	-0.5127	-0.6430	-1.70
4.020	-0.5145	-0.6412	-1.70
4.025	-0.5158	-0.6397	-1.71
4.030	-0.5174	-0.6383	-1.71
4.035	-0.5191	-0.6364	-1.71
4.040	-0.5205	-0.6347	-1.71
4.045	-0.5219	-0.6331	-1.72
4.050	-0.5233	-0.6314	-1.72
4.055	-0.5247	-0.6298	-1.73
4.060	-0.5259	-0.6282	-1.73
4.065	-0.5273	-0.6268	-1.73
4.070	-0.5285	-0.6251	-1.74
4.075	-0.5296	-0.6238	-1.74
4.080	-0.5309	-0.6225	-1.74
4.085	-0.5322	-0.6210	-1.75
4.090	-0.5331	-0.6198	-1.75
4.095	-0.5340	-0.6185	-1.75
4.100	-0.5353	-0.6170	-1.76
4.105	-0.5364	-0.6159	-1.76
4.110	-0.5375	-0.6148	-1.76
4.115	-0.5389	-0.6138	-1.76
4.120	-0.5399	-0.6128	-1.76
4.125	-0.5416	-0.6113	-1.76
4.130	-0.5426	-0.6103	-1.76
4.135	-0.5438	-0.6089	-1.76
4.140	-0.5450	-0.6077	-1.76
4.145	-0.5458	-0.6060	-1.77
4.150	-0.5468	-0.6043	-1.78
4.155	-0.5475	-0.6029	-1.78
4.160	-0.5480	-0.6016	-1.79
4.165	-0.5480	-0.6004	-1.80
4.170	-0.5481	-0.5994	-1.81
4.175	-0.5476	-0.5984	-1.82
4.180	-0.5479	-0.5979	-1.82
4.185	-0.5474	-0.5971	-1.83
4.190	-0.5476	-0.5965	-1.83
4.195	-0.5473	-0.5966	-1.83
4.200	-0.5471	-0.5963	-1.84
4.205	-0.5469	-0.5962	-1.84
4.210	-0.5468	-0.5961	-1.84
4.215	-0.5468	-0.5960	-1.84
4.220	-0.5464	-0.5962	-1.84
4.225	-0.5463	-0.5960	-1.85
4.230	-0.5460	-0.5964	-1.85

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.235	-0.5459	-0.5966	-1.84
4.240	-0.5456	-0.5967	-1.85
4.245	-0.5452	-0.5971	-1.85
4.250	-0.5450	-0.5976	-1.84
4.255	-0.5447	-0.5980	-1.84
4.260	-0.5442	-0.5982	-1.84
4.265	-0.5438	-0.5988	-1.84
4.270	-0.5437	-0.5992	-1.84
4.275	-0.5431	-0.5996	-1.84
4.280	-0.5431	-0.6000	-1.84
4.285	-0.5425	-0.6007	-1.84
4.290	-0.5423	-0.6010	-1.84
4.295	-0.5418	-0.6015	-1.83
4.300	-0.5418	-0.6022	-1.83
4.305	-0.5416	-0.6026	-1.83
4.310	-0.5413	-0.6032	-1.83
4.315	-0.5410	-0.6037	-1.82
4.320	-0.5406	-0.6043	-1.82
4.325	-0.5405	-0.6051	-1.82
4.330	-0.5401	-0.6056	-1.81
4.335	-0.5400	-0.6064	-1.81
4.340	-0.5399	-0.6072	-1.80
4.345	-0.5396	-0.6082	-1.80
4.350	-0.5395	-0.6089	-1.79
4.355	-0.5390	-0.6098	-1.79
4.360	-0.5386	-0.6107	-1.78
4.365	-0.5381	-0.6117	-1.78
4.370	-0.5378	-0.6124	-1.78
4.375	-0.5370	-0.6135	-1.77
4.380	-0.5367	-0.6143	-1.77
4.385	-0.5362	-0.6152	-1.77
4.390	-0.5353	-0.6161	-1.76
4.395	-0.5347	-0.6168	-1.76
4.400	-0.5343	-0.6175	-1.76
4.405	-0.5337	-0.6180	-1.76
4.410	-0.5330	-0.6186	-1.76
4.415	-0.5322	-0.6190	-1.76
4.420	-0.5321	-0.6192	-1.76
4.425	-0.5315	-0.6193	-1.77
4.430	-0.5311	-0.6200	-1.76
4.435	-0.5308	-0.6199	-1.77
4.440	-0.5307	-0.6200	-1.77
4.445	-0.5304	-0.6206	-1.76
4.450	-0.5304	-0.6205	-1.76
4.455	-0.5306	-0.6214	-1.75
4.460	-0.5297	-0.6218	-1.76

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.465	-0.5295	-0.6227	-1.75
4.470	-0.5292	-0.6239	-1.74
4.475	-0.5289	-0.6248	-1.74
4.480	-0.5285	-0.6263	-1.73
4.485	-0.5285	-0.6275	-1.72
4.490	-0.5279	-0.6293	-1.71
4.495	-0.5279	-0.6306	-1.70
4.500	-0.5274	-0.6315	-1.69
4.505	-0.5270	-0.6326	-1.69
4.510	-0.5266	-0.6327	-1.69
4.515	-0.5259	-0.6331	-1.69
4.520	-0.5255	-0.6337	-1.69
4.525	-0.5246	-0.6335	-1.70
4.530	-0.5248	-0.6328	-1.70
4.535	-0.5244	-0.6322	-1.71
4.540	-0.5246	-0.6318	-1.71
4.545	-0.5245	-0.6307	-1.72
4.550	-0.5249	-0.6304	-1.72
4.555	-0.5251	-0.6297	-1.72
4.560	-0.5255	-0.6293	-1.73
4.565	-0.5255	-0.6286	-1.73
4.570	-0.5258	-0.6277	-1.74
4.575	-0.5263	-0.6268	-1.74
4.580	-0.5264	-0.6261	-1.75
4.585	-0.5269	-0.6250	-1.75
4.590	-0.5273	-0.6245	-1.75
4.595	-0.5275	-0.6237	-1.76
4.600	-0.5282	-0.6227	-1.76
4.605	-0.5283	-0.6218	-1.77
4.610	-0.5289	-0.6209	-1.77
4.615	-0.5295	-0.6196	-1.78
4.620	-0.5299	-0.6189	-1.78
4.625	-0.5305	-0.6178	-1.78
4.630	-0.5307	-0.6172	-1.79
4.635	-0.5309	-0.6161	-1.80
4.640	-0.5309	-0.6151	-1.80
4.645	-0.5312	-0.6143	-1.81
4.650	-0.5314	-0.6136	-1.81
4.655	-0.5318	-0.6126	-1.82
4.660	-0.5317	-0.6118	-1.82
4.665	-0.5317	-0.6109	-1.83
4.670	-0.5317	-0.6102	-1.84
4.675	-0.5315	-0.6095	-1.84
4.680	-0.5313	-0.6086	-1.85
4.685	-0.5309	-0.6083	-1.86
4.690	-0.5307	-0.6076	-1.87

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.695	-0.5305	-0.6068	-1.87
4.700	-0.5301	-0.6063	-1.88
4.705	-0.5299	-0.6054	-1.89
4.710	-0.5300	-0.6048	-1.89
4.715	-0.5299	-0.6039	-1.90
4.720	-0.5299	-0.6028	-1.91
4.725	-0.5303	-0.6021	-1.91
4.730	-0.5310	-0.6007	-1.92
4.735	-0.5316	-0.6000	-1.92
4.740	-0.5325	-0.5987	-1.92
4.745	-0.5329	-0.5975	-1.93
4.750	-0.5337	-0.5957	-1.94
4.755	-0.5348	-0.5945	-1.94
4.760	-0.5354	-0.5933	-1.95
4.765	-0.5361	-0.5924	-1.95
4.770	-0.5363	-0.5916	-1.95
4.775	-0.5361	-0.5908	-1.96
4.780	-0.5363	-0.5898	-1.97
4.785	-0.5361	-0.5891	-1.98
4.790	-0.5351	-0.5887	-1.99
4.795	-0.5345	-0.5880	-2.00
4.800	-0.5337	-0.5874	-2.01
4.805	-0.5329	-0.5868	-2.02
4.810	-0.5320	-0.5858	-2.03
4.815	-0.5312	-0.5858	-2.04
4.820	-0.5301	-0.5853	-2.05
4.825	-0.5298	-0.5852	-2.05
4.830	-0.5291	-0.5850	-2.06
4.835	-0.5287	-0.5849	-2.07
4.840	-0.5284	-0.5849	-2.07
4.845	-0.5285	-0.5848	-2.07
4.850	-0.5283	-0.5851	-2.07
4.855	-0.5284	-0.5856	-2.06
4.860	-0.5286	-0.5860	-2.06
4.865	-0.5292	-0.5859	-2.05
4.870	-0.5293	-0.5867	-2.05
4.875	-0.5298	-0.5874	-2.04
4.880	-0.5302	-0.5880	-2.03
4.885	-0.5308	-0.5885	-2.02
4.890	-0.5315	-0.5892	-2.01
4.895	-0.5319	-0.5897	-2.00
4.900	-0.5325	-0.5899	-2.00
4.905	-0.5329	-0.5904	-1.99
4.910	-0.5335	-0.5903	-1.99
4.915	-0.5341	-0.5908	-1.98
4.920	-0.5348	-0.5910	-1.97

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.925	-0.5352	-0.5911	-1.97
4.930	-0.5356	-0.5913	-1.96
4.935	-0.5361	-0.5916	-1.96
4.940	-0.5366	-0.5917	-1.95
4.945	-0.5370	-0.5920	-1.95
4.950	-0.5371	-0.5922	-1.94
4.955	-0.5377	-0.5923	-1.94
4.960	-0.5381	-0.5925	-1.93
4.965	-0.5384	-0.5928	-1.93
4.970	-0.5387	-0.5931	-1.92
4.975	-0.5392	-0.5933	-1.92
4.980	-0.5397	-0.5936	-1.91
4.985	-0.5399	-0.5939	-1.91
4.990	-0.5405	-0.5944	-1.90
4.995	-0.5408	-0.5946	-1.90
5.000	-0.5414	-0.5949	-1.89
5.005	-0.5418	-0.5951	-1.89
5.010	-0.5425	-0.5951	-1.88
5.015	-0.5426	-0.5953	-1.88
5.020	-0.5431	-0.5954	-1.87
5.025	-0.5432	-0.5956	-1.87
5.030	-0.5437	-0.5950	-1.87
5.035	-0.5438	-0.5955	-1.87
5.040	-0.5439	-0.5948	-1.87
5.045	-0.5438	-0.5952	-1.87
5.050	-0.5436	-0.5945	-1.88
5.055	-0.5436	-0.5949	-1.88
5.060	-0.5433	-0.5943	-1.88
5.065	-0.5438	-0.5945	-1.88
5.070	-0.5434	-0.5941	-1.88
5.075	-0.5444	-0.5940	-1.88
5.080	-0.5440	-0.5941	-1.88
5.085	-0.5452	-0.5936	-1.87
5.090	-0.5452	-0.5935	-1.87
5.095	-0.5464	-0.5933	-1.87
5.100	-0.5468	-0.5929	-1.87
5.105	-0.5477	-0.5922	-1.87
5.110	-0.5484	-0.5921	-1.86
5.115	-0.5493	-0.5913	-1.86
5.120	-0.5500	-0.5909	-1.86
5.125	-0.5512	-0.5900	-1.86
5.130	-0.5520	-0.5898	-1.85
5.135	-0.5527	-0.5888	-1.86
5.140	-0.5538	-0.5879	-1.86
5.145	-0.5552	-0.5870	-1.85
5.150	-0.5559	-0.5862	-1.85

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.155	-0.5573	-0.5846	-1.85
5.160	-0.5585	-0.5832	-1.86
5.165	-0.5595	-0.5817	-1.86
5.170	-0.5609	-0.5804	-1.86
5.175	-0.5623	-0.5785	-1.87
5.180	-0.5632	-0.5766	-1.87
5.185	-0.5643	-0.5752	-1.88
5.190	-0.5655	-0.5727	-1.89
5.195	-0.5665	-0.5710	-1.89
5.200	-0.5676	-0.5685	-1.90
5.205	-0.5687	-0.5665	-1.91
5.210	-0.5695	-0.5645	-1.92
5.215	-0.5706	-0.5623	-1.93
5.220	-0.5715	-0.5600	-1.94
5.225	-0.5728	-0.5574	-1.95
5.230	-0.5736	-0.5553	-1.96
5.235	-0.5745	-0.5531	-1.97
5.240	-0.5755	-0.5509	-1.97
5.245	-0.5764	-0.5489	-1.98
5.250	-0.5772	-0.5466	-1.99
5.255	-0.5784	-0.5443	-2.00
5.260	-0.5792	-0.5421	-2.01
5.265	-0.5803	-0.5401	-2.02
5.270	-0.5814	-0.5381	-2.02
5.275	-0.5822	-0.5360	-2.03
5.280	-0.5829	-0.5341	-2.04
5.285	-0.5837	-0.5320	-2.05
5.290	-0.5848	-0.5303	-2.05
5.295	-0.5856	-0.5283	-2.06
5.300	-0.5863	-0.5267	-2.07
5.305	-0.5874	-0.5250	-2.07
5.310	-0.5878	-0.5233	-2.08
5.315	-0.5886	-0.5217	-2.09
5.320	-0.5894	-0.5198	-2.09
5.325	-0.5900	-0.5182	-2.10
5.330	-0.5905	-0.5167	-2.11
5.335	-0.5910	-0.5150	-2.11
5.340	-0.5915	-0.5133	-2.12
5.345	-0.5925	-0.5116	-2.13
5.350	-0.5926	-0.5101	-2.14
5.355	-0.5933	-0.5083	-2.14
5.360	-0.5937	-0.5067	-2.15
5.365	-0.5946	-0.5051	-2.16
5.370	-0.5950	-0.5034	-2.16
5.375	-0.5959	-0.5014	-2.17
5.380	-0.5967	-0.5001	-2.17

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.385	-0.5972	-0.4983	-2.18
5.390	-0.5981	-0.4971	-2.18
5.395	-0.5988	-0.4959	-2.19
5.400	-0.5999	-0.4944	-2.19
5.405	-0.6007	-0.4932	-2.19
5.410	-0.6017	-0.4920	-2.19
5.415	-0.6026	-0.4909	-2.19
5.420	-0.6033	-0.4895	-2.19
5.425	-0.6046	-0.4884	-2.19
5.430	-0.6051	-0.4874	-2.19
5.435	-0.6057	-0.4866	-2.19
5.440	-0.6066	-0.4858	-2.19
5.445	-0.6072	-0.4848	-2.19
5.450	-0.6079	-0.4839	-2.19
5.455	-0.6084	-0.4831	-2.19
5.460	-0.6093	-0.4825	-2.19
5.465	-0.6098	-0.4816	-2.19
5.470	-0.6104	-0.4811	-2.19
5.475	-0.6109	-0.4803	-2.19
5.480	-0.6117	-0.4797	-2.19
5.485	-0.6125	-0.4788	-2.19
5.490	-0.6131	-0.4782	-2.19
5.495	-0.6139	-0.4775	-2.18
5.500	-0.6147	-0.4769	-2.18
5.505	-0.6154	-0.4764	-2.18
5.510	-0.6168	-0.4762	-2.17
5.515	-0.6177	-0.4761	-2.16
5.520	-0.6186	-0.4760	-2.15
5.525	-0.6193	-0.4760	-2.15
5.530	-0.6203	-0.4760	-2.14
5.535	-0.6210	-0.4761	-2.13
5.540	-0.6217	-0.4762	-2.12
5.545	-0.6221	-0.4763	-2.12
5.550	-0.6229	-0.4765	-2.11
5.555	-0.6236	-0.4766	-2.10
5.560	-0.6243	-0.4768	-2.10
5.565	-0.6250	-0.4768	-2.09
5.570	-0.6259	-0.4768	-2.08
5.575	-0.6266	-0.4770	-2.08
5.580	-0.6273	-0.4771	-2.07
5.585	-0.6281	-0.4770	-2.06
5.590	-0.6290	-0.4771	-2.05
5.595	-0.6299	-0.4771	-2.04
5.600	-0.6309	-0.4772	-2.04
5.605	-0.6319	-0.4774	-2.03
5.610	-0.6330	-0.4775	-2.02

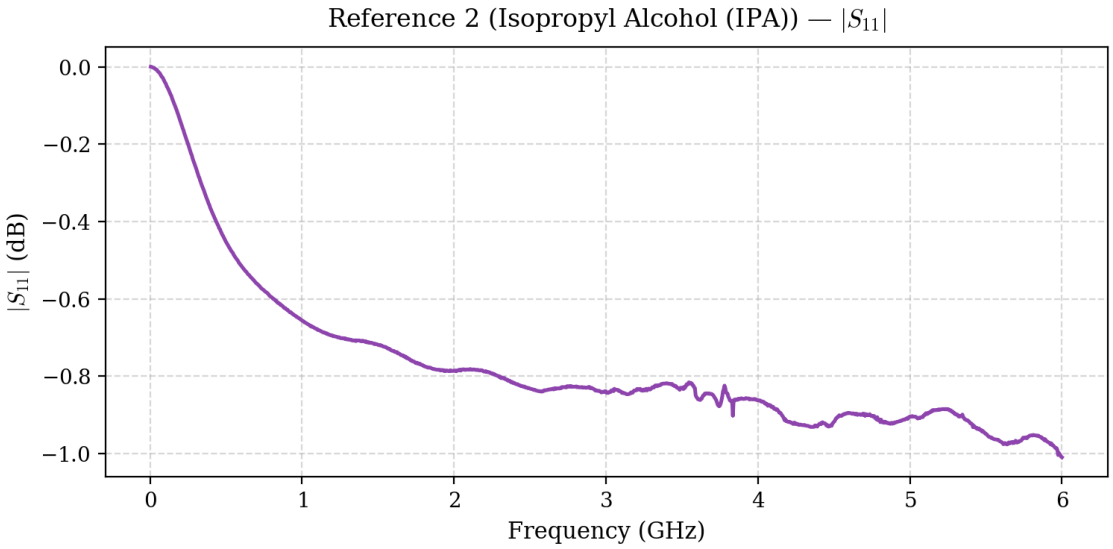
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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.615	-0.6337	-0.4779	-2.01
5.620	-0.6348	-0.4780	-2.00
5.625	-0.6356	-0.4785	-1.99
5.630	-0.6364	-0.4787	-1.98
5.635	-0.6370	-0.4789	-1.97
5.640	-0.6376	-0.4791	-1.96
5.645	-0.6382	-0.4794	-1.96
5.650	-0.6389	-0.4794	-1.95
5.655	-0.6393	-0.4795	-1.95
5.660	-0.6399	-0.4795	-1.94
5.665	-0.6404	-0.4797	-1.94
5.670	-0.6410	-0.4798	-1.93
5.675	-0.6415	-0.4797	-1.93
5.680	-0.6424	-0.4796	-1.92
5.685	-0.6421	-0.4794	-1.92
5.690	-0.6425	-0.4791	-1.92
5.695	-0.6430	-0.4791	-1.92
5.700	-0.6430	-0.4789	-1.92
5.705	-0.6436	-0.4786	-1.92
5.710	-0.6435	-0.4784	-1.92
5.715	-0.6439	-0.4784	-1.92
5.720	-0.6442	-0.4784	-1.91
5.725	-0.6441	-0.4781	-1.92
5.730	-0.6445	-0.4775	-1.92
5.735	-0.6446	-0.4774	-1.91
5.740	-0.6447	-0.4767	-1.92
5.745	-0.6453	-0.4768	-1.91
5.750	-0.6450	-0.4758	-1.92
5.755	-0.6450	-0.4754	-1.92
5.760	-0.6448	-0.4748	-1.93
5.765	-0.6454	-0.4741	-1.93
5.770	-0.6450	-0.4735	-1.94
5.775	-0.6453	-0.4721	-1.94
5.780	-0.6448	-0.4717	-1.95
5.785	-0.6452	-0.4705	-1.95
5.790	-0.6452	-0.4698	-1.96
5.795	-0.6450	-0.4685	-1.97
5.800	-0.6448	-0.4671	-1.98
5.805	-0.6448	-0.4661	-1.99
5.810	-0.6443	-0.4649	-2.00
5.815	-0.6444	-0.4637	-2.00
5.820	-0.6445	-0.4623	-2.01
5.825	-0.6443	-0.4613	-2.02
5.830	-0.6442	-0.4598	-2.03
5.835	-0.6441	-0.4584	-2.04
5.840	-0.6438	-0.4569	-2.05

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.845	-0.6441	-0.4555	-2.06
5.850	-0.6441	-0.4544	-2.07
5.855	-0.6438	-0.4530	-2.08
5.860	-0.6440	-0.4514	-2.09
5.865	-0.6441	-0.4504	-2.09
5.870	-0.6442	-0.4487	-2.10
5.875	-0.6443	-0.4477	-2.11
5.880	-0.6446	-0.4463	-2.11
5.885	-0.6447	-0.4450	-2.12
5.890	-0.6447	-0.4439	-2.13
5.895	-0.6447	-0.4429	-2.13
5.900	-0.6449	-0.4419	-2.14
5.905	-0.6451	-0.4410	-2.14
5.910	-0.6452	-0.4404	-2.15
5.915	-0.6449	-0.4393	-2.15
5.920	-0.6449	-0.4381	-2.16
5.925	-0.6450	-0.4373	-2.17
5.930	-0.6447	-0.4361	-2.18
5.935	-0.6446	-0.4351	-2.18
5.940	-0.6448	-0.4341	-2.19
5.945	-0.6444	-0.4334	-2.20
5.950	-0.6445	-0.4322	-2.20
5.955	-0.6448	-0.4312	-2.21
5.960	-0.6451	-0.4303	-2.21
5.965	-0.6454	-0.4296	-2.21
5.970	-0.6460	-0.4286	-2.21
5.975	-0.6468	-0.4279	-2.21
5.980	-0.6469	-0.4282	-2.20
5.985	-0.6475	-0.4276	-2.20
5.990	-0.6480	-0.4277	-2.20
5.995	-0.6486	-0.4279	-2.19
6.000	-0.6488	-0.4281	-2.19

3.4 Reference 2 (Isopropyl Alcohol (IPA))



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.005	1.0001	-0.0016	0.00
0.010	1.0000	-0.0028	0.00
0.015	0.9999	-0.0040	-0.00
0.020	0.9998	-0.0053	-0.00
0.025	0.9997	-0.0064	-0.00
0.030	0.9996	-0.0077	-0.00
0.035	0.9993	-0.0089	-0.01
0.040	0.9992	-0.0102	-0.01
0.045	0.9991	-0.0112	-0.01
0.050	0.9986	-0.0125	-0.01
0.055	0.9985	-0.0137	-0.01
0.060	0.9982	-0.0148	-0.01
0.065	0.9980	-0.0161	-0.02
0.070	0.9975	-0.0172	-0.02
0.075	0.9970	-0.0183	-0.02
0.080	0.9968	-0.0194	-0.03
0.085	0.9965	-0.0206	-0.03
0.090	0.9959	-0.0216	-0.03
0.095	0.9955	-0.0225	-0.04
0.100	0.9950	-0.0237	-0.04
0.105	0.9946	-0.0248	-0.04
0.110	0.9939	-0.0257	-0.05
0.115	0.9935	-0.0268	-0.05
0.120	0.9931	-0.0278	-0.06
0.125	0.9925	-0.0287	-0.06
0.130	0.9919	-0.0297	-0.07

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.135	0.9914	-0.0305	-0.07
0.140	0.9910	-0.0315	-0.07
0.145	0.9902	-0.0325	-0.08
0.150	0.9896	-0.0334	-0.09
0.155	0.9890	-0.0341	-0.09
0.160	0.9883	-0.0351	-0.10
0.165	0.9878	-0.0358	-0.10
0.170	0.9871	-0.0366	-0.11
0.175	0.9866	-0.0374	-0.11
0.180	0.9858	-0.0382	-0.12
0.185	0.9852	-0.0388	-0.12
0.190	0.9844	-0.0397	-0.13
0.195	0.9837	-0.0405	-0.14
0.200	0.9831	-0.0412	-0.14
0.205	0.9824	-0.0418	-0.15
0.210	0.9817	-0.0424	-0.15
0.215	0.9809	-0.0433	-0.16
0.220	0.9804	-0.0438	-0.16
0.225	0.9795	-0.0444	-0.17
0.230	0.9788	-0.0449	-0.18
0.235	0.9782	-0.0455	-0.18
0.240	0.9775	-0.0461	-0.19
0.245	0.9768	-0.0468	-0.19
0.250	0.9762	-0.0473	-0.20
0.255	0.9754	-0.0477	-0.21
0.260	0.9746	-0.0482	-0.21
0.265	0.9740	-0.0487	-0.22
0.270	0.9733	-0.0493	-0.22
0.275	0.9728	-0.0496	-0.23
0.280	0.9718	-0.0502	-0.24
0.285	0.9712	-0.0505	-0.24
0.290	0.9705	-0.0509	-0.25
0.295	0.9699	-0.0514	-0.25
0.300	0.9692	-0.0517	-0.26
0.305	0.9686	-0.0521	-0.26
0.310	0.9679	-0.0525	-0.27
0.315	0.9672	-0.0527	-0.28
0.320	0.9665	-0.0530	-0.28
0.325	0.9660	-0.0535	-0.29
0.330	0.9652	-0.0538	-0.29
0.335	0.9645	-0.0541	-0.30
0.340	0.9639	-0.0543	-0.31
0.345	0.9634	-0.0545	-0.31
0.350	0.9627	-0.0548	-0.32
0.355	0.9622	-0.0552	-0.32
0.360	0.9615	-0.0554	-0.33

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.365	0.9609	-0.0556	-0.33
0.370	0.9603	-0.0558	-0.34
0.375	0.9597	-0.0560	-0.34
0.380	0.9592	-0.0563	-0.35
0.385	0.9585	-0.0564	-0.35
0.390	0.9581	-0.0566	-0.36
0.395	0.9573	-0.0569	-0.36
0.400	0.9568	-0.0570	-0.37
0.405	0.9564	-0.0572	-0.37
0.410	0.9557	-0.0572	-0.38
0.415	0.9553	-0.0574	-0.38
0.420	0.9548	-0.0576	-0.39
0.425	0.9543	-0.0577	-0.39
0.430	0.9538	-0.0579	-0.39
0.435	0.9532	-0.0580	-0.40
0.440	0.9528	-0.0581	-0.40
0.445	0.9524	-0.0583	-0.41
0.450	0.9519	-0.0584	-0.41
0.455	0.9513	-0.0584	-0.42
0.460	0.9510	-0.0585	-0.42
0.465	0.9504	-0.0586	-0.42
0.470	0.9500	-0.0587	-0.43
0.475	0.9495	-0.0589	-0.43
0.480	0.9491	-0.0589	-0.44
0.485	0.9487	-0.0590	-0.44
0.490	0.9482	-0.0590	-0.44
0.495	0.9478	-0.0592	-0.45
0.500	0.9474	-0.0592	-0.45
0.505	0.9470	-0.0592	-0.46
0.510	0.9466	-0.0592	-0.46
0.515	0.9462	-0.0594	-0.46
0.520	0.9458	-0.0594	-0.47
0.525	0.9455	-0.0594	-0.47
0.530	0.9451	-0.0595	-0.47
0.535	0.9448	-0.0596	-0.48
0.540	0.9445	-0.0596	-0.48
0.545	0.9442	-0.0597	-0.48
0.550	0.9437	-0.0597	-0.49
0.555	0.9434	-0.0598	-0.49
0.560	0.9430	-0.0599	-0.49
0.565	0.9427	-0.0599	-0.49
0.570	0.9424	-0.0600	-0.50
0.575	0.9422	-0.0599	-0.50
0.580	0.9419	-0.0600	-0.50
0.585	0.9414	-0.0599	-0.51
0.590	0.9412	-0.0600	-0.51

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.595	0.9409	-0.0601	-0.51
0.600	0.9406	-0.0601	-0.51
0.605	0.9403	-0.0602	-0.52
0.610	0.9400	-0.0602	-0.52
0.615	0.9398	-0.0602	-0.52
0.620	0.9395	-0.0602	-0.52
0.625	0.9392	-0.0603	-0.53
0.630	0.9390	-0.0602	-0.53
0.635	0.9388	-0.0603	-0.53
0.640	0.9385	-0.0604	-0.53
0.645	0.9382	-0.0604	-0.54
0.650	0.9380	-0.0605	-0.54
0.655	0.9378	-0.0604	-0.54
0.660	0.9375	-0.0604	-0.54
0.665	0.9373	-0.0604	-0.54
0.670	0.9370	-0.0605	-0.55
0.675	0.9368	-0.0605	-0.55
0.680	0.9366	-0.0607	-0.55
0.685	0.9363	-0.0606	-0.55
0.690	0.9361	-0.0606	-0.56
0.695	0.9358	-0.0606	-0.56
0.700	0.9356	-0.0606	-0.56
0.705	0.9355	-0.0607	-0.56
0.710	0.9352	-0.0608	-0.56
0.715	0.9350	-0.0608	-0.57
0.720	0.9349	-0.0608	-0.57
0.725	0.9345	-0.0608	-0.57
0.730	0.9343	-0.0607	-0.57
0.735	0.9341	-0.0609	-0.57
0.740	0.9340	-0.0608	-0.58
0.745	0.9338	-0.0609	-0.58
0.750	0.9336	-0.0610	-0.58
0.755	0.9334	-0.0610	-0.58
0.760	0.9332	-0.0610	-0.58
0.765	0.9331	-0.0610	-0.58
0.770	0.9329	-0.0611	-0.58
0.775	0.9327	-0.0610	-0.59
0.780	0.9325	-0.0612	-0.59
0.785	0.9322	-0.0612	-0.59
0.790	0.9320	-0.0611	-0.59
0.795	0.9318	-0.0613	-0.59
0.800	0.9317	-0.0613	-0.60
0.805	0.9315	-0.0614	-0.60
0.810	0.9312	-0.0614	-0.60
0.815	0.9312	-0.0614	-0.60
0.820	0.9310	-0.0615	-0.60

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.825	0.9307	-0.0615	-0.60
0.830	0.9307	-0.0615	-0.60
0.835	0.9303	-0.0617	-0.61
0.840	0.9301	-0.0617	-0.61
0.845	0.9301	-0.0616	-0.61
0.850	0.9299	-0.0617	-0.61
0.855	0.9298	-0.0619	-0.61
0.860	0.9295	-0.0618	-0.62
0.865	0.9294	-0.0621	-0.62
0.870	0.9293	-0.0621	-0.62
0.875	0.9291	-0.0620	-0.62
0.880	0.9287	-0.0620	-0.62
0.885	0.9286	-0.0621	-0.62
0.890	0.9286	-0.0622	-0.62
0.895	0.9283	-0.0623	-0.63
0.900	0.9282	-0.0621	-0.63
0.905	0.9281	-0.0623	-0.63
0.910	0.9278	-0.0624	-0.63
0.915	0.9277	-0.0625	-0.63
0.920	0.9274	-0.0624	-0.63
0.925	0.9275	-0.0625	-0.63
0.930	0.9273	-0.0626	-0.64
0.935	0.9270	-0.0627	-0.64
0.940	0.9269	-0.0627	-0.64
0.945	0.9267	-0.0628	-0.64
0.950	0.9266	-0.0627	-0.64
0.955	0.9265	-0.0628	-0.64
0.960	0.9262	-0.0629	-0.65
0.965	0.9262	-0.0630	-0.65
0.970	0.9260	-0.0629	-0.65
0.975	0.9258	-0.0630	-0.65
0.980	0.9258	-0.0630	-0.65
0.985	0.9256	-0.0631	-0.65
0.990	0.9254	-0.0632	-0.65
0.995	0.9253	-0.0633	-0.65
1.000	0.9251	-0.0633	-0.66
1.005	0.9250	-0.0634	-0.66
1.010	0.9249	-0.0636	-0.66
1.015	0.9246	-0.0636	-0.66
1.020	0.9245	-0.0636	-0.66
1.025	0.9244	-0.0636	-0.66
1.030	0.9243	-0.0637	-0.66
1.035	0.9241	-0.0638	-0.66
1.040	0.9240	-0.0638	-0.67
1.045	0.9239	-0.0639	-0.67
1.050	0.9237	-0.0639	-0.67

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.055	0.9235	-0.0640	-0.67
1.060	0.9235	-0.0640	-0.67
1.065	0.9233	-0.0640	-0.67
1.070	0.9233	-0.0642	-0.67
1.075	0.9232	-0.0643	-0.67
1.080	0.9230	-0.0643	-0.67
1.085	0.9229	-0.0643	-0.68
1.090	0.9228	-0.0644	-0.68
1.095	0.9226	-0.0645	-0.68
1.100	0.9226	-0.0646	-0.68
1.105	0.9223	-0.0647	-0.68
1.110	0.9223	-0.0647	-0.68
1.115	0.9222	-0.0647	-0.68
1.120	0.9222	-0.0648	-0.68
1.125	0.9220	-0.0648	-0.68
1.130	0.9219	-0.0649	-0.68
1.135	0.9218	-0.0651	-0.69
1.140	0.9216	-0.0650	-0.69
1.145	0.9217	-0.0651	-0.69
1.150	0.9216	-0.0650	-0.69
1.155	0.9215	-0.0652	-0.69
1.160	0.9213	-0.0653	-0.69
1.165	0.9214	-0.0654	-0.69
1.170	0.9212	-0.0654	-0.69
1.175	0.9210	-0.0655	-0.69
1.180	0.9210	-0.0657	-0.69
1.185	0.9210	-0.0655	-0.69
1.190	0.9208	-0.0656	-0.69
1.195	0.9208	-0.0657	-0.69
1.200	0.9208	-0.0657	-0.69
1.205	0.9206	-0.0658	-0.70
1.210	0.9206	-0.0658	-0.70
1.215	0.9205	-0.0659	-0.70
1.220	0.9205	-0.0658	-0.70
1.225	0.9204	-0.0659	-0.70
1.230	0.9204	-0.0660	-0.70
1.235	0.9202	-0.0661	-0.70
1.240	0.9202	-0.0661	-0.70
1.245	0.9203	-0.0661	-0.70
1.250	0.9201	-0.0663	-0.70
1.255	0.9200	-0.0663	-0.70
1.260	0.9200	-0.0663	-0.70
1.265	0.9200	-0.0664	-0.70
1.270	0.9199	-0.0664	-0.70
1.275	0.9199	-0.0665	-0.70
1.280	0.9198	-0.0665	-0.70

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.285	0.9199	-0.0666	-0.70
1.290	0.9197	-0.0666	-0.70
1.295	0.9197	-0.0667	-0.70
1.300	0.9196	-0.0669	-0.71
1.305	0.9197	-0.0669	-0.70
1.310	0.9196	-0.0669	-0.71
1.315	0.9196	-0.0669	-0.71
1.320	0.9194	-0.0670	-0.71
1.325	0.9195	-0.0671	-0.71
1.330	0.9193	-0.0671	-0.71
1.335	0.9193	-0.0672	-0.71
1.340	0.9193	-0.0671	-0.71
1.345	0.9192	-0.0672	-0.71
1.350	0.9193	-0.0674	-0.71
1.355	0.9191	-0.0673	-0.71
1.360	0.9193	-0.0674	-0.71
1.365	0.9193	-0.0675	-0.71
1.370	0.9193	-0.0675	-0.71
1.375	0.9193	-0.0675	-0.71
1.380	0.9192	-0.0676	-0.71
1.385	0.9191	-0.0677	-0.71
1.390	0.9190	-0.0679	-0.71
1.395	0.9192	-0.0680	-0.71
1.400	0.9192	-0.0680	-0.71
1.405	0.9190	-0.0680	-0.71
1.410	0.9190	-0.0681	-0.71
1.415	0.9189	-0.0681	-0.71
1.420	0.9190	-0.0683	-0.71
1.425	0.9187	-0.0682	-0.71
1.430	0.9187	-0.0684	-0.71
1.435	0.9187	-0.0683	-0.71
1.440	0.9187	-0.0685	-0.71
1.445	0.9186	-0.0687	-0.71
1.450	0.9185	-0.0687	-0.71
1.455	0.9186	-0.0689	-0.71
1.460	0.9185	-0.0691	-0.71
1.465	0.9184	-0.0691	-0.71
1.470	0.9183	-0.0691	-0.72
1.475	0.9182	-0.0692	-0.72
1.480	0.9182	-0.0692	-0.72
1.485	0.9181	-0.0695	-0.72
1.490	0.9181	-0.0695	-0.72
1.495	0.9180	-0.0696	-0.72
1.500	0.9179	-0.0698	-0.72
1.505	0.9179	-0.0699	-0.72
1.510	0.9178	-0.0699	-0.72

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.515	0.9178	-0.0701	-0.72
1.520	0.9177	-0.0703	-0.72
1.525	0.9176	-0.0704	-0.72
1.530	0.9175	-0.0704	-0.72
1.535	0.9174	-0.0705	-0.72
1.540	0.9173	-0.0707	-0.72
1.545	0.9172	-0.0709	-0.72
1.550	0.9172	-0.0709	-0.72
1.555	0.9171	-0.0710	-0.73
1.560	0.9169	-0.0712	-0.73
1.565	0.9168	-0.0714	-0.73
1.570	0.9166	-0.0714	-0.73
1.575	0.9167	-0.0716	-0.73
1.580	0.9165	-0.0715	-0.73
1.585	0.9163	-0.0718	-0.73
1.590	0.9162	-0.0719	-0.73
1.595	0.9161	-0.0721	-0.73
1.600	0.9161	-0.0721	-0.73
1.605	0.9160	-0.0723	-0.74
1.610	0.9158	-0.0723	-0.74
1.615	0.9157	-0.0725	-0.74
1.620	0.9157	-0.0727	-0.74
1.625	0.9157	-0.0729	-0.74
1.630	0.9153	-0.0730	-0.74
1.635	0.9153	-0.0731	-0.74
1.640	0.9152	-0.0731	-0.74
1.645	0.9150	-0.0734	-0.74
1.650	0.9148	-0.0736	-0.75
1.655	0.9148	-0.0736	-0.75
1.660	0.9147	-0.0738	-0.75
1.665	0.9147	-0.0740	-0.75
1.670	0.9143	-0.0741	-0.75
1.675	0.9143	-0.0743	-0.75
1.680	0.9143	-0.0743	-0.75
1.685	0.9139	-0.0744	-0.75
1.690	0.9141	-0.0746	-0.75
1.695	0.9138	-0.0746	-0.75
1.700	0.9137	-0.0748	-0.75
1.705	0.9137	-0.0749	-0.76
1.710	0.9134	-0.0750	-0.76
1.715	0.9132	-0.0750	-0.76
1.720	0.9132	-0.0754	-0.76
1.725	0.9130	-0.0753	-0.76
1.730	0.9130	-0.0754	-0.76
1.735	0.9129	-0.0756	-0.76
1.740	0.9128	-0.0757	-0.76

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.745	0.9125	-0.0760	-0.77
1.750	0.9124	-0.0759	-0.77
1.755	0.9125	-0.0761	-0.77
1.760	0.9123	-0.0762	-0.77
1.765	0.9123	-0.0763	-0.77
1.770	0.9121	-0.0764	-0.77
1.775	0.9122	-0.0766	-0.77
1.780	0.9120	-0.0766	-0.77
1.785	0.9119	-0.0768	-0.77
1.790	0.9118	-0.0768	-0.77
1.795	0.9117	-0.0769	-0.77
1.800	0.9115	-0.0770	-0.77
1.805	0.9115	-0.0772	-0.77
1.810	0.9113	-0.0772	-0.78
1.815	0.9113	-0.0774	-0.78
1.820	0.9112	-0.0774	-0.78
1.825	0.9111	-0.0776	-0.78
1.830	0.9110	-0.0777	-0.78
1.835	0.9110	-0.0779	-0.78
1.840	0.9110	-0.0780	-0.78
1.845	0.9109	-0.0781	-0.78
1.850	0.9108	-0.0780	-0.78
1.855	0.9107	-0.0782	-0.78
1.860	0.9106	-0.0780	-0.78
1.865	0.9106	-0.0781	-0.78
1.870	0.9106	-0.0782	-0.78
1.875	0.9105	-0.0784	-0.78
1.880	0.9104	-0.0786	-0.78
1.885	0.9104	-0.0787	-0.78
1.890	0.9104	-0.0788	-0.78
1.895	0.9103	-0.0789	-0.78
1.900	0.9103	-0.0790	-0.78
1.905	0.9103	-0.0790	-0.78
1.910	0.9101	-0.0791	-0.79
1.915	0.9102	-0.0793	-0.78
1.920	0.9101	-0.0793	-0.79
1.925	0.9102	-0.0795	-0.78
1.930	0.9100	-0.0794	-0.79
1.935	0.9100	-0.0796	-0.79
1.940	0.9099	-0.0797	-0.79
1.945	0.9100	-0.0799	-0.79
1.950	0.9100	-0.0799	-0.79
1.955	0.9100	-0.0800	-0.79
1.960	0.9099	-0.0800	-0.79
1.965	0.9098	-0.0802	-0.79
1.970	0.9099	-0.0803	-0.79

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.975	0.9100	-0.0804	-0.79
1.980	0.9099	-0.0804	-0.79
1.985	0.9099	-0.0804	-0.79
1.990	0.9098	-0.0806	-0.79
1.995	0.9099	-0.0806	-0.79
2.000	0.9099	-0.0808	-0.79
2.005	0.9099	-0.0807	-0.79
2.010	0.9099	-0.0809	-0.79
2.015	0.9097	-0.0810	-0.79
2.020	0.9099	-0.0811	-0.79
2.025	0.9100	-0.0812	-0.79
2.030	0.9099	-0.0812	-0.79
2.035	0.9101	-0.0814	-0.78
2.040	0.9101	-0.0814	-0.78
2.045	0.9099	-0.0816	-0.79
2.050	0.9101	-0.0816	-0.78
2.055	0.9102	-0.0816	-0.78
2.060	0.9101	-0.0819	-0.78
2.065	0.9102	-0.0818	-0.78
2.070	0.9100	-0.0821	-0.78
2.075	0.9100	-0.0821	-0.78
2.080	0.9101	-0.0822	-0.78
2.085	0.9101	-0.0824	-0.78
2.090	0.9101	-0.0824	-0.78
2.095	0.9100	-0.0825	-0.78
2.100	0.9100	-0.0828	-0.78
2.105	0.9102	-0.0829	-0.78
2.110	0.9100	-0.0828	-0.78
2.115	0.9101	-0.0830	-0.78
2.120	0.9099	-0.0831	-0.78
2.125	0.9100	-0.0833	-0.78
2.130	0.9100	-0.0835	-0.78
2.135	0.9100	-0.0835	-0.78
2.140	0.9099	-0.0837	-0.78
2.145	0.9098	-0.0839	-0.78
2.150	0.9098	-0.0840	-0.78
2.155	0.9099	-0.0842	-0.78
2.160	0.9097	-0.0843	-0.78
2.165	0.9097	-0.0844	-0.78
2.170	0.9096	-0.0846	-0.79
2.175	0.9097	-0.0847	-0.78
2.180	0.9096	-0.0849	-0.79
2.185	0.9096	-0.0850	-0.79
2.190	0.9094	-0.0852	-0.79
2.195	0.9093	-0.0855	-0.79
2.200	0.9092	-0.0856	-0.79

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.205	0.9092	-0.0857	-0.79
2.210	0.9093	-0.0859	-0.79
2.215	0.9091	-0.0860	-0.79
2.220	0.9090	-0.0862	-0.79
2.225	0.9090	-0.0864	-0.79
2.230	0.9089	-0.0866	-0.79
2.235	0.9089	-0.0869	-0.79
2.240	0.9087	-0.0869	-0.79
2.245	0.9086	-0.0872	-0.79
2.250	0.9084	-0.0874	-0.79
2.255	0.9085	-0.0874	-0.79
2.260	0.9084	-0.0877	-0.79
2.265	0.9083	-0.0879	-0.79
2.270	0.9081	-0.0880	-0.80
2.275	0.9080	-0.0882	-0.80
2.280	0.9079	-0.0884	-0.80
2.285	0.9078	-0.0886	-0.80
2.290	0.9077	-0.0888	-0.80
2.295	0.9077	-0.0888	-0.80
2.300	0.9073	-0.0889	-0.80
2.305	0.9075	-0.0894	-0.80
2.310	0.9071	-0.0893	-0.81
2.315	0.9072	-0.0895	-0.80
2.320	0.9070	-0.0898	-0.81
2.325	0.9069	-0.0901	-0.81
2.330	0.9070	-0.0903	-0.80
2.335	0.9068	-0.0904	-0.81
2.340	0.9066	-0.0904	-0.81
2.345	0.9066	-0.0908	-0.81
2.350	0.9065	-0.0910	-0.81
2.355	0.9062	-0.0912	-0.81
2.360	0.9062	-0.0913	-0.81
2.365	0.9062	-0.0914	-0.81
2.370	0.9059	-0.0916	-0.81
2.375	0.9057	-0.0916	-0.82
2.380	0.9058	-0.0920	-0.81
2.385	0.9055	-0.0921	-0.82
2.390	0.9056	-0.0923	-0.82
2.395	0.9054	-0.0926	-0.82
2.400	0.9053	-0.0929	-0.82
2.405	0.9051	-0.0929	-0.82
2.410	0.9050	-0.0931	-0.82
2.415	0.9050	-0.0933	-0.82
2.420	0.9050	-0.0935	-0.82
2.425	0.9047	-0.0936	-0.82
2.430	0.9047	-0.0936	-0.82

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.435	0.9046	-0.0938	-0.82
2.440	0.9044	-0.0940	-0.83
2.445	0.9044	-0.0942	-0.83
2.450	0.9044	-0.0943	-0.83
2.455	0.9042	-0.0945	-0.83
2.460	0.9042	-0.0946	-0.83
2.465	0.9042	-0.0947	-0.83
2.470	0.9040	-0.0948	-0.83
2.475	0.9040	-0.0949	-0.83
2.480	0.9039	-0.0951	-0.83
2.485	0.9039	-0.0952	-0.83
2.490	0.9037	-0.0953	-0.83
2.495	0.9037	-0.0955	-0.83
2.500	0.9036	-0.0954	-0.83
2.505	0.9035	-0.0957	-0.83
2.510	0.9034	-0.0958	-0.83
2.515	0.9034	-0.0959	-0.83
2.520	0.9033	-0.0961	-0.83
2.525	0.9031	-0.0962	-0.84
2.530	0.9032	-0.0964	-0.84
2.535	0.9031	-0.0964	-0.84
2.540	0.9030	-0.0965	-0.84
2.545	0.9029	-0.0967	-0.84
2.550	0.9028	-0.0968	-0.84
2.555	0.9028	-0.0968	-0.84
2.560	0.9028	-0.0970	-0.84
2.565	0.9027	-0.0970	-0.84
2.570	0.9027	-0.0970	-0.84
2.575	0.9027	-0.0971	-0.84
2.580	0.9026	-0.0971	-0.84
2.585	0.9028	-0.0972	-0.84
2.590	0.9028	-0.0972	-0.84
2.595	0.9029	-0.0975	-0.84
2.600	0.9029	-0.0976	-0.84
2.605	0.9029	-0.0976	-0.84
2.610	0.9029	-0.0977	-0.84
2.615	0.9030	-0.0979	-0.84
2.620	0.9031	-0.0981	-0.83
2.625	0.9030	-0.0982	-0.83
2.630	0.9031	-0.0984	-0.83
2.635	0.9032	-0.0981	-0.83
2.640	0.9031	-0.0985	-0.83
2.645	0.9031	-0.0985	-0.83
2.650	0.9032	-0.0987	-0.83
2.655	0.9030	-0.0987	-0.83
2.660	0.9032	-0.0990	-0.83

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.665	0.9033	-0.0989	-0.83
2.670	0.9031	-0.0988	-0.83
2.675	0.9032	-0.0993	-0.83
2.680	0.9032	-0.0993	-0.83
2.685	0.9032	-0.0995	-0.83
2.690	0.9032	-0.0995	-0.83
2.695	0.9032	-0.0997	-0.83
2.700	0.9033	-0.0999	-0.83
2.705	0.9032	-0.0999	-0.83
2.710	0.9035	-0.1001	-0.83
2.715	0.9035	-0.1002	-0.83
2.720	0.9035	-0.1003	-0.83
2.725	0.9034	-0.1003	-0.83
2.730	0.9034	-0.1004	-0.83
2.735	0.9035	-0.1005	-0.83
2.740	0.9034	-0.1005	-0.83
2.745	0.9037	-0.1008	-0.83
2.750	0.9036	-0.1007	-0.83
2.755	0.9036	-0.1008	-0.83
2.760	0.9037	-0.1008	-0.83
2.765	0.9035	-0.1010	-0.83
2.770	0.9037	-0.1010	-0.83
2.775	0.9034	-0.1011	-0.83
2.780	0.9035	-0.1011	-0.83
2.785	0.9035	-0.1013	-0.83
2.790	0.9035	-0.1012	-0.83
2.795	0.9035	-0.1015	-0.83
2.800	0.9035	-0.1015	-0.83
2.805	0.9035	-0.1017	-0.83
2.810	0.9033	-0.1018	-0.83
2.815	0.9034	-0.1020	-0.83
2.820	0.9033	-0.1020	-0.83
2.825	0.9032	-0.1020	-0.83
2.830	0.9032	-0.1021	-0.83
2.835	0.9033	-0.1022	-0.83
2.840	0.9032	-0.1024	-0.83
2.845	0.9032	-0.1027	-0.83
2.850	0.9031	-0.1026	-0.83
2.855	0.9031	-0.1027	-0.83
2.860	0.9031	-0.1030	-0.83
2.865	0.9030	-0.1031	-0.83
2.870	0.9030	-0.1033	-0.83
2.875	0.9030	-0.1033	-0.83
2.880	0.9030	-0.1034	-0.83
2.885	0.9031	-0.1036	-0.83
2.890	0.9032	-0.1036	-0.83

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.895	0.9028	-0.1036	-0.83
2.900	0.9028	-0.1037	-0.83
2.905	0.9029	-0.1038	-0.83
2.910	0.9027	-0.1039	-0.83
2.915	0.9026	-0.1041	-0.83
2.920	0.9025	-0.1041	-0.83
2.925	0.9023	-0.1042	-0.84
2.930	0.9021	-0.1042	-0.84
2.935	0.9022	-0.1045	-0.84
2.940	0.9020	-0.1045	-0.84
2.945	0.9019	-0.1049	-0.84
2.950	0.9019	-0.1051	-0.84
2.955	0.9020	-0.1049	-0.84
2.960	0.9018	-0.1057	-0.84
2.965	0.9016	-0.1053	-0.84
2.970	0.9014	-0.1056	-0.84
2.975	0.9016	-0.1058	-0.84
2.980	0.9015	-0.1060	-0.84
2.985	0.9015	-0.1060	-0.84
2.990	0.9017	-0.1063	-0.84
2.995	0.9016	-0.1063	-0.84
3.000	0.9016	-0.1066	-0.84
3.005	0.9012	-0.1064	-0.84
3.010	0.9012	-0.1069	-0.84
3.015	0.9011	-0.1068	-0.84
3.020	0.9013	-0.1071	-0.84
3.025	0.9014	-0.1073	-0.84
3.030	0.9014	-0.1071	-0.84
3.035	0.9018	-0.1073	-0.84
3.040	0.9017	-0.1073	-0.84
3.045	0.9019	-0.1075	-0.84
3.050	0.9015	-0.1076	-0.84
3.055	0.9022	-0.1075	-0.83
3.060	0.9019	-0.1074	-0.84
3.065	0.9020	-0.1072	-0.84
3.070	0.9020	-0.1072	-0.84
3.075	0.9019	-0.1071	-0.84
3.080	0.9019	-0.1071	-0.84
3.085	0.9016	-0.1072	-0.84
3.090	0.9017	-0.1070	-0.84
3.095	0.9016	-0.1070	-0.84
3.100	0.9014	-0.1070	-0.84
3.105	0.9011	-0.1072	-0.84
3.110	0.9012	-0.1073	-0.84
3.115	0.9010	-0.1073	-0.84
3.120	0.9010	-0.1074	-0.84

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.125	0.9009	-0.1076	-0.84
3.130	0.9008	-0.1077	-0.85
3.135	0.9006	-0.1081	-0.85
3.140	0.9007	-0.1081	-0.85
3.145	0.9006	-0.1084	-0.85
3.150	0.9006	-0.1085	-0.85
3.155	0.9007	-0.1088	-0.85
3.160	0.9007	-0.1088	-0.85
3.165	0.9010	-0.1090	-0.84
3.170	0.9009	-0.1090	-0.84
3.175	0.9009	-0.1092	-0.84
3.180	0.9012	-0.1095	-0.84
3.185	0.9014	-0.1098	-0.84
3.190	0.9015	-0.1093	-0.84
3.195	0.9015	-0.1096	-0.84
3.200	0.9015	-0.1097	-0.84
3.205	0.9019	-0.1097	-0.83
3.210	0.9018	-0.1095	-0.83
3.215	0.9018	-0.1094	-0.83
3.220	0.9019	-0.1090	-0.83
3.225	0.9019	-0.1095	-0.83
3.230	0.9015	-0.1094	-0.84
3.235	0.9016	-0.1096	-0.84
3.240	0.9014	-0.1098	-0.84
3.245	0.9015	-0.1097	-0.84
3.250	0.9017	-0.1103	-0.83
3.255	0.9014	-0.1103	-0.84
3.260	0.9016	-0.1104	-0.83
3.265	0.9018	-0.1106	-0.83
3.270	0.9014	-0.1106	-0.84
3.275	0.9016	-0.1108	-0.83
3.280	0.9018	-0.1109	-0.83
3.285	0.9017	-0.1116	-0.83
3.290	0.9015	-0.1111	-0.83
3.295	0.9016	-0.1115	-0.83
3.300	0.9019	-0.1115	-0.83
3.305	0.9021	-0.1116	-0.83
3.310	0.9021	-0.1117	-0.83
3.315	0.9020	-0.1120	-0.83
3.320	0.9023	-0.1120	-0.83
3.325	0.9026	-0.1121	-0.82
3.330	0.9023	-0.1125	-0.83
3.335	0.9023	-0.1126	-0.83
3.340	0.9025	-0.1127	-0.82
3.345	0.9024	-0.1128	-0.82
3.350	0.9024	-0.1130	-0.82

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.355	0.9026	-0.1129	-0.82
3.360	0.9026	-0.1131	-0.82
3.365	0.9028	-0.1131	-0.82
3.370	0.9028	-0.1135	-0.82
3.375	0.9029	-0.1131	-0.82
3.380	0.9029	-0.1132	-0.82
3.385	0.9029	-0.1135	-0.82
3.390	0.9029	-0.1135	-0.82
3.395	0.9029	-0.1136	-0.82
3.400	0.9029	-0.1138	-0.82
3.405	0.9028	-0.1137	-0.82
3.410	0.9029	-0.1137	-0.82
3.415	0.9028	-0.1139	-0.82
3.420	0.9027	-0.1138	-0.82
3.425	0.9027	-0.1138	-0.82
3.430	0.9027	-0.1139	-0.82
3.435	0.9027	-0.1140	-0.82
3.440	0.9024	-0.1139	-0.82
3.445	0.9023	-0.1143	-0.82
3.450	0.9020	-0.1147	-0.83
3.455	0.9021	-0.1145	-0.83
3.460	0.9020	-0.1148	-0.83
3.465	0.9016	-0.1149	-0.83
3.470	0.9016	-0.1151	-0.83
3.475	0.9014	-0.1152	-0.83
3.480	0.9012	-0.1153	-0.83
3.485	0.9009	-0.1154	-0.84
3.490	0.9014	-0.1162	-0.83
3.495	0.9013	-0.1166	-0.83
3.500	0.9011	-0.1166	-0.83
3.505	0.9014	-0.1165	-0.83
3.510	0.9010	-0.1167	-0.83
3.515	0.9015	-0.1171	-0.83
3.520	0.9012	-0.1172	-0.83
3.525	0.9016	-0.1181	-0.83
3.530	0.9021	-0.1180	-0.82
3.535	0.9020	-0.1180	-0.82
3.540	0.9022	-0.1179	-0.82
3.545	0.9027	-0.1177	-0.82
3.550	0.9023	-0.1175	-0.82
3.555	0.9026	-0.1175	-0.82
3.560	0.9026	-0.1169	-0.82
3.565	0.9025	-0.1167	-0.82
3.570	0.9020	-0.1167	-0.82
3.575	0.9020	-0.1163	-0.82
3.580	0.9016	-0.1155	-0.83

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.585	0.9014	-0.1153	-0.83
3.590	0.8994	-0.1153	-0.85
3.595	0.8989	-0.1157	-0.85
3.600	0.8986	-0.1162	-0.86
3.605	0.8982	-0.1166	-0.86
3.610	0.8981	-0.1171	-0.86
3.615	0.8979	-0.1183	-0.86
3.620	0.8977	-0.1189	-0.86
3.625	0.8980	-0.1191	-0.86
3.630	0.8979	-0.1198	-0.86
3.635	0.8985	-0.1198	-0.85
3.640	0.8989	-0.1196	-0.85
3.645	0.8989	-0.1197	-0.85
3.650	0.8993	-0.1197	-0.85
3.655	0.8992	-0.1199	-0.85
3.660	0.8994	-0.1197	-0.84
3.665	0.8994	-0.1197	-0.84
3.670	0.8993	-0.1196	-0.85
3.675	0.8995	-0.1195	-0.84
3.680	0.8993	-0.1194	-0.85
3.685	0.8993	-0.1193	-0.85
3.690	0.8993	-0.1194	-0.85
3.695	0.8993	-0.1194	-0.85
3.700	0.8992	-0.1190	-0.85
3.705	0.8990	-0.1189	-0.85
3.710	0.8987	-0.1190	-0.85
3.715	0.8984	-0.1187	-0.86
3.720	0.8980	-0.1185	-0.86
3.725	0.8977	-0.1188	-0.86
3.730	0.8970	-0.1189	-0.87
3.735	0.8964	-0.1194	-0.87
3.740	0.8960	-0.1198	-0.88
3.745	0.8958	-0.1207	-0.88
3.750	0.8958	-0.1217	-0.88
3.755	0.8963	-0.1227	-0.87
3.760	0.8970	-0.1235	-0.86
3.765	0.8980	-0.1239	-0.85
3.770	0.8995	-0.1235	-0.84
3.775	0.9001	-0.1229	-0.83
3.780	0.9011	-0.1229	-0.82
3.785	0.9007	-0.1210	-0.83
3.790	0.8996	-0.1202	-0.84
3.795	0.8994	-0.1202	-0.84
3.800	0.8993	-0.1202	-0.85
3.805	0.8984	-0.1200	-0.85
3.810	0.8980	-0.1203	-0.86

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.815	0.8973	-0.1207	-0.86
3.820	0.8974	-0.1206	-0.86
3.825	0.8970	-0.1210	-0.87
3.830	0.8970	-0.1214	-0.87
3.835	0.8927	-0.1242	-0.90
3.840	0.8971	-0.1227	-0.86
3.845	0.8973	-0.1228	-0.86
3.850	0.8970	-0.1230	-0.86
3.855	0.8973	-0.1231	-0.86
3.860	0.8972	-0.1233	-0.86
3.865	0.8971	-0.1236	-0.86
3.870	0.8974	-0.1236	-0.86
3.875	0.8972	-0.1239	-0.86
3.880	0.8973	-0.1238	-0.86
3.885	0.8973	-0.1242	-0.86
3.890	0.8974	-0.1244	-0.86
3.895	0.8973	-0.1244	-0.86
3.900	0.8973	-0.1245	-0.86
3.905	0.8972	-0.1249	-0.86
3.910	0.8972	-0.1247	-0.86
3.915	0.8972	-0.1252	-0.86
3.920	0.8973	-0.1254	-0.86
3.925	0.8972	-0.1256	-0.86
3.930	0.8972	-0.1258	-0.86
3.935	0.8971	-0.1258	-0.86
3.940	0.8971	-0.1263	-0.86
3.945	0.8968	-0.1265	-0.86
3.950	0.8968	-0.1267	-0.86
3.955	0.8969	-0.1269	-0.86
3.960	0.8968	-0.1271	-0.86
3.965	0.8967	-0.1273	-0.86
3.970	0.8967	-0.1275	-0.86
3.975	0.8968	-0.1279	-0.86
3.980	0.8966	-0.1280	-0.86
3.985	0.8966	-0.1282	-0.86
3.990	0.8964	-0.1284	-0.86
3.995	0.8963	-0.1288	-0.86
4.000	0.8962	-0.1290	-0.86
4.005	0.8962	-0.1293	-0.86
4.010	0.8961	-0.1297	-0.86
4.015	0.8961	-0.1298	-0.86
4.020	0.8958	-0.1300	-0.87
4.025	0.8956	-0.1301	-0.87
4.030	0.8957	-0.1306	-0.87
4.035	0.8956	-0.1309	-0.87
4.040	0.8954	-0.1310	-0.87

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.045	0.8950	-0.1311	-0.87
4.050	0.8950	-0.1317	-0.87
4.055	0.8947	-0.1319	-0.87
4.060	0.8947	-0.1323	-0.87
4.065	0.8942	-0.1324	-0.88
4.070	0.8944	-0.1326	-0.87
4.075	0.8942	-0.1330	-0.88
4.080	0.8943	-0.1334	-0.88
4.085	0.8939	-0.1334	-0.88
4.090	0.8937	-0.1338	-0.88
4.095	0.8938	-0.1342	-0.88
4.100	0.8936	-0.1343	-0.88
4.105	0.8931	-0.1346	-0.88
4.110	0.8930	-0.1350	-0.88
4.115	0.8930	-0.1353	-0.88
4.120	0.8926	-0.1355	-0.89
4.125	0.8928	-0.1358	-0.89
4.130	0.8925	-0.1362	-0.89
4.135	0.8921	-0.1366	-0.89
4.140	0.8917	-0.1365	-0.89
4.145	0.8917	-0.1369	-0.89
4.150	0.8914	-0.1375	-0.90
4.155	0.8910	-0.1378	-0.90
4.160	0.8907	-0.1378	-0.90
4.165	0.8904	-0.1383	-0.91
4.170	0.8900	-0.1384	-0.91
4.175	0.8894	-0.1382	-0.91
4.180	0.8897	-0.1385	-0.91
4.185	0.8891	-0.1387	-0.92
4.190	0.8890	-0.1390	-0.92
4.195	0.8889	-0.1392	-0.92
4.200	0.8888	-0.1390	-0.92
4.205	0.8888	-0.1395	-0.92
4.210	0.8887	-0.1399	-0.92
4.215	0.8886	-0.1394	-0.92
4.220	0.8885	-0.1399	-0.92
4.225	0.8884	-0.1395	-0.92
4.230	0.8881	-0.1401	-0.92
4.235	0.8881	-0.1402	-0.92
4.240	0.8884	-0.1405	-0.92
4.245	0.8882	-0.1406	-0.92
4.250	0.8880	-0.1407	-0.92
4.255	0.8881	-0.1409	-0.92
4.260	0.8878	-0.1411	-0.93
4.265	0.8879	-0.1409	-0.92
4.270	0.8877	-0.1413	-0.93

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.275	0.8877	-0.1414	-0.93
4.280	0.8877	-0.1415	-0.93
4.285	0.8876	-0.1417	-0.93
4.290	0.8876	-0.1418	-0.93
4.295	0.8875	-0.1418	-0.93
4.300	0.8875	-0.1420	-0.93
4.305	0.8875	-0.1421	-0.93
4.310	0.8874	-0.1422	-0.93
4.315	0.8873	-0.1423	-0.93
4.320	0.8873	-0.1423	-0.93
4.325	0.8872	-0.1424	-0.93
4.330	0.8872	-0.1425	-0.93
4.335	0.8871	-0.1426	-0.93
4.340	0.8870	-0.1427	-0.93
4.345	0.8869	-0.1430	-0.93
4.350	0.8868	-0.1431	-0.93
4.355	0.8868	-0.1432	-0.93
4.360	0.8868	-0.1434	-0.93
4.365	0.8868	-0.1437	-0.93
4.370	0.8868	-0.1436	-0.93
4.375	0.8871	-0.1440	-0.93
4.380	0.8871	-0.1441	-0.93
4.385	0.8869	-0.1445	-0.93
4.390	0.8867	-0.1444	-0.93
4.395	0.8870	-0.1447	-0.93
4.400	0.8872	-0.1448	-0.93
4.405	0.8874	-0.1453	-0.92
4.410	0.8874	-0.1451	-0.92
4.415	0.8873	-0.1450	-0.92
4.420	0.8877	-0.1450	-0.92
4.425	0.8878	-0.1449	-0.92
4.430	0.8875	-0.1451	-0.92
4.435	0.8874	-0.1454	-0.92
4.440	0.8874	-0.1454	-0.92
4.445	0.8873	-0.1455	-0.92
4.450	0.8868	-0.1454	-0.93
4.455	0.8870	-0.1457	-0.93
4.460	0.8866	-0.1456	-0.93
4.465	0.8866	-0.1453	-0.93
4.470	0.8868	-0.1454	-0.93
4.475	0.8868	-0.1450	-0.93
4.480	0.8870	-0.1447	-0.93
4.485	0.8868	-0.1451	-0.93
4.490	0.8874	-0.1452	-0.92
4.495	0.8876	-0.1454	-0.92
4.500	0.8878	-0.1456	-0.92

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.505	0.8880	-0.1456	-0.92
4.510	0.8884	-0.1461	-0.91
4.515	0.8887	-0.1461	-0.91
4.520	0.8888	-0.1463	-0.91
4.525	0.8891	-0.1461	-0.90
4.530	0.8891	-0.1464	-0.91
4.535	0.8895	-0.1465	-0.90
4.540	0.8893	-0.1462	-0.90
4.545	0.8893	-0.1464	-0.90
4.550	0.8893	-0.1463	-0.90
4.555	0.8895	-0.1465	-0.90
4.560	0.8896	-0.1465	-0.90
4.565	0.8898	-0.1465	-0.90
4.570	0.8897	-0.1467	-0.90
4.575	0.8898	-0.1466	-0.90
4.580	0.8899	-0.1466	-0.90
4.585	0.8899	-0.1468	-0.90
4.590	0.8899	-0.1468	-0.90
4.595	0.8900	-0.1470	-0.90
4.600	0.8900	-0.1469	-0.90
4.605	0.8899	-0.1472	-0.90
4.610	0.8899	-0.1471	-0.90
4.615	0.8898	-0.1474	-0.90
4.620	0.8897	-0.1473	-0.90
4.625	0.8898	-0.1477	-0.90
4.630	0.8897	-0.1476	-0.90
4.635	0.8895	-0.1478	-0.90
4.640	0.8896	-0.1480	-0.90
4.645	0.8894	-0.1478	-0.90
4.650	0.8896	-0.1481	-0.90
4.655	0.8893	-0.1481	-0.90
4.660	0.8894	-0.1482	-0.90
4.665	0.8894	-0.1484	-0.90
4.670	0.8894	-0.1487	-0.90
4.675	0.8892	-0.1488	-0.90
4.680	0.8892	-0.1486	-0.90
4.685	0.8892	-0.1487	-0.90
4.690	0.8895	-0.1487	-0.90
4.695	0.8891	-0.1488	-0.90
4.700	0.8894	-0.1490	-0.90
4.705	0.8888	-0.1496	-0.90
4.710	0.8892	-0.1495	-0.90
4.715	0.8892	-0.1494	-0.90
4.720	0.8890	-0.1496	-0.90
4.725	0.8890	-0.1493	-0.90
4.730	0.8890	-0.1493	-0.90

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.735	0.8893	-0.1492	-0.90
4.740	0.8891	-0.1493	-0.90
4.745	0.8887	-0.1495	-0.90
4.750	0.8888	-0.1494	-0.90
4.755	0.8888	-0.1497	-0.90
4.760	0.8884	-0.1500	-0.91
4.765	0.8887	-0.1497	-0.90
4.770	0.8882	-0.1501	-0.91
4.775	0.8878	-0.1507	-0.91
4.780	0.8876	-0.1506	-0.91
4.785	0.8882	-0.1509	-0.91
4.790	0.8874	-0.1508	-0.91
4.795	0.8874	-0.1511	-0.91
4.800	0.8874	-0.1512	-0.91
4.805	0.8868	-0.1510	-0.92
4.810	0.8875	-0.1513	-0.91
4.815	0.8870	-0.1509	-0.92
4.820	0.8867	-0.1511	-0.92
4.825	0.8870	-0.1508	-0.92
4.830	0.8866	-0.1512	-0.92
4.835	0.8870	-0.1510	-0.92
4.840	0.8869	-0.1508	-0.92
4.845	0.8867	-0.1507	-0.92
4.850	0.8867	-0.1508	-0.92
4.855	0.8867	-0.1503	-0.92
4.860	0.8866	-0.1505	-0.92
4.865	0.8868	-0.1499	-0.92
4.870	0.8865	-0.1503	-0.92
4.875	0.8867	-0.1503	-0.92
4.880	0.8868	-0.1504	-0.92
4.885	0.8865	-0.1504	-0.92
4.890	0.8867	-0.1505	-0.92
4.895	0.8866	-0.1504	-0.92
4.900	0.8868	-0.1505	-0.92
4.905	0.8868	-0.1505	-0.92
4.910	0.8868	-0.1507	-0.92
4.915	0.8869	-0.1508	-0.92
4.920	0.8870	-0.1509	-0.92
4.925	0.8870	-0.1510	-0.92
4.930	0.8870	-0.1512	-0.92
4.935	0.8870	-0.1513	-0.92
4.940	0.8872	-0.1514	-0.92
4.945	0.8871	-0.1516	-0.92
4.950	0.8872	-0.1517	-0.91
4.955	0.8874	-0.1518	-0.91
4.960	0.8873	-0.1520	-0.91

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.965	0.8873	-0.1520	-0.91
4.970	0.8875	-0.1522	-0.91
4.975	0.8876	-0.1522	-0.91
4.980	0.8876	-0.1526	-0.91
4.985	0.8874	-0.1528	-0.91
4.990	0.8877	-0.1529	-0.91
4.995	0.8878	-0.1533	-0.91
5.000	0.8879	-0.1534	-0.90
5.005	0.8879	-0.1537	-0.90
5.010	0.8878	-0.1540	-0.91
5.015	0.8879	-0.1542	-0.90
5.020	0.8877	-0.1543	-0.91
5.025	0.8877	-0.1549	-0.90
5.030	0.8876	-0.1551	-0.91
5.035	0.8876	-0.1549	-0.90
5.040	0.8875	-0.1552	-0.91
5.045	0.8873	-0.1551	-0.91
5.050	0.8871	-0.1551	-0.91
5.055	0.8872	-0.1547	-0.91
5.060	0.8874	-0.1548	-0.91
5.065	0.8873	-0.1544	-0.91
5.070	0.8875	-0.1545	-0.91
5.075	0.8876	-0.1545	-0.91
5.080	0.8876	-0.1545	-0.91
5.085	0.8876	-0.1543	-0.91
5.090	0.8881	-0.1543	-0.90
5.095	0.8882	-0.1543	-0.90
5.100	0.8883	-0.1542	-0.90
5.105	0.8887	-0.1542	-0.90
5.110	0.8886	-0.1544	-0.90
5.115	0.8884	-0.1545	-0.90
5.120	0.8888	-0.1545	-0.89
5.125	0.8887	-0.1546	-0.90
5.130	0.8889	-0.1546	-0.89
5.135	0.8889	-0.1546	-0.89
5.140	0.8892	-0.1547	-0.89
5.145	0.8891	-0.1550	-0.89
5.150	0.8894	-0.1550	-0.89
5.155	0.8893	-0.1552	-0.89
5.160	0.8893	-0.1553	-0.89
5.165	0.8894	-0.1554	-0.89
5.170	0.8893	-0.1555	-0.89
5.175	0.8893	-0.1557	-0.89
5.180	0.8893	-0.1556	-0.89
5.185	0.8894	-0.1558	-0.89
5.190	0.8895	-0.1558	-0.89

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.195	0.8893	-0.1559	-0.89
5.200	0.8894	-0.1559	-0.89
5.205	0.8894	-0.1560	-0.89
5.210	0.8894	-0.1562	-0.89
5.215	0.8895	-0.1563	-0.89
5.220	0.8894	-0.1564	-0.89
5.225	0.8894	-0.1566	-0.89
5.230	0.8893	-0.1566	-0.89
5.235	0.8894	-0.1569	-0.88
5.240	0.8892	-0.1571	-0.89
5.245	0.8893	-0.1570	-0.89
5.250	0.8893	-0.1573	-0.89
5.255	0.8892	-0.1575	-0.89
5.260	0.8889	-0.1574	-0.89
5.265	0.8890	-0.1574	-0.89
5.270	0.8888	-0.1577	-0.89
5.275	0.8885	-0.1579	-0.89
5.280	0.8885	-0.1580	-0.89
5.285	0.8882	-0.1580	-0.89
5.290	0.8881	-0.1585	-0.89
5.295	0.8878	-0.1583	-0.90
5.300	0.8877	-0.1587	-0.90
5.305	0.8875	-0.1588	-0.90
5.310	0.8874	-0.1593	-0.90
5.315	0.8873	-0.1593	-0.90
5.320	0.8868	-0.1596	-0.90
5.325	0.8870	-0.1595	-0.90
5.330	0.8868	-0.1598	-0.90
5.335	0.8868	-0.1602	-0.90
5.340	0.8868	-0.1603	-0.90
5.345	0.8872	-0.1606	-0.90
5.350	0.8866	-0.1604	-0.91
5.355	0.8863	-0.1606	-0.91
5.360	0.8859	-0.1610	-0.91
5.365	0.8858	-0.1607	-0.91
5.370	0.8857	-0.1612	-0.91
5.375	0.8855	-0.1609	-0.91
5.380	0.8855	-0.1612	-0.91
5.385	0.8853	-0.1615	-0.92
5.390	0.8851	-0.1614	-0.92
5.395	0.8847	-0.1616	-0.92
5.400	0.8846	-0.1617	-0.92
5.405	0.8844	-0.1623	-0.92
5.410	0.8842	-0.1622	-0.93
5.415	0.8834	-0.1628	-0.93
5.420	0.8838	-0.1628	-0.93

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.425	0.8834	-0.1633	-0.93
5.430	0.8833	-0.1630	-0.93
5.435	0.8830	-0.1633	-0.93
5.440	0.8831	-0.1636	-0.93
5.445	0.8827	-0.1635	-0.94
5.450	0.8827	-0.1640	-0.94
5.455	0.8828	-0.1641	-0.94
5.460	0.8824	-0.1642	-0.94
5.465	0.8823	-0.1641	-0.94
5.470	0.8822	-0.1647	-0.94
5.475	0.8818	-0.1646	-0.94
5.480	0.8819	-0.1648	-0.94
5.485	0.8816	-0.1649	-0.94
5.490	0.8815	-0.1652	-0.95
5.495	0.8812	-0.1652	-0.95
5.500	0.8811	-0.1653	-0.95
5.505	0.8809	-0.1654	-0.95
5.510	0.8806	-0.1656	-0.95
5.515	0.8804	-0.1657	-0.96
5.520	0.8801	-0.1660	-0.96
5.525	0.8801	-0.1659	-0.96
5.530	0.8800	-0.1662	-0.96
5.535	0.8798	-0.1663	-0.96
5.540	0.8798	-0.1665	-0.96
5.545	0.8795	-0.1668	-0.96
5.550	0.8794	-0.1668	-0.96
5.555	0.8793	-0.1669	-0.96
5.560	0.8792	-0.1671	-0.96
5.565	0.8792	-0.1671	-0.96
5.570	0.8792	-0.1672	-0.96
5.575	0.8791	-0.1673	-0.97
5.580	0.8791	-0.1674	-0.96
5.585	0.8789	-0.1676	-0.97
5.590	0.8787	-0.1676	-0.97
5.595	0.8785	-0.1679	-0.97
5.600	0.8784	-0.1679	-0.97
5.605	0.8782	-0.1681	-0.97
5.610	0.8780	-0.1685	-0.97
5.615	0.8776	-0.1687	-0.98
5.620	0.8776	-0.1691	-0.98
5.625	0.8778	-0.1693	-0.97
5.630	0.8775	-0.1697	-0.98
5.635	0.8775	-0.1700	-0.98
5.640	0.8773	-0.1702	-0.98
5.645	0.8774	-0.1706	-0.98
5.650	0.8777	-0.1709	-0.97

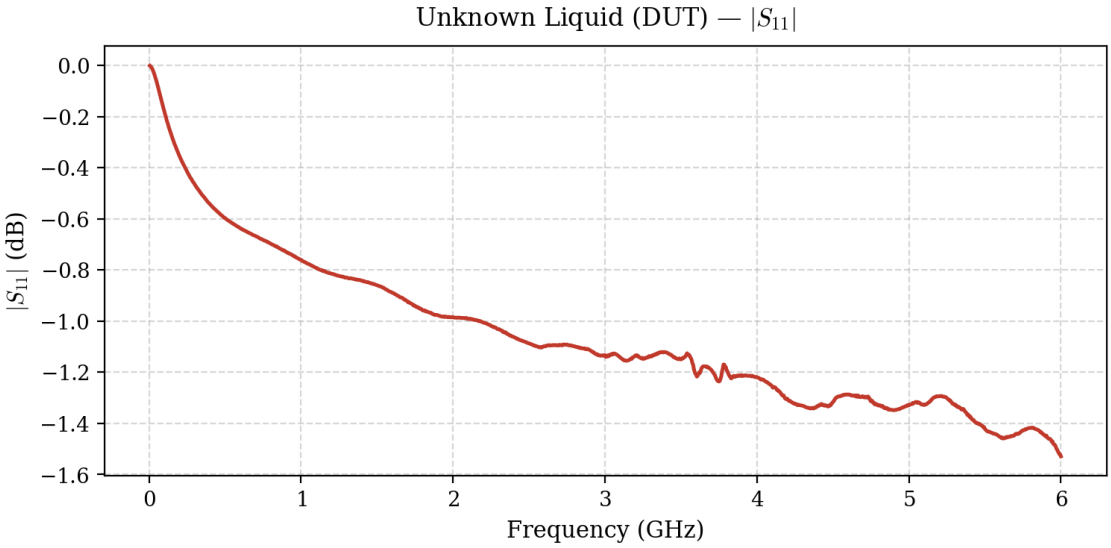
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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.655	0.8773	-0.1712	-0.97
5.660	0.8776	-0.1713	-0.97
5.665	0.8776	-0.1712	-0.97
5.670	0.8775	-0.1721	-0.97
5.675	0.8773	-0.1720	-0.97
5.680	0.8775	-0.1724	-0.97
5.685	0.8772	-0.1726	-0.97
5.690	0.8773	-0.1732	-0.97
5.695	0.8777	-0.1734	-0.97
5.700	0.8772	-0.1737	-0.97
5.705	0.8770	-0.1741	-0.97
5.710	0.8773	-0.1740	-0.97
5.715	0.8771	-0.1742	-0.97
5.720	0.8773	-0.1747	-0.97
5.725	0.8772	-0.1752	-0.97
5.730	0.8772	-0.1754	-0.97
5.735	0.8774	-0.1754	-0.97
5.740	0.8777	-0.1759	-0.96
5.745	0.8780	-0.1759	-0.96
5.750	0.8779	-0.1765	-0.96
5.755	0.8778	-0.1764	-0.96
5.760	0.8777	-0.1768	-0.96
5.765	0.8779	-0.1768	-0.96
5.770	0.8776	-0.1770	-0.96
5.775	0.8779	-0.1771	-0.96
5.780	0.8779	-0.1776	-0.96
5.785	0.8781	-0.1779	-0.95
5.790	0.8781	-0.1781	-0.95
5.795	0.8781	-0.1781	-0.95
5.800	0.8781	-0.1786	-0.95
5.805	0.8780	-0.1785	-0.95
5.810	0.8781	-0.1788	-0.95
5.815	0.8781	-0.1789	-0.95
5.820	0.8780	-0.1790	-0.95
5.825	0.8779	-0.1793	-0.95
5.830	0.8779	-0.1795	-0.95
5.835	0.8778	-0.1795	-0.95
5.840	0.8777	-0.1797	-0.95
5.845	0.8777	-0.1801	-0.95
5.850	0.8775	-0.1800	-0.96
5.855	0.8774	-0.1802	-0.96
5.860	0.8772	-0.1805	-0.96
5.865	0.8772	-0.1805	-0.96
5.870	0.8768	-0.1807	-0.96
5.875	0.8767	-0.1809	-0.96
5.880	0.8766	-0.1811	-0.96

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Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
5.885	0.8762	-0.1812	-0.97
5.890	0.8764	-0.1817	-0.96
5.895	0.8760	-0.1818	-0.97
5.900	0.8758	-0.1818	-0.97
5.905	0.8757	-0.1820	-0.97
5.910	0.8754	-0.1823	-0.97
5.915	0.8753	-0.1825	-0.97
5.920	0.8751	-0.1828	-0.97
5.925	0.8747	-0.1829	-0.98
5.930	0.8749	-0.1833	-0.97
5.935	0.8744	-0.1833	-0.98
5.940	0.8744	-0.1835	-0.98
5.945	0.8743	-0.1831	-0.98
5.950	0.8739	-0.1838	-0.98
5.955	0.8737	-0.1835	-0.99
5.960	0.8734	-0.1838	-0.99
5.965	0.8735	-0.1840	-0.99
5.970	0.8729	-0.1842	-0.99
5.975	0.8716	-0.1840	-1.00
5.980	0.8722	-0.1839	-1.00
5.985	0.8715	-0.1848	-1.00
5.990	0.8713	-0.1851	-1.01
5.995	0.8709	-0.1854	-1.01
6.000	0.8707	-0.1854	-1.01

3.5 Unknown Liquid (DUT)



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.005	0.9999	-0.0028	-0.00
0.010	0.9996	-0.0055	-0.00
0.015	0.9992	-0.0083	-0.01
0.020	0.9985	-0.0109	-0.01
0.025	0.9978	-0.0134	-0.02
0.030	0.9969	-0.0158	-0.03
0.035	0.9958	-0.0182	-0.03
0.040	0.9947	-0.0205	-0.04
0.045	0.9936	-0.0225	-0.05
0.050	0.9924	-0.0245	-0.06
0.055	0.9910	-0.0263	-0.08
0.060	0.9897	-0.0282	-0.09
0.065	0.9883	-0.0296	-0.10
0.070	0.9870	-0.0312	-0.11
0.075	0.9856	-0.0325	-0.12
0.080	0.9842	-0.0339	-0.13
0.085	0.9829	-0.0352	-0.14
0.090	0.9814	-0.0363	-0.16
0.095	0.9802	-0.0374	-0.17
0.100	0.9790	-0.0383	-0.18
0.105	0.9776	-0.0392	-0.19
0.110	0.9764	-0.0401	-0.20
0.115	0.9750	-0.0409	-0.21
0.120	0.9740	-0.0417	-0.22
0.125	0.9729	-0.0424	-0.23
0.130	0.9716	-0.0431	-0.24
0.135	0.9706	-0.0437	-0.25
0.140	0.9694	-0.0443	-0.26
0.145	0.9684	-0.0449	-0.27
0.150	0.9674	-0.0455	-0.28
0.155	0.9665	-0.0460	-0.29
0.160	0.9656	-0.0463	-0.29
0.165	0.9646	-0.0469	-0.30
0.170	0.9637	-0.0473	-0.31
0.175	0.9630	-0.0478	-0.32
0.180	0.9620	-0.0482	-0.33
0.185	0.9612	-0.0486	-0.33
0.190	0.9604	-0.0490	-0.34
0.195	0.9595	-0.0494	-0.35
0.200	0.9588	-0.0498	-0.35
0.205	0.9580	-0.0502	-0.36
0.210	0.9573	-0.0506	-0.37
0.215	0.9565	-0.0509	-0.37
0.220	0.9558	-0.0514	-0.38
0.225	0.9552	-0.0516	-0.39
0.230	0.9545	-0.0518	-0.39

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
0.235	0.9538	-0.0523	-0.40
0.240	0.9532	-0.0526	-0.40
0.245	0.9526	-0.0530	-0.41
0.250	0.9518	-0.0533	-0.42
0.255	0.9512	-0.0537	-0.42
0.260	0.9506	-0.0539	-0.43
0.265	0.9501	-0.0542	-0.43
0.270	0.9493	-0.0545	-0.44
0.275	0.9489	-0.0547	-0.44
0.280	0.9484	-0.0549	-0.45
0.285	0.9477	-0.0554	-0.45
0.290	0.9472	-0.0557	-0.46
0.295	0.9468	-0.0559	-0.46
0.300	0.9462	-0.0563	-0.46
0.305	0.9457	-0.0565	-0.47
0.310	0.9453	-0.0568	-0.47
0.315	0.9446	-0.0571	-0.48
0.320	0.9442	-0.0573	-0.48
0.325	0.9437	-0.0577	-0.49
0.330	0.9434	-0.0579	-0.49
0.335	0.9428	-0.0582	-0.49
0.340	0.9424	-0.0584	-0.50
0.345	0.9419	-0.0587	-0.50
0.350	0.9415	-0.0589	-0.51
0.355	0.9410	-0.0592	-0.51
0.360	0.9407	-0.0594	-0.51
0.365	0.9403	-0.0597	-0.52
0.370	0.9398	-0.0600	-0.52
0.375	0.9395	-0.0601	-0.52
0.380	0.9389	-0.0606	-0.53
0.385	0.9386	-0.0608	-0.53
0.390	0.9381	-0.0611	-0.54
0.395	0.9379	-0.0614	-0.54
0.400	0.9375	-0.0615	-0.54
0.405	0.9371	-0.0617	-0.55
0.410	0.9368	-0.0620	-0.55
0.415	0.9364	-0.0623	-0.55
0.420	0.9360	-0.0624	-0.56
0.425	0.9357	-0.0628	-0.56
0.430	0.9354	-0.0630	-0.56
0.435	0.9350	-0.0633	-0.56
0.440	0.9348	-0.0635	-0.57
0.445	0.9344	-0.0636	-0.57
0.450	0.9341	-0.0640	-0.57
0.455	0.9338	-0.0642	-0.57
0.460	0.9335	-0.0645	-0.58

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.465	0.9332	-0.0647	-0.58
0.470	0.9328	-0.0648	-0.58
0.475	0.9325	-0.0651	-0.59
0.480	0.9323	-0.0654	-0.59
0.485	0.9320	-0.0656	-0.59
0.490	0.9317	-0.0659	-0.59
0.495	0.9314	-0.0662	-0.60
0.500	0.9312	-0.0663	-0.60
0.505	0.9309	-0.0666	-0.60
0.510	0.9306	-0.0668	-0.60
0.515	0.9304	-0.0671	-0.60
0.520	0.9301	-0.0674	-0.61
0.525	0.9299	-0.0675	-0.61
0.530	0.9297	-0.0677	-0.61
0.535	0.9295	-0.0681	-0.61
0.540	0.9293	-0.0683	-0.61
0.545	0.9290	-0.0685	-0.62
0.550	0.9288	-0.0688	-0.62
0.555	0.9285	-0.0690	-0.62
0.560	0.9284	-0.0691	-0.62
0.565	0.9281	-0.0694	-0.62
0.570	0.9279	-0.0696	-0.63
0.575	0.9277	-0.0699	-0.63
0.580	0.9274	-0.0701	-0.63
0.585	0.9273	-0.0704	-0.63
0.590	0.9271	-0.0705	-0.63
0.595	0.9269	-0.0708	-0.63
0.600	0.9266	-0.0710	-0.64
0.605	0.9264	-0.0713	-0.64
0.610	0.9263	-0.0715	-0.64
0.615	0.9259	-0.0717	-0.64
0.620	0.9259	-0.0720	-0.64
0.625	0.9256	-0.0723	-0.64
0.630	0.9255	-0.0725	-0.65
0.635	0.9252	-0.0728	-0.65
0.640	0.9251	-0.0729	-0.65
0.645	0.9250	-0.0733	-0.65
0.650	0.9247	-0.0734	-0.65
0.655	0.9245	-0.0737	-0.65
0.660	0.9243	-0.0739	-0.66
0.665	0.9242	-0.0741	-0.66
0.670	0.9240	-0.0744	-0.66
0.675	0.9238	-0.0747	-0.66
0.680	0.9236	-0.0748	-0.66
0.685	0.9234	-0.0751	-0.66
0.690	0.9233	-0.0753	-0.66

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.695	0.9232	-0.0755	-0.67
0.700	0.9230	-0.0757	-0.67
0.705	0.9228	-0.0759	-0.67
0.710	0.9227	-0.0762	-0.67
0.715	0.9224	-0.0764	-0.67
0.720	0.9223	-0.0767	-0.67
0.725	0.9220	-0.0770	-0.67
0.730	0.9217	-0.0771	-0.68
0.735	0.9216	-0.0774	-0.68
0.740	0.9216	-0.0777	-0.68
0.745	0.9214	-0.0778	-0.68
0.750	0.9211	-0.0781	-0.68
0.755	0.9210	-0.0784	-0.68
0.760	0.9208	-0.0786	-0.69
0.765	0.9207	-0.0788	-0.69
0.770	0.9205	-0.0791	-0.69
0.775	0.9203	-0.0793	-0.69
0.780	0.9202	-0.0794	-0.69
0.785	0.9199	-0.0797	-0.69
0.790	0.9198	-0.0801	-0.69
0.795	0.9196	-0.0802	-0.70
0.800	0.9194	-0.0805	-0.70
0.805	0.9192	-0.0808	-0.70
0.810	0.9189	-0.0810	-0.70
0.815	0.9188	-0.0812	-0.70
0.820	0.9187	-0.0813	-0.70
0.825	0.9184	-0.0817	-0.71
0.830	0.9183	-0.0819	-0.71
0.835	0.9180	-0.0821	-0.71
0.840	0.9181	-0.0823	-0.71
0.845	0.9177	-0.0826	-0.71
0.850	0.9176	-0.0828	-0.71
0.855	0.9173	-0.0830	-0.71
0.860	0.9171	-0.0833	-0.72
0.865	0.9169	-0.0836	-0.72
0.870	0.9167	-0.0839	-0.72
0.875	0.9164	-0.0841	-0.72
0.880	0.9163	-0.0844	-0.72
0.885	0.9161	-0.0846	-0.72
0.890	0.9161	-0.0849	-0.72
0.895	0.9158	-0.0851	-0.73
0.900	0.9156	-0.0854	-0.73
0.905	0.9154	-0.0856	-0.73
0.910	0.9151	-0.0859	-0.73
0.915	0.9151	-0.0861	-0.73
0.920	0.9147	-0.0864	-0.74

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.925	0.9146	-0.0867	-0.74
0.930	0.9145	-0.0869	-0.74
0.935	0.9143	-0.0873	-0.74
0.940	0.9141	-0.0874	-0.74
0.945	0.9138	-0.0876	-0.74
0.950	0.9137	-0.0880	-0.74
0.955	0.9133	-0.0881	-0.75
0.960	0.9133	-0.0885	-0.75
0.965	0.9131	-0.0887	-0.75
0.970	0.9128	-0.0889	-0.75
0.975	0.9126	-0.0892	-0.75
0.980	0.9124	-0.0894	-0.76
0.985	0.9122	-0.0897	-0.76
0.990	0.9121	-0.0900	-0.76
0.995	0.9117	-0.0902	-0.76
1.000	0.9117	-0.0905	-0.76
1.005	0.9115	-0.0908	-0.76
1.010	0.9112	-0.0910	-0.76
1.015	0.9110	-0.0911	-0.77
1.020	0.9108	-0.0916	-0.77
1.025	0.9107	-0.0918	-0.77
1.030	0.9104	-0.0920	-0.77
1.035	0.9102	-0.0923	-0.77
1.040	0.9100	-0.0925	-0.77
1.045	0.9098	-0.0929	-0.78
1.050	0.9097	-0.0929	-0.78
1.055	0.9096	-0.0932	-0.78
1.060	0.9093	-0.0936	-0.78
1.065	0.9091	-0.0936	-0.78
1.070	0.9090	-0.0940	-0.78
1.075	0.9087	-0.0943	-0.78
1.080	0.9086	-0.0945	-0.79
1.085	0.9084	-0.0947	-0.79
1.090	0.9082	-0.0950	-0.79
1.095	0.9080	-0.0952	-0.79
1.100	0.9078	-0.0954	-0.79
1.105	0.9078	-0.0957	-0.79
1.110	0.9074	-0.0960	-0.80
1.115	0.9073	-0.0963	-0.80
1.120	0.9071	-0.0966	-0.80
1.125	0.9069	-0.0967	-0.80
1.130	0.9068	-0.0969	-0.80
1.135	0.9067	-0.0971	-0.80
1.140	0.9064	-0.0974	-0.80
1.145	0.9065	-0.0976	-0.80
1.150	0.9062	-0.0978	-0.80

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.155	0.9062	-0.0982	-0.80
1.160	0.9060	-0.0983	-0.81
1.165	0.9058	-0.0986	-0.81
1.170	0.9056	-0.0988	-0.81
1.175	0.9055	-0.0990	-0.81
1.180	0.9054	-0.0992	-0.81
1.185	0.9054	-0.0996	-0.81
1.190	0.9052	-0.0997	-0.81
1.195	0.9051	-0.0999	-0.81
1.200	0.9050	-0.1002	-0.81
1.205	0.9048	-0.1002	-0.82
1.210	0.9048	-0.1005	-0.82
1.215	0.9047	-0.1008	-0.82
1.220	0.9045	-0.1009	-0.82
1.225	0.9044	-0.1012	-0.82
1.230	0.9042	-0.1013	-0.82
1.235	0.9041	-0.1016	-0.82
1.240	0.9040	-0.1018	-0.82
1.245	0.9039	-0.1020	-0.82
1.250	0.9038	-0.1022	-0.82
1.255	0.9037	-0.1024	-0.82
1.260	0.9036	-0.1026	-0.82
1.265	0.9035	-0.1026	-0.83
1.270	0.9034	-0.1030	-0.83
1.275	0.9034	-0.1031	-0.83
1.280	0.9032	-0.1034	-0.83
1.285	0.9031	-0.1036	-0.83
1.290	0.9029	-0.1037	-0.83
1.295	0.9030	-0.1038	-0.83
1.300	0.9029	-0.1042	-0.83
1.305	0.9028	-0.1045	-0.83
1.310	0.9026	-0.1045	-0.83
1.315	0.9025	-0.1047	-0.83
1.320	0.9025	-0.1050	-0.83
1.325	0.9026	-0.1051	-0.83
1.330	0.9024	-0.1053	-0.83
1.335	0.9023	-0.1056	-0.83
1.340	0.9022	-0.1057	-0.83
1.345	0.9022	-0.1060	-0.83
1.350	0.9020	-0.1060	-0.84
1.355	0.9020	-0.1062	-0.84
1.360	0.9018	-0.1065	-0.84
1.365	0.9018	-0.1067	-0.84
1.370	0.9017	-0.1066	-0.84
1.375	0.9016	-0.1070	-0.84
1.380	0.9016	-0.1073	-0.84

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.385	0.9015	-0.1074	-0.84
1.390	0.9015	-0.1076	-0.84
1.395	0.9013	-0.1077	-0.84
1.400	0.9011	-0.1080	-0.84
1.405	0.9011	-0.1082	-0.84
1.410	0.9009	-0.1084	-0.84
1.415	0.9009	-0.1086	-0.84
1.420	0.9007	-0.1086	-0.85
1.425	0.9006	-0.1088	-0.85
1.430	0.9005	-0.1092	-0.85
1.435	0.9004	-0.1094	-0.85
1.440	0.9004	-0.1096	-0.85
1.445	0.9002	-0.1099	-0.85
1.450	0.9000	-0.1100	-0.85
1.455	0.8999	-0.1103	-0.85
1.460	0.8999	-0.1105	-0.85
1.465	0.8996	-0.1106	-0.85
1.470	0.8996	-0.1108	-0.85
1.475	0.8994	-0.1112	-0.86
1.480	0.8994	-0.1113	-0.86
1.485	0.8992	-0.1117	-0.86
1.490	0.8990	-0.1118	-0.86
1.495	0.8989	-0.1121	-0.86
1.500	0.8988	-0.1124	-0.86
1.505	0.8986	-0.1125	-0.86
1.510	0.8985	-0.1127	-0.86
1.515	0.8983	-0.1130	-0.86
1.520	0.8982	-0.1133	-0.86
1.525	0.8981	-0.1135	-0.86
1.530	0.8979	-0.1138	-0.87
1.535	0.8977	-0.1141	-0.87
1.540	0.8975	-0.1142	-0.87
1.545	0.8974	-0.1146	-0.87
1.550	0.8971	-0.1148	-0.87
1.555	0.8968	-0.1150	-0.88
1.560	0.8967	-0.1153	-0.88
1.565	0.8965	-0.1157	-0.88
1.570	0.8963	-0.1159	-0.88
1.575	0.8959	-0.1162	-0.88
1.580	0.8957	-0.1164	-0.88
1.585	0.8956	-0.1167	-0.88
1.590	0.8954	-0.1171	-0.89
1.595	0.8952	-0.1173	-0.89
1.600	0.8950	-0.1175	-0.89
1.605	0.8949	-0.1179	-0.89
1.610	0.8947	-0.1181	-0.89

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.615	0.8944	-0.1184	-0.89
1.620	0.8943	-0.1187	-0.89
1.625	0.8940	-0.1189	-0.90
1.630	0.8938	-0.1192	-0.90
1.635	0.8936	-0.1195	-0.90
1.640	0.8934	-0.1197	-0.90
1.645	0.8932	-0.1202	-0.90
1.650	0.8928	-0.1203	-0.91
1.655	0.8927	-0.1207	-0.91
1.660	0.8926	-0.1211	-0.91
1.665	0.8925	-0.1214	-0.91
1.670	0.8922	-0.1215	-0.91
1.675	0.8919	-0.1218	-0.91
1.680	0.8916	-0.1222	-0.92
1.685	0.8915	-0.1225	-0.92
1.690	0.8912	-0.1227	-0.92
1.695	0.8910	-0.1230	-0.92
1.700	0.8908	-0.1233	-0.92
1.705	0.8903	-0.1234	-0.93
1.710	0.8904	-0.1239	-0.93
1.715	0.8900	-0.1241	-0.93
1.720	0.8898	-0.1244	-0.93
1.725	0.8896	-0.1246	-0.93
1.730	0.8894	-0.1248	-0.93
1.735	0.8892	-0.1252	-0.93
1.740	0.8887	-0.1255	-0.94
1.745	0.8888	-0.1258	-0.94
1.750	0.8884	-0.1260	-0.94
1.755	0.8883	-0.1263	-0.94
1.760	0.8879	-0.1266	-0.95
1.765	0.8879	-0.1270	-0.94
1.770	0.8877	-0.1272	-0.95
1.775	0.8874	-0.1273	-0.95
1.780	0.8874	-0.1276	-0.95
1.785	0.8870	-0.1279	-0.95
1.790	0.8870	-0.1281	-0.95
1.795	0.8867	-0.1282	-0.95
1.800	0.8863	-0.1285	-0.96
1.805	0.8862	-0.1287	-0.96
1.810	0.8862	-0.1291	-0.96
1.815	0.8862	-0.1295	-0.96
1.820	0.8857	-0.1296	-0.96
1.825	0.8856	-0.1299	-0.96
1.830	0.8856	-0.1300	-0.96
1.835	0.8854	-0.1304	-0.96
1.840	0.8852	-0.1305	-0.97

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.845	0.8851	-0.1308	-0.97
1.850	0.8850	-0.1309	-0.97
1.855	0.8849	-0.1311	-0.97
1.860	0.8846	-0.1313	-0.97
1.865	0.8844	-0.1315	-0.97
1.870	0.8844	-0.1316	-0.97
1.875	0.8840	-0.1319	-0.98
1.880	0.8839	-0.1321	-0.98
1.885	0.8839	-0.1324	-0.98
1.890	0.8839	-0.1327	-0.98
1.895	0.8838	-0.1328	-0.98
1.900	0.8837	-0.1330	-0.98
1.905	0.8835	-0.1331	-0.98
1.910	0.8834	-0.1332	-0.98
1.915	0.8832	-0.1334	-0.98
1.920	0.8832	-0.1337	-0.98
1.925	0.8830	-0.1339	-0.98
1.930	0.8829	-0.1341	-0.98
1.935	0.8829	-0.1343	-0.98
1.940	0.8829	-0.1346	-0.98
1.945	0.8827	-0.1347	-0.98
1.950	0.8828	-0.1348	-0.98
1.955	0.8828	-0.1350	-0.98
1.960	0.8827	-0.1352	-0.98
1.965	0.8825	-0.1355	-0.98
1.970	0.8825	-0.1356	-0.98
1.975	0.8826	-0.1359	-0.98
1.980	0.8824	-0.1361	-0.99
1.985	0.8823	-0.1360	-0.99
1.990	0.8824	-0.1362	-0.98
1.995	0.8823	-0.1362	-0.99
2.000	0.8824	-0.1365	-0.98
2.005	0.8822	-0.1368	-0.99
2.010	0.8821	-0.1370	-0.99
2.015	0.8820	-0.1371	-0.99
2.020	0.8820	-0.1373	-0.99
2.025	0.8820	-0.1374	-0.99
2.030	0.8819	-0.1375	-0.99
2.035	0.8820	-0.1379	-0.99
2.040	0.8820	-0.1379	-0.99
2.045	0.8818	-0.1381	-0.99
2.050	0.8818	-0.1383	-0.99
2.055	0.8817	-0.1384	-0.99
2.060	0.8817	-0.1387	-0.99
2.065	0.8816	-0.1388	-0.99
2.070	0.8816	-0.1390	-0.99

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.075	0.8814	-0.1391	-0.99
2.080	0.8815	-0.1392	-0.99
2.085	0.8815	-0.1396	-0.99
2.090	0.8815	-0.1397	-0.99
2.095	0.8811	-0.1397	-0.99
2.100	0.8811	-0.1400	-0.99
2.105	0.8811	-0.1401	-0.99
2.110	0.8811	-0.1404	-0.99
2.115	0.8809	-0.1406	-0.99
2.120	0.8809	-0.1410	-0.99
2.125	0.8807	-0.1412	-0.99
2.130	0.8807	-0.1413	-0.99
2.135	0.8805	-0.1415	-1.00
2.140	0.8804	-0.1417	-1.00
2.145	0.8803	-0.1419	-1.00
2.150	0.8803	-0.1424	-1.00
2.155	0.8802	-0.1425	-1.00
2.160	0.8801	-0.1428	-1.00
2.165	0.8797	-0.1431	-1.00
2.170	0.8798	-0.1433	-1.00
2.175	0.8795	-0.1435	-1.00
2.180	0.8795	-0.1438	-1.00
2.185	0.8793	-0.1441	-1.00
2.190	0.8792	-0.1443	-1.00
2.195	0.8790	-0.1446	-1.00
2.200	0.8787	-0.1450	-1.01
2.205	0.8787	-0.1451	-1.01
2.210	0.8785	-0.1456	-1.01
2.215	0.8784	-0.1458	-1.01
2.220	0.8781	-0.1463	-1.01
2.225	0.8780	-0.1463	-1.01
2.230	0.8778	-0.1466	-1.01
2.235	0.8777	-0.1470	-1.01
2.240	0.8774	-0.1473	-1.02
2.245	0.8773	-0.1475	-1.02
2.250	0.8770	-0.1478	-1.02
2.255	0.8768	-0.1481	-1.02
2.260	0.8767	-0.1485	-1.02
2.265	0.8763	-0.1489	-1.02
2.270	0.8763	-0.1492	-1.02
2.275	0.8761	-0.1495	-1.02
2.280	0.8760	-0.1499	-1.02
2.285	0.8756	-0.1502	-1.03
2.290	0.8754	-0.1504	-1.03
2.295	0.8751	-0.1508	-1.03
2.300	0.8751	-0.1513	-1.03

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.305	0.8749	-0.1518	-1.03
2.310	0.8746	-0.1519	-1.03
2.315	0.8744	-0.1523	-1.04
2.320	0.8742	-0.1525	-1.04
2.325	0.8741	-0.1529	-1.04
2.330	0.8735	-0.1530	-1.04
2.335	0.8735	-0.1535	-1.04
2.340	0.8732	-0.1539	-1.05
2.345	0.8732	-0.1545	-1.04
2.350	0.8730	-0.1548	-1.05
2.355	0.8724	-0.1550	-1.05
2.360	0.8724	-0.1553	-1.05
2.365	0.8722	-0.1556	-1.05
2.370	0.8718	-0.1560	-1.05
2.375	0.8715	-0.1562	-1.06
2.380	0.8715	-0.1566	-1.06
2.385	0.8713	-0.1568	-1.06
2.390	0.8710	-0.1572	-1.06
2.395	0.8709	-0.1576	-1.06
2.400	0.8707	-0.1578	-1.06
2.405	0.8705	-0.1582	-1.06
2.410	0.8702	-0.1585	-1.07
2.415	0.8700	-0.1587	-1.07
2.420	0.8699	-0.1591	-1.07
2.425	0.8696	-0.1594	-1.07
2.430	0.8695	-0.1597	-1.07
2.435	0.8693	-0.1600	-1.07
2.440	0.8690	-0.1602	-1.07
2.445	0.8689	-0.1604	-1.07
2.450	0.8687	-0.1607	-1.08
2.455	0.8687	-0.1609	-1.08
2.460	0.8685	-0.1612	-1.08
2.465	0.8683	-0.1614	-1.08
2.470	0.8681	-0.1617	-1.08
2.475	0.8680	-0.1620	-1.08
2.480	0.8679	-0.1622	-1.08
2.485	0.8676	-0.1623	-1.08
2.490	0.8675	-0.1625	-1.09
2.495	0.8673	-0.1628	-1.09
2.500	0.8671	-0.1630	-1.09
2.505	0.8671	-0.1632	-1.09
2.510	0.8668	-0.1635	-1.09
2.515	0.8667	-0.1637	-1.09
2.520	0.8666	-0.1639	-1.09
2.525	0.8663	-0.1642	-1.09
2.530	0.8661	-0.1643	-1.10

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.535	0.8660	-0.1646	-1.10
2.540	0.8658	-0.1648	-1.10
2.545	0.8658	-0.1650	-1.10
2.550	0.8656	-0.1653	-1.10
2.555	0.8655	-0.1655	-1.10
2.560	0.8653	-0.1656	-1.10
2.565	0.8651	-0.1656	-1.10
2.570	0.8650	-0.1657	-1.10
2.575	0.8652	-0.1658	-1.10
2.580	0.8649	-0.1659	-1.10
2.585	0.8650	-0.1659	-1.10
2.590	0.8652	-0.1660	-1.10
2.595	0.8652	-0.1662	-1.10
2.600	0.8654	-0.1664	-1.10
2.605	0.8654	-0.1666	-1.10
2.610	0.8652	-0.1669	-1.10
2.615	0.8655	-0.1673	-1.10
2.620	0.8655	-0.1673	-1.10
2.625	0.8656	-0.1675	-1.09
2.630	0.8655	-0.1677	-1.09
2.635	0.8654	-0.1678	-1.10
2.640	0.8655	-0.1681	-1.09
2.645	0.8654	-0.1681	-1.10
2.650	0.8654	-0.1682	-1.09
2.655	0.8654	-0.1683	-1.09
2.660	0.8655	-0.1687	-1.09
2.665	0.8654	-0.1687	-1.09
2.670	0.8652	-0.1692	-1.09
2.675	0.8652	-0.1693	-1.09
2.680	0.8653	-0.1694	-1.09
2.685	0.8652	-0.1696	-1.09
2.690	0.8650	-0.1697	-1.10
2.695	0.8650	-0.1700	-1.10
2.700	0.8652	-0.1703	-1.09
2.705	0.8650	-0.1704	-1.09
2.710	0.8650	-0.1705	-1.09
2.715	0.8650	-0.1707	-1.09
2.720	0.8649	-0.1708	-1.09
2.725	0.8651	-0.1710	-1.09
2.730	0.8652	-0.1711	-1.09
2.735	0.8651	-0.1712	-1.09
2.740	0.8651	-0.1713	-1.09
2.745	0.8650	-0.1714	-1.09
2.750	0.8651	-0.1716	-1.09
2.755	0.8649	-0.1716	-1.09
2.760	0.8647	-0.1717	-1.09

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.765	0.8647	-0.1719	-1.09
2.770	0.8646	-0.1720	-1.10
2.775	0.8646	-0.1721	-1.09
2.780	0.8645	-0.1723	-1.10
2.785	0.8643	-0.1723	-1.10
2.790	0.8643	-0.1725	-1.10
2.795	0.8641	-0.1726	-1.10
2.800	0.8641	-0.1727	-1.10
2.805	0.8639	-0.1728	-1.10
2.810	0.8638	-0.1731	-1.10
2.815	0.8637	-0.1732	-1.10
2.820	0.8635	-0.1734	-1.10
2.825	0.8636	-0.1736	-1.10
2.830	0.8634	-0.1739	-1.10
2.835	0.8631	-0.1740	-1.11
2.840	0.8632	-0.1742	-1.10
2.845	0.8630	-0.1744	-1.11
2.850	0.8628	-0.1748	-1.11
2.855	0.8628	-0.1751	-1.11
2.860	0.8628	-0.1755	-1.11
2.865	0.8627	-0.1754	-1.11
2.870	0.8624	-0.1757	-1.11
2.875	0.8625	-0.1757	-1.11
2.880	0.8622	-0.1759	-1.11
2.885	0.8623	-0.1762	-1.11
2.890	0.8623	-0.1762	-1.11
2.895	0.8620	-0.1763	-1.11
2.900	0.8618	-0.1765	-1.11
2.905	0.8616	-0.1766	-1.11
2.910	0.8613	-0.1768	-1.12
2.915	0.8611	-0.1769	-1.12
2.920	0.8609	-0.1774	-1.12
2.925	0.8606	-0.1774	-1.12
2.930	0.8605	-0.1775	-1.12
2.935	0.8601	-0.1780	-1.13
2.940	0.8598	-0.1782	-1.13
2.945	0.8596	-0.1786	-1.13
2.950	0.8597	-0.1788	-1.13
2.955	0.8592	-0.1791	-1.13
2.960	0.8591	-0.1794	-1.13
2.965	0.8590	-0.1797	-1.13
2.970	0.8588	-0.1801	-1.14
2.975	0.8588	-0.1803	-1.13
2.980	0.8587	-0.1806	-1.14
2.985	0.8588	-0.1810	-1.13
2.990	0.8586	-0.1811	-1.14

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.995	0.8585	-0.1816	-1.13
3.000	0.8588	-0.1815	-1.13
3.005	0.8580	-0.1818	-1.14
3.010	0.8582	-0.1818	-1.14
3.015	0.8580	-0.1823	-1.14
3.020	0.8583	-0.1830	-1.13
3.025	0.8581	-0.1825	-1.14
3.030	0.8585	-0.1834	-1.13
3.035	0.8585	-0.1833	-1.13
3.040	0.8585	-0.1832	-1.13
3.045	0.8588	-0.1836	-1.13
3.050	0.8585	-0.1834	-1.13
3.055	0.8588	-0.1836	-1.13
3.060	0.8590	-0.1835	-1.13
3.065	0.8588	-0.1833	-1.13
3.070	0.8588	-0.1831	-1.13
3.075	0.8587	-0.1831	-1.13
3.080	0.8584	-0.1832	-1.13
3.085	0.8583	-0.1829	-1.13
3.090	0.8580	-0.1830	-1.14
3.095	0.8578	-0.1830	-1.14
3.100	0.8575	-0.1828	-1.14
3.105	0.8572	-0.1831	-1.14
3.110	0.8570	-0.1832	-1.15
3.115	0.8567	-0.1834	-1.15
3.120	0.8564	-0.1835	-1.15
3.125	0.8562	-0.1838	-1.15
3.130	0.8561	-0.1841	-1.15
3.135	0.8559	-0.1844	-1.15
3.140	0.8558	-0.1846	-1.16
3.145	0.8557	-0.1851	-1.15
3.150	0.8557	-0.1854	-1.15
3.155	0.8558	-0.1855	-1.15
3.160	0.8559	-0.1859	-1.15
3.165	0.8560	-0.1862	-1.15
3.170	0.8563	-0.1864	-1.15
3.175	0.8560	-0.1868	-1.15
3.180	0.8566	-0.1870	-1.14
3.185	0.8566	-0.1873	-1.14
3.190	0.8568	-0.1871	-1.14
3.195	0.8571	-0.1871	-1.14
3.200	0.8574	-0.1869	-1.13
3.205	0.8575	-0.1869	-1.13
3.210	0.8572	-0.1868	-1.14
3.215	0.8575	-0.1865	-1.13
3.220	0.8571	-0.1865	-1.14

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.225	0.8569	-0.1864	-1.14
3.230	0.8566	-0.1863	-1.14
3.235	0.8565	-0.1865	-1.14
3.240	0.8563	-0.1866	-1.15
3.245	0.8565	-0.1869	-1.14
3.250	0.8561	-0.1869	-1.15
3.255	0.8559	-0.1875	-1.15
3.260	0.8559	-0.1878	-1.15
3.265	0.8563	-0.1876	-1.14
3.270	0.8562	-0.1883	-1.14
3.275	0.8562	-0.1884	-1.14
3.280	0.8563	-0.1887	-1.14
3.285	0.8563	-0.1888	-1.14
3.290	0.8567	-0.1890	-1.14
3.295	0.8565	-0.1893	-1.14
3.300	0.8565	-0.1895	-1.14
3.305	0.8564	-0.1897	-1.14
3.310	0.8568	-0.1898	-1.13
3.315	0.8567	-0.1900	-1.14
3.320	0.8569	-0.1904	-1.13
3.325	0.8568	-0.1905	-1.13
3.330	0.8571	-0.1907	-1.13
3.335	0.8570	-0.1909	-1.13
3.340	0.8572	-0.1912	-1.13
3.345	0.8571	-0.1911	-1.13
3.350	0.8575	-0.1915	-1.12
3.355	0.8574	-0.1916	-1.12
3.360	0.8577	-0.1916	-1.12
3.365	0.8576	-0.1915	-1.12
3.370	0.8577	-0.1918	-1.12
3.375	0.8577	-0.1918	-1.12
3.380	0.8575	-0.1918	-1.12
3.385	0.8578	-0.1917	-1.12
3.390	0.8577	-0.1921	-1.12
3.395	0.8576	-0.1920	-1.12
3.400	0.8576	-0.1922	-1.12
3.405	0.8575	-0.1920	-1.12
3.410	0.8573	-0.1923	-1.12
3.415	0.8570	-0.1924	-1.13
3.420	0.8570	-0.1923	-1.13
3.425	0.8567	-0.1924	-1.13
3.430	0.8565	-0.1925	-1.13
3.435	0.8564	-0.1928	-1.13
3.440	0.8561	-0.1929	-1.13
3.445	0.8558	-0.1931	-1.14
3.450	0.8554	-0.1930	-1.14

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.455	0.8553	-0.1934	-1.14
3.460	0.8550	-0.1938	-1.14
3.465	0.8549	-0.1940	-1.14
3.470	0.8546	-0.1943	-1.15
3.475	0.8543	-0.1948	-1.15
3.480	0.8541	-0.1952	-1.15
3.485	0.8541	-0.1956	-1.15
3.490	0.8542	-0.1961	-1.15
3.495	0.8538	-0.1961	-1.15
3.500	0.8537	-0.1968	-1.15
3.505	0.8538	-0.1966	-1.15
3.510	0.8540	-0.1978	-1.14
3.515	0.8539	-0.1977	-1.15
3.520	0.8542	-0.1979	-1.14
3.525	0.8542	-0.1986	-1.14
3.530	0.8543	-0.1994	-1.14
3.535	0.8550	-0.1989	-1.13
3.540	0.8557	-0.1984	-1.13
3.545	0.8552	-0.1984	-1.13
3.550	0.8554	-0.1974	-1.13
3.555	0.8553	-0.1971	-1.13
3.560	0.8547	-0.1966	-1.14
3.565	0.8542	-0.1965	-1.14
3.570	0.8537	-0.1958	-1.15
3.575	0.8526	-0.1950	-1.16
3.580	0.8518	-0.1952	-1.17
3.585	0.8500	-0.1954	-1.19
3.590	0.8489	-0.1953	-1.20
3.595	0.8480	-0.1961	-1.21
3.600	0.8472	-0.1974	-1.21
3.605	0.8463	-0.1984	-1.22
3.610	0.8464	-0.1998	-1.21
3.615	0.8466	-0.2007	-1.21
3.620	0.8472	-0.2016	-1.20
3.625	0.8470	-0.2018	-1.20
3.630	0.8477	-0.2027	-1.19
3.635	0.8486	-0.2029	-1.18
3.640	0.8490	-0.2031	-1.18
3.645	0.8494	-0.2030	-1.18
3.650	0.8495	-0.2024	-1.18
3.655	0.8495	-0.2024	-1.18
3.660	0.8495	-0.2025	-1.18
3.665	0.8494	-0.2023	-1.18
3.670	0.8492	-0.2021	-1.18
3.675	0.8495	-0.2017	-1.18
3.680	0.8491	-0.2017	-1.18

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.685	0.8490	-0.2017	-1.18
3.690	0.8488	-0.2014	-1.19
3.695	0.8486	-0.2014	-1.19
3.700	0.8483	-0.2013	-1.19
3.705	0.8479	-0.2012	-1.20
3.710	0.8475	-0.2011	-1.20
3.715	0.8470	-0.2009	-1.20
3.720	0.8466	-0.2010	-1.21
3.725	0.8460	-0.2012	-1.21
3.730	0.8451	-0.2012	-1.22
3.735	0.8444	-0.2018	-1.23
3.740	0.8437	-0.2025	-1.23
3.745	0.8433	-0.2035	-1.23
3.750	0.8429	-0.2046	-1.24
3.755	0.8427	-0.2058	-1.23
3.760	0.8433	-0.2072	-1.23
3.765	0.8440	-0.2083	-1.22
3.770	0.8458	-0.2088	-1.20
3.775	0.8471	-0.2087	-1.19
3.780	0.8487	-0.2089	-1.17
3.785	0.8492	-0.2065	-1.17
3.790	0.8486	-0.2054	-1.18
3.795	0.8486	-0.2048	-1.18
3.800	0.8477	-0.2041	-1.19
3.805	0.8472	-0.2037	-1.20
3.810	0.8464	-0.2039	-1.20
3.815	0.8457	-0.2040	-1.21
3.820	0.8451	-0.2044	-1.22
3.825	0.8445	-0.2047	-1.22
3.830	0.8440	-0.2051	-1.22
3.835	0.8440	-0.2071	-1.22
3.840	0.8439	-0.2074	-1.22
3.845	0.8443	-0.2077	-1.21
3.850	0.8443	-0.2079	-1.21
3.855	0.8439	-0.2080	-1.22
3.860	0.8441	-0.2084	-1.22
3.865	0.8439	-0.2088	-1.22
3.870	0.8442	-0.2089	-1.21
3.875	0.8440	-0.2091	-1.21
3.880	0.8440	-0.2093	-1.21
3.885	0.8441	-0.2096	-1.21
3.890	0.8440	-0.2096	-1.21
3.895	0.8441	-0.2101	-1.21
3.900	0.8437	-0.2101	-1.21
3.905	0.8439	-0.2104	-1.21
3.910	0.8437	-0.2107	-1.21

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.915	0.8436	-0.2110	-1.21
3.920	0.8437	-0.2113	-1.21
3.925	0.8436	-0.2119	-1.21
3.930	0.8435	-0.2118	-1.21
3.935	0.8433	-0.2122	-1.21
3.940	0.8432	-0.2125	-1.21
3.945	0.8431	-0.2129	-1.21
3.950	0.8430	-0.2133	-1.21
3.955	0.8429	-0.2138	-1.21
3.960	0.8429	-0.2139	-1.21
3.965	0.8425	-0.2144	-1.22
3.970	0.8427	-0.2145	-1.21
3.975	0.8424	-0.2149	-1.22
3.980	0.8422	-0.2152	-1.22
3.985	0.8420	-0.2155	-1.22
3.990	0.8418	-0.2160	-1.22
3.995	0.8418	-0.2163	-1.22
4.000	0.8415	-0.2168	-1.22
4.005	0.8413	-0.2173	-1.22
4.010	0.8411	-0.2178	-1.22
4.015	0.8408	-0.2180	-1.22
4.020	0.8405	-0.2185	-1.22
4.025	0.8401	-0.2186	-1.23
4.030	0.8401	-0.2191	-1.23
4.035	0.8399	-0.2195	-1.23
4.040	0.8397	-0.2201	-1.23
4.045	0.8394	-0.2205	-1.23
4.050	0.8391	-0.2208	-1.23
4.055	0.8387	-0.2212	-1.24
4.060	0.8384	-0.2217	-1.24
4.065	0.8384	-0.2217	-1.24
4.070	0.8379	-0.2222	-1.24
4.075	0.8376	-0.2228	-1.24
4.080	0.8371	-0.2232	-1.25
4.085	0.8370	-0.2235	-1.25
4.090	0.8366	-0.2238	-1.25
4.095	0.8362	-0.2242	-1.25
4.100	0.8358	-0.2245	-1.26
4.105	0.8351	-0.2250	-1.26
4.110	0.8353	-0.2254	-1.26
4.115	0.8352	-0.2258	-1.26
4.120	0.8348	-0.2268	-1.26
4.125	0.8344	-0.2269	-1.26
4.130	0.8337	-0.2274	-1.27
4.135	0.8332	-0.2279	-1.27
4.140	0.8333	-0.2283	-1.27

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.145	0.8328	-0.2284	-1.27
4.150	0.8323	-0.2294	-1.28
4.155	0.8318	-0.2300	-1.28
4.160	0.8312	-0.2303	-1.28
4.165	0.8306	-0.2305	-1.29
4.170	0.8304	-0.2310	-1.29
4.175	0.8303	-0.2315	-1.29
4.180	0.8296	-0.2315	-1.30
4.185	0.8289	-0.2319	-1.30
4.190	0.8286	-0.2322	-1.30
4.195	0.8284	-0.2326	-1.31
4.200	0.8285	-0.2332	-1.30
4.205	0.8279	-0.2332	-1.31
4.210	0.8275	-0.2334	-1.31
4.215	0.8275	-0.2339	-1.31
4.220	0.8272	-0.2341	-1.31
4.225	0.8269	-0.2344	-1.31
4.230	0.8265	-0.2347	-1.32
4.235	0.8264	-0.2348	-1.32
4.240	0.8262	-0.2351	-1.32
4.245	0.8259	-0.2351	-1.32
4.250	0.8258	-0.2356	-1.32
4.255	0.8257	-0.2357	-1.32
4.260	0.8253	-0.2359	-1.33
4.265	0.8251	-0.2362	-1.33
4.270	0.8251	-0.2364	-1.33
4.275	0.8249	-0.2366	-1.33
4.280	0.8249	-0.2367	-1.33
4.285	0.8248	-0.2369	-1.33
4.290	0.8246	-0.2372	-1.33
4.295	0.8245	-0.2373	-1.33
4.300	0.8244	-0.2374	-1.33
4.305	0.8243	-0.2376	-1.33
4.310	0.8241	-0.2377	-1.33
4.315	0.8240	-0.2378	-1.33
4.320	0.8238	-0.2380	-1.34
4.325	0.8236	-0.2379	-1.34
4.330	0.8234	-0.2383	-1.34
4.335	0.8233	-0.2384	-1.34
4.340	0.8231	-0.2386	-1.34
4.345	0.8232	-0.2387	-1.34
4.350	0.8228	-0.2391	-1.34
4.355	0.8228	-0.2395	-1.34
4.360	0.8229	-0.2395	-1.34
4.365	0.8229	-0.2399	-1.34
4.370	0.8225	-0.2402	-1.34

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.375	0.8227	-0.2404	-1.34
4.380	0.8228	-0.2405	-1.34
4.385	0.8226	-0.2410	-1.34
4.390	0.8228	-0.2411	-1.34
4.395	0.8232	-0.2411	-1.33
4.400	0.8231	-0.2414	-1.33
4.405	0.8232	-0.2414	-1.33
4.410	0.8236	-0.2416	-1.33
4.415	0.8234	-0.2415	-1.33
4.420	0.8237	-0.2417	-1.33
4.425	0.8239	-0.2417	-1.32
4.430	0.8236	-0.2415	-1.33
4.435	0.8234	-0.2416	-1.33
4.440	0.8235	-0.2417	-1.33
4.445	0.8233	-0.2413	-1.33
4.450	0.8233	-0.2417	-1.33
4.455	0.8233	-0.2418	-1.33
4.460	0.8232	-0.2417	-1.33
4.465	0.8230	-0.2416	-1.33
4.470	0.8229	-0.2414	-1.33
4.475	0.8234	-0.2413	-1.33
4.480	0.8233	-0.2415	-1.33
4.485	0.8234	-0.2416	-1.33
4.490	0.8239	-0.2420	-1.32
4.495	0.8241	-0.2420	-1.32
4.500	0.8246	-0.2422	-1.32
4.505	0.8247	-0.2426	-1.31
4.510	0.8253	-0.2426	-1.31
4.515	0.8256	-0.2429	-1.30
4.520	0.8260	-0.2430	-1.30
4.525	0.8262	-0.2429	-1.30
4.530	0.8265	-0.2430	-1.30
4.535	0.8265	-0.2431	-1.29
4.540	0.8265	-0.2429	-1.30
4.545	0.8266	-0.2432	-1.29
4.550	0.8266	-0.2431	-1.29
4.555	0.8268	-0.2431	-1.29
4.560	0.8268	-0.2433	-1.29
4.565	0.8269	-0.2434	-1.29
4.570	0.8272	-0.2434	-1.29
4.575	0.8270	-0.2433	-1.29
4.580	0.8272	-0.2434	-1.29
4.585	0.8272	-0.2435	-1.29
4.590	0.8271	-0.2436	-1.29
4.595	0.8272	-0.2434	-1.29
4.600	0.8271	-0.2436	-1.29

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.605	0.8271	-0.2437	-1.29
4.610	0.8270	-0.2439	-1.29
4.615	0.8269	-0.2439	-1.29
4.620	0.8266	-0.2443	-1.29
4.625	0.8265	-0.2441	-1.29
4.630	0.8265	-0.2445	-1.29
4.635	0.8263	-0.2447	-1.29
4.640	0.8265	-0.2448	-1.29
4.645	0.8261	-0.2451	-1.29
4.650	0.8263	-0.2453	-1.29
4.655	0.8257	-0.2455	-1.30
4.660	0.8261	-0.2456	-1.29
4.665	0.8261	-0.2457	-1.29
4.670	0.8255	-0.2456	-1.30
4.675	0.8256	-0.2459	-1.30
4.680	0.8255	-0.2460	-1.30
4.685	0.8254	-0.2460	-1.30
4.690	0.8255	-0.2465	-1.30
4.695	0.8256	-0.2465	-1.29
4.700	0.8253	-0.2466	-1.30
4.705	0.8250	-0.2467	-1.30
4.710	0.8254	-0.2467	-1.30
4.715	0.8250	-0.2469	-1.30
4.720	0.8249	-0.2473	-1.30
4.725	0.8248	-0.2472	-1.30
4.730	0.8253	-0.2471	-1.29
4.735	0.8251	-0.2467	-1.30
4.740	0.8245	-0.2473	-1.30
4.745	0.8239	-0.2469	-1.31
4.750	0.8233	-0.2472	-1.31
4.755	0.8235	-0.2477	-1.31
4.760	0.8231	-0.2477	-1.31
4.765	0.8224	-0.2483	-1.32
4.770	0.8225	-0.2482	-1.32
4.775	0.8217	-0.2485	-1.33
4.780	0.8216	-0.2491	-1.32
4.785	0.8216	-0.2494	-1.32
4.790	0.8211	-0.2500	-1.33
4.795	0.8211	-0.2499	-1.33
4.800	0.8208	-0.2499	-1.33
4.805	0.8208	-0.2505	-1.33
4.810	0.8209	-0.2499	-1.33
4.815	0.8206	-0.2499	-1.33
4.820	0.8204	-0.2499	-1.33
4.825	0.8207	-0.2500	-1.33
4.830	0.8203	-0.2499	-1.34

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.835	0.8200	-0.2495	-1.34
4.840	0.8202	-0.2496	-1.34
4.845	0.8198	-0.2496	-1.34
4.850	0.8199	-0.2494	-1.34
4.855	0.8201	-0.2490	-1.34
4.860	0.8196	-0.2491	-1.34
4.865	0.8193	-0.2493	-1.35
4.870	0.8195	-0.2494	-1.34
4.875	0.8195	-0.2491	-1.35
4.880	0.8193	-0.2492	-1.35
4.885	0.8193	-0.2490	-1.35
4.890	0.8193	-0.2493	-1.35
4.895	0.8191	-0.2494	-1.35
4.900	0.8191	-0.2495	-1.35
4.905	0.8190	-0.2495	-1.35
4.910	0.8191	-0.2495	-1.35
4.915	0.8191	-0.2497	-1.35
4.920	0.8192	-0.2498	-1.35
4.925	0.8192	-0.2500	-1.35
4.930	0.8193	-0.2500	-1.34
4.935	0.8193	-0.2503	-1.34
4.940	0.8193	-0.2504	-1.34
4.945	0.8194	-0.2505	-1.34
4.950	0.8196	-0.2508	-1.34
4.955	0.8196	-0.2509	-1.34
4.960	0.8196	-0.2512	-1.34
4.965	0.8198	-0.2514	-1.34
4.970	0.8198	-0.2516	-1.34
4.975	0.8197	-0.2519	-1.33
4.980	0.8200	-0.2521	-1.33
4.985	0.8199	-0.2524	-1.33
4.990	0.8201	-0.2528	-1.33
4.995	0.8198	-0.2530	-1.33
5.000	0.8201	-0.2533	-1.33
5.005	0.8199	-0.2538	-1.33
5.010	0.8200	-0.2538	-1.33
5.015	0.8200	-0.2540	-1.33
5.020	0.8201	-0.2543	-1.32
5.025	0.8203	-0.2549	-1.32
5.030	0.8202	-0.2547	-1.32
5.035	0.8203	-0.2552	-1.32
5.040	0.8204	-0.2555	-1.32
5.045	0.8206	-0.2554	-1.32
5.050	0.8204	-0.2559	-1.32
5.055	0.8200	-0.2560	-1.32
5.060	0.8202	-0.2563	-1.32

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.065	0.8201	-0.2563	-1.32
5.070	0.8200	-0.2561	-1.32
5.075	0.8197	-0.2561	-1.32
5.080	0.8195	-0.2558	-1.33
5.085	0.8195	-0.2555	-1.33
5.090	0.8196	-0.2551	-1.33
5.095	0.8195	-0.2549	-1.33
5.100	0.8196	-0.2546	-1.33
5.105	0.8200	-0.2541	-1.33
5.110	0.8199	-0.2540	-1.33
5.115	0.8200	-0.2540	-1.33
5.120	0.8204	-0.2538	-1.32
5.125	0.8206	-0.2538	-1.32
5.130	0.8209	-0.2538	-1.32
5.135	0.8210	-0.2539	-1.32
5.140	0.8214	-0.2538	-1.31
5.145	0.8215	-0.2540	-1.31
5.150	0.8220	-0.2542	-1.31
5.155	0.8222	-0.2544	-1.30
5.160	0.8224	-0.2547	-1.30
5.165	0.8226	-0.2548	-1.30
5.170	0.8227	-0.2550	-1.30
5.175	0.8227	-0.2550	-1.30
5.180	0.8229	-0.2554	-1.29
5.185	0.8228	-0.2556	-1.29
5.190	0.8227	-0.2558	-1.29
5.195	0.8227	-0.2559	-1.29
5.200	0.8226	-0.2562	-1.29
5.205	0.8227	-0.2564	-1.29
5.210	0.8225	-0.2566	-1.29
5.215	0.8226	-0.2569	-1.29
5.220	0.8222	-0.2572	-1.30
5.225	0.8221	-0.2573	-1.30
5.230	0.8221	-0.2576	-1.30
5.235	0.8218	-0.2578	-1.30
5.240	0.8218	-0.2579	-1.30
5.245	0.8215	-0.2582	-1.30
5.250	0.8212	-0.2583	-1.30
5.255	0.8208	-0.2585	-1.30
5.260	0.8206	-0.2587	-1.31
5.265	0.8201	-0.2588	-1.31
5.270	0.8201	-0.2591	-1.31
5.275	0.8197	-0.2591	-1.31
5.280	0.8191	-0.2597	-1.32
5.285	0.8186	-0.2595	-1.32
5.290	0.8185	-0.2599	-1.32

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.295	0.8181	-0.2601	-1.33
5.300	0.8178	-0.2605	-1.33
5.305	0.8177	-0.2613	-1.33
5.310	0.8171	-0.2612	-1.33
5.315	0.8168	-0.2616	-1.33
5.320	0.8169	-0.2615	-1.33
5.325	0.8162	-0.2622	-1.34
5.330	0.8163	-0.2624	-1.34
5.335	0.8158	-0.2632	-1.34
5.340	0.8155	-0.2630	-1.34
5.345	0.8151	-0.2630	-1.35
5.350	0.8152	-0.2636	-1.34
5.355	0.8148	-0.2636	-1.35
5.360	0.8149	-0.2639	-1.34
5.365	0.8143	-0.2643	-1.35
5.370	0.8138	-0.2640	-1.35
5.375	0.8135	-0.2643	-1.36
5.380	0.8131	-0.2644	-1.36
5.385	0.8121	-0.2652	-1.37
5.390	0.8124	-0.2651	-1.37
5.395	0.8119	-0.2651	-1.37
5.400	0.8112	-0.2655	-1.38
5.405	0.8112	-0.2661	-1.37
5.410	0.8105	-0.2662	-1.38
5.415	0.8099	-0.2665	-1.38
5.420	0.8097	-0.2670	-1.39
5.425	0.8094	-0.2675	-1.39
5.430	0.8087	-0.2677	-1.39
5.435	0.8085	-0.2678	-1.39
5.440	0.8081	-0.2684	-1.40
5.445	0.8077	-0.2685	-1.40
5.450	0.8077	-0.2689	-1.40
5.455	0.8074	-0.2690	-1.40
5.460	0.8072	-0.2697	-1.40
5.465	0.8068	-0.2696	-1.40
5.470	0.8069	-0.2698	-1.40
5.475	0.8064	-0.2703	-1.41
5.480	0.8062	-0.2700	-1.41
5.485	0.8060	-0.2703	-1.41
5.490	0.8054	-0.2704	-1.42
5.495	0.8054	-0.2707	-1.41
5.500	0.8047	-0.2707	-1.42
5.505	0.8043	-0.2709	-1.43
5.510	0.8039	-0.2713	-1.43
5.515	0.8035	-0.2717	-1.43
5.520	0.8032	-0.2718	-1.43

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.525	0.8031	-0.2721	-1.43
5.530	0.8030	-0.2724	-1.43
5.535	0.8027	-0.2728	-1.43
5.540	0.8024	-0.2730	-1.44
5.545	0.8023	-0.2732	-1.44
5.550	0.8020	-0.2735	-1.44
5.555	0.8018	-0.2736	-1.44
5.560	0.8017	-0.2739	-1.44
5.565	0.8016	-0.2740	-1.44
5.570	0.8015	-0.2742	-1.44
5.575	0.8013	-0.2744	-1.44
5.580	0.8010	-0.2747	-1.45
5.585	0.8007	-0.2748	-1.45
5.590	0.8003	-0.2749	-1.45
5.595	0.8001	-0.2752	-1.45
5.600	0.7998	-0.2755	-1.45
5.605	0.7995	-0.2757	-1.46
5.610	0.7991	-0.2763	-1.46
5.615	0.7992	-0.2767	-1.45
5.620	0.7986	-0.2771	-1.46
5.625	0.7987	-0.2777	-1.46
5.630	0.7984	-0.2781	-1.46
5.635	0.7984	-0.2785	-1.46
5.640	0.7986	-0.2790	-1.45
5.645	0.7984	-0.2790	-1.46
5.650	0.7984	-0.2796	-1.45
5.655	0.7983	-0.2800	-1.45
5.660	0.7982	-0.2803	-1.45
5.665	0.7981	-0.2806	-1.45
5.670	0.7982	-0.2812	-1.45
5.675	0.7981	-0.2815	-1.45
5.680	0.7979	-0.2823	-1.45
5.685	0.7979	-0.2825	-1.45
5.690	0.7977	-0.2828	-1.45
5.695	0.7979	-0.2833	-1.45
5.700	0.7981	-0.2838	-1.44
5.705	0.7980	-0.2842	-1.44
5.710	0.7980	-0.2845	-1.44
5.715	0.7978	-0.2848	-1.44
5.720	0.7977	-0.2852	-1.44
5.725	0.7979	-0.2854	-1.44
5.730	0.7979	-0.2858	-1.44
5.735	0.7982	-0.2860	-1.43
5.740	0.7981	-0.2867	-1.43
5.745	0.7983	-0.2871	-1.43
5.750	0.7983	-0.2874	-1.43

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.755	0.7982	-0.2876	-1.43
5.760	0.7983	-0.2880	-1.42
5.765	0.7985	-0.2883	-1.42
5.770	0.7983	-0.2884	-1.42
5.775	0.7984	-0.2887	-1.42
5.780	0.7985	-0.2890	-1.42
5.785	0.7984	-0.2894	-1.42
5.790	0.7985	-0.2895	-1.42
5.795	0.7984	-0.2898	-1.42
5.800	0.7984	-0.2899	-1.42
5.805	0.7982	-0.2902	-1.42
5.810	0.7983	-0.2904	-1.42
5.815	0.7983	-0.2906	-1.42
5.820	0.7981	-0.2907	-1.42
5.825	0.7977	-0.2910	-1.42
5.830	0.7976	-0.2911	-1.42
5.835	0.7974	-0.2913	-1.42
5.840	0.7972	-0.2914	-1.42
5.845	0.7970	-0.2916	-1.43
5.850	0.7968	-0.2919	-1.43
5.855	0.7965	-0.2921	-1.43
5.860	0.7962	-0.2921	-1.43
5.865	0.7957	-0.2925	-1.43
5.870	0.7957	-0.2926	-1.43
5.875	0.7951	-0.2929	-1.44
5.880	0.7949	-0.2931	-1.44
5.885	0.7944	-0.2934	-1.44
5.890	0.7942	-0.2938	-1.44
5.895	0.7938	-0.2937	-1.45
5.900	0.7933	-0.2944	-1.45
5.905	0.7935	-0.2946	-1.45
5.910	0.7926	-0.2949	-1.46
5.915	0.7924	-0.2950	-1.46
5.920	0.7920	-0.2953	-1.46
5.925	0.7919	-0.2958	-1.46
5.930	0.7917	-0.2957	-1.46
5.935	0.7912	-0.2961	-1.46
5.940	0.7906	-0.2958	-1.47
5.945	0.7905	-0.2962	-1.47
5.950	0.7897	-0.2961	-1.48
5.955	0.7893	-0.2962	-1.48
5.960	0.7893	-0.2961	-1.48
5.965	0.7886	-0.2963	-1.49
5.970	0.7874	-0.2970	-1.50
5.975	0.7869	-0.2972	-1.50
5.980	0.7862	-0.2971	-1.51

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.985	0.7857	-0.2972	-1.51
5.990	0.7849	-0.2977	-1.52
5.995	0.7847	-0.2978	-1.52
6.000	0.7835	-0.2989	-1.53